

Review

A Review of Sustainability in Hot Asphalt Production: Greenhouse Gas Emissions and Energy Consumption

Yancheng Liu ^{1,2,*} , Zhengyi Liu ¹ , Youwei Zhu ² and Haitao Zhang ¹ 

¹ School of Civil Engineering and Transportation, Northeast Forestry University, Harbin 150040, China; liuzhengyi@nefu.edu.cn (Z.L.); zht6781@163.com (H.Z.)

² Longjian Road & Bridge Co., Ltd., Harbin 150040, China; zhuyouwei2024@163.com

* Correspondence: yanchengliu@nefu.edu.cn

Abstract: This study conducted a thorough analysis of energy consumption and greenhouse gas (GHG) emissions in the production of hot mix asphalt. The primary sources of energy usage in asphalt mixes are the heating of aggregates, asphalt, and burners, with aggregate heating accounting for a remarkable 97% of the total energy consumption. The results indicate that low-temperature asphalt mixes offer significant benefits over conventional hot mix asphalt in terms of energy efficiency and GHG emissions, with reductions in carbon dioxide emissions ranging from 18% to 36% and energy savings between 15% and 87%. Additionally, the use of recycled asphalt pavement led to a 12% decrease in carbon dioxide emissions and a 15% reduction in energy consumption. The study further explored the effects of various fuel types on emissions, revealing that replacing fuel oil with natural gas can effectively diminish the carbon footprint of the production process. By optimizing production temperatures and selecting cleaner fuel alternatives, this research highlights the potential for considerable energy savings and emission reductions within the asphalt production sector. These strategies not only promote sustainable road construction practices but also play a vital role in environmental protection and climate change mitigation, advocating for the adoption of innovative technologies in asphalt pavement production.

Keywords: hot mix asphalt; energy consumption; clean energy; recycled material; low-carbon pathway



Citation: Liu, Y.; Liu, Z.; Zhu, Y.; Zhang, H. A Review of Sustainability in Hot Asphalt Production: Greenhouse Gas Emissions and Energy Consumption. *Appl. Sci.* **2024**, *14*, 10246. <https://doi.org/10.3390/app142210246>

Academic Editors: Giuseppe Lacidogna and Paulo Santos

Received: 9 September 2024

Revised: 26 October 2024

Accepted: 3 November 2024

Published: 7 November 2024



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1. Introduction

The increasing volume of vehicular traffic on highways inherently leads to heavier representative vehicles, thereby placing a growing burden on highway bridges. It is imperative to construct high-quality pavements that can accommodate the demands of increasingly heavy traffic. A roadway consists of four distinct layers: the road base, subgrade, intermediate layer, and pavement. The pavement primarily comprises cement concrete, asphalt concrete, asphalt-crushed stone, and gravel. The type and thickness of the pavement materials are determined based on the anticipated traffic density [1–3]. Asphalt concrete pavements represent approximately 90% of the global total, while cement concrete pavements account for around 10% [4]. Figure 1 illustrates the global production of asphalt mixtures [5,6].

Asphalt pavement is composed of asphalt, which is derived from petroleum, and bonded together with aggregate. This product offers numerous advantages, including reduced fuel consumption, low noise levels, excellent skid resistance, and minimal vehicle wear [7–9]. Moreover, it can be adapted to meet varying performances and environmental specifications. The production of asphalt concrete requires significant quantities of high-quality stone and petroleum asphalt, along with substantial amounts of diesel, heavy oil, and other fuels. Furthermore, the high-temperature production and construction processes result in the release of certain harmful gases [10–12]. The introduction of more stringent environmental regulations has led to the closure of numerous stone mines, thereby reducing

the availability of high-quality stone. Additionally, the cost of road construction has risen considerably due to the increasing price of asphalt and the escalating expenses associated with road development. The rise in road mileage and density has led to a corresponding increase in environmental pollution. Consequently, there is a pressing need to reduce energy consumption and greenhouse gas emissions [13,14].

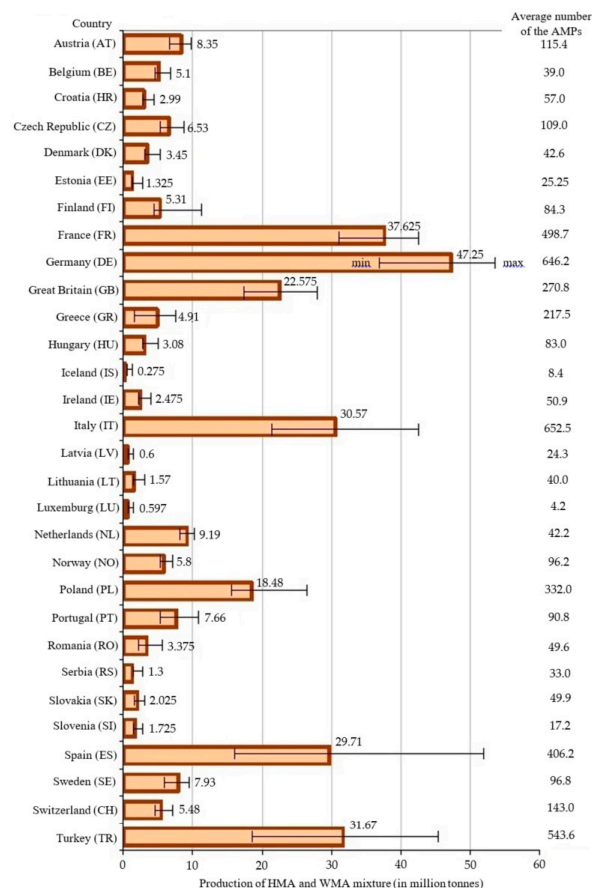


Figure 1. Annual production of asphalt mixtures [5,6].

The production of asphalt mixtures necessitates a considerable input of heat, which is derived from non-renewable energy sources. A reduction in energy consumption will result in a concomitant decrease in carbon emissions. This is one of the principal objectives of energy conservation and emission reduction. It has been established that between 60% and 85% of carbon emissions generated during the construction phase originate from earthworks activities [15,16]. Thus, advancements in technology in this area are crucial for ensuring the sustainable development of road infrastructure.

Although existing studies have focused on the environmental impacts of asphalt mixtures, there is still a lack of systematic analysis regarding the various factors in the production process, particularly concerning the specific effects of different heating technologies and fuel types on energy consumption and carbon emissions. This paper aims to fill this research gap by thoroughly investigating how these factors influence the production efficiency and environmental sustainability of asphalt mixtures. The study analyzed the energy consumption and carbon emissions associated with asphalt mixture production, particularly in the context of hot mix asphalt production. Furthermore, the effects of aggregate moisture content, combustion type, and energy source on energy consumption and carbon emissions were evaluated. The energy consumption and carbon emissions of asphalt mixes were assessed for different heating technologies, green raw materials, and heating energy types. The primary objective of this study was to evaluate and analyze the main sources of energy consumption and greenhouse gas emissions in order to identify effective measures

for reducing environmental impact. The specific goals were as follows: (i) assessing energy consumption: analyzing the energy demands of different asphalt mixture production technologies (such as hot mix and warm mix) to identify high-energy consumption stages; (ii) analyzing environmental impact: reviewing the resource utilization, emissions, and potential environmental impacts associated with the production of asphalt mixtures, and emphasizing the importance of effective management and technological innovation in achieving energy conservation and sustainable development; (iii) proposing improvement measures: investigating how to reduce carbon emissions through the enhancement of raw materials, heating technologies, and the use of clean energy sources (such as substituting natural gas for fuel oil), with the aim of identifying construction materials that minimize their impact on the human environment, thereby promoting human health and contributing to the green development of asphalt concrete. This research holds significant importance in the field of road engineering.

The following outlines the structure of this paper. Section 2 introduces the research approach and methods employed in this study. Section 3 provides foundational knowledge on road types, emphasizing the importance of assessing the energy consumption and greenhouse gas emissions associated with asphalt road surfaces. Next, Section 4 clarifies the impact of temperature on the energy consumption and gas emissions of asphalt mixtures. In Section 5, a detailed overview of the energy requirements and gas emissions during the different production processes of asphalt mixtures is presented. Section 6 reviews the energy consumption of asphalt and highlights the necessity of implementing measures to reduce the carbon footprint for sustainable development. Subsequently, Section 7 discusses low-carbon green development of road asphalt from the perspectives of recycled materials (recycled asphalt, recycled shredded rubber, and biochar) and clean energy. Finally, Section 8 summarizes the main conclusions of this investigation and proposes recommendations for future research directions.

2. Methodology

This study employed a research methodology comprising three parts: (i) a comprehensive review of asphalt mixtures from the perspectives of energy consumption and gas emissions; (ii) an exploration of low-carbon development technologies; and (iii) the conclusions of this research.

First, we defined the research focus as the energy consumption and gas emissions of asphalt mixtures, categorizing it into five areas: pavement types, temperature effects, production processes, energy consumption, and low-carbon development. This framework ensures the relevance and systematic nature of the literature review. We systematically reviewed the existing literature using the Web of Science and Scopus databases, conducting an initial analysis of high-quality studies published between 2010 and 2024 related to the energy consumption and gas emissions of asphalt pavements. The keywords employed in the search included “energy consumption” + “asphalt”, “gas emissions” + “asphalt”, “energy consumption” + “pavement”, and “green development” + “asphalt”. This process was supplemented by manual searches. The selection strategy identified 103 high-quality articles (i.e., peer-reviewed indexed research articles). These works enabled the authors to critically examine the latest technologies to address the research questions. Ultimately, five sections were established to analyze and categorize the energy consumption and greenhouse gas emission characteristics of asphalt mixtures. Through a detailed analysis of each research aspect, key insights were derived, as outlined below:

- Pavement types: analyzing the impact of different pavement types on energy consumption and emissions, highlighting their significance.
- Temperature effects: investigating the influence of temperature on the performance and production efficiency of asphalt mixtures, noting the energy consumption and emissions issues associated with high-temperature production.

- Production processes: providing a detailed description of the production process of asphalt mixtures, emphasizing the energy consumption and emission characteristics at each stage.
- Energy consumption: evaluating the overall energy consumption of asphalt mixtures, identifying high-energy consumption segments, and exploring methods to reduce energy usage.

Additionally, we assessed green technologies and material recycling in the context of recycled materials and clean energy, focusing on their roles in reducing carbon emissions and achieving sustainable development. Finally, we distilled conclusions and recommendations that summarize the main findings across all aspects, emphasizing the importance of asphalt mixtures in environmental impact assessments and proposing future research directions that encourage the exploration of new technologies and materials (see also Figure 2).

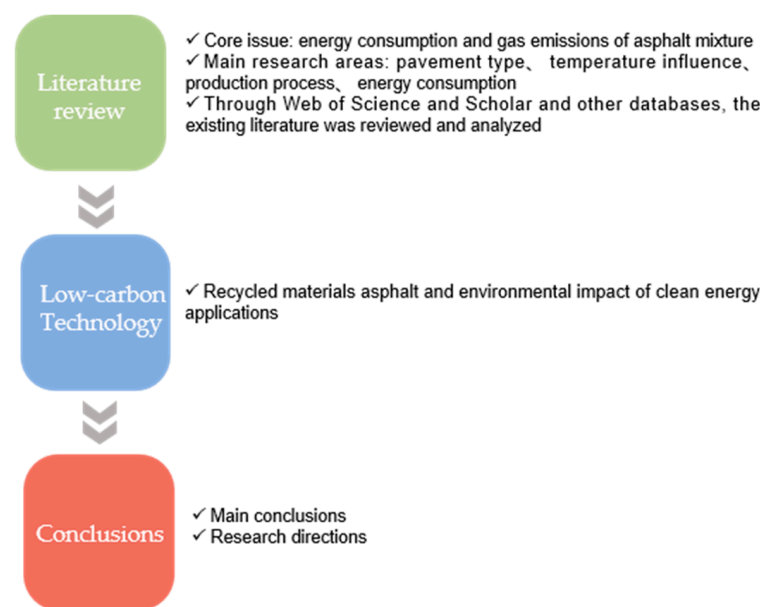


Figure 2. Research methodology.

3. Road Pavement Types

A road is defined as a layered structure comprising a range of road construction materials, laid on top of a road base layer for vehicular traffic. The functions of a road include the carriage of vehicle loads, the resistance of wheel friction, and the maintenance of continuity of the road surface. Accordingly, the pavement must possess elevated tensile strength, optimal structural stability, superior leveling, adequate anti-slip properties, and minimal dust generation during vehicle traversal [17–20]. The objective is to minimize damage to the pavement and vehicle components, ensure clear visibility, and reduce environmental pollution. The mechanical properties of the pavement determine its classification as rigid, flexible, or semi-rigid. The flexural strength of rigid pavements, such as those constructed from Portland cement concrete, is high due to the plate action that occurs under vehicle loads. In contrast, flexible pavements have low flexural strength and rely mainly on compressive and shear strengths to withstand vehicle loads. Repeated loads result in residual deformation, as is the case with asphalt and gravel pavements [21,22]. As an intermediate type between rigid and flexible pavements, semi-rigid pavements combine the advantages of both, exhibiting good flexural strength while its structural design allows it to withstand a certain degree of deformation. Semi-rigid pavements typically utilize cement-stabilized materials as a base layer, enhancing their load-bearing capacity and reducing the damage caused by repeated loading to some extent. Consequently, semi-rigid pavements provide

more stable and durable performance under varying traffic conditions, making them a significant choice in modern road engineering [23–25].

As shown in Figure 3, the asphalt concrete pavement structure comprises multiple layers, including a hot mix asphalt surface layer, a pavement base, a sub-base, and a road base [26,27]. The surface layer serves as a structural component, providing resilience against repeated wheel loads and the effects of natural elements. It may consist of one to three sub-layers. In accordance with usage requirements, the surface layer should feature a non-slip, wear-resistant, dense, and stable asphalt layer. Hot mix asphalt is a combination of various aggregates and asphalt cement, created by heating asphalt to high temperatures. The grading of the particles used in the mix determines the classification of hot mix asphalt into four types: dense-graded, semi-open-graded, open-graded, and intermittent-graded. Rigid pavements are constructed using hydraulic cement concrete, which comprises cement (specifically Portland cement and supplementary cementitious materials such as fly ash and ground granulated blast furnace slag), sand, stone, water, and admixtures [28]. Unlike asphalt mixtures, Portland cement mixtures do not require the aggregates to be dried before mixing, as additional moisture can be incorporated into the mixture design [29]. Consequently, the energy consumed in the production of Portland cement mixes, aside from the energy required for the production of raw materials, is primarily utilized for the transportation of these materials, with a minor proportion allocated for the operation of the aggregate crushing plant. In contrast, the production and transportation of asphalt mixes necessitate an evaluation of energy consumption and greenhouse gas emissions. A significant number of researchers have investigated the high performance and durability of flexible pavements while emphasizing their implications for green sustainability. They have highlighted the importance of utilizing recycled or waste materials as aggregates or components wherever feasible, aiming to reduce oil consumption [8,30–32].

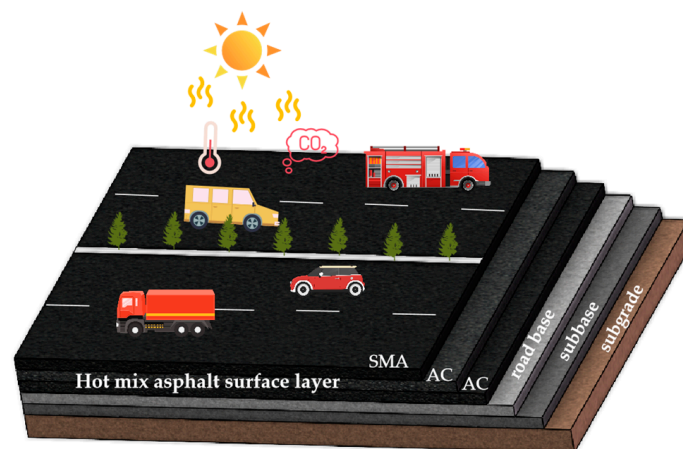


Figure 3. The asphalt concrete pavement structure.

4. Effect of Temperature on Asphalt Mixtures

Asphalt mixtures are composite materials, commonly referred to as hot mix asphalt or conventional mix asphalt. These mixtures primarily consist of aggregates (both sand and coarse aggregates), fillers, and asphalt cement [33]. In some cases, additives such as binder mobilisers [34], modifiers [35,36] or fibers [37,38] are added to improve the product properties. Asphalt mixtures can be classified based on several criteria, including production temperature, mineral composition, and the incorporation of recycled materials. They are widely used in the construction and maintenance of highways, urban roads, airports, and parking lots. Figure 4 illustrates the categorization of hot mix asphalt according to its production temperature [39]. Cold mix asphalt is produced at temperatures below 60 °C, while semi-temperature mixing is typically conducted between 70 °C and 95 °C under conditions below 100 °C. Warm mix asphalt is produced at temperatures ranging from

110 °C to 140 °C, whereas hot mix asphalt is produced at temperatures between 150 °C and 170 °C [40].

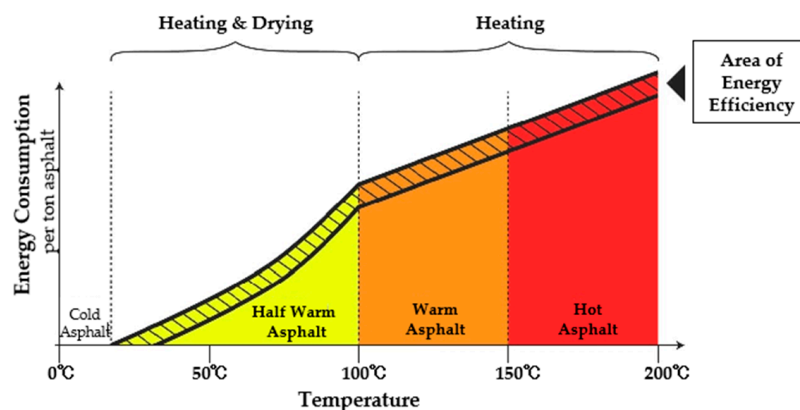


Figure 4. Classification by temperature range and potential energy savings [39].

It is common practice in hot mix asphalt production and paving operations to emit complex hydrocarbon aerosols, vapors, and gaseous mixtures consisting of combustion products, volatile organic compounds (VOCs), and polycyclic aromatic hydrocarbons (PAHs) into the atmosphere [13,41–43]. This occurs during all phases of the production and paving process. The use of low-temperature blended asphalt represents a novel and sustainable technology that significantly decreases production and application temperatures. This approach encompasses a range of blended asphalts, including cold mix, semi-temperature mix, and warm mix asphalt. The innovative asphalt production plant effectively reduces harmful emissions and fumes, thereby improving working conditions for those employed at the plant, including operators and other personnel [44,45].

Autelitano et al. [46] discovered that incorporating organic waxes and other additives into warm mix asphalt mixtures utilized in road construction could potentially reduce the required production temperature by approximately 30 °C, thus minimizing asphalt emissions. This outcome is particularly significant when temperatures approach or exceed the melting point of waxes (110 °C). A 10 °C reduction in asphalt production temperature has been found to result in a 1 L reduction in fuel consumption and a 1 kg/ton reduction in equivalent CO₂ emissions, according to data from the World Bank [47]. The use of warm mix asphalt mixtures enhances processability and extends transportation distances, with these benefits being in addition to those associated with lower production temperatures. Furthermore, the reduction in energy consumption that results from lower production temperatures impacts overall production costs [48,49].

Using reclaimed asphalt is another effective strategy for reducing energy consumption and pollutant emissions [50]. The term “Reclaimed Asphalt Pavement (RAP)” refers to the product of a pavement maintenance program or the full-depth demolition and rehabilitation of a pavement [51]. In recent years, RAP has been recognized as an important source of recycled material for pavement construction. Recycled asphalt mixes and aggregates are supplied to asphalt production plants for reuse. This practice has the dual benefit of reducing energy consumption in the production process and limiting the extraction of virgin aggregate and waste. Among the numerous RAP hybrid technologies, the parallel drum dryer has demonstrated superior environmental compatibility and energy efficiency, establishing it as a leading contender in this field. The primary factors influencing fuel consumption are the temperature and moisture content of the RAP mixture [52,53].

5. Asphalt Mixture Production

The primary function of an asphalt mixing plant is to produce substantial quantities of asphalt, which is then utilized in the construction of motorways, local roads, and other road projects. The principal components of an asphalt mixing plant include a batching

system, which adjusts the proportion of raw materials to meet the specifications of the construction project; a drying system, which eliminates moisture from the raw materials; a vibrating screen, which pulverizes large particles of the raw materials; a weighing and mixing system, which ensures the accuracy of the production process; and a dust collection system, which minimizes the emission of pollutants during production. The categorization of mixing plants depends on their production capacity, with classifications including small, medium, and large; portable, semi-permanent, and permanent facilities [54].

Figure 5 illustrates the asphalt mixing process flow with emission sources. Prior to incorporation into the mixer, the asphalt mix mineral powder is stored, transported, de-dusted on two occasions, and metered with precision. Aggregates are moved from the warehouse to the drying room, where they are heated to a temperature range of 150–170 °C. They are then screened, stored, and accurately dosed before entering the mixing system. Following the melting, dewatering, and elevation of the asphalt to the requisite production standards, the material undergoes a heating process, is held at a constant temperature, and is measured with great precision. It is then introduced into the mixing system. The selection of suitable additives follows; these additives are stored, transported, and measured accurately before being added to the mixing system. The production team then initiates the mixing process, and once the internal composition of the mixing system has been verified, the final asphalt product is transported to the finished silo for use [55].

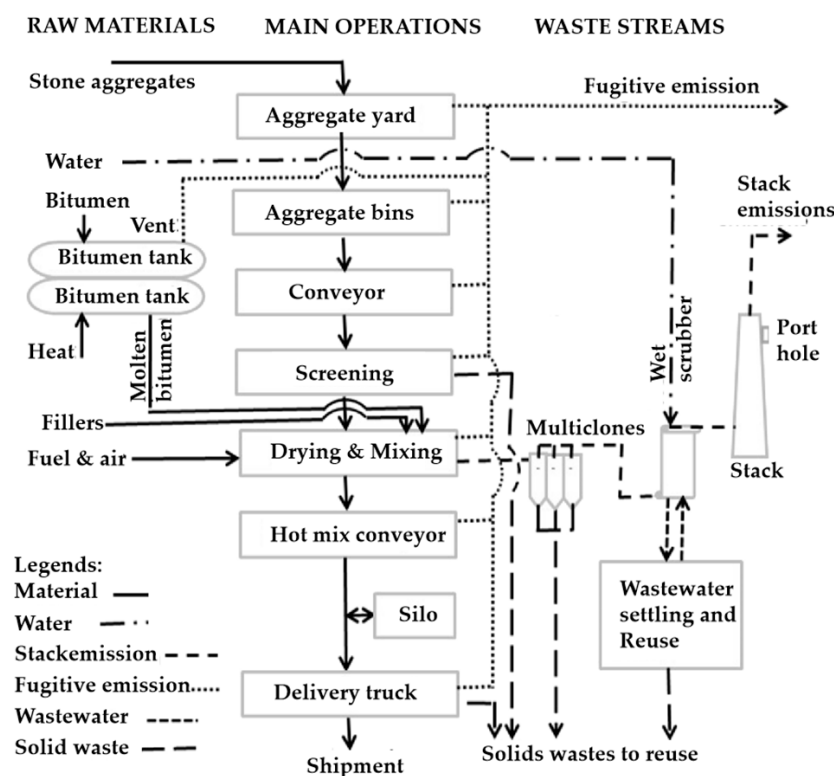


Figure 5. Emissions from hot mix asphalt plants and their effect on ambient air quality [55].

The fuel used for heating and drying materials in the production process is the primary source of emissions, accounting for 80% of total CO₂ emissions, while electricity contributes the remaining 20%. The vast majority (97%) of thermal energy is utilized for heating and drying aggregates. Therefore, managing the moisture content of these aggregates is critical for energy conservation. At a moisture content of 4%, the energy required for drying aggregates increases by 60% [56]. Furthermore, the removal of 1% moisture from a mineral mixture necessitates an input of 7.34 kWh of thermal energy [33], as illustrated in Table 1.

Table 1. Heating and drying of materials in the hot mix asphalt production process as studied in previous research.

Reference	Year and Area of Analysis	Result
Androjić and Kaluđer [57]	2016 Effect of variable insulation thickness on the use of solar panels.	Solar aggregate piles preheat the mineral mix and reduce moisture content, reducing the energy required to produce warm mix asphalt.
Androjić et al. [6]	2017 Use of the stockpiles of solar aggregates—influence of the variable surface area of the stockpiles of solar aggregates exposed to the sun.	Influence of the variable solar surface area of a solar pile on the heat accumulation results of mineral mixtures in different seasons (summer–autumn). For every 10 °C increase in temperature, a potential energy saving of 3.7 kWh can be calculated.
Androjić et al. [58]	2019 Utilization of solar aggregate stockpiles—storage of recycled materials.	Influence of time spent in sunlight, type of material stored, method of storage, and time stored.
Androjić et al. [39]	2019 Predicting the temperature of stored materials—using solar thermal storage.	Temperature prediction of stored materials using solar aggregate reserve storage. Lower temperatures were predicted for mixtures stored at 0 °C air temperature and higher values at 30 °C air temperature.
Androjić et al. [33]	2019 The energy consumption of asphalt mixes is significantly influenced by the moisture content of the aggregates.	Removing 1% water from a mineral mixture requires 7.34 kWh of thermal energy.
Gruber and Hofko [59]	2023 Effect of aggregate moisture content on greenhouse gas emissions.	Total greenhouse gas emissions are 47.1 kg CO ₂ e/t for natural aggregates with a moisture content of 5% (0%: 36.6 kg CO ₂ e/t).
Ferrotti et al. [60]	2024 Unit drying/heating reduces fuel oil consumption.	Drying/heating units cut PM, Nox, and VOC emissions by around 18 percent, 22 percent, and 35!percent, respectively.

The solar aggregate inventory method is employed to decrease moisture content [6]. The primary objective of designing solar energy storage systems is to harness high-quality thermal energy from solar radiation when mineral mixtures are exposed. Figure 6 presents an experiment conducted in Croatia to analyze the impact of varying sunlight on the thermal storage performance of mineral mixtures in solar aggregate silos across different seasons (summer and autumn). Three solar aggregate inventory test models, along with one reference model simulating uncovered and unprepared surface storage conditions, were constructed. It was observed that utilizing solar energy aggregate inventory can lead to the preheating of aggregates and a reduction in their moisture content. This approach has the potential to improve energy efficiency and thermal storage performance while addressing challenges associated with variable sunlight conditions throughout the seasons. Androjić et al. [39] discuss the potential energy savings and the ability to predict the temperature of aggregate piles to achieve sustainability using an artificial neural network (ANN) model. The modeling results indicate that the ANN can successfully predict the temperature of aggregate piles during production and storage as part of the entire asphalt mixture production process.

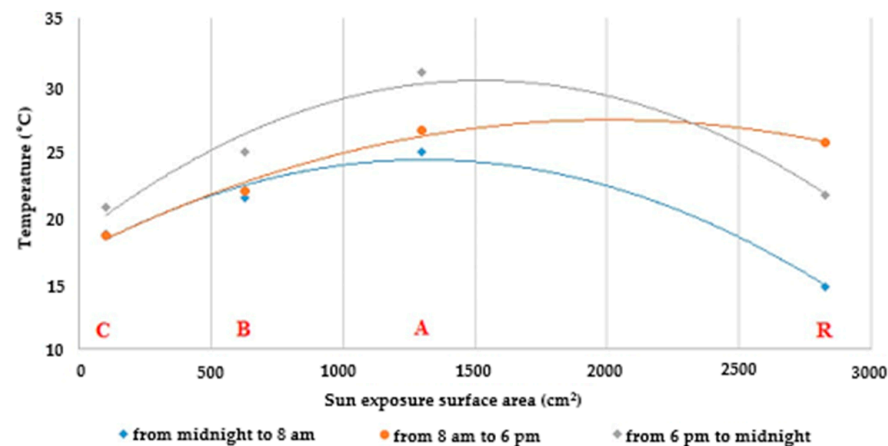


Figure 6. Relationship between the exposed ceiling surface area and the temperature of the mineral mixture. The ceiling glass window areas are as follows: Model A: 0.36×0.36 m; Model B: 0.25×0.25 m; Model C: 0.10×0.10 m; Model R: 0.6 m in diameter [62].

6. Asphalt Mixture Energy Consumption

All stages of road development, from construction to operation, result in the consumption of significant amounts of ecosystem goods and services, as well as the generation of waste and emissions. The energy consumption of asphalt in the manufacturing process primarily depends on diesel and electricity usage during the operation of the asphalt plant. The transportation of asphalt materials, plant operations, transportation of mixtures to the construction site, and placement on the road surface all rely on diesel fuel produced by refineries. Electricity is supplied by the generation and distribution sectors and is primarily consumed during the operation of hot mix asphalt mixing plants [54,61]. Generally, the two main stages of energy consumption during pavement construction are material production and construction. The most commonly used paving material is hot mix asphalt, which involves heating the aggregate and asphalt at high temperatures, resulting in substantial energy consumption. In many modified asphalt mixtures, the asphalt is continuously stirred and heated during the modification process. Higher temperatures not only significantly increase energy consumption but also lead to the release of smoke during mixing and laying [62]. A considerable amount of CO_2 , CH_4 , NO_2 , and other greenhouse gases are emitted concurrently with energy consumption. The carbon emission evaluation model can assess the carbon emission status of existing asphalt pavement construction, providing a quantitative description of current emissions. This model also serves as a reference for transforming asphalt pavement construction from a high-carbon emission mode to a low-carbon emission mode. The selection of carbon emission evaluation indicators is based on investigations and calculations at each stage, with the objective of accurately reflecting the project's actual situation being the chosen index [63]. Peng et al. [42] introduced a calculation model for energy consumption emissions to break down carbon emissions into two components: energy consumption emissions in the pavement construction process and volatile emissions from the asphalt mixture itself. As shown in Figure 7a, their findings indicate that aggregate heating and asphalt heating are the primary sources of carbon emissions resulting from energy consumption, accounting for an astonishing 92.08% of total carbon emissions generated during this stage. Additionally, Figure 7b depicts that the rolling and spreading processes are the main sources of carbon emissions from asphalt mixture volatilization, accounting for a significant 98.53% of total emissions in this category. Hence, these two processes are identified as the primary sources of carbon emissions stemming from asphalt mixture volatilization. Cao et al. [64] established a quantitative model for greenhouse gas emissions during the construction of asphalt pavements for different energy types, and evaluated carbon emissions during the construction period. The results showed that the high-emission links during the construction of asphalt pavements were aggregate heating, asphalt heating, and asphalt mixture mixing, accounting for 64.66%, 14.63%, and

13.21% of the total energy consumption and greenhouse gas emissions, respectively. In summary, the construction of asphalt pavements necessitates substantial quantities of materials, technology, and fuel, leading to significant CO₂ emissions. Given the pressing need to reduce these emissions, it is imperative to lower the carbon emission levels associated with asphalt pavements and to implement effective mitigation materials and technologies. Such measures will not only contribute to the achievement of sustainability goals but also mitigate the ecological impact of asphalt pavement construction and maintenance.

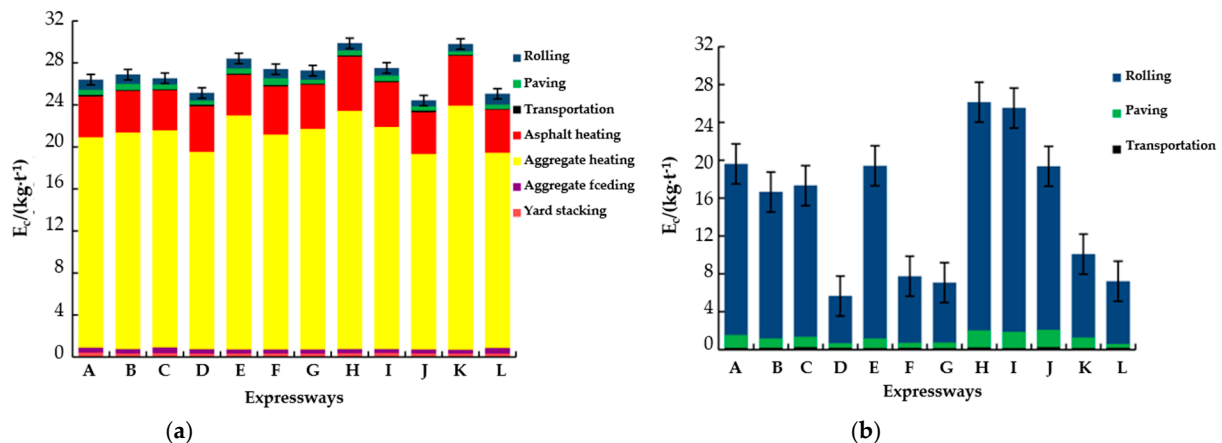


Figure 7. Carbon emissions of 12 expressways [42]: (a) carbon emissions from energy consumption; (b) asphalt mixture carbon emissions. (A: Jingqin Expressway; B: Jishan Expressway; C: Huangyan Expressway; D: Sancha Expressway; E: Xixian Expressway; F: Pengteng Expressway; G: Liaoyu Expressway; H: G3014; I: G310; J: Yanyan Expressway; K: West Copper Expressway; and L: Binhai Expressway).

7. Low-Carbon Green Development of Asphalt Mixtures

Low-carbon and green development in asphalt pavement construction is a crucial aspect of alleviating the pressure on environmental resources. Global warming, primarily driven by greenhouse gases such as carbon dioxide, poses a critical environmental issue. It is essential to modify raw materials and implement emission reduction measures in heating technology, as well as to modify raw materials, heating energy types, and other aspects to reduce greenhouse gas emissions from energy consumption during the construction process [65]. In response to this challenge, many countries have committed to achieving net-zero emissions by 2050, with over 125 nations establishing long-term reduction targets that reflect a global commitment to addressing environmental change [66]. While traditional asphalt is widely used due to its durability and cost-effectiveness, its production and usage contribute to greenhouse gas emissions and biodiversity loss [2,67]. Consequently, the utilization of recycled waste and innovative materials, such as recycled materials, has become essential. These new materials not only reduce dependence on conventional materials but also provide viable solutions for achieving low-carbon emissions [59,68].

In light of the pressing need to reduce emissions, it is crucial to lower the carbon footprint of asphalt pavements. This not only aids in meeting sustainable development goals but also mitigates the ecological impact of construction and maintenance activities. Sustainable asphalt pavement practices offer significant advantages: the use of recycled materials can reduce greenhouse gas emissions, decrease maintenance costs over the long term, and enhance pavement durability [69,70]. The adoption of green low-carbon technologies, such as the utilization of recycled materials and the integration of clean energy sources, will further advance the sustainable development of asphalt pavements and contribute to the construction of more environmentally friendly transportation infrastructure.

7.1. Recycled Material

The application of recycled materials (such as recycled asphalt pavement, biochar, and crumb rubber) and advanced technologies in asphalt pavement can significantly reduce negative environmental impacts, including greenhouse gas emissions and the depletion of fossil fuel resources [49]. This approach also promotes effective waste management and decreases the need for dumping sites.

7.1.1. Recycled Asphalt

Reusing milling material from old asphalt pavement in the construction of new roads can significantly reduce waste disposal, which typically consumes land resources. This practice minimizes the need for mining new materials, conserves energy, and facilitates the recycling of materials.

The reutilization of milling material derived from old asphalt pavement offers numerous environmental benefits. It not only alleviates the land requirements associated with waste disposal but also reduces the need for extracting new materials. This approach conserves energy and promotes material recycling while minimizing overall environmental impact. Giani et al. [71] conducted a study analyzing the environmental benefits of using recycled asphalt pavement. Their research demonstrated that when recycled asphalt pavement (RAP) is mixed with new asphalt, it can decrease carbon emissions by up to 6.8%. Furthermore, the combination of RAP and warm mix asphalt (WMA) technology has resulted in a 12% reduction in CO₂ emissions and a 15% decrease in energy consumption throughout the road's life cycle. Gulotta et al. [72] also demonstrated the substantial benefits of combining recycled asphalt pavement (RAP) with warm mix asphalt (WMA) in mitigating CO₂ emissions. This approach not only enhances the sustainability of asphalt mixtures but also contributes to the overall reduction in the carbon footprint associated with road construction and maintenance. Furthermore, Wang et al. [73] investigated the carbon emissions associated with the construction of foam asphalt cold-recycled pavement. Their findings revealed that when producing 1 ton of asphalt mixtures, the foam asphalt cold regeneration technology significantly curtails energy consumption, saving 28.984 kg of standard coal and resulting in a decrease of 71.206 kg in CO₂ emissions. This is a remarkable 47.1% reduction compared to the emissions generated during the production of asphalt-stabilized gravel, highlighting the significant energy-saving benefits of this technology. Feng et al. conducted a comprehensive evaluation of the sustainability of recycled asphalt pavements (RAPs) from three key perspectives: cost, energy, and environmental impact. The study found that, particularly in the context of thin asphalt overlays, the use of RAP represents a more economically efficient and environmentally feasible alternative [74].

The combination of 100% recycled asphalt with crumb rubber and waste engine oil (WEO) is considered the most environmentally friendly type of asphalt. This mixture results in a 49% reduction in carbon dioxide (CO₂) emissions, a 97% reduction in volatile organic compounds (VOCs), a 97% reduction in nitrogen oxides (NO_x), a 22% reduction in methane (CH₄), and a 39% reduction in sulfur dioxide (SO₂) emissions. The production energy consumption of the R100-CRM-x-WEO mixture is 52% lower than that of the original mixture, with 99% of the materials derived from buried waste, significantly improving environmental quality [75]. Additionally, the use of recycled asphalt as a base layer can reduce phosphorus compound emissions in water by 45 g/m³, outperforming hot mix asphalt (HMA). In terms of ecotoxicity and human non-carcinogenic toxicity, the indicators for the CMRAJ mixture are 60% lower than those for HMA. The incorporation of recycled asphalt pavements (RAPs), non-conforming fly ash (OFA), and food waste bio-oil (FWBO) further reduces environmental impacts [76]. In summary, the introduction of sustainable asphalt mixtures and innovative material combinations not only effectively reduces the emissions of various pollutants but also promotes the widespread application of environmentally friendly asphalt, significantly enhancing the environmental benefits of road construction.

7.1.2. Recycled Shredded Rubber

Crumb rubber, a recycled material derived from discarded tires, has the potential to become a sustainable asphalt pavement when mixed with asphalt mix through both wet and dry processes. This innovative approach to waste material utilization offers a promising future for sustainable road construction. Tushar et al. [77] investigated the environmental impact of utilizing crumb rubber (CR) particles as a polymer additive to enhance asphalt performance. Their findings revealed that recycling waste tire rubber can significantly reduce carbon dioxide (CO₂) emissions by 71.9% compared to landfilling or incineration. When adding 2% CR to the asphalt mixtures, the resulting reduction in carbon emissions is 23.20%. These findings emphasize the importance of recycling waste materials and the potential of sustainable asphalt pavements. Furthermore, Wang et al. [78] investigated the performance of crumb rubber-modified asphalt in 1-ton asphalt mixtures. As illustrated in Table 2, their findings revealed that this sustainable pavement material resulted in a reduction in carbon emissions ranging from 18% to 36%, while also achieving energy savings between 15% and 87%. These impressive figures demonstrate the significant environmental benefits of using recycled tire rubber in asphalt mixtures. Wang et al. [79] compared the performance of clastic rubber modified asphalt with SBS modified asphalt, specifically focusing on their respective energy-saving and emission reduction capabilities. The results of their study revealed that the detrital rubber modifier can significantly reduce energy consumption by 86.24–89.50% and CO₂ emissions by 89.5%. This finding highlights the potential of using the CR modifier as a sustainable replacement for SBS modifier in asphalt mixtures. Moghimi et al. modified warm mix asphalt with different levels of crumb rubber (10%, 15%, and 20%) and demonstrated that 10% crumb rubber-modified asphalt exhibited optimal rutting, moisture sensitivity, and fatigue resistance, and a 22% increase in tensile strength ratio relative to the 10% specimen [80]. The reduction in emissions and optimization of performance can be primarily attributed to the balance between the physical properties of crumb rubber materials and their chemical interactions. On the one hand, the incorporation of crumb rubber enhances the elasticity and crack resistance of asphalt, thereby improving the durability and lifespan of the pavement. On the other hand, the characteristics of rubber contribute to the reduction in asphalt volatility at elevated temperatures, leading to decreased emissions of volatile organic compounds (VOCs) and other pollutants. Consequently, the synergistic effects of this material achieve a favorable balance between environmental benefits and performance enhancement. However, the scarcity of shredded rubber, an important renewable resource in the recycling of end-of-life vehicles, has become increasingly pronounced. This scarcity not only arises from the inherent limitations of the resource and the challenges associated with its recycling, but is also profoundly influenced by factors such as the recycling rates of old vehicles and variations in legislative frameworks across countries. Increasing the recycling rates of end-of-life vehicles can significantly increase the supply of shredded rubber. However, notable disparities in the legislation, technological capabilities, and policy incentives related to end-of-life vehicle recycling among different countries result in inconsistent recycling rates. Countries such as those in Europe have established relatively sophisticated recycling systems, implementing a series of effective policies and regulations to promote the recovery of waste tires [81]. In contrast, although China has the highest levels of rubber production and consumption, there are no standardized technical and quality criteria for the recycled products of waste rubber [82–84]. On the other hand, differences in the comprehensiveness of the legal frameworks for end-of-life vehicle recycling, the strength of policy incentives, and the robustness of regulatory enforcement across nations also directly affect the effective recovery and reuse of shredded rubber resources, thereby exacerbating their scarcity [85–87]. Therefore, addressing these challenges is crucial for maximizing the potential of crumb rubber in sustainable asphalt pavement applications and ensuring a stable supply of this valuable resource.

Table 2. Energy and heavy oil consumption of different mix CRMAs [78].

Asphalt Mixtures		Total Energy Consumption (MJ)	Energy Conserved (%)	Heavy Oil Consumption (kg)	Heavy Oil Conserved (kg)
Hot mixing	CRMA	300.89	—	7.65	—
	Sa-CRMA	238.62	20.69	6.07	1.58
Warm mixing	Ev-DAT CRMA	192.33	36.08	4.89	2.67
	TOR-CRMA	245.83	18.30	6.25	1.40
	Asp-CRMA	245.87	18.28	6.25	1.40

7.1.3. Biochar

In recent years, there has been an increasing focus within the pavement industry on reducing carbon footprints and providing new strategies to address climate change. Biochar, as a renewable resource, has garnered significant research interest due to its potential in soil enhancement and carbon sequestration. Biochar is a carbon-rich material produced through the pyrolysis of biomass, primarily derived from agricultural residues (such as rice husks and corn stalks), forestry waste (such as branches and bark), and food processing by-products (such as fruit pits and vegetable scraps) [88–90]. Researchers have explored the use of biochar as an additive in asphalt mixtures to reduce overall carbon emissions. Zhou et al. [91] found that the energy consumption of biochar-modified bioasphalt (BMBA) is only 25% of that of conventional asphalt, with energy savings of 25% and 38% during the production and preparation stages, respectively. The greenhouse gas (GHG) emissions reported in this study range from 0.16 to 0.19 kg CO₂ equivalent per kg of BMBA. By partially substituting traditional mineral fillers with biochar, it is possible to effectively reduce the carbon intensity of materials while enhancing the performance and durability of asphalt. However, the production process of asphalt pavements inevitably releases volatile organic compounds (VOCs), posing health risks to construction workers [92]. To address this issue, some researchers have begun to investigate the application of biochar for the removal of VOCs from asphalt. Studies have shown that biochar can reduce VOC emissions by approximately 50%, with its adsorption efficacy closely related to the type of biochar used [93,94]. For instance, biochar derived from straw or wood primarily interacts with VOCs through chemical adsorption, whereas biochar derived from pig manure relies more on physical adsorption [95]. By increasing the specific surface area, pore volume, and number of surface chemical functional groups of biochar while simultaneously reducing pore size, the adsorption capacity of biochar in asphalt can be significantly enhanced [96]. Despite the promising potential of biochar in VOC removal, research on the synergistic effects between biochar's efficiency in VOC elimination and the enhancement of asphalt performance remains relatively scarce. Furthermore, the annual production of 373 million tons of biochar from agricultural waste has the potential to sequester approximately 500 million tons of carbon dioxide each year, equivalent to 1.5% of the global annual CO₂ emissions [97,98]. These findings suggest that biochar, as an alternative material, can provide effective solutions for the sustainable development of the pavement industry while simultaneously achieving the dual benefits of environmental and economic improvements.

7.2. Clean Energy

In the production of asphalt mixture equipment, heavy oil is typically used as the primary source of heating energy. However, through technological advancements, cleaner energy sources can be identified and implemented as substitutes for heavy oil. By finding a fuel that is equally effective in heating the aggregate but with lower pollution and emissions, we can take a significant step toward the development of a cleaner society. This approach aligns with the direction of environmental protection and sustainable development. Peng et al. [42] compared the energy emission reduction effects of using coal and natural gas for heating asphalt during the processes of aggregate and asphalt heating. The results

indicated that natural gas significantly reduced carbon emissions by 40.92% compared to coal. Moreover, natural gas was also effective in reducing carbon emissions by 27.68% during the aggregate heating process. These findings demonstrate the potential of using natural gas as a cleaner alternative to coal in these high-carbon emission processes. As illustrated in Figure 8, Almeida et al. [99] conducted a comparative study on the CO₂ emissions of high-modulus asphalt concrete and crude oil asphalt concrete. They found that converting fossil fuels to natural gas can significantly reduce energy consumption while simultaneously decreasing CO₂ emissions and lowering the release of other harmful gases and particulates. Notably, the fuel-related portion constituted only 36% of the total emissions. Paranhos et al. [100] employed a regression model to identify the optimal mixing conditions for reducing energy consumption and emissions from hot mix asphalt (HMA). Their findings revealed that CO₂ emissions from natural gas plants were only half of those from fuel oil plants, while nitrogen oxide emissions were reduced by a factor of nine, and CO₂ emissions were diminished by a factor of eighteen. Furthermore, small plants were found to have an advantage over large plants in terms of NO_x and CO₂ emissions. Therefore, changing fuel types clearly helps to reduce carbon emissions and improve energy efficiency in asphalt production processes. Abdalla et al. [101] demonstrated that there are significant environmental advantages to be gained from switching from asphalt-mixed BTS fuel to natural gas in place of heavy fuel oil. The impact on the different environmental impacts of operating an asphalt mixing plant with natural gas compared to the baseline scenario with HFO is shown in Figure 9. Using natural gas instead of HFO is associated with a sustained reduction in environmental impact, as shown by negative relative values indicating a lower risk of environmental impact from AS1 compared to baseline. The most significant relative reductions were observed for acidification, ozone depletion and effects on the respiratory tract in the LC80-A and the LC100-A. Switching the fuel type to natural gas, recognized as the most environmentally friendly fuel, not only significantly reduces carbon emissions but also enhances the energy efficiency of asphalt production. This underscores the importance of transitioning to cleaner energy sources to mitigate the environmental impacts associated with asphalt pavements.

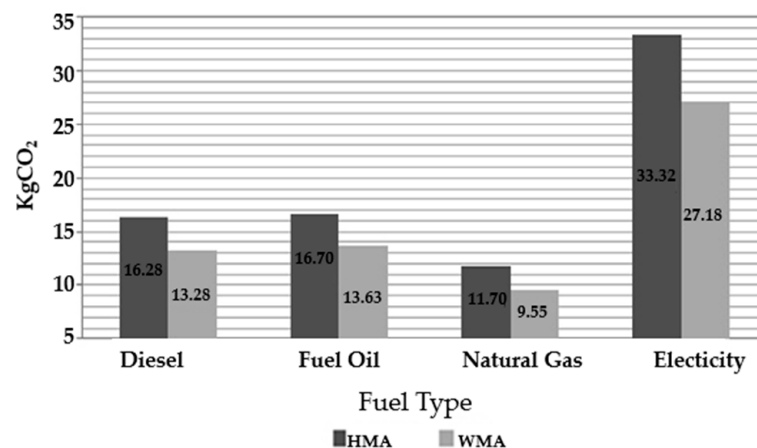


Figure 8. CO₂ emissions from fuel combustion during synthetic HMA production [99].

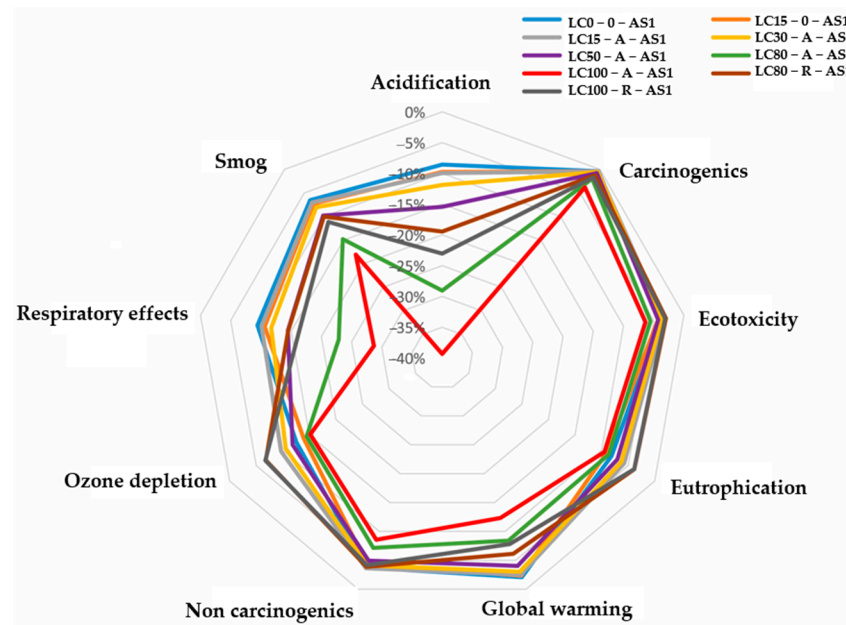


Figure 9. The variation of the LCIA results of utilizing natural gas in the asphalt mixture production as an alternative scenario relative to the baseline scenario [101].

8. Conclusions

This paper examined the use of asphalt pavement for construction purposes, focusing on its impact on energy consumption and carbon dioxide emissions. Over recent decades, there has been a considerable increase in the demand for road infrastructure, accompanied by growing concerns about its detrimental effects on the environment. Consequently, efforts are being made to mitigate the adverse effects of road construction. Based on a review of the relevant literature, the following conclusions can be drawn:

Energy consumption in asphalt mixtures predominantly occurs during three distinct production stages: the heating of aggregates, the heating of asphalt, and the mixing of the asphalt mixtures. The combustion of fuel in asphalt production is the primary source of greenhouse gas emissions. Indeed, aggregate heating accounts for 97% of the energy consumed during this process. Therefore, it is crucial to consider the role of moisture content in the aggregates used. The temperature at which the asphalt mixture is produced directly impacts both manufacturing costs and energy consumption, subsequently affecting greenhouse gas emissions. The use of low-temperature asphalt pavement, compared to hot mix asphalt, results in significant reductions in both energy consumption and greenhouse gas emissions. This, in turn, leads to improved working conditions for plant operators and workers, representing a new and sustainable technology with a positive environmental impact.

The consumption of asphalt during the transportation of materials, the operation of the plant, the transport of mixtures to the construction site, and the placement of these mixtures on the road surface result in considerable environmental impacts, including the production of greenhouse gases such as CO₂, CH₄, and NO₂. The adverse effects of asphalt plant emissions on the ecological environment can be effectively reduced by modifying raw materials, improving heating technology, utilizing alternative heating energy sources, and implementing green emission reduction measures. The use of recycled waste significantly reduces gas emissions, conserves energy, and facilitates material recycling. Additionally, using natural gas instead of fuel oil provides a comparable heating effect for aggregates while significantly reducing carbon emissions. This represents a fundamental transition toward cleaner energy sources that will lead to a sustainable society. The construction of green roads is not only a goal for future infrastructure but also an essential requirement for safeguarding our living environment.

Future research directions should emphasize the application of artificial intelligence (AI) technologies to enhance carbon dioxide emission assessment. By leveraging AI for the modeling and optimization of sustainable low-carbon technologies, significant improvements can be made in evaluating emissions and energy demands associated with various strategies. This approach not only facilitates independent analyses but also strengthens the integration of multiple assessment methods, ultimately leading to more robust and standardized evaluation protocols that adequately account for inherent uncertainties.

Additionally, in the study of sustainable asphalt pavements, analyzing the energy consumption and toxic substance emissions associated with recycled materials is crucial. A thorough exploration of the life cycle impacts of recycled asphalt pavements (RAPs), bioasphalts, and waste tire materials is essential for understanding their environmental implications. The production of these materials often entails significant energy consumption and harmful emissions, making it necessary to assess their environmental impact throughout their entire life cycle. Notably, hazardous substances such as sulfur and zinc present in waste rubber may leach into soil and water, threatening the sustainability of ecological resources. Therefore, prioritizing research on material recovery and ecological protection strategies is essential to mitigate these potential risks.

Author Contributions: Resources and data curation, Y.L.; writing—original draft preparation and writing—review and editing, Z.L.; visualization, Y.Z.; supervision and project administration, H.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Conflicts of Interest: Authors Yancheng Liu and Zhu Youwei were employed by the company Longjian Road & Bridge Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Review

A Review of Carbon Footprint Reduction in Construction Industry, from Design to Operation

Banu Sizirici ¹, Yohanna Fseha ¹ , Chung-Suk Cho ^{1,*}, Ibrahim Yildiz ² and Young-Ji Byon ¹ 

¹ Department of Civil Infrastructure and Environmental Engineering, Khalifa University, Abu Dhabi P.O. Box 127788, United Arab Emirates; banu.yildiz@ku.ac.ae (B.S.); 100049403@ku.ac.ae (Y.F.); youngji.byon@ku.ac.ae (Y.-J.B.)

² Department of Chemistry, Khalifa University, Abu Dhabi P.O. Box 127788, United Arab Emirates; ibrahim.yildiz@ku.ac.ae

* Correspondence: chung.cho@ku.ac.ae

Abstract: Construction is among the leading industries/activities contributing the largest carbon footprint. This review paper aims to promote awareness of the sources of carbon footprint in the construction industry, from design to operation and management during manufacturing, transportation, construction, operations, maintenance and management, and end-of-life deconstruction phases. In addition, it summarizes the latest studies on carbon footprint reduction strategies in different phases of construction by the use of alternative additives in building materials, improvements in design, recycling construction waste, promoting the utility of alternative water resources, and increasing efficiencies of water technologies and other building systems. It was reported that the application of alternative additives/materials or techniques/systems can reduce up to 90% of CO₂ emissions at different stages in the construction and building operations. Therefore, this review can be beneficial at the stage of conceptualization, design, and construction to assist clients and stakeholders in selecting materials and systems; consequently, it promotes consciousness of the environmental impacts of fabrication, transportation, and operation.

Keywords: embodied carbon; recycled asphalt; recycled aggregate; construction waste materials; alternative additives; alternative water resources



Citation: Sizirici, B.; Fseha, Y.; Cho, C.-S.; Yildiz, I.; Byon, Y.-J. A Review of Carbon Footprint Reduction in Construction Industry, from Design to Operation. *Materials* **2021**, *14*, 6094. <https://doi.org/10.3390/ma14206094>

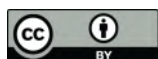
Academic Editor: Alessandro P. Fantilli

Received: 9 September 2021

Accepted: 12 October 2021

Published: 15 October 2021

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1. Introduction

This paper aims to bring attention to the carbon footprint in the construction industry (building, maintaining, and deconstructing the structures), since the construction industry is listed as the single largest global consumer of resources [1,2]. In the European Union, building construction consumes 40% of materials and 40% of primary energy, and generates 40% of waste annually [1]. Globally, in developed and developing countries, buildings contributes to 33% of the greenhouse gas (GHG) emissions and 40% of the global energy consumption which stem from the usage of the equipment, the manufacturing of building materials and transportation [3,4]. The total CO₂ emission of the construction sector was 5.7 billion tons which made up 23% of the emissions of global economic activity in 2009 [5]. Globally, the urban population is predicted to exceed six billion in 2045, and this could lead to more construction in the future.

According to the 4th Assessment Report of the Intergovernmental Panel on Climate Change (IPCC), GHG emissions from buildings contributed 8.6 billion t-CO₂-e in 2004. It is predicted that it could reach up to 15.6 billion t-CO₂-e by 2030, creating an increase of 26% CO₂ which accounts for 30–40% of the total GHG emissions [6]. It is necessary to take action to reduce GHGs resulted from construction activities. Hence, it is vital to implement policies that focus on GHG emissions mitigation. Such schemes are broadly classified into two approaches: (1) indirect pricing such as regulations and (2) direct pricing such as carbon taxes and emission trading schemes (ETS) [7].

Regulations such as building codes can effectively reduce GHG emissions if enforced well enough, and can ensure new buildings incorporate designs that are both cost and energy effective [8]. Required codes, including the European Union's zero energy mandate by 2021, Australia's NatHERS 5-star standard, volunteer certificates such as Leadership in Energy and Environmental Design (LEED) which is required for all new federal government construction projects and renovations in the USA but voluntary for private construction, and the Building Research Establishment Environmental Assessment Method (BREEAM), would force designers and contractors to reconsider material usage that has a high embodied carbon content and also to rethink way they conduct their operations [9,10].

Another instrument for the mitigation of GHG emission is the carbon tax. Carbon taxes are simpler to design, have relatively low administration costs, and are attractive to stakeholders in the building sector due to their familiarity with the tax mechanism [11,12]. Carbon taxes encourage industry and the general public to help reduce GHG emissions by using energy efficiently and opting for cleaner, renewable sources of energy which in turn leads to innovations in technology and processes [13]. In terms of ETS, the cumulative amount of GHG emissions mitigated can be quantified with ETS and emission permits can be distributed for free or auctioned off [7,14]. As both energy supply and demand have equal weights, an ETS can be especially useful in the construction industry, thereby, encouraging the use of technologies that are energy efficient [15].

Studies have shown that a variety of factors slow down the move towards a carbon neutral construction industry. A study conducted in Singapore and Hong Kong found that lack of awareness, education, incentives, and high initial costs are the obstacles to such a move [16]. In another study that focused on commercial buildings in the Chinese cities of Beijing and Shanghai, the barriers were identified to be lack of regulations and financial incentives, ineffective monitoring, and lack of awareness around energy saving [17]. Therefore, this paper aims to bring attention/awareness where carbon footprint resulting from design to operation/management phases, such as manufacturing, transportation, construction, operation and maintenance, and end-of-life deconstruction in construction industry. If these sources are well identified, it will be helpful to reduce GHGs at the stage of conceptualization, design, construction, and management via selecting material, system, operation and management having less carbon footprint, which will promote environmental consciousness in whole construction operations.

There are many studies focused on CO₂ reduction at different phases in the construction industry. However, there is no other study focusing on carbon reduction in all stages from design to operation and management phases with emphasis on manufacturing, transportation, construction, operation and maintenance, and end-of-life deconstruction comprehensively. Therefore, this paper reviewed a variety of the latest techniques for reducing the carbon footprint of each phase such as the use of alternative additives in building materials, improvements in the design, recycling of construction waste, promoting the use of alternative water resources, increasing the efficiency of water technologies, and building novel systems to improve the sustainability of the construction industry.

2. Carbon Footprint of Mining, Manufacturing, and Materials Transporting in the Construction Industry and GHG Reduction

Construction process undergoes several phases, starting with production of materials (non-metallic minerals, oil, cement mortar, iron, steel, concrete) and material transportation which contributes 82–96% of the total CO₂ emissions through the construction period as shown in Figure 1 [18–21].

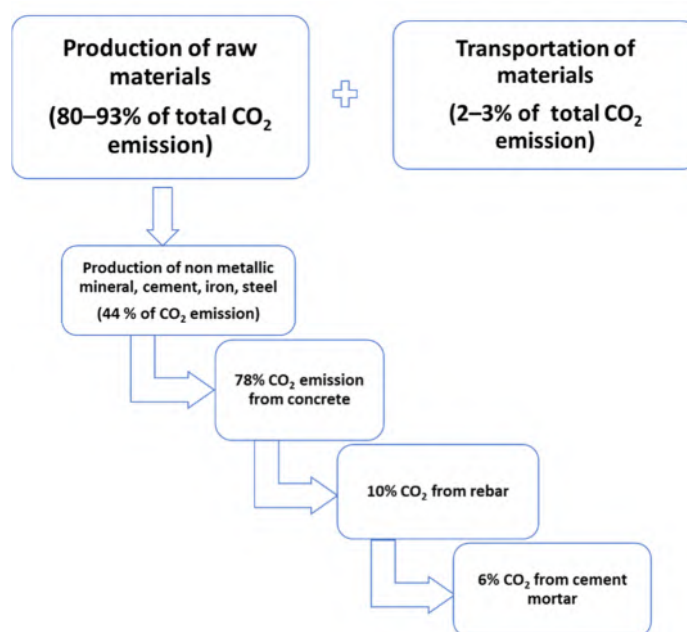


Figure 1. CO₂ emission from different phases in the construction industry.

A study showed that carbon footprint of urban buildings increased from 8.95 million tons in 2005 to 13.57 million tons in 2009, and that 45% of CO₂ resulted from building material production whereas 40% of CO₂ resulted from building energy in Xiamen, China [22]. Another study indicated that life-cycle carbon emission of a five-story brick-concrete residential building in Nanjing city of PR China was 1807.31 t, and 90% of CO₂ were emitted at the stage of construction materials preparation and the stage of building operation [23].

2.1. Carbon Footprint of Limestone Quarrying

Limestone is one of the largest produced crushed rocks which is the basic component of construction materials, such as aggregate, lime, cement, and building stones for the construction industry [24]. The energy required for lime quarrying is associated with the machine fuel, diesel, and electricity that are needed for the limestone processing. The machines used together with their energy requirements and CO₂ emissions are listed in Figure 2. A study found that the main cause of resource depletion in limestone quarrying was the use of diesel fuel in the transportation process, and that based on the GHG Protocol the GHGs emission was found to be 3.13 kg CO₂ eq. per ton crushed rock product. This study suggested the adoption of alternative renewable energies such as solar, thermal, and biodiesel which will have significant impact on the reduction of GHG emissions (0.21 Mt-CO₂ eq. annually) [25].

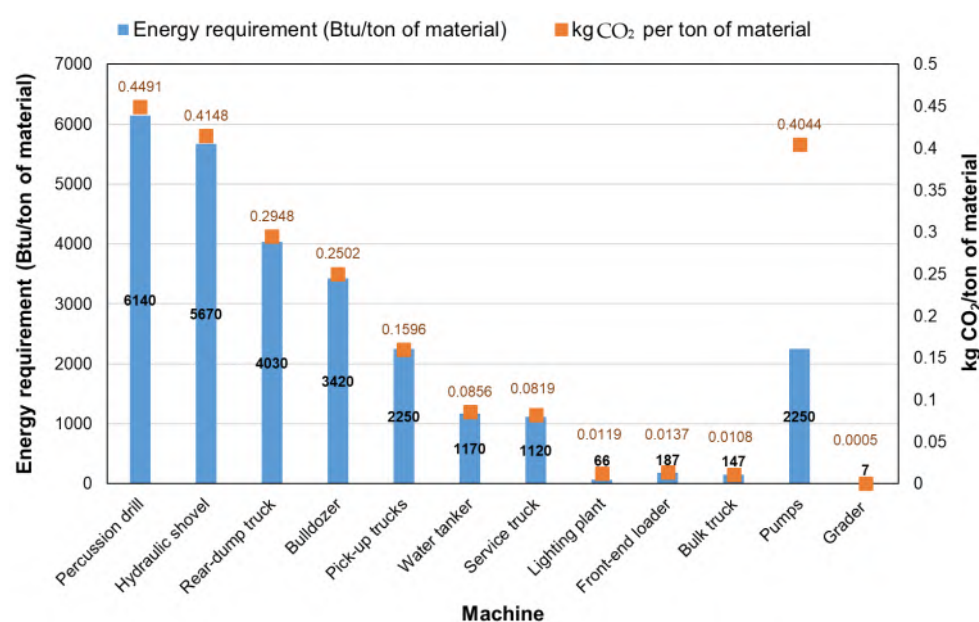


Figure 2. The machines used together with their energy requirements and CO₂ emissions in limestone processing. Adapted from [24].

2.2. Carbon Footprint of Cement and Concrete Manufacturing

Globally, cement manufacturing accounts for 5% of CO₂ emissions [26]. It has been reported that manufacturing of 1 kg of Portland clinker releases nearly 1 kg of CO₂ to the atmosphere. The calcination process that takes place in the cement kiln contributes nearly 0.55 kg CO₂ per kg of cement clinker [27]. Concrete's typical composition is 34% sand, 12% Portland cement, 48% crushed stone, and 6% water. Since the cement percentage is relatively small in concrete, it is considered non-energy intensive compared to other construction materials [24]. CO₂ emission rate during the production of concrete is between 347 and 351 kg of CO₂-e/m³ [28]. According to Solís-Guzmán, Cement II/AL 32.5 N in two four-story blocks of flats (a total of 107 dwellings with total area of 10,243.69 m²) gives 148,180 kg CO₂ eq/year and concrete HA25/B/40 gives 312,596.55 kg CO₂ eq/year during one year of construction process [1]. A study in China reported that 1 km Portland cement concrete pavement construction gives 8215.31 CO₂e tons in which raw material production accounts for 92.7%, concrete manufacturing phase accounts for 7.2% and onsite pavement construction phase accounts for 0.1% of the total GHG emissions [26]. The energy consumption on-site and CO₂ emissions from the production of cement/concrete annually are listed in Table 1. The United States was the third largest producer of cement globally with 50–55 million metric tons (Mt) of CO₂eq emissions which is equivalent to 4% of the total GHG emissions in the country in 2012. These numbers are expected to increase further as the production of cement grows [29].

Table 1. On-site energy consumption and CO₂ emissions from cement/concrete production annually. Adapted from [24,26].

Activity	Cement		Concrete	
	Energy Use/Ton (Btu)	CO ₂ Emissions (Ton)/Ton of Material	Energy Use/Ton (Btu)	CO ₂ Emissions (Ton)/Ton of Material
Quarrying and crushing	4.29×10^4	4.05×10^{-3}	1.61×10^5	1.44×10^{-2}
Cement manufacturing	Raw grinding	9.39×10^4		
	Kiln fuels	4.62×10^6		
	Reactions			
	Finish milling	2.71×10^5		
Concrete production	Blending/mixing		3.54×10^5	6.36×10^{-2}
	Transportation		6.97×10^5	5.10×10^{-2}

2.3. Carbon Footprint of Asphalt Production and Construction

Asphalt is a substance used as binder for pavement materials. The energy consumption for asphalt binder production includes the extraction of crude oil, transportation, and the refining process. The energy consumption for asphalt binders is 4900 MJ per ton, and the corresponding GHG emissions is 285 kg CO₂ per ton [30]. Heating aggregates account for 67% CO₂ emission, asphalt heating accounts for 14% CO₂, and mixing process accounts for 12% of total carbon emissions [31]. A case study in China reported that 20 km long asphalt pavement construction emitted 52,264,916.06 kg CO₂-e, which includes raw materials production accounting for 43% of total GHG emissions, mixing accounting for 54% of total GHG emissions, and transportation, laying, compacting, and curing phase accounting for 3% of total GHG emission [32].

2.4. Carbon Footprint of Steel Production

Steel production starts with the reaction between iron ore and a reducing agent, coking coal, in the blast furnaces producing melted iron which is converted to steel in a later stage. The reaction of iron ore with carbon is the major contributor of CO₂ emission in the steel production corresponding to 70–80% of the total CO₂ emissions [25]. Globally, steel manufacturing accounts for 6% of CO₂ emissions [33]. Globally, steel manufacturing accounts for 6% of CO₂ emissions [26]. According to Solis Guzmán [1], Steel B 500S in two four-story blocks of flats (a total of 107 dwellings with total area of 10,243.69 m²) gives 281,898.38 kg CO₂ eq/year during one year of construction process. Table 2 shows the energy type and consumption, and CO₂ emissions associated with steel production in the integrated steel making and secondary steel making stages.

Table 2. The energy type and consumption and the CO₂ emissions associated with steel production. Adapted from [34].

Activity	Energy Type and Consumption	Final Energy (MBtu/Ton)	CO ₂ Emission (Ton)/Ton of Material
Primary steel making			
Sinter making	26 PJ fuel and 2 PJ electricity	0.264	0.009
Coke making	74 PJ fuel and 2 PJ electricity	0.718	0.007
Iron making	676 PJ fuel and 4 PJ electricity	6.421	0.120
Steel making (Basic oxygen furnace)	19 PJ fuel and 6 PJ electricity	0.236	0.005
Casting	15 PJ fuel and 11 PJ electricity	0.236	0.010
Hot rolling	157 PJ fuel and 34 PJ electricity	1.803	0.041
Cold rolling and finishing	43 PJ fuel and 15 PJ electricity	0.548	0.014
Boilers	167 PJ fuel and 0 electricity	1.577	0.085
Co-generation (integrated steel making)	101 PJ fuel and -22 PJ electricity	0.746	0.004
Secondary steel making			
Steel making using electric arc furnace	6 PJ fuel and 62 PJ electricity	0.642	0.031
Casting	1 PJ fuel and 4 PJ electricity	0.028	0.002
Hot rolling	102 PJ fuel and 22 PJ electricity	1.171	0.026
Cold rolling and finishing	Not required	none	-
Boilers	42 PJ fuel and 0 PJ electricity	0.397	0.026
Co-generation	11 PJ fuel and -2 PJ electricity	0.085	0.0004

2.5. GHG Reduction in Materials and Chemicals

Alternative additives or recycled concrete waste materials can be used in common construction materials such as cement, concrete, asphalt, and clay to reduce environmental impact in the construction industry.

2.5.1. Cement and Concrete Additives

Cement manufacturing requires energy; therefore, it is recommended to substitute the clinker content partially with industrial by-products. It is safe to substitute the clinker content by 30% (by weight of total binder) without compromising the strength or performance [35–38]. Also, high energy milling can be done to blend constituents to increase their reactivity and to increase their surface area, both of which can help improve the compres-

sive strength development [39]. Recent studies have shown that regular Portland Cement can be replaced with alkali-activated slag mortars. These alkali-activated slags (AAS) can help reduce environmental impacts greatly since the production of AAS results in low energy consumption and lower energy consumption leading to lower CO₂ emissions [40].

The admixtures used for alkali-activated slag were Peramin SRA 40 (SRA), polymer polyethylene glycol (PEG), and polypropylene glycol (PPG) [41]. In order to reduce the CO₂ emissions, alternative clinker chemistries can be used as well as changing cement production methods in favor of more energy efficient technologies which result in reduction of 374 kg CO₂/t clinker and totaling annual 224,540 t-CO₂ emission release [42]. Table 3 reports CO₂ emission reductions from some alternative technologies and materials in cement manufacturing. For instance, fluidized bed kilns can be used instead of conventional rotary kilns to burn raw materials into powder using a new technology called granulation control/hot self-granulation; such a change can lower energy consumption by 10–15% and reduce NO_x emissions to 0.77 kg per ton of clinker as compared to 2.1–2.6 kg per ton of clinker for conventional kilns [43]. Oxy-fuel technologies have emerged as a promising candidate for CO₂ capture in new cement kilns by using pure oxygen for fuel burning. Due to reduced fuel combustion, oxy-fuel technology reduces CO₂ emissions by 454–726 kg CO₂ per ton of cement. However, due to increased electricity usage, CO₂ emissions increase slightly by 50–68 kg of CO₂ per ton of cement [44].

Table 3. Reduction in CO₂ emissions from alternative technology/materials in cement production. Adapted from [42].

Technology/Material	Alternative	Reduction in CO ₂ Emissions
Cement production methods	Fluidized bed kiln; high activation grinding	20 to 30 kg CO ₂ / ton product
Changes in raw material	Calcareous oil shale, steel slag	60 kg CO ₂ /ton of clinker
	Carbide slag	374 kg CO ₂ /ton of clinker
Emerging alternative cement products	Novacem cement	750 kg CO ₂ /ton product
	Geopolymer cement	300 kg CO ₂ /ton product
	Calera cement manufacturing	500 kg CO ₂ /ton of product
Carbon capture technologies	Concrete curing	120 kg CO ₂ /ton product
	Carbonate looping	370 to 500 kg CO ₂ / ton product
Fuel technologies	Oxygen enrichment and Oxy-fuel	404 to 676 kg CO ₂ /ton cement
Post-combustion carbon capture	Absorption	690 to 725 CO ₂ /ton clinker
Industrial recycling	CO ₂ from cement process into high-energy algal biomass	1800 kg of CO ₂ will be utilized per ton of dry algal biomass produced

In order to mitigate the impact of concrete on the environment, its physical and mechanical properties such as strength, durability and light weight can be enhanced. For instance, lightweight concretes (LWCs) with high volume of additives such as fly ash or silica fume, which reduces the overall structural volume to withstand load, reduces CO₂ emissions by 30–50% as compared to conventional concrete and improves mechanical properties of LWC [45]. Demolition waste such as old tires, crushed glass, and various materials from the incineration process can be granulated and cast into concrete as fillers [27]. According to a study, a sustainable Ultra High-Performance Fiber Reinforced Cement Composite (UHPFRCC) was produced using silica flour, blast-furnace slag cement, silica fume, superplasticizers, wollastonite, and steel fibers [46].

Another study stated that pulverized fuel ash (PFA) and high calcium wood ash (HCWA) were reused as concrete materials and HCWA:PFA of 50:50 and 40:60 provide the optimal flexural and compressive strength [47]. Titanium dioxide (TiO₂) can be used as both an additive and as a coating layer. It was found that photocatalytic concrete containing TiO₂ was effective to remove NO_x in urban streets [48]. According to a study, the carbon footprint of a building with 4020 m² gross area and 5633 tons of total weight was 14,229 tons of CO₂-e; in particular, this building contributed to 42% of the total emissions during both productions of material and construction stages [49]. A fractional replacement of cement in concrete with fly-ash together with the use of ground granulated blast furnace slag and the use of natural aggregates with recycle crushed aggregate can reduce up to 3.8% (10.5 kg

CO₂-e) in comparison to the conventional concrete mixture during the life cycle of the building [50].

2.5.2. Asphalt Additives

Asphalt is used in most road and pavement construction, and it is a considerable contributor to GHGs in construction industry [31]. There are several additives that can assist in reducing GHG emissions, such as Sasobit, which also can reduce mixture viscosity and lower conventional mix temperature. Recent studies compared Warm Mix Asphalt (WMA) and Hot Mix Asphalt (HMA) in terms of their emission profiles. It was determined that mixture containing Sasobit additives with WMA produces the lowest CO₂ emissions which ranges from 450 ppm to 550 ppm while HMA produces 700 ppm to 750 ppm of CO₂ [41]. Another additive for the production of WMA is synthetic zeolite. It reduces the viscosity and increases asphalt mixtures' workability. Furthermore, by allowing stronger coatings of bitumen on aggregates, it improves the bonding [51]. To improve the bonding of aggregates with bitumen at low temperature, zeolite can be doped with Ca (OH)₂ which would also control the emission of CO₂ [52]. Studies have shown that with the addition of 6% of additive by weight, mixing temperatures of asphalt mixtures reduced from 180 °C HMA to 120 °C WMA which in turn reduces the CO₂ emissions from 7500 ppm to 500 ppm [53].

2.5.3. Clay Additives

Fired and unfired clay bricks are used in the construction industry. However, fired clay bricks require a large amount of energy for their production [54]. In order to lessen the environmental impacts and achieve sustainable building industry development, unfired clay bricks are more suitable than fired bricks. Unfired clay bricks are composed of clay soils and a binder such as lime or cement [55]. Calcium-based binder such as lime and cement increases carbon in the air, due to high energy consumption during manufacturing; furthermore, the rocks naturally change CaCO₃ into CaO which further releases CO₂ [56]. Various additives have been tested, and it was found that MgO can be a potential alternative to calcium-based binders. MgO has some similar attributes of CaO, however MgO has the ability to immobilize heavy metals in contaminated soil. In addition, magnesite is used in manufacturing refractory products [57]. The reduction of CO₂ emissions for unfired clay bricks were estimated 9.96 kg CO₂-e per fu (functional unit) [50].

2.5.4. Recycled Aggregate Concrete

In addition to crushed concrete, recycled aggregate concrete (RAC) consists of materials such as bricks, metals, tiles, and other materials including plastic, wood, glass, and paper [4]. RAC has inferior durability and mechanical properties as compared to conventional concrete. However, desirable RAC properties can be obtained by using admixtures such as silica fume, GGBS, fly ash, and meta-kaolin, and by modifying mixing procedures [4]. In a study, it was found that RAC together with industrial wastes such as fly ash, silica fume, blast slag, etc. can improve concrete's carbon footprint and provide great benefits [58]. Chetumal Institute of Technology in Mexico studied the influence of the fine and normal/recycled coarse aggregates on carbon footprint [59]. The result showed that recycled coarse aggregates contributes 39% of CO₂-e, fine aggregate contributes 19% of CO₂-e and normal coarse aggregate contributes 42% of CO₂-e. The study concluded that increasing recycled aggregates may help reduce 22,343 tons of CO₂-e annually in the region of Mexico alone. A study found that 100% reclaimed and recycled aggregates, which is called Pixelcrete, reduced the content of Portland cement (60% less than the conventional concrete) in office building, and led to 73.8 t-CO₂-e reduction in GHG emission [60].

2.5.5. Recycled Asphalt

Reclaimed asphalt pavement (RAP) is used to describe re-used asphalt containing pavement. In 2010, 62.1 million tons of RAP was used in asphalt pavements [61]. The

RAP could be used in three different categories of production: either as hot mix asphalt, or cold mix asphalt, or as aggregates. The RAP is generated through removal of asphalt pavement by either milling the surface using a milling machine or full depth removal. The recycling process includes both hot and cold mix asphalt and can be done in recycling plant or in place [62]. In a study, it was found that virgin asphalt produces 132 kg CO₂-e per t. In addition, 1/3 of this value was contributed by the energy intensive processes such as heating and drying; therefore, several studies were carried out to determine the factors that affect the reduction of CO₂ emissions including the RAP [28]. It was reported that RAP mix resulted in 5.5% reduction of carbon content, and it enhanced the reduction by 14% when larger aggregates sizes were used. By using RAP, embodied carbon content dropped to an average of 84.35 kg CO₂-e per t [3].

3. Carbon Footprint of On-Site Construction Processes and GHGs Reduction

Carbon footprints are resulted during manufacturing, transportation, and installation processes of ground foundation, wood/steel/concrete framed construction at on-site construction activities. The amount of CO₂ released from a concrete-steel residential tower in the Tehran Metropolitan City was 13,076,390,236 kg CO₂-e, and the amount of CO₂ emissions in 1 m² of Gross Floor Area (GFA) was 435,879.67 kg CO₂-e/m², of which 83% was related to the emissions from transportation of materials and 14% was related to construction wastes and 3% was related to on-site construction process [20]. A prefabricated wood-frame multi-story building in Quebec City produced a total of embodied carbon emissions of 275 kg CO₂-e, which was 25% less than traditional buildings built with steel or concrete. The fabrication phase of building material contributed the most (75%) to the carbon emissions, while transportation (13%), construction (1%), and waste management (11%) contributed 25% [50].

A study found that the embodied carbon of a 3-bedroom semi-detached house constructed using offsite panelized timber frame was approximately 35 t-CO₂ (82% of the total embodied carbon is embodied in the materials incorporated in the building, 2% of the total embodied carbon resulted from transportation of the materials from point of distribution to site and the rest resulted from waste materials exported from the site and energy used onsite), and an equivalent home constructed using traditional masonry construction was 52 t-CO₂. Using modern methods in construction resulted in a 34% reduction in embodied carbon [63]. The overall CO₂ emissions from the 1008 m wastewater pipeline project in China were calculated in tons through the entire construction period; the results were found to be 452.81 tons, 61.32 tons, and 6.59 tons from transportation phase, material manufacturing phase, and installation phase, respectively [64]. The global warming and energy consumption of 1 m² of hoarding construction using large amounts of steel products and concrete in the construction site resulted in 3 tons of CO₂ eq GWP and 39 GJ of non-renewable energy consumption [65]. Another study showed that home building with ready mix concrete results in 40% less CO₂ emissions and less fuel consumption per lot by changing concrete slab size from 3000 ft² to 1500 ft². In addition, choosing the closest ready mix concrete plant saves 46 gallons of diesel and eliminates 1020 lb of CO₂ emissions per lot in Greater Phoenix Arizona area [66].

Enhancement of energy efficiency and optimization of construction machines can reduce direct carbon emissions in construction industry [67]. Oil and electricity consumption during the on-site construction contribute to carbon footprint of construction industry. According to this study, the sources of CO₂ emission from the on-site construction are as follows: reinforced concrete work produced 44.1 t-CO₂ (23.9% of the total CO₂ emissions), earthworks produced 39.1 t-CO₂ (21.2% of the total CO₂ emissions), ground heat construction (close loop) produced 31.9 t-CO₂ (16.7% of the total CO₂ emissions), foundation work (PHC PILE) produced 26.7 t-CO₂ (14.4% of the total CO₂ emissions), and ground heat construction (open loop) produced 16.6 t-CO₂ (8.5% of the total CO₂ emissions) of 84.6% of the total CO₂ at the on-site construction phase. Furthermore, electricity consumption of concrete works on-site accounts for 41.9% of the total electricity used during the construction,

resulting 14.1% (13,279 kWh) of the total electricity usage during building operations [21]. A case study has shown that on an average 99.8% of carbon present in the fossil fuel consumed by an excavator is released into the atmosphere as CO₂ [5,67]. Additionally, emission factors during idling times contribute to overall average emission factors.

A study showed that the total CO₂ emission increased during engine idling of non-road diesel construction equipment was considered although during the idle the time fuel use and CO₂ emissions are between 1/3 to 1/5 of the non-idle time. During idling time, 2.7 kg CO₂/liter was produced at a diesel fuel consumption rate of 03.7 L/h [68]. According to the EPA (2005), operators should take the equipment needs into consideration, including the time required for warm-ups and cool-downs. An operational efficiency system that is commonly accepted and used to estimate equipment productivity is 50 min = h (83%), which indicates 50 min of non-idle time and 10 min of idle time per hour. Equipment such as backhoes and bulldozers have equipment productivity ranging from 80% to 85%. However, off-road trucks have equipment productivity of 41% considering that a large part of their time is spent cycle idling, mainly loading and offloading of cargo. If off-road truck average operational efficiency increased from 40% to 50% by reducing idle time by only 6 min/h, the hourly fuel use and CO₂ emissions can be reduced 10% [68].

A case study of a construction project in USA involved a roadway construction of an 18.8-mile highway requiring 184 pieces of machinery categorized into 35 equipment types, with idle time assumed to be 6 h per day for 7 days per week for this machinery. It was shown that the net total emission was 179,055 Mt-CO₂-e during a period of 2.5 years (71,609 Mt-CO₂-e per year), of which 40,023 Mt-CO₂-e/km was contributed by the constructed roadway [69]. Amount of CO₂ resulted from idling time can be reduced using different technologies such as direct-fire heaters, auxiliary power units (APU), thermal storage systems, on-board batteries, and automatic engine shut-off devices [70]. According to a study, direct fired heaters can reduce NO_x and CO₂ emissions by 99% and 94–96%, respectively, since heat is transferred directly to the heat exchanger from the combustion flame resulting in less fuel usage than diesel engines [71].

4. Carbon Footprint of Construction and Demolition Waste Generation and GHG Reduction

Construction demolition waste (CDW) stems from construction, renovation, and demolition workplaces which include (i) excavation materials, (ii) road building and maintenance materials, (iii) demolition materials, and (iv) other worksite waste materials, (e.g., unpainted, non-treated wood scrap, unpainted, non-treated wood pallets, plastic, packaging), land clearing, and development activities [72]. Construction waste is increasing in volume and affecting the environment adversely [73]. Over 80% of CDW is composed of excavated earth in construction works. Mixed CDW contains the remaining of materials and packaging. [74] A 3-bedroom modular timber frame semi-detached house with 83 m² internal floor area produced 17 m³ of waste (excavated inert materials, waste and unused construction materials, and other waste) totaling 4.9 t-CO₂ equating to 109 kgCO₂ per m². Timber and packaging contributed to 33% and 31% of the total waste, respectively [63].

When a building reaches the end of its service life, it is demolished; the process is responsible for an emission of 0.004 to 0.01 kg CO₂ per kg of the concrete material. This figure depends on the type of reinforcement and structure used, in addition to the general working conditions on the site during demolition [27]. A situ-concrete type building was being demolished in Korea; it required total energy consumption of 51.5 MJ/m² from diesel fuel to demolish it; thus, the level of CO₂ emitted during demolition was 10.3 kg-CO₂/10 m². In consideration of the CO₂ that is emitted during the transportation of the demolition debris, 24.4 Kg-CO₂/10 m², 26.3 kg-CO₂, and 17.6 kg-CO₂ were obtained for a single-family house, a flat, and a multi-family house, respectively [75]. Waste transportation consumes energy which leads to CO₂ emission. According to study, during the construction period, 530 tons of waste generated and during the transportation of this waste 527 L of diesel oil consumed totaling 1.4 t-CO₂ emission from the waste transportation phase [21].

Waste materials generated from the construction industry (concrete and concrete rubble, construction ceramics, timber and wood, glass, plastics, steel, iron, aluminum, excavated soil, and Styrofoam) or from general life can be recycled as alternative construction materials [61]. During demolition, interior finishing from buildings can be reused or recycled. To look after the environment and determine the recycling and reuse values of CDW, the waste management must be planned via volume and composition determination [76]. Concrete blocks can be crushed so that they can be used for landscaping or landfilled. The fiber generated from the carpet waste can be used in fiber reinforced concrete (FCR) and fiber reinforced soil as well. The fiber improved several mechanical properties of the concrete such as toughness, strength in tension, fatigue strength, and durability, while it reduced possible cracks and defects [77]. Waste materials can act as substitutes of concrete components; it is estimated that plastic and glass can replace fine aggregates in concrete mixes by up to 20%, while waste concrete could make up for 20% of the coarse aggregate mixes in concrete [78].

Recycling one kg of aluminum as building demolition waste can contribute to emission reduction of 20.07 kg CO₂-e [79]. Demolition debris that contains steel is separated so that the steel can be sold to scrap dealers. The economically not valuable waste can be sent to dump sites [80]. When the waste steel from hoarding construction is recycled as steel scraps, 281 kg CO₂-e/m GHGs emission can be reduced [65]. New asphalt can be used from asphalt removed from road that is refurbished. The landscaping clearing wastes can be used as well. A portion of waste glass can be used in place of fine aggregate in asphalt paving mixtures (glassphalt) [81]. Reusing wood waste in production of particleboard reduced embodied carbon emissions up to 14.6% (−28.6 kg CO₂-e/m²) [50].

5. Carbon Footprint during Operational Stage and GHGs Reduction

Over the full cycle, building operations contribute to the CO₂ balance when in service [82]. Carbon emission during operational stage of a building was a major contributor, accounting for 85.4% of the total emission followed by the construction stage which accounted for 12.6% of total emissions [83]. A high-rise residential housing block in Hong Kong demonstrated that GHG emission was estimated to about 213.03 t-CO₂-e/flat and 4980 kg CO₂-e/m², of which 85.82% was stemming from the operating energy, 12.69% from materials, 1.14% from renovation, 0.28% from end-of-life of the building, and 0.07% from other factors [84]. The energy consumption per area of the buildings from urban, national, and global scales are 3.03 GJ/m², 4.27 GJ/m² and 0.44 GJ/m² which correspond to 0.40 t-CO₂-e/m², 0.14 t-CO₂-e/m² and 0.04 t-CO₂-e/m² greenhouse gas emissions, respectively, based on hybrid systems analysis combining input–output analysis and process analysis in China [85].

In order to contribute to CO₂ reduction, new technologies were implemented in buildings. According to a study, low-carbon strategies, such as increased energy efficiency design for new buildings and energy-saving retrofit for existing buildings would decrease energy consumption by 2.98% with a carbon emission reduction of 3.15 million t-CO₂-e [22]. Choosing correct materials, systems, and technologies which are listed in following sections at the phase of design and materials selection, will reduce energy consumption and CO₂ emissions during operational stage of the buildings.

5.1. Alternate Water Resources for Water Reuses

Reusing water in a typical office building is estimated to conserve about 75% of the indoor potable water [86]. The average water saving of a green building was estimated to reach 37.6% with applying water efficiency technologies [87]. The rise of the water savings will reduce energy consumption and CO₂ emissions [88]. The passive irrigation system has two stages: collecting water when it rains and supplying water in drought conditions [89]. Water flow in the system is natural under gravity or capillarization method [90]. A 250-room hotel in Birmingham, UK, with the rainwater recycling system saved up to 780 m³ of potable water per year [91]. According a comparative simulation

model, gravity fed rainwater harvesting system for a high rise building in Mexico saved up to 8.5% of GHG [92]. Graywater is the water produced by bathroom, laundry machines, sinks, showers, and bathtubs [93]. Treated graywater can be reused for landscape irrigation and toilets [86].

Efficiency of water use can be improved by graywater recycling systems for flushing of toilets by dual piping, which will contribute to reducing urban water demand from 10% to 25% [94]. NH Campo de Gibraltar hotel substitutes 20% of potable water with filtered and treated grey water from showers, which resulted in a 20% reduction in annual water bill [91]. Blackwater comes from toilets and kitchens. Blackwater reuse showed a positive response from people who used automated or remotely controlled systems by the installer. Another study reported that it is costly and has poor process design [95]. Condensate recovery reuses water produced by air conditioning (AC) systems [95]. AC condensate can be used in flushing toilets, irrigation, cooling towers, roof cleaning, green roofs, and spray cooling [96].

Examples of water reuse and alternative water supplies include water conserving toilets, waterless toilets, waterless urinals, alternative shower and faucet fixtures (alternative controls, self-powering, low flow), water efficient appliances, and alternative landscaping (high efficiency irrigation, water conserving plant selection) [95]. Some statistical studies showed that water technologies increase water efficiency. For instance, urinals and commercial dishwashers showed the greatest reductions of water use, while showers and commercial toilets showed the least savings [88]. In the same manner, wastewater centralized reuse system (WWCRS) require more energy for treatment which leads to higher CO₂ emissions, while the greywater decentralized reuse system (GWDRS) requires less energy (11.8–37.5%) than WWCRS consumed [97]. A constructed wetlands system treats wastewater in a building so that it can be used in low-flow toilets and urinals, which reduces the water use in total by a percentage higher than 60% [86].

5.2. Heating, Ventilation and Air Conditioning

Heating, ventilation, and air conditioning (HVAC) systems of buildings consume about 40–60% of total energy taking into consideration the embodied energy which stems from the production of the building [98]. Owing to their large thermal mass, concrete and other heavy weight materials positively impact the energy consumption of buildings; for a heavy weight building (based on concrete frame), energy needed for heating/cooling/ventilation is 10 MJ/m² resulting in 1.3 CO₂/m²; and for light-weight building, (based on plaster boards stud walls), it is 20 MJ/m² resulting in 2.6 CO₂/m² in Northern Europe [27].

Equipment sizing and selection, pipe/duct sizing, energy performance analysis, system optimization, real-time performance optimization, control analysis, control optimization, and simulation and programming for HVAC systems can reduce energy consumption and increase the comfort of residents [99,100]. According to a study, using a high energy performance air conditioner resulted in 7664.4 t-CO₂-e reduction in an office building in Nanhaiyiki 3, China; 451.5 t-CO₂-e reduction in a Pixel building in Australia during the life cycle of the buildings. In the same fashion, using natural ventilation and lighting resulted in a 5687.6 t-CO₂-e reduction in Nanhaiyiki 3, China; 4649.8 t-CO₂-e reduction in the Pixel building in Australia during the life cycle of the buildings [60].

5.3. Other Building Systems and Technologies

There are various technologies and systems that can be applied to enhance the efficiency of buildings and decrease CO₂ emissions. Such innovations include: windows and building surfaces with tunable optical properties; high-efficiency heat pumps; highly efficient lighting devices; thin insulating materials; improved software for analyzing building design and operations; inexpensive, energy harvesting sensors and controls; optimized control strategies; and interoperable building communication systems [101]. A study was conducted to compare different systems in a building, and it found that systems like in-

terior lights (−150%), mechanical ventilation (−25%), and pumps (−11%) had the least energy savings whereas systems like interior fans (100%), heat rejection units (56%) and receptacle equipment (33%) had the highest energy savings. The negative values show that the systems are less efficient when compared to the baseline [82]. In another study, it was found that using renewable energies such as a solar photovoltaic system, wind turbine, and anaerobic digester resulted in 1204.1 t-CO₂-e reduction in an office building in Australia, and using renewable energy such as a solar photovoltaic system, a solar thermal water system, and a ground source heat pump resulted in 2871.6 t-CO₂-e reduction in an office building in China during the life cycle of the buildings [60]. Expanded polystyrene (EPS), cellulose, and elastomer as insulation and sealing materials resulted in an average 3.5 kg CO₂-e/kg emission, some insulation materials such as sheep's wool could reduce its impact up to 98% [50].

6. Discussion

Globally, in the developed and developing countries, the whole process of construction and building operations contributes to 33% of greenhouse gas (GHG) emissions and 40% of global energy consumption, stemming from the usage of the equipment, transportation, and the manufacturing of building materials. The urban population is increasing, which leads to more construction in the future and increased GHGs emissions [6]. Therefore, new policies are required for mitigation of GHG emissions. Regulations such as building codes can effectively reduce GHG emissions if enforced well enough and can ensure new buildings incorporate designs that are both cost and energy effective. However, regulations alone can result in extra costs for the governments, and they should be designed to cover all aspects of GHG emission activities [7]. Moreover, this policy instrument has been widely criticized for being inflexible, complex and for not taking into consideration differences in technology and geography [102].

On the other hand, a carbon tax is simpler to design, has relatively low administration costs and is attractive to stakeholders in the building sector due to their familiarity with the tax mechanism. The revenues earned from carbon tax can be redistributed to other policy instruments such as incentives [7]. However, establishing an appropriate tax rate can be a challenging task for governments as it involves complete knowledge of costs of mitigation, the growth of the economy, progresses in technology and other factors which need to be taken into consideration. Moreover, due to opposition from the public and also to avoid pressuring the construction industry intensively, governments could also face problems in establishing a deterring tax rate that can reduce GHG emissions [102].

The cumulative amount of GHG emissions mitigated can be quantified with ETS and emission permits can be distributed for free or auctioned off. However, there are concerns of market failures and regulatory based loop holes because the construction sectors lacks proper GHG accounting [12]. It is necessary to move beyond the debate of policy instruments in order to be able to pinpoint the factors that are actually slowing the move to a carbon neutral construction industry. One of the common cited barriers to carbon reduction schemes in the construction industry is the incremental cost associated with it [103,104]. Studies have shown that building contractors and developers often overestimate the cost associated with energy efficiency [105]. For example, in Germany, new buildings with very little heating requirements can be constructed with an extra cost of no more than 5–12%, while, in Northern China, a building project was able to achieve reductions of 65% in heating consumption with an extra cost of no more than 8% without compromising thermal comfort [103,106]. Therefore, correct estimations are important for cost estimations.

Other cited barriers to carbon reduction in construction industry were the skills and knowledge gap of not only the designers and contractors but also of the end users, i.e., the occupants of the buildings [104,107]. As a conclusion, each instrument has some limitations; therefore, a variety of economic, environmental, political, and social factors need to be taken into consideration [7]. Training and education should be emphasized as important ways to

reduce GHG emissions in the construction industry by enabling behavioral changes within organizations. In this context, identifying sources of the carbon footprint at the different stages during construction and showing possible carbon reduction technologies/systems and techniques as summarized in Table 4 will be helpful for awareness and to fulfill the knowledge gap at the design stage from clients to designers and contractors.

Table 4. Summary of findings.

Building Operations	CO ₂ Emission	Reduction Material/Techniques	CO ₂ Reduction	References
Limestone quarrying	3.13 kg CO ₂ -e per ton crushed rock product	Application of alternative/ renewable energy such as solar thermal and biodiesel as compared to acquiring energy needs for quarrying from the grid or natural gas	More than 81% reduction in GHG emissions annually	[24]
Portland clinker manufacturing	nearly 1 kg of CO ₂ per one kg of Portland clinker ^(b)	Alternative clinker substitution—use of calcium carbide residue in replacement of limestone partially	374 kg CO ₂ /ton of clinker annually, or more than 37% reduction in CO ₂ emissions per ton of clinker	[26,39]
Asphalt	0.05 ppm of CO ₂ per ton per year for conventional asphalt production	Sasobit additives with Warm Mix Asphalt	0.003 ppm to 0.004 ppm of CO ₂ per ton or more than 94% reduction in CO ₂ emissions	[38,51]
		Sasobit additives with Hot Mix Asphalt	0.005 ppm to 0.0054 ppm of CO ₂ /ton, or more than 90% reduction in CO ₂ emissions	[38]
Concrete	132 kg CO ₂ equivalent /ton of virgin asphalt produced	Reclaimed asphalt pavement	Dropped to average of 84.35 kg CO ₂ equivalent/ton, or more than 36% reduction in CO ₂ emissions	[3,56]
	5 w/c were between 347 and 351 kg of CO ₂ -e/m ³	Recycled coarse aggregates	Reduce 0.03 tons of CO ₂ -e/m ³	[56,58]
	293 kg of CO ₂ -e/m ³	Fractional replacement of cement in concrete with fly-ash and ground granulated blast furnace slag and natural aggregates with recycle crushed aggregate	Reductions of up to 3.8% (10.5 kg CO ₂ -e/m ³)	[47]
	(a) During idling, at a fuel consumption rate of 0.84 gal/hour, 2.7 kg CO ₂ /liter was produced	Reducing idling time by using direct fired heaters instead of diesel engines	Direct fired heaters can reduce NOx and CO ₂ emissions by 99% and 94–96% respectively during idling time	[70]
Onsite construction process	Traditional building with steel products or concrete produces 366 kg CO ₂ -e/m ² total embodied carbon emissions	Using prefabricated wood instead of steel or concrete	25% reduction in total GHG emissions	[47,64]
Construction, demolition waste	3-bedroom semi-detached house constructed using traditional masonry construction produces 405 kg CO ₂ /m ²	using offsite panelized timber frame and modern methods of construction	34% reduction in total embodied carbon emissions	[61]
	0.004 to 0.01 kg CO ₂ per kg of the demolition waste	Recycling building demolition waste such as aluminum	20.07 kg CO ₂ -e per kg of aluminum recycled	[26,78]
		Recycling waste steel from hoarding construction as steel scraps	281 kg CO ₂ -e per m, or about 8% reduction in CO ₂ emissions	[64]
		Reusing wood waste into production-use of particleboard	14.6% reduction in CO ₂ emissions (−28.6 kg CO ₂ -e/m ²)	[47]
Building's operations when in service	Account for 85.4% of the total emissions of a building's life cycle	High energy performance air-conditioner	19 kg CO ₂ -eq/m ²	[47,48,82]
		Utilization of renewable energy such as a solar photovoltaic system, solar thermal water system, and a ground source heat pump	4.6 kg CO ₂ -eq/m ²	
		Use of natural ventilation and lighting	9.1 kg CO ₂ -eq/m ²	[26]
		Use of sheep's wool as insulation material in buildings	98% reduction in GHG emissions	
		Applying large thermal mass, concrete, and other heavy weight materials for reduction of HVAC energy	50% reduction in CO ₂ emissions/m ²	[26]
		Rainwater harvesting system	8.5% reduction of GHG	[91]

7. Conclusions

GHG emissions mitigation can be achieved by indirect pricing such as regulations and direct pricing such as carbon tax and emission trading schemes (ETS). However, regulations can be inflexible, complex, and may not take into consideration differences in technology and geography. In addition, ETS can be complex because the construction sector lacks proper GHG accounting. Therefore, increasing the awareness, education, and incentives can lessen the carbon footprint of construction industry. Consequently, we aimed to increase awareness of the carbon footprint sources in construction and building operations during manufacturing, transportation, construction, operations/management, and end-of-life deconstruction. As a result, various carbon reduction techniques/systems were identified. It was found that mining and manufacturing of materials and chemicals contributed to high energy usage and 90% of the total CO₂ emissions. Therefore,

- Testing different blends of cement with addition of alternative additives such as alkali-activated slag mortars or fly ash in concrete;
- Changing cement production methods;
- Addition of Sasobit or reclaimed asphalt pavement in asphalt mixtures;
- Recycling building wastes such as concrete aggregate and recycled asphalt in common construction materials;
- Conversion from the wet process to the dry process in concrete manufacturing;
- Substitution of lower carbon content fuels for coal, coke, and petroleum coke;
- Alternate options in terms of vehicle type, engine power, truck capacity, and fuel type to improve the fuel efficiency in the construction vehicles;
- Reducing idle time by using direct fired heaters, auxiliary power units (APU), thermal storage systems, on-board batteries, and automatic engine shut-off devices;
- Applications of alternate water resources for water reuse purposes;
- Switching to efficient HVAC systems; and
- Utilization of different building operations/systems will lessen energy consumption and reduce GHG emissions up to 90% in different stages in construction industry.

This review can be useful at the stage of conceptualization, design, and construction to assist clients and stakeholders in selecting materials and systems. There is large scope for further research on how to decrease carbon footprint in construction. Some of the areas that require attention include:

- improving recyclable waste materials such as glass, rubber crumbs, etc., as construction materials;
- developing decision making tools for effective carbon footprinting;
- creating inventory databases for Life Cycle Assessment for each alternative material's embodied carbon value.

Author Contributions: Conceptualization, B.S. and Y.F.; formal analysis, B.S. and Y.F.; resources, Y.F., C.-S.C. and I.Y.; writing—original draft preparation, B.S. and Y.F.; writing—review and editing, C.-S.C., I.Y. and Y.-J.B.; project administration, B.S.; funding acquisition, B.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Khalifa University, KUIRF L1 210045 grant.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Acknowledgments: The authors would like to extend gratitude to the following students for their assistance: Tethkar Alhammadi, Madeya Al Mehairbi, Fatmah Aldhanhani, and Fatima Almusharrek.

Conflicts of Interest: The authors declare no conflict of interest.

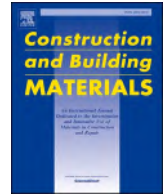
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Effect of warm mix asphalt (WMA) technologies on the moisture resistance of asphalt mixtures

Mayank Sukhija^{a,*}, Nikhil Saboo^b, Agnivesh Pani^a

^a Department of Civil Engineering, Indian Institute of Technology (BHU), Varanasi, India

^b Department of Civil Engineering, Indian Institute of Technology Roorkee, India

ARTICLE INFO

Keywords:

Moisture damage
Warm mix asphalt
Hot mix asphalt, Bond strength
Tensile strength ratio
Bond strength ratio

ABSTRACT

Moisture damage has been identified as a major concern for asphalt pavements. This concern becomes more critical when the asphalt mixtures are prepared at lower mixing and compaction temperatures using technologies such as warm mix asphalt (WMA). In this study, the effect of different WMA technologies on the moisture resistance of asphalt mixtures was studied using a variety of experimental investigations including bond strength ratio (BSR), percentage coating, retained Marshall stability (RMS), and tensile strength ratio (TSR). Two aggregate sources (granite and dolomite), two base asphalt binders (VG30 and PMB40), and five WMA additives categorized under organic, chemical, and foaming technologies were used. Moisture damage in WMA mixtures was found to be a function of aggregate type and base asphalt binder. Rediset, a chemical WMA additive, showed higher moisture resistance in comparison to other technologies. Appreciable correlations were found between BSR and TSR (R^2 ranged from 0.61 to 0.72). This study suggests a minimum BSR of 0.7 for satisfactory performance against moisture damage.

1. Introduction

Moisture damage is one of the premature and complex forms of failure common in asphalt pavement [1,2]. Asphalt binder, in an asphalt mix, holds the mineral aggregates together and acts as an adhesive sealant against moisture [3]. Various mechanisms have been proposed to explain the occurrence of moisture damage [4]. Broadly, the ingress of moisture in an asphalt mixture leads to the occurrence of cohesive failure (failure within the asphalt binder) and/or adhesive failure (bond failure between asphalt binder and aggregate) [5]. Identifying these failure modes in an asphalt mixture is crucial to understand the compatibility between the mineral aggregates and the asphalt binder, and assessing the effect of asphalt binder modification, if any, on the strength characteristics. Water ingress often breaks the adhesive bond between the asphalt binder and aggregates, resulting in the separation/removal of the asphalt binder film from the aggregate surface [6]. This phenomenon is generally referred to as stripping (or adhesive failure). In cohesive failure, though the bond between the aggregate and bitumen is not broken, the thickness of the asphalt binder film reduces due to failure within the bitumen structure [7,8]. With time, further reduction in the asphalt binder film will expose the aggregate surface

and damage the asphalt mixture [8].

In addition to the individual material properties and interaction between them, the production temperature of the asphalt mixture also influences its performance against moisture [9]. During production, the mixture of aggregates and asphalt binder is heated to high temperature (typically greater than 150 °C), which removes the residual moisture from the surface of the aggregates and facilitates improved bonding between the two. There is a belief that technologies that helps in the reduction of production temperature of asphalt mixture, for example, warm mix asphalt (WMA), may lead to improper drying of aggregates, thereby increasing the susceptibility of the mix to moisture damage. WMA is produced by adding certain additives in the asphalt binder, or the asphalt mixture [10,11]. These additives can be broadly divided into three categories: a) Organic based, b) Chemical based, and c) Foaming based [12]. Organic and chemical based additives are usually added to the virgin asphalt binder, while foaming additives can be added to the asphalt binder, or directly to the asphalt mixture [13]. These technologies finally improve the workability of the mixture, and are able to provide binder coating, consistency, and compaction characteristics similar to the HMA, but at reduced temperatures [14]. Organic based additives generally reduce the viscosity of the base binder, which leads

* Corresponding author.

E-mail addresses: mayanksukhija.rs.civ18@itbhu.ac.in (M. Sukhija), nikhil.saboo@ce.iitr.ac.in (N. Saboo), agniveshpani@gmail.com (A. Pani).

<https://doi.org/10.1016/j.conbuildmat.2023.130589>

Received 21 November 2022; Received in revised form 3 January 2023; Accepted 28 January 2023

Available online 7 February 2023

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Table 1

Optimum dosage of WMA additives and corresponding mixing and compaction temperatures of HMA and WMA mixtures.

Base Bitumen Binder	WMA Additives	Dosage depending on Aggregate Type	
		Granite (MT, CT)	Dolomite (MT, CT)
VG30	No additive (VG30)	0 % (163, 148)	0 % (163, 148)
	Sasobit (S)	3 % (146, 128)	2 % (148, 132)
	Sasobit Redux (SR)	1.35 % (144, 120)	1.35 % (143, 129)
	Cecabase (C)	0.5 % (146, 111)	0.2 % (148, 133)
	Rediset (R)	0.4 % (152, 133)	0.6 % (143, 130)
	Aspha-min (Am)	0.3 % (147, 119)	0.3 % (139, 123)
PMB40	No additive (PMB40)	0 % (175, 153)	0 % (174, 166)
	Sasobit (PS)	2 % (162, 138)	2 % (163, 154)
	Sasobit Redux (PSR)	1.35 % (158, 135)	1.35 % (161, 151)
	Cecabase (PC)	0.5 % (152, 129)	0.5 % (161, 152)
	Rediset (PR)	0.6 % (150, 127)	0.6 % (159, 150)
	Aspha-min (PAm)	0.3 % (156, 134)	0.3 % (159, 150)

Note: MT and CT refer to mixing and compaction temperatures in °C.

to improvement in workability. They are waxes or fatty amides, which melts in the temperature range of 80–120 °C [15]. Below the melting point they crystallize and form a lattice structure, which helps in improving the stiffness of the base binder [16]. Chemical additives are mostly surfactants, emulsifiers, polymers, or hybrid products, which reduces the frictional forces between the binder and mineral aggregates, thereby facilitating appropriate coating at reduced production temperatures [17]. In foaming technologies, water is added either to the hot asphalt binder using artificial nozzles, or to the asphalt mixture in the form of zeolite. They temporarily increase the volume of bitumen, which helps in coating the aggregate particles and improves the workability of the mixture [18,19].

In the literature, various test methods are available for assessing the resistance of the asphalt mixture against moisture damage. Mehrara and Khodaii [20] and Kakar et al. [21] have comprehensively reviewed these test methods. In general, the laboratory evaluation of moisture damage can be grouped into two test domains: (a) test on loose asphalt mixture or mixture components, and (b) tests on compacted asphalt mixture. While a variety of test methods have evolved over a period of time, based on the popularity and ease of testing under each category, some

typical test methods can be identified. For loose mixture, boiling water test (BWT) and pneumatic adhesion test (PAT) are more commonly used. The results from these test methods are interpreted in terms of percentage coating (of the asphalt binder over the mineral aggregates) and bond strength ratio (BSR), respectively. For evaluating the moisture damage in compacted asphalt mixture, tensile strength ratio (TSR) and retained Marshall stability (RMS) values are typically adopted. The details of these test procedures and the associated parameter(s) for quantification of moisture damage are discussed in later sections. Except PAT, none of these test methods can differentiate between the occurrence of cohesive and/or adhesive failure. However, there are no available specifications on the minimum desired value of BSR. Therefore, there is a need to establish a specification limit for BSR; this is attempted in the present study using the test results.

Several studies [11, 17–19] have reported the effect of moisture damage in WMA mixtures using these test methods. Mirzababaei [22] carried out boiling water test (BWT) on HMA and WMA mixtures and stated that aggregate source and gradation significantly influence the resistance against stripping. WMA mixtures were found to exhibit lower stripping than the control mixes. On the contrary, Ai et al. [23] reported unsatisfactory performance of WMA mixtures against stripping resistance. In BWT, more than 80 % stripping was observed in WMA mixtures. It has to be noted that different additives were evaluated in both the preceding studies. Therefore, it can be inferred that the stripping resistance of WMA mixtures is a function of the WMA technology. Yousefi et al. [24] and Zou et al. [25] also stated that moisture susceptibility in WMA mixtures depends on the type of WMA technology. Habal et al. [26] studied the moisture resistance of three different WMA additives with two aggregate substrates using a pneumatic adhesion test

Table 2

Mix design attributes of conventional HMA mixtures.

Binder Type	VG30		PMB40		Specification
Aggregate Type	Granite	Dolomite	Granite	Dolomite	Limits [40]
OBC, %	5.8	5.6	6	5.9	Min. 5.4
G _{mb}	2.407	2.408	2.357	2.403	–
G _{mm}	2.509	2.506	2.460	2.507	–
Air voids, %	4.07	3.91	4.19	4.13	3–5
VMA, %	14.05	13.83	15.58	14.27	Min. 12
VFB, %	71.07	71.73	73.13	71.05	65–75
Stability, kN	14.86	14.42	17.76	17.49	>9
Flow, mm	3.60	4.13	4.15	3.76	2–4

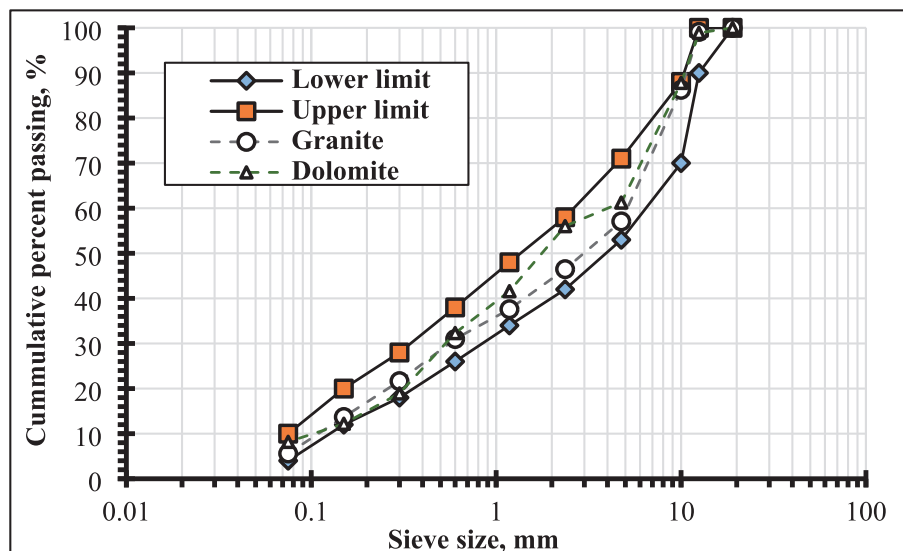


Fig. 1. Aggregate gradation chosen in the present study.

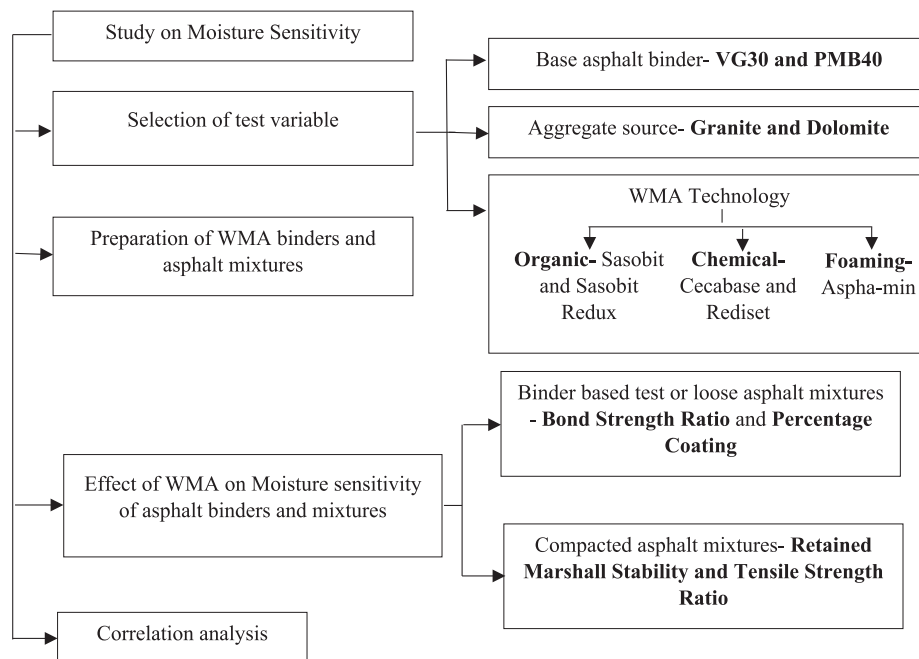


Fig. 2. Research framework.

(PAT). Results indicated that Sasobit improved the moisture resistance when bonded with limestone aggregates, whereas asphalt binders prepared with Rediset and Advera performed relatively similar to the base asphalt binder. In the case of granite as the aggregate source, Advera showed superior performance against moisture damage, followed by Rediset and Sasobit. Thus, it can be stated that the moisture sensitivity of WMA depends on the combined effect of warm mix technology and aggregate source. Experiments conducted by Roy et al. [27] revealed that the presence of residual foaming water (in foaming technologies) at the interface of asphalt binder and aggregate matrix erodes the asphalt binder and weakens the bond strength. Liu et al. [28] found that addition of warm mix additives increases moisture resistance in terms of tensile strength ratio (TSR). The addition of 3 % Sasobit increased the value of TSR from 79 % (control mixture) to 86 %. On the other hand, Rahman et al. [29] showed that WMA mixtures are more prone to moisture damage in comparison to HMA mixtures. The reason behind the poor moisture resistance of WMA mixtures is speculated to be the improper drying of the mineral aggregates which leads to an inadequate coating of WMA binder over the surface of the aggregates [30]. Interestingly, few studies [31,32] have reported that WMA (prepared with Aspha-min and Evotherm) and HMA mixtures show similar performance. WMA mixtures, in a few cases [33,34], failed to meet the minimum threshold requirement of moisture resistance. From the literature review it can be concluded that, in general, foaming-based WMA technologies show poor resistance against moisture damage, whereas the inclusion of chemical and organic additives improves the moisture resistance of the asphalt mixture.

2. Research gaps and objectives

While several studies [35–38] have demonstrated that the performance of WMA (produced using different technologies) in terms of workability, and rutting and fatigue characteristics are comparable to (or even better than) HMA, performance in terms of moisture resistance has not been explored to a satisfactory extent. Comparing the results from different available test methods is crucial to identify the sensitivity of the test method on the ranking of different asphalt mixtures and to understand the effect of the test methods in quantifying the occurrence of cohesive and adhesive failure in asphalt mixtures. Moreover, results

on moisture resistance of WMA, considering the effect of base binder (used to prepare WMA), aggregate type, and test methods used for quantification of moisture damage, results on moisture resistance of WMA are not available. This forms the motivation for the present work.

The present study intends to cover the following objectives:

1. To study and compare the effect of different WMA additives on the moisture sensitivity of asphalt mixtures using different test methods.
2. To define specification limits of BSR based on its correlation with other parameters.

3. Materials and mix design

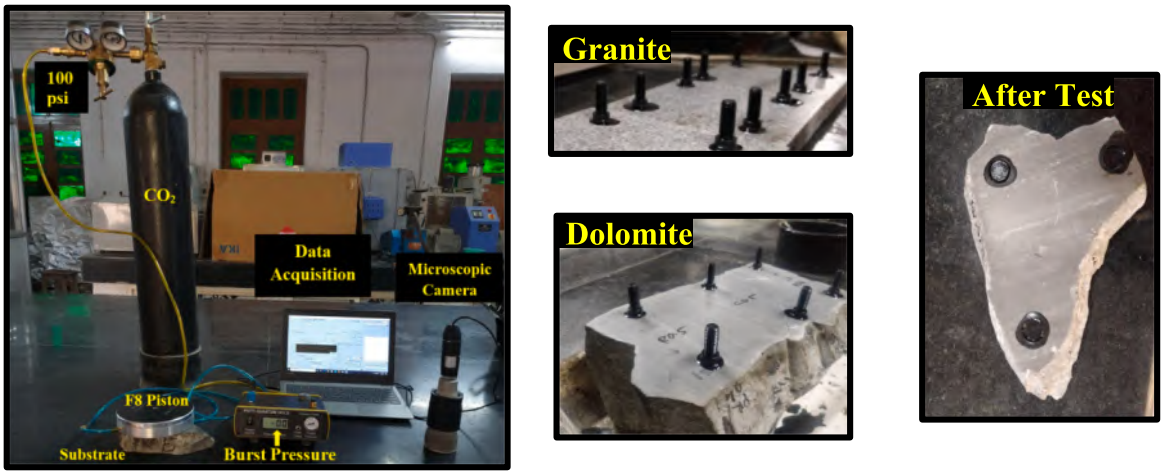
3.1. Asphalt binders and mineral aggregates

Two base asphalt binders, viscosity graded (VG), VG30, and polymer modified binder (PMB), PMB40, were used in this experimental study. Two different aggregates sources (granite and dolomite) were used to prepare asphalt mixtures.

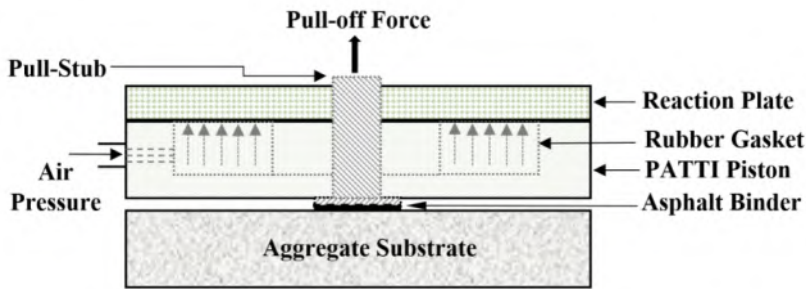
3.2. WMA additives

Five different WMA additives categorized under various WMA technologies (organic, chemical, and foaming) were adopted in this study. These included Sasobit (S- 1 %, 2 %, 3 %), Sasobit Redux (SR- 0.7 %, 1.35 %, 2 %), Cecabase (C- 0.2 %, 0.35 %, 0.5 %), Rediset (R- 0.4 %, 0.5 %, 0.6 %), and Aspha-min (Am- 0.3 %). All the WMA additives, except Am, were added by weight of asphalt binder. The quantity of Am was added by weight of the total mix, as per manufacturer's recommendations.

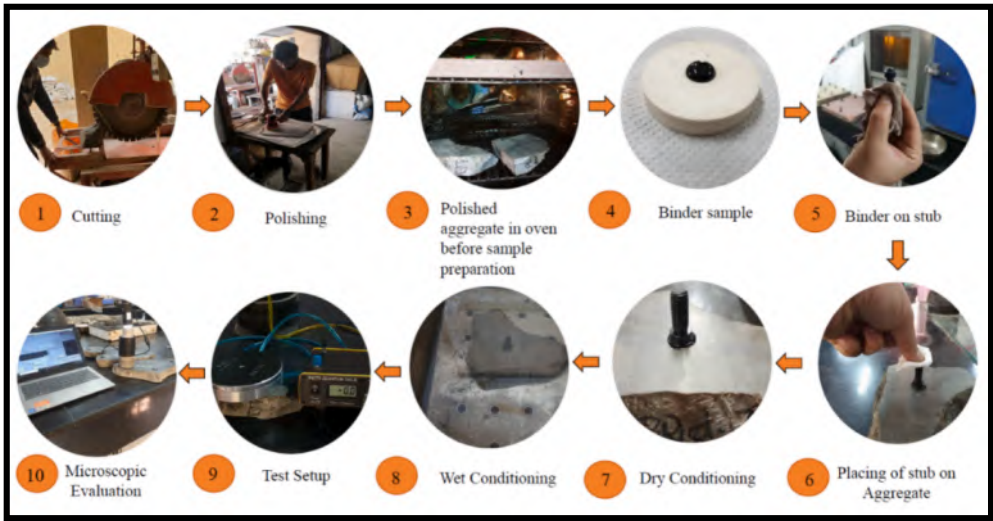
WMA binders were prepared by adding the WMA additive to the base asphalt binder using a high shear mixer operated at 500 rpm. Before adding the WMA additive, the base asphalt binder was heated to attain sufficient fluidity. In the case of VG30, mixing was done for 30 min under a controlled temperature range of 130–140 °C. For PMB40, mixing was done in two stages. Firstly, PMB40 was stirred for 15 min to ensure homogenous dispersion of the polymer within the asphalt binder. This was followed by addition of WMA additive and blending for 30 min at a temperature range of 140–150 °C. Control binders (base asphalt



(a)



(b)



(c)

Fig. 3. Bond strength mechanism (a) Pneumatic piston setup with different testing components, (b) Schematic representation of BS tester, and (c) Stepwise procedure to determine BS and BSR.

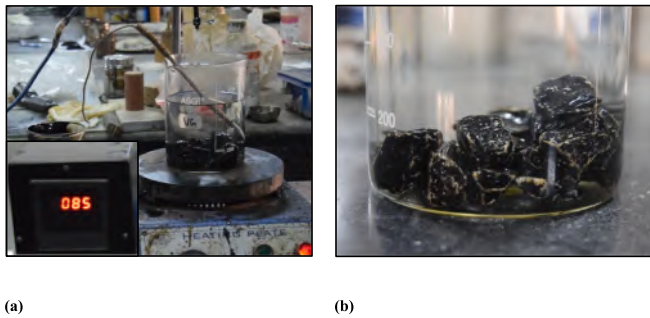


Fig. 4. Boiling water test (a) Boiling of the loose asphalt mixture at 85 °C, and (b) Representation of stripping of asphalt binder from the aggregates after BWT.

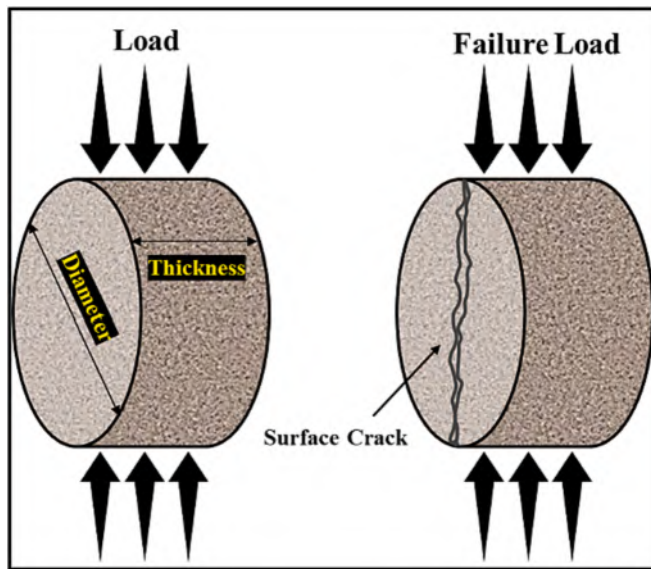


Fig. 5. ITS test loading representation.

binders) were also subjected to a similar heating procedure for an appropriate comparison of results. It should be noted that since Asphamin was directly added to the asphalt mixture, preparation of a modified binder was not required.

Further, an experimental program was designed to assess the appropriate mixing and compaction temperature of WMA mixtures. A novel workability-based approach was used in this direction. The details of the developed procedure have been published elsewhere [38] and are not discussed here for brevity. Next, the optimum dosage corresponding to each WMA additive was determined by comparing the air voids of WMA and HMA mixtures. The minimum dosage of WMA at which the air voids in the WMA mixture are similar or within 10 % variability of the air voids of HMA was considered as the optimum dosage. In this study, Marshall mix design was done (as per MS-2 [39]) on bituminous concrete (BC-2) gradation, which is a typical dense-graded wearing course used in Indian highways. The optimum dosage of different WMA additives and the mixing and compaction temperature of WMA (at the optimum dosage) and HMA mixtures are presented in Table 1. The aggregate gradation of BC-2 and the volumetric details of asphalt mixtures produced at the optimum dosage are presented in Fig. 1 and Table 2, respectively. It should be noted that the optimum WMA additive dosage is a function of aggregate source (granite and dolomite), base asphalt binder (VG30 and PMB40), and WMA type (S, SR, C, and R). Notably, the manufacturer recommended a single dosage of Am; thus, the same dosage was considered as the optimum, irrespective of the aggregate source and base asphalt binder. For each test, three replicates

per sample were used, and the average results are shown here.

3.3. Experimental design

In this study, different experiments were conducted to understand the effect of WMA on the moisture sensitivity of the asphalt mixture. These test methods included the pneumatic adhesion test (PAT), boiling water test (BWT), retained Marshall stability (RMS), and tensile strength ratio (TSR). The research framework is briefed in Fig. 2. The details of these test methods are provided in the subsequent sections.

3.3.1. Pneumatic Adhesion Test (PAT)

PAT was carried out using Pneumatic Adhesion Tensile Testing Instrument (PATTI) to determine the bond strength (BS) between the asphalt binder and aggregate particles. The test setup includes a CO₂ cylinder, pneumatic piston, reaction base plate, pull-out stub, and microscopic camera. Fig. 3a shows the pneumatic piston setup and images of representative test samples used in the present study. The schematic representation of the BS tester is shown in Fig. 3b.

The aggregate substrates were prepared by cutting larger-sized aggregates and smoothening the surface using extra-fine #280 silicon carbide (SiC) grit (average size: 39–45 μm). The prepared aggregate substrates were cleaned and dried using a draft oven at a controlled temperature of 150 °C for a minimum of 30 min as per AASHTO T361 [41]. The oven-dried aggregates and the stub were heated to their respective mixing temperature (as shown in Table 1). The asphalt binder, on the other hand, was also heated and poured into a silicone mold, which was further cooled. The asphalt binder was attached to the preheated stub and pressed against the aggregate (preheated) surface. It should be noted that the pressing process is subjective; therefore, proper care should be taken while pressing the stub on the aggregate substrate (in terms of pressing time and pressure). The prepared arrangement was kept undisturbed for 24 h at room temperature to allow proper adherence between the asphalt binder and aggregate substrate. As per the test procedure, an upward pressure is applied on the pull-off stub to detach the stub from the aggregate substrate (as shown in Fig. 3b). The value of upward pressure, typically known as burst pressure (BP), was used to measure the BS as per AASHTO T361 [41]. Equation (1) was used to calculate the BS between the asphalt binder and aggregate surface. All the samples were tested at a fixed air pressure (100 psi) to eliminate any variability in the results. Failure occurs when the applied force exceeds the adhesive/cohesive strength.

$$BS = \frac{(BP \times A_g) - C}{A_{ps}} \quad (1)$$

where BS and BP are bond strength (kPa) and burst pressure (kPa), respectively; A_g denotes the contact area of gasket combined with reaction plate (mm²); A_{ps} refers to the area of pull stub (mm²), and C is the value of piston constant. The value of A_g , A_{ps} , and C is dependent on the piston configuration and is usually provided by the manufacturer.

The test was carried out on dry and moisture-exposed specimens to study moisture sensitivity. Moisture resistance was assessed in terms of bond strength ratio (BSR) (as shown in Equation (2)), which is the ratio of BS in wet (exposing the sample to moisture for 24 h at 40 °C) and dry conditions. A higher BSR value indicate higher resistance against moisture damage. Fig. 3c illustrates the stepwise procedure followed in the present study to evaluate the BS and BSR.

$$BSR = \frac{BS_{Wet}}{BS_{Dry}} \quad (2)$$

3.3.2. Boiling water test (BWT)

BWT was performed to qualitatively determine the adhesive bond characteristics at the interface of asphalt binder and aggregate after moisture exposure. ASTM D3625 [42] was followed to determine the

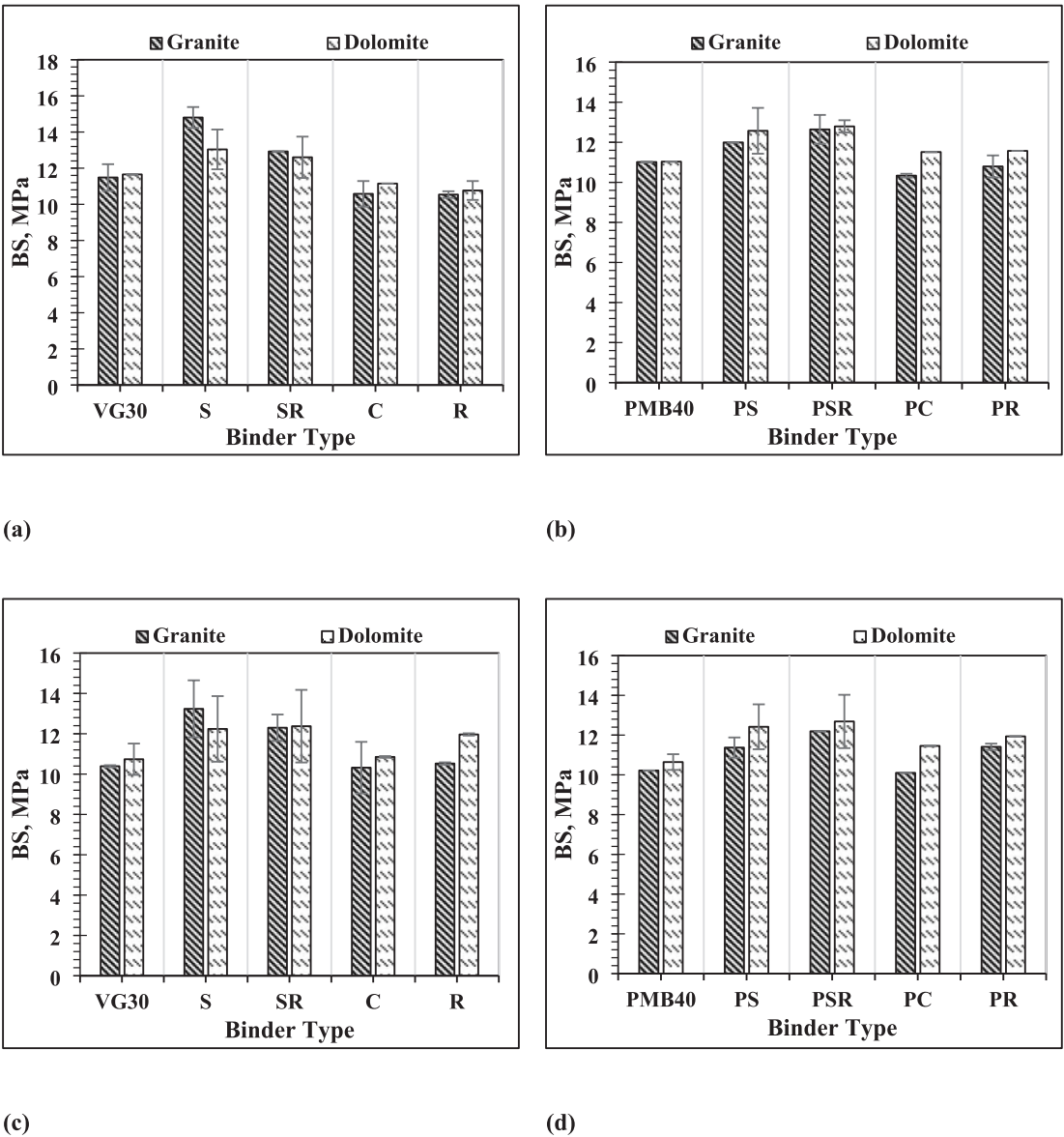


Fig. 6. Bond strength values (a) Dry conditioned samples with VG30 as base binder, (b) Dry conditioned samples with PMB40 as base binder, (c) Wet conditioned samples with VG30 as base binder, and (d) Wet conditioned samples with PMB40 as base binder.

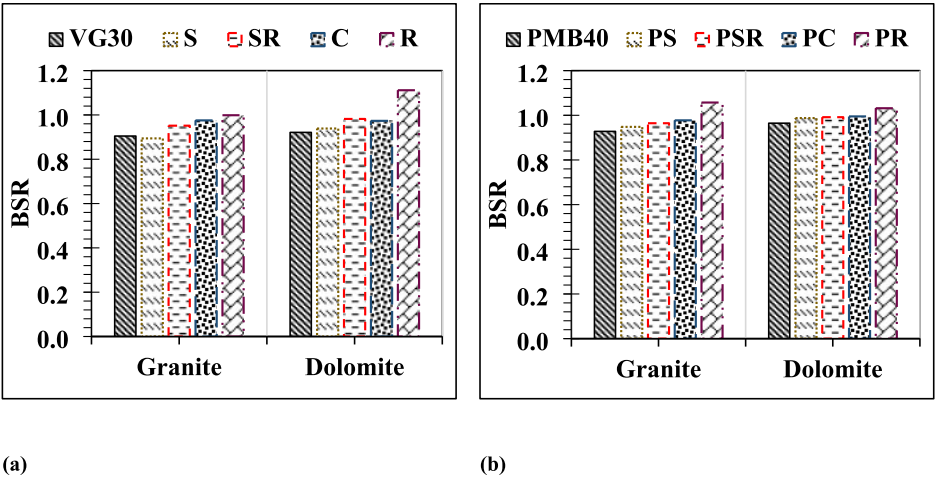
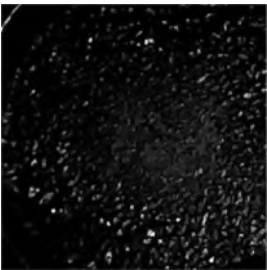
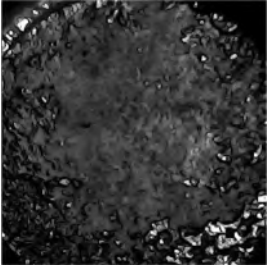
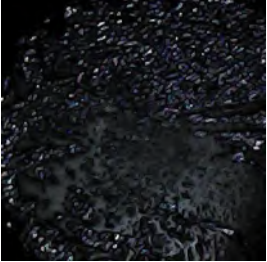


Fig. 7. Variation in BSR with the addition of WMA additives in different base asphalt binders (a) VG30 and (b) PMB40.

Table 3
Failure patterns and interpretation criteria.

Failure	Interpretation Criteria	Remarks	Example
Cohesive (C)	More than 50 % of the aggregate surface remains coated with the asphalt binder.	Applied stress exceeds the cohesive strength of the asphalt binder.	
Adhesive (A)	More than 50 % of the aggregate surface is exposed.	Applied stress is higher than the interfacial BS between asphalt binder and aggregate matrix,	
Hybrid (A/C)	Half of the aggregate surface is covered with asphalt binder, and half of the aggregate surface is exposed.	Neither purely cohesive nor purely adhesive.	

percentage of stripping/coating over the aggregates. As per the standard procedure, 200 g of aggregates (passing 20 mm and retained on a 12.5 mm standard sieve) were mixed with 5 % asphalt binder (by weight of the aggregates). The loose asphalt mixture was transferred to a glass beaker filled with 500 ml of boiled water. The sample was subjected to continuous boiling for 10 min at a temperature of 80–85 °C, as shown in Fig. 4. The sample was further allowed to cool down at room temperature. The water was decanted, and the loose asphalt mixture was spread over a white paper towel to visualize the amount of stripping. It is a simple qualitative method to determine the moisture sensitivity of loose asphalt mixtures. The asphalt mixture which fails to meet the minimum coating requirement of 85 % is anticipated to have a higher likelihood of stripping [43–45].

3.3.3. Retained Marshall stability (RMS)

In this test method, Marshall stability was determined for dry and wet-conditioned samples as per ASTM D6927 [46]. Marshall samples prepared at the predetermined OBC were placed in a temperature-controlled water bath at 60°C for 30–40 min. The specimens were further placed in the Marshall stability machine, where it was subjected to a uniformly (50.8 mm/minute) increasing load. The maximum load in kN, defined as the Marshall stability, was noted. This value is defined as unconditioned Marshall stability. Additional specimens were prepared and conditioned in the water bath at a temperature of 60°C for 24 h prior to the measurement of stability as described previously. Stability values obtained for these specimens are denoted as the conditioned Marshall stability. RMS (Equation (3)) is defined as the ratio of Marshall stability of conditioned and unconditioned specimens.

$$RMS = \frac{\text{Marshall Stability}_{\text{conditioned}}}{\text{Marshall Stability}_{\text{unconditioned}}} \times 100 \quad (3)$$

3.3.4. Tensile strength ratio (TSR)

The modified Lottman test is one of the most popular methods to evaluate the effect of moisture on compacted asphalt mixtures. In this test, the indirect tensile strength (ITS) of compacted asphalt mixtures (prepared at the OBC with 7 % air voids) in dry and wet conditioned states was determined following AASHTO T283 [47]. Test samples were loaded (at a rate of 50.8 mm/minute) using a standard rectangular strip (Fig. 5) along the diametric plane. The load at failure (P_{max}) was used in Equation (4) to calculate the ITS value. The unconditioned compacted specimens were tested after immersing the samples in a water bath at 25 °C for 2 h. Moisture conditioning was done by placing the samples underwater for 24 h at 60 °C, followed by further placement at 25 °C for 2 h. The ratio of ITS of conditioned and unconditioned samples is defined as Tensile strength ratio (TSR) (Equation (5)), which should be typically greater than 80 % to ensure resistance to moisture damage.

$$ITS = \frac{2000P_{max}}{\pi td} \quad (4)$$

$$TSR = \frac{ITS_{\text{conditioned}}}{ITS_{\text{unconditioned}}} \times 100 \quad (5)$$

where, ITS is the indirect tensile strength in (kPa), P_{max} is the peak load (N), t is the thickness of the specimen (mm), and d is the diameter of the specimen (mm).

4. Results and discussion

4.1. Effect of warm mix technologies on moisture resistance of asphalt mixtures

This section presents the results and analysis related to the moisture resistance of WMA and HMA mixtures obtained from different test methods:

4.1.1. Pneumatic adhesion test (PAT)

Fig. 6 (a–d) presents the influence of WMA additives on the BS values of different asphalt binder–aggregate combinations in dry and wet conditions. It was found that WMA binders prepared with organic additives (S and SR) showed higher values of BS than the control binders in dry as well as wet conditions. This is attributed to the presence of crystallized wax, which increases the stiffness of the binder at the ambient temperature range [48]. The effect of Rediset and Cecabase was not found to be significant in terms of BS values. In both dry and wet conditions, the BS values of binders (except Sasobit) with dolomite was higher than granite. This indicates that calcareous aggregates offer better binding with asphalt binder in comparison to siliceous aggregates. The extent of change in BS of different asphalt binders is also dependent on the aggregate source. For example, in the dry state, the percentage increase (with respect to the base binder, VG30) in BS of WMA binders prepared with S and SR was found to be 29 % and 13 %, respectively, while the corresponding increase was 12 % and 8 % with dolomite. On the other hand, in PMB40, the percentage increase with SR is higher than S. Attributed to the higher stiffness, the dry BS of polymer-modified blends were higher than VG30 blends. However, in the wet state, their BS was found to be comparable.

Moisture sensitivity of the asphalt binders was quantified in terms of BSR, which is the ratio of BS value in wet and dry conditions. Fig. 7 shows the effect of WMA additives on the BSR of different asphalt binder–aggregate combinations. In general, the BSR of all the WMA blends were higher than the values of the control binders. This observation led to an important conclusion that the reduction in production temperatures of WMA does not deteriorate their moisture resistance. The extent of improvement in BSR was greater for chemical WMAs than organic WMAs. Among all the WMA additives, Rediset showed superior moisture resistance, resulting in approximately 7–12 % higher BSR,

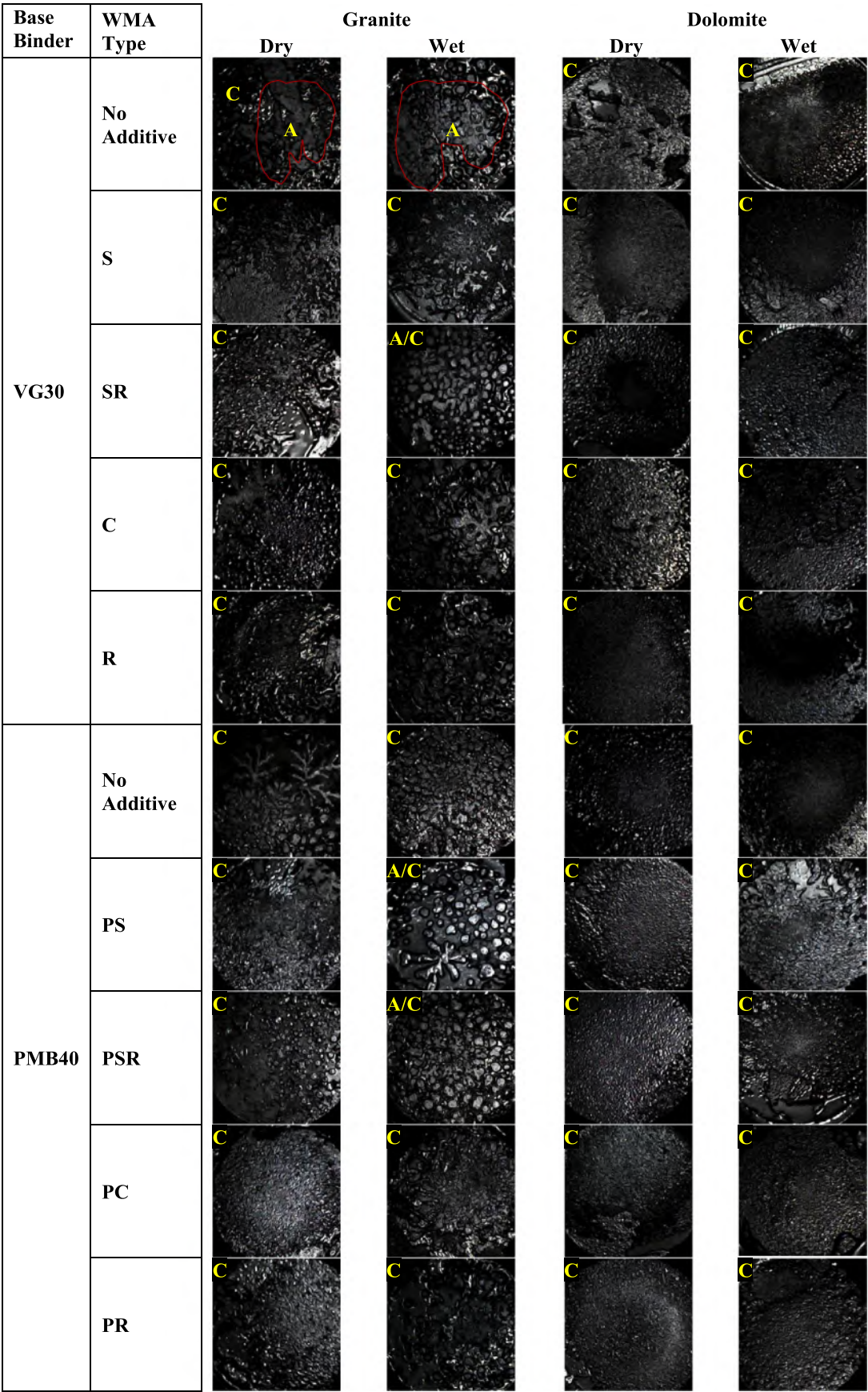
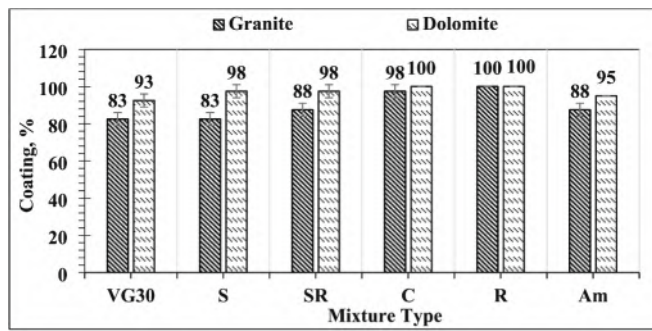
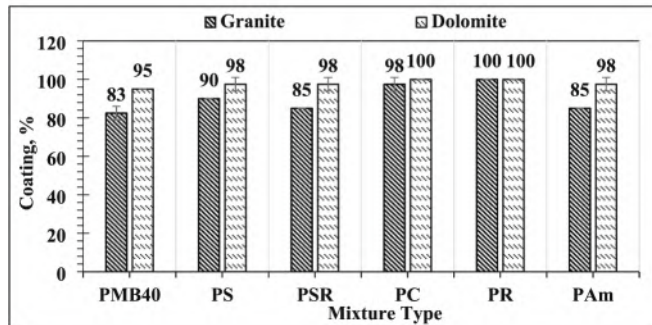


Fig. 8. Failure patterns of different asphalt binders.



(a)



(b)

Fig. 9. Variation in coating over the aggregates with the addition of WMA additives in different base asphalt binders (a) VG30 and (b) PMB40.

followed by Cecabase (3–8 %), Sasobit Redux (3–7 %), and Sasobit (0–2 %). Chemical-based WMA additives act as surface active agents, which improves the binder coating, and reduces the chances of moisture damage at the interface of asphalt binder and aggregate [13,14].

Images of the failure surface obtained in PAT can help identify the failure mode (cohesive, adhesive or hybrid). AASHTO TP91 [49] has outlined the criteria for interpreting the results, as described in Table 3. Such identification is important to understand the compatibility between the mineral aggregates and asphalt binder. Fig. 8 presents the failure pattern of different asphalt binder and aggregate combinations under dry and wet-conditioned states. Under the dry-conditioned state, more than 50 % of the asphalt binder was coated over the aggregates for all the combinations, thus indicating cohesive failure. In the dry state, the bond at the interface of mineral aggregate and asphalt binder is strong and thus the failure takes place due to the loss of material integrity. It is usually anticipated that the bond between aggregate and binder weakens after moisture conditioning, which may lead to adhesive failure. Interestingly, it was found that none of the WMA binders failed in adhesion even after moisture conditioning. This indicates that WMA additives promote improvement in the bonding between aggregate and asphalt binder. Organic WMAs were found to fail in either cohesive or hybrid mode, while chemical WMAs, irrespective of any combination, failed in cohesion. This observation also confirms the antistripping characteristics of chemical-based WMA additives, which is due to the strong bond between cationic surfactant and minerals present in aggregate particles (here, silica in granite and calcium in dolomite) [54, 55]. However, it was found that aggregate source also affect the failure pattern. In dolomite aggregates, cohesive failure was observed, irrespective of the base asphalt binder, WMA technology, and conditioning process. The cohesive failure is attributed to the formation of calcium salts when the asphalt binder reacts with the active minerals in dolomite aggregates, which is very stable even in the presence of moisture.

4.1.2. BWT results

Fig. 9 present the values of percentage coating obtained for different samples. Irrespective of the base binder and type of aggregates, WMA binders showed better performance than the base asphalt binders. The qualitative inferences from the analysis of BWT are in agreement with the PAT results, where chemical WMA additives outperformed other additives. Though the effect of base binder was not significant, dolomite aggregates showed higher resistance to moisture damage than granite. The disagreement in the ranking of the binders between BWT and PAT is attributed to the differences in the simulation procedure of moisture ingress. Moreover, the subjective assessment of percentage coating in BWT may lead to variation in the results.

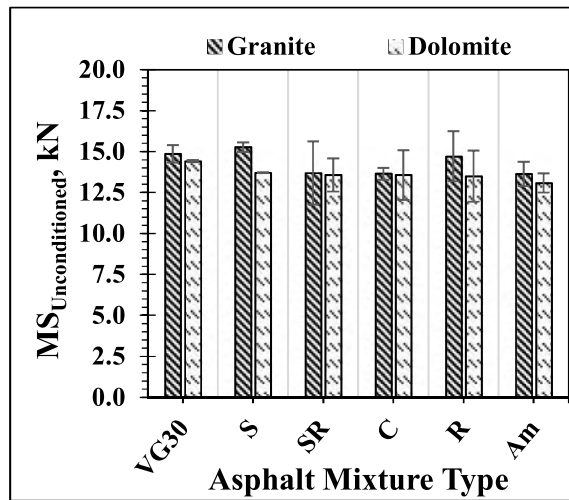
4.1.3. Discussion on retained marshal stability

The unconditioned and conditioned Marshall stability along with the RMS values are presented in Fig. 10. Unlike PAT and BWT, RMS is obtained by testing compacted mixtures. Hence the results are expected to depend on other parameters, such as air-voids, aggregate gradation in the mixture, etc. Owing to the higher stiffness, the unconditioned and conditioned Marshall stability of PMB samples were higher than VG30 samples. Marshall stability values of WMA samples (except Sasobit with granite aggregates) were within ± 10 % variation of the respective control mixture. Thus, this test method could not make strong inferences about the interaction between aggregate and binder. RMS values, which quantify the moisture damage in the asphalt mixture, indicated that all the mixtures provided sufficient resistance against moisture damage, with a value greater than 90 %. Cecabase with PMB40 and dolomite showed the highest RMS (99.11 %), while Sasobit with granite and VG30 displayed the lowest value (92.55 %). Lower RMS for Sasobit indicates relatively higher moisture damage, which may be attributed to the reduction in adhesion between the aggregates and Sasobit inclusive asphalt binder [50]. The relative values between different samples were found to be in close proximity, with no significant difference.

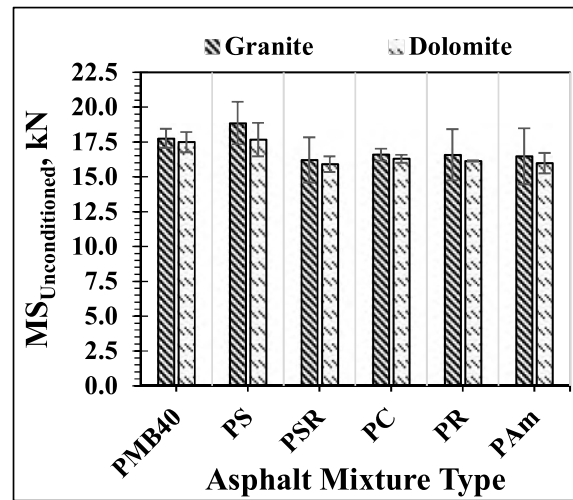
4.1.4. Indirect tensile strength and tensile strength ratio

As indicated previously, TSR ratio is calculated using the unconditioned and conditioned values of ITS. Besides quantifying the moisture sensitivity, ITS value also describes the tensile strength properties in asphalt mixtures. Fig. 11 presents the variation in unconditioned and conditioned ITS values of different asphalt mixtures along with the calculated values of TSR. In contrast to Marshall stability, the ITS values of mixtures produced with dolomite aggregates were higher than those with granite aggregates. Better adhesive bond offered by dolomite increases the strain tolerance in asphalt mixtures, which must have contributed to the increase in ITS. As is expected, the ITS of PMB mixtures were higher than VG30 mixtures. This is attributed to the higher stiffness of PMB in comparison to VG30. While the ITS values of organic WMA mixtures were comparable with the control mixtures, ITS (unconditioned as well as conditioned) of chemical and foaming based WMA mixtures were found to be lower. Considering the variability, this reduction in the average values is not statistically significant (at 95 % confidence level). The effect of Sasobit was found to be binder specific. Addition of Sasobit in VG30 increased the ITS value; however, this effect was not similar in PMB40. The crystallized wax in Sasobit increases the stiffness of VG30 at ambient temperature. However, in PMB40, the polymeric network is more dominant, and the effect of crystallization is not evident.

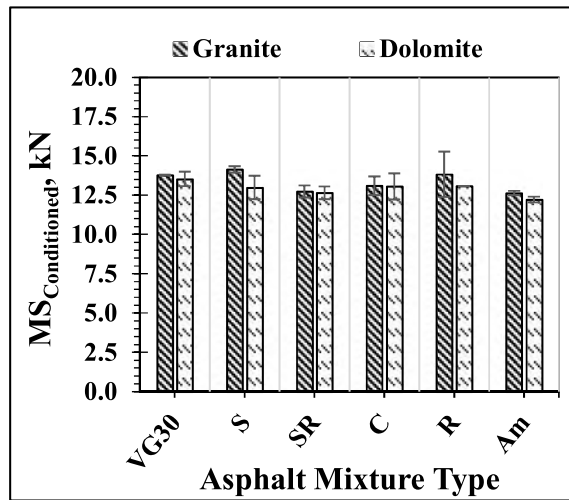
The TSR values of all the tested mixtures were found to be greater than 80 %, which is usually considered as a specification limit to control moisture damage in asphalt mixtures. The TSR in chemical WMAs were found to range between 98 and 100 %, while organic additives showed comparable results with control mixtures with an average TSR of 92 %. Foaming technology reduced moisture resistance in the control mixtures, though the value was higher than the minimum specification requirement.



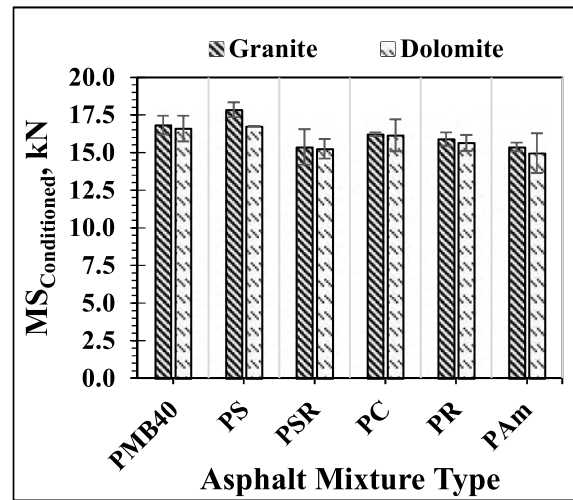
(a)



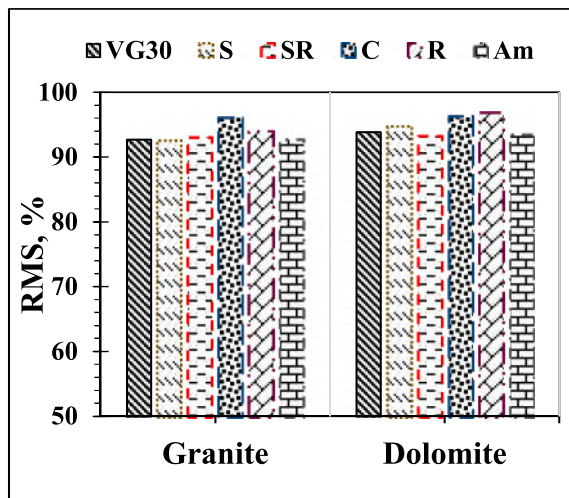
(b)



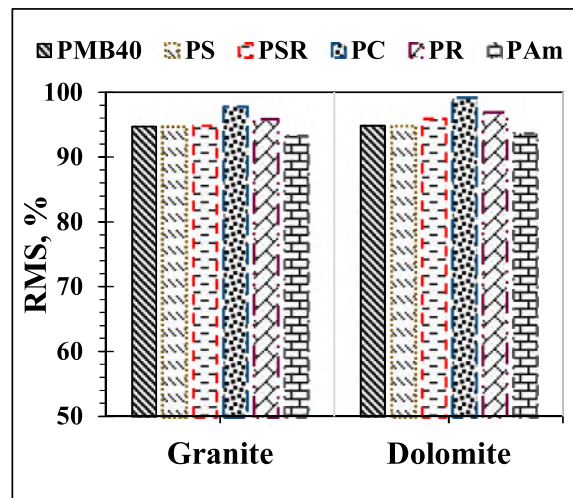
(c)



(d)



(e)



(f)

Fig. 10. Variation in MS for WMA mixtures prepared using different base asphalt binders (a) VG30 in unconditioned state (b) PMB40 in unconditioned state, (c) VG30 in conditioned state, (d) PMB40 in conditioned state, (e) RMS when VG30 is base asphalt binder, and (f) RMS when PMB40 is base asphalt binder.

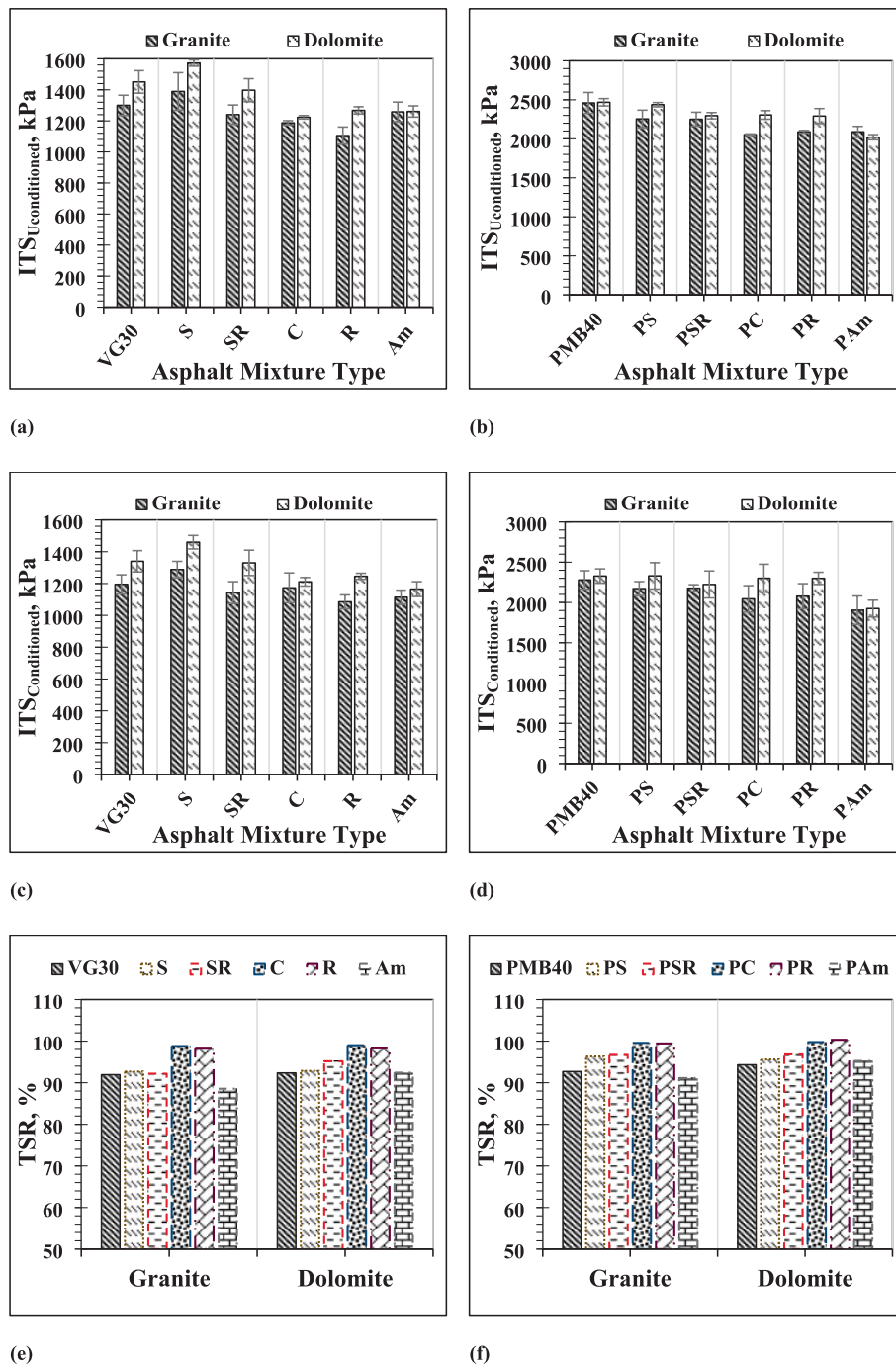


Fig. 11. Variation in ITS for WMA mixtures prepared using different base asphalt binders (a) VG30 in unconditioned state (b) PMB40 in unconditioned state, (c) VG30 in conditioned state, (d) PMB40 in conditioned state, (e) TSR when VG30 is base asphalt binder, and (f) TSR when PMB40 is base asphalt binder.

4.2. Correlation analysis

As shown in previous sections, different test methods may depict variations in the moisture resistance of asphalt mixtures. Also, due to the differences in loading condition, conditioning period and temperature, test temperature, and failure mechanism, the assessment of cohesive/adhesive damage, and the interaction between binder, aggregates and moisture, vary considerably. The results of the present study contradict the general belief that lower production temperatures of WMA mixtures lead to reduced resistance against moisture damage. Amongst all the test methods, PAT was able to clearly identify the effect of base asphalt binder and aggregate source on the moisture resistance of control and

WMA specimens. Past studies [51,52] have also recommended using BSR as a surrogate measure of moisture resistance, in lieu of other methods such as BWT, RMS, and TSR. BSR being a relatively new parameter for the quantification of moisture damage, with no existing specification on its limiting values, it is imperative to understand its correlation with existing popular methodologies, and develop minimum BSR criteria for use in specifications.

The correlations of BSR with other test methods used in this study are shown in Fig. 12(a-d). The correlations are shown for different material combinations, viz., (1) granite and VG30 (GVG), (2) granite and PMB40 (GP), (3) dolomite and VG30 (DVG), and (4) dolomite and PMB40 (DP). Results of Aspha-min are not shown as it was not evaluated in PAT (since

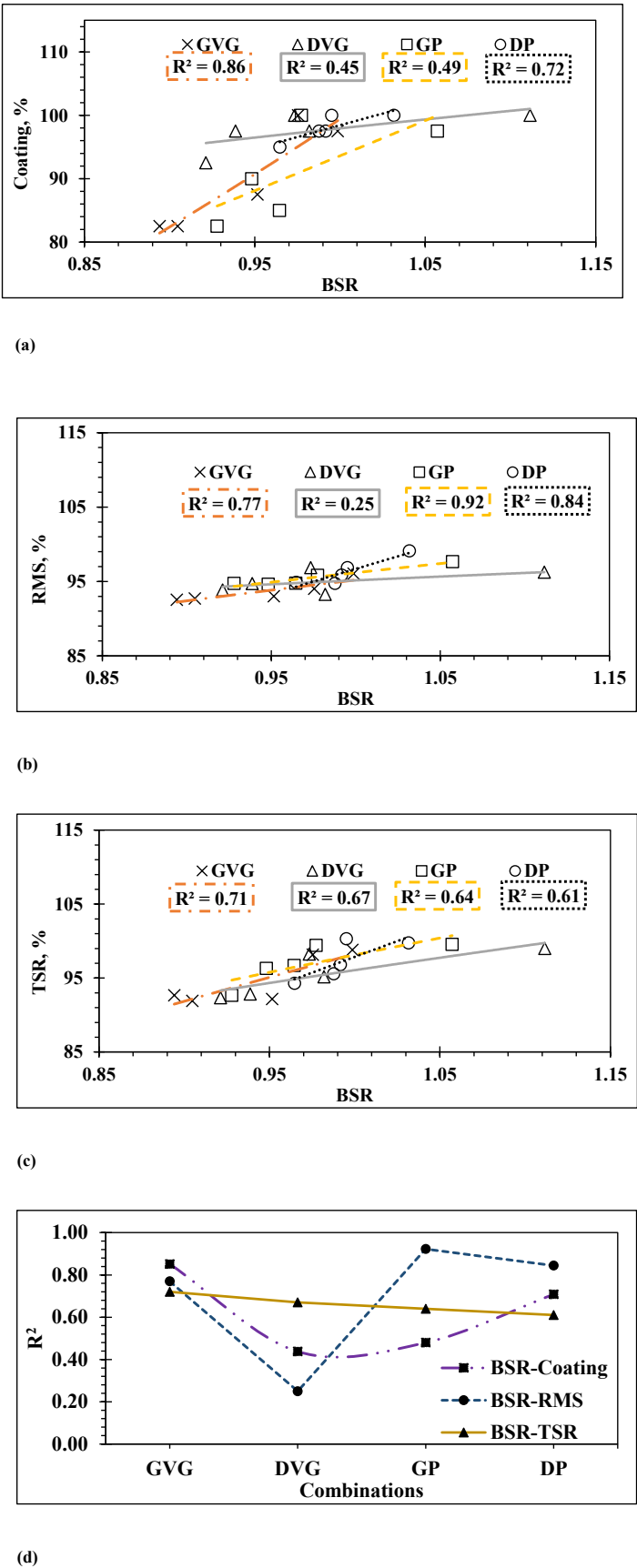


Fig. 12. Correlation analysis (a) BSR-Coating, (b) BSR-RMS, (c) BSR-TSR, and (d) Variation in R^2 for different test combinations.

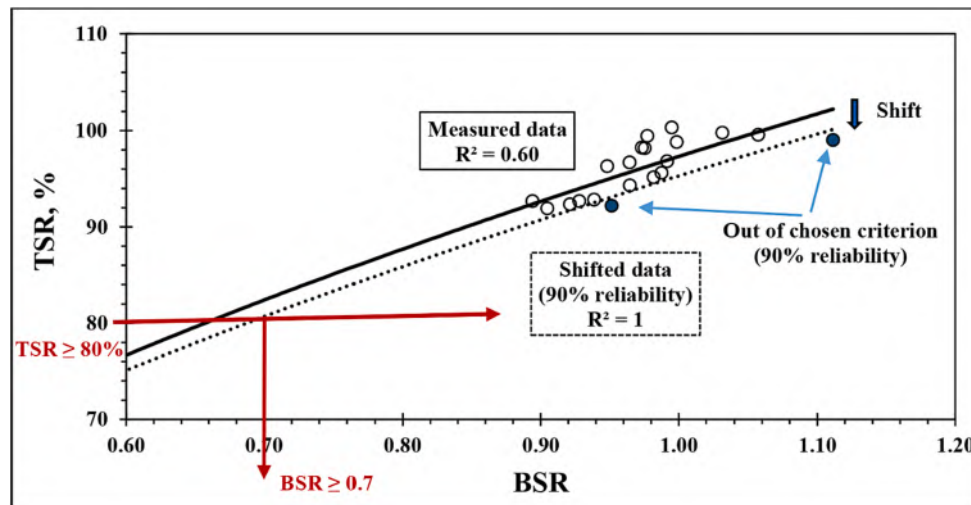


Fig. 13. Correlation between BSR-TSR for establishing the limiting BSR value.

the additive was directly added in the asphalt mixture). Though a positive correlation was obtained between the test methods, for example, higher values of coating percentage, RMS, and TSR lead to higher BSR values, significant variations in coefficient of determination (R^2) were obtained corresponding to different test methods. R^2 between BSR and other test parameters, i.e., RMS, TSR, and coating percentage, for a particular combination, say GP, was calculated as 0.92 (very strong), 0.63 (fair), and 0.48 (weak), respectively. These discrepancies might be attributed to the variation in adopted test procedures. The correlation also depended on the combination of asphalt binder and aggregate source. For example, as shown in Fig. 12b, the correlation between BSR-RMS for DVG was found to be weak ($R^2 = 0.25$), whereas the combination comprising the same aggregate source with PMB40 (DP) showed a strong correlation ($R^2 = 0.84$). Similar variations in R^2 were observed between other pairs of test parameters (BSR-Coating and BSR-TSR).

Fig. 12d presents the variations in R^2 between BSR and other selected test parameters (coating, RMS, and TSR), corresponding to different combinations of asphalt binder and aggregate type. As can be seen, a consistent correlation between BSR and TSR was found, which varied from 0.61 to 0.72. Other test methods showed higher variation in the values of R^2 . Therefore, TSR was considered for developing specification limits of BSR. This is shown in Fig. 13. The variation between TSR and BSR was modelled using a power law. For obtaining 90 % reliability, the curve was shifted vertically downwards such that 10 % of the data (2 out of 20 in this case) fall below this curve. Next, a horizontal line was plotted intersecting the minimum threshold value of TSR (≥ 80 %), as specified by the MoRTH [53] (and various other specifications). The horizontal line crossed the shifted curve at a BSR value of approximately 0.7. Therefore, the asphalt mixture having $BSR \geq 0.7$ is expected to be considered as a moisture-resistant mix and vice-versa. This criterion is independent of WMA technology, aggregate type, and base asphalt binder. Further, more variations in chosen materials (aggregate, asphalt binder, WMA technology, or any other modifier/additive) are required to confirm the use of the developed criteria.

5. Summary and conclusion

In this study, a comprehensive evaluation of moisture damage in WMA mixtures was taken up using various test methods. Two base binders, two aggregate sources, and five WMA additives were employed in the experimental study. First, a comparative analysis was carried out to understand the effect of organic, chemical and foaming based warm mix technologies on the moisture susceptibility of loose and compacted asphalt mixtures. Cohesive and adhesive failure modes were identified using the results of pneumatic adhesion test. Further, a correlation study

was done to assess the consistency in ranking of different mixtures. Finally, the correlation between TSR and BSR was used to develop a specification limit of BSR.

Overall, it can be concluded that despite lower production temperatures than conventional HMA, WMA mixture provide sufficient resistance against moisture damage, irrespective of the test method (PAT, BWT, RMS, and TSR). Amongst the various WMA additives used in the study, Rediset showed superior performance. Not necessarily adhesive failure occurs after moisture conditioning. WMA binders (such as SR, PS, and PSR) with granite aggregates showed a hybrid failure mode (adhesive/cohesive) after moisture conditioning, whereas purely cohesive failure was observed with dolomite aggregates. These observations were independent of aggregate type, base asphalt binder, WMA technology, and conditioning process. Asphalt mixtures with dolomite aggregates exhibit higher moisture resistance than granite aggregates. Relative ranking of asphalt mixtures maynot be similar if different test methods are used. Except the correlation between BSR and TSR, all the other parameters showed variability in correlations with each other. This study recommends using a BSR of 0.7 (or higher) for satisfactory performance against moisture-induced damage.

Since the proposed BSR criteria is based on the combination of limited WMA additives, aggregate source, and base asphalt binders, more variations are required to validate its use in the future. Further, there is a need to explore for an appropriate test method for the evaluation of moisture damage in asphalt mixture that can help to identify the failure mode, and the effect of binder and mixture variables.

CRediT authorship contribution statement

Mayank Sukhija: Conceptualization, Writing – original draft, Investigation, Data curation. **Nikhil Saboo:** Data curation, Writing – original draft, Conceptualization, Supervision. **Agnivesh Pani:** Supervision, Writing – review & editing.

Declaration of Competing Interest

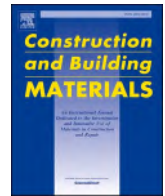
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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A comprehensive review of moisture damage in asphalt mixtures: Mechanisms, evaluation methods, and mitigation strategies

Pratik Neupane^a, Shenghua Wu^{b,*}

^a Department of Civil, Coastal, and Environmental Engineering, University of South Alabama, 150 Student Services Drive, Mobile, AL 36688, USA

^b Department of Civil, Coastal, and Environmental Engineering, University of South Alabama, 150 Student Services Drive, Shelby Hall 3116, Mobile, AL 36688, USA

ARTICLE INFO

Keywords:

Moisture damage
Asphalt mixtures
Pavement durability
HMA
WMA
CMA
Sustainable materials
Testing methods
Mitigation strategies
Nanomaterials
Machine learning
Climate resilience
Drainage systems
Advanced technologies
Infrastructure sustainability

ABSTRACT

The durability of asphalt pavements faces increasing threats from moisture-induced damage, especially under the impacts of extreme weather events such as heavy precipitation and flooding. This comprehensive review synthesizes existing research to uncover the underlying mechanisms of moisture damage in asphalt mixtures, while critically evaluating laboratory testing methods and advanced mitigation strategies. The study uniquely integrates state-of-the-art technologies, such as nanomaterials and machine learning applications, offering innovative solutions for enhancing moisture resistance and durability. By exploring the vulnerabilities of various asphalt types—hot mix asphalt (HMA), warm mix asphalt (WMA), and cold mix asphalt (CMA)—this paper provides a nuanced understanding of their respective susceptibilities and improvement potentials. Actionable insights include recommendations for optimizing mix designs, employing sustainable materials, and advancing evaluation methodologies to bridge laboratory findings with field performance. This work paves the way for resilient and sustainable pavement infrastructure, with future research focusing on field validation, climate adaptation, and the integration of cutting-edge technologies to mitigate the evolving challenges of moisture damage.

1. Introduction

Moisture-induced damage is a critical issue in asphalt pavements, significantly impacting their durability and longevity. Asphalt mixtures consist of asphalt binders, aggregates, and occasionally modifiers. The primary role of the binder is to securely hold the aggregates together and provide a protective barrier against moisture infiltration [1]. However, moisture can penetrate the asphalt mixture, weakening the bond between the binder and aggregate, and leading to a deterioration of the pavement structure—a process commonly known as stripping. This weakening can compromise the structural integrity of asphalt, resulting in rutting, cracking, and potholes, ultimately necessitating frequent maintenance and repairs.

Moisture damage in asphalt mixtures can manifest as two primary types of failures: adhesive and cohesive. Adhesive failure occurs when the bond between the binder and aggregate is compromised, leading to a clean aggregate surface after failure [2], as illustrated in Fig. 1. In contrast, cohesive failure refers to the breakdown within the binder or

binder-filler mastic itself, leading to visible distress [3], also shown in Fig. 1. Numerous studies have explored various definitions and interpretations of moisture damage, as documented in [4]. The severity of moisture damage is influenced by factors such as the type of aggregates and asphalt, mixture design, construction methods, environmental conditions, traffic volumes, and the properties of antistripping additives summarized in Table 1 [5]. Furthermore, moisture damage results from complex thermodynamic, chemical, physical, and mechanical processes, including water permeability, capillary rise, and vapor diffusion [6]. This multifaceted nature of moisture damage affects the mechanical properties of the mastic and the adhesion between mastic and aggregates [7].

Efforts to understand moisture susceptibility in asphalt mixtures date back to the 1920s, with researchers continuously seeking reliable tests to differentiate between well-performing and poorly-performing mixtures [8]. Traditional methods, such as core extraction, are often time-consuming and costly, prompting a shift toward non-destructive methods like Ground Penetrating Radar (GPR) and Rolling Wheel

* Corresponding author.

E-mail addresses: pn2327@jagmail.southalabama.edu (P. Neupane), shenghuawu@southalabama.edu (S. Wu).

<https://doi.org/10.1016/j.conbuildmat.2025.140740>

Received 7 November 2024; Received in revised form 12 January 2025; Accepted 5 March 2025

Available online 11 March 2025

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Deflectometer (RWD) for early damage detection [9]. Although these methods have higher initial costs, they can reduce long-term repair expenses by identifying moisture-related damage before significant structural issues arise. For instance, studies by the Louisiana Transportation Research Center (LTRC) indicate that moisture damage can reduce the service life of asphalt concrete overlays by approximately 4.6 years compared to unaffected sections, leading to increased maintenance needs [10]. This deterioration is reflected in a 23 % reduction in the Pavement Condition Index (PCI) and a 0.5 % decrease in cost-effectiveness [10]. Rehabilitation strategies such as AC overlays, chip seals, and micro-surfacing can mitigate moisture damage if underlying issues are addressed. Failure to do so, however, often results in premature pavement failures and escalated costs [11].

Moisture-induced damage not only impacts pavement performance but also imposes substantial financial burdens. In the United States, damages such as stripping and pothole formation have led to significant expenses. For example, pothole-related repairs cost drivers around \$26.5 billion in 2021, with each repair averaging approximately \$600 [12]. With the exacerbating effects of climate change, maintenance demands are expected to increase, potentially adding up to \$19 billion in extra annual costs by 2040 [13,14]. Addressing these challenges requires a focus on early detection and preventative measures to mitigate long-term repair costs.

Moreover, extreme weather conditions such as heavy rainfall and flooding have a considerable impact on pavement performance. Flooding is among the most expensive natural disasters in the U.S., with average annual losses ranging from \$5 to \$8.2 billion [15]. These costs can increase significantly when considering the broader economic impacts, including infrastructure repairs and productivity losses [16]. In addition to economic challenges, moisture-induced damage poses safety risks, including hydroplaning, reduced friction, and structural weakening, which elevate the likelihood of accidents during wet conditions. Approximately 24 % of road sections in the continental USA are prone to hydroplaning annually, with states like Texas, Florida, and Alabama experiencing up to 25 high-risk events (HREs) each year [17]. The risk is particularly high on roads with elevated traffic volumes and speed limits over 60 mph, where even a thin water film can cause vehicles to lose control [17]. Further studies show that weakened pavement integrity due to moisture damage can increase accident rates by over 20 % when rut depths exceed 7.6 mm [18]. Analysis from the Fatality Analysis Reporting System (FARS) reveals that more than 114,960 fatal crashes occurred under rainy conditions between 1980 and 2017, with about 38 % of these incidents occurring in regions susceptible to hydroplaning [17]. Additionally, rough pavements, characterized by high International Roughness Index (IRI) values, are associated with a 95.7 % increase in accident rates compared to smoother surfaces [19]. Improving pavement materials, such as using polymer-modified asphalt, can mitigate these issues by reducing rutting by 15–20 % and potentially lowering hydroplaning-related incidents by 10 % [18]. These safety improvements could save around \$320 million annually by reducing accidents and vehicle wear [17].

Table 1
Causes of moisture susceptibility (adopted from [6]).

Cause	Description
Aggregate	Shape, texture, porosity, cleanliness, moisture
Asphalt Binder	Film thickness, composition, aging
Production and Construction	Segregation, improper drying, compaction
Traffic and Environmental Conditions	Water, temperature, freeze-thaw cycles
Pavement Design Considerations	Drainage, material selection

The impact of moisture damage is also evident in the structural aspects of pavement layers, as illustrated in Fig. 2. Asphalt pavement systems consist of overlay, base, and subgrade soil layers. Extreme weather conditions, such as high and low temperatures, heavy rainfall, and flooding, significantly affect these layers. For example, temperature fluctuations can cause thermal cracking in the overlay, while moisture infiltration from rainfall can lead to rutting, weaken the base, and result in pothole formation. A high-water table can further reduce subgrade strength, compromising the overall structural integrity of the pavement system.

Fig. 3 shows an instance of weakened shoulder support following heavy precipitation after Hurricane Ian in Florida, leading to additional damage to the asphalt pavement due to the stress from heavy trucks.

Given the evolving challenges posed by climate change and advancements in evaluation methods, a comprehensive review of moisture-induced damage in asphalt mixtures is both timely and necessary. This review paper offers a unique contribution to the body of knowledge on moisture-induced damage in asphalt mixtures by addressing several critical gaps in the literature. Unlike previous

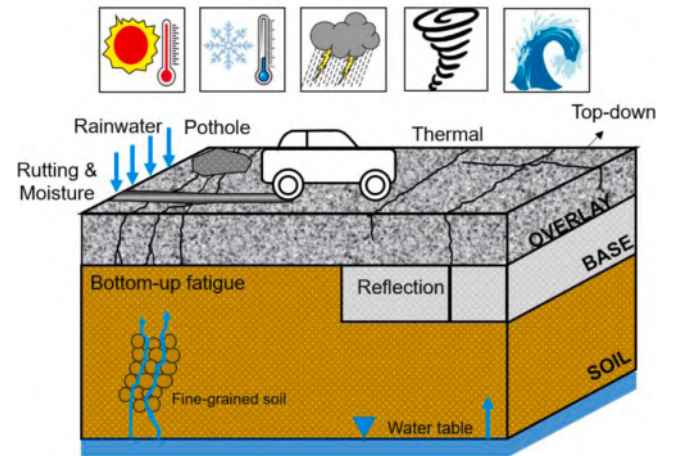


Fig. 2. Cross-section of asphalt pavement showing weather and traffic impacts.

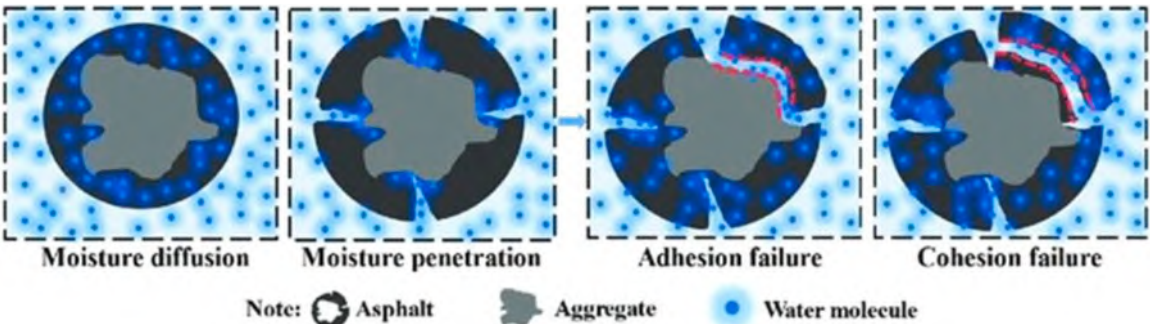


Fig. 1. Moisture-induced damage process diagram of asphalt pavement [20].



Fig. 3. Road damage in Florida after Hurricane Ian in 2022 (photo taken during a post-disaster survey).

reviews, which predominantly focus on isolated mechanisms or specific evaluation methods, this work synthesizes a holistic perspective by integrating diverse findings into a comprehensive framework. Key distinctions include the identification and analysis of emerging technologies, such as nanomaterial applications and innovative evaluation techniques like machine learning integration, which have not been extensively reviewed in past studies. Furthermore, this paper introduces novel categorizations of damage mechanisms and mitigation strategies, emphasizing practical applications for sustainable pavement design. By addressing limitations in prior research, such as the absence of field-validation data for advanced laboratory tests, this review bridges the gap between theoretical advancements and their practical implications. In doing so, it equips researchers and practitioners with actionable

insights into the evolving challenges posed by moisture damage, particularly in the context of climate resilience and infrastructure sustainability.

2. Objective and research approach

The objective of this paper is to provide a comprehensive review of the effects of moisture damage in asphalt mixtures, critically evaluate the existing laboratory test methods for assessing moisture susceptibility, examine the potential of various additives and fillers in mitigating moisture damage, and identify key challenges and recommendations for future research to promote sustainable asphalt pavement construction. The review structure of this paper is shown in Fig. 4.

3. Methodology

3.1. Keywords and databases

The keywords include “moisture damage”, “asphalt mixtures”, “pavement durability”, “HMA”, “WMA”, “CMA”, “testing methods”, “additives”, “nanomaterials”, and “sustainability”. These keywords were combined to search relevant publications from multiple databases. The investigated databases include “ScienceDirect”, “Scopus”, “Web of Science”, “Google Scholar”, “Nature”, “Science”, “IOPscience”, and “IEEE Xplore”. Advanced settings for searching include (1) Article types: review articles, research articles, book chapters, and short communications. (2) Language: English.

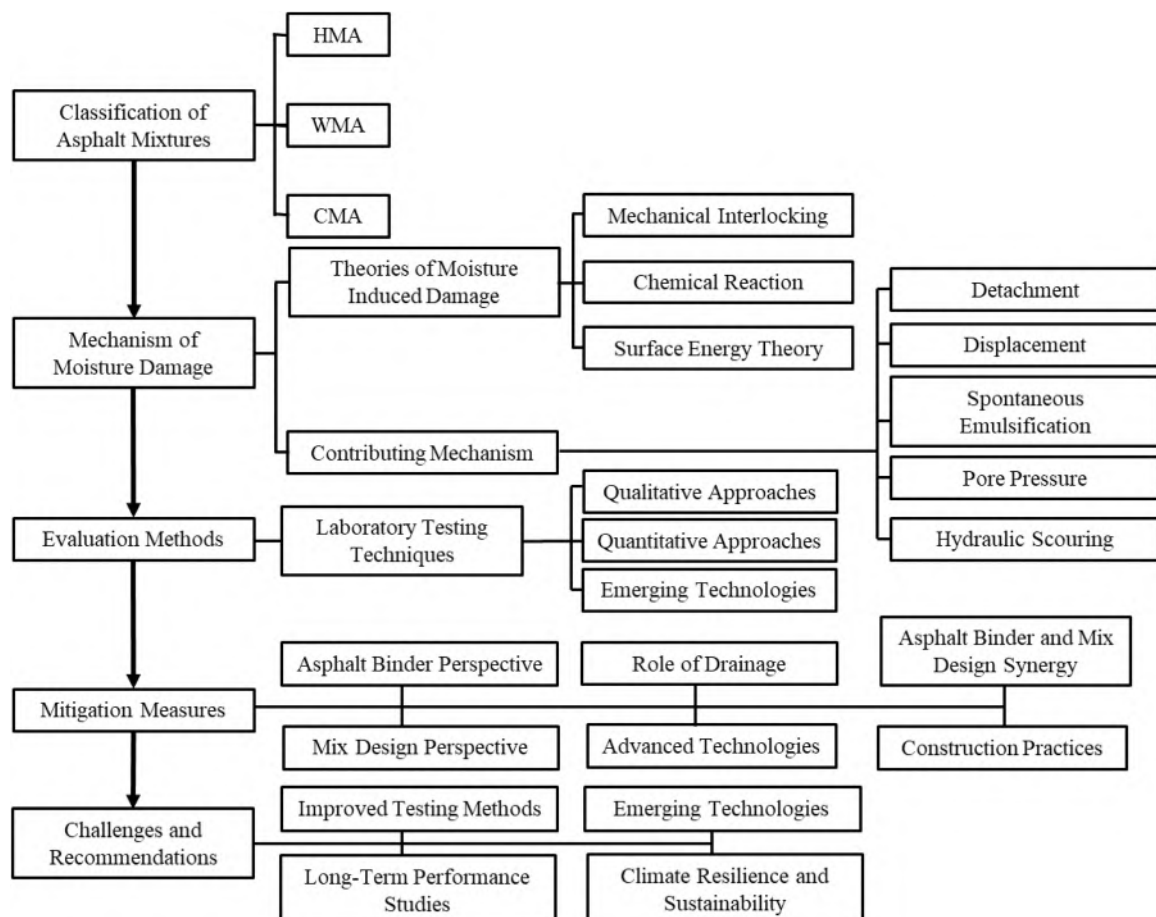


Fig. 4. Review structure.

3.2. Statistics of relevant publications

The number of publications related to asphalt mixtures, moisture damage, additives, and evaluation methods has increased from 2000 to 2024, as shown in Fig. 5(b). The graph in Fig. 5(a) shows the number of publications from different countries suggesting that this problem exists in many countries, and they are determined to address this challenge. The graph in Fig. 5(b) shows a rapid rise, indicating that researchers have been putting more effort into finding solutions to prevent moisture damage in asphalt mixtures and develop sustainable answers.

4. Comparison of asphalt mixtures by moisture susceptibility

Moisture damage costs the country billions of dollars in pavement maintenance and rehabilitation [21]. An annual cost of approximately 109 billion dollars comes around due to driving on bad roads in the USA [22]. Fig. 6 demonstrates a direct relationship between the increase in production temperature and the corresponding rise in energy consumption and greenhouse gas (GHG) emissions. Additionally, the processes contributing to these increases (heating, vaporization, drying) are highlighted on the right side of Fig. 6.

Table 2 provides a breakdown of the essential elements found in asphalt, outlining their functions and how their specific attributes impact the mechanical properties and overall performance, and the properties of these mixtures are described in Table 3. An overview of different asphalt mixture types is shortly defined in Table 4.

4.1. Hot mix asphalt

Hot mix asphalt is the composite material used in pavements, consisting of stone aggregates, sand, asphalt binder, and sometimes additives [35]. It is typically produced in high temperatures typically ranging from 300 °F to 350 °F. Moisture damage in HMA usually occurs due to the breakdown of adhesion between the binder and aggregate, or the internal breakdown of the binder itself [3]. Also, moisture can cause significant reductions in indirect tensile strength (ITS) and tensile strength ratio (TSR) in HMA. Treatments like polymeric coatings can improve adhesion between the asphalt binder and aggregates, such as HDPE and LDPE [3]. The long-term effect of moisture in HMA can be accelerated by freeze-thaw cycles and research suggests that multiple freeze-thaw cycles can lead to a higher rate of adhesive failure and a decrease in ITS [36].

4.2. Warm mix asphalt

Warm-mix asphalt is manufactured and spread at a lower temperature than HMA. WMA is typically produced between 100°C-140°C as

compared to HMA at 150°C-190°C [32]. Comparative research between HMA and WMA suggests that although WMA generally has good performance, it may be more prone to moisture damage because of the lower production temperatures [36]. WMA analysis should include tests like volumetrics, modulus, rut resistance, moisture susceptibility, and dust proportion same as that of HMA [38]. Different additives can be used to reduce moisture-induced damage in WMA, but they must be used in the proper composition. If not used correctly, improper composition can result in increased aggregate content and reduced binder viscosity, leading to poor adhesion and cohesion. A study performed by [37] demonstrated that using a high amount of natural zeolite increased the aggregate content, making the asphalt more vulnerable to moisture.

4.3. Cold mix asphalt

Since the world is seeking sustainable solutions, engineers have been working on developing paving materials with low carbon footprints. HMA has always been one of the most desirable paving technologies due to its superior performance. However, due to its limitations, such as an increased carbon footprint and higher energy consumption, engineers have been looking for more sustainable alternatives [34]. CMA is one such alternative used in road construction to reduce carbon footprints. This mixture is manufactured and applied at lower temperatures (ambient temperature) compared to HMA, thereby requiring less energy to reduce bitumen viscosity than traditional HMA methods [39]. CMA also notably reduces factory fumes and emissions, thereby enhancing working conditions for production staff and operators [39]. Apart from all these advantages moisture damage is one of the huge problems associated with CMA. Adhesion failure is the most common failure in CMA [34,40]. Since water is used in the preparation of CMA and the aggregates are not heated, moisture can affect the interaction between the binder and aggregates. This often leads to reduced adhesion, which can result in various forms of distress in CMA pavements, such as rutting, fatigue cracking, and stripping [34,41,42]. However, due to it being sustainable different additives and modifiers have been used to mitigate these drawbacks. A study conducted by [43] shows that Crumb rubber and cement have been somehow efficient in mitigating these drawbacks in certain aspects. More study is needed to be done in this sector as CMA could be one of the sustainable solutions to the environmental problems caused by other traditional mixes.

5. Mechanism of moisture damage

5.1. Theories related to moisture-induced damage in asphalt pavement

Numerous theories have been discussed to define moisture-induced

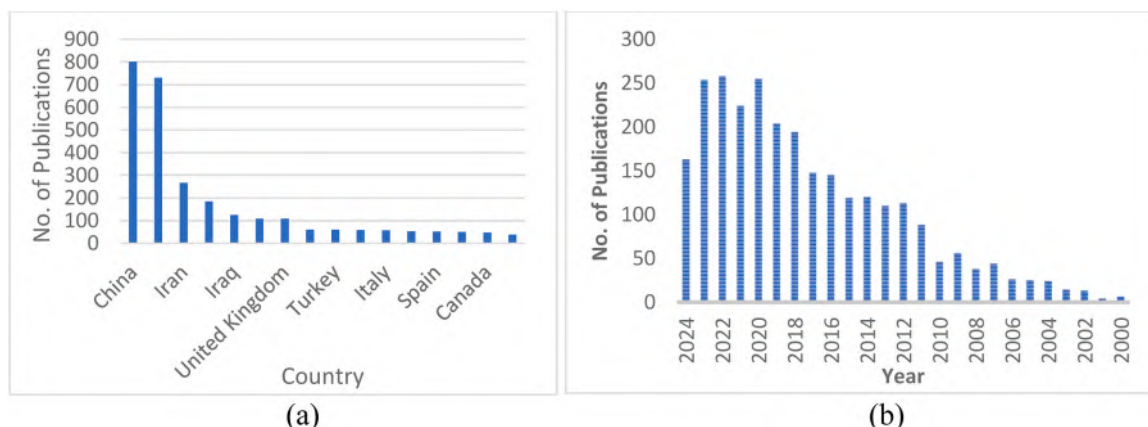


Fig. 5. Number of publications from 2000 to 2024: (a) from different countries, and (b) in each year.

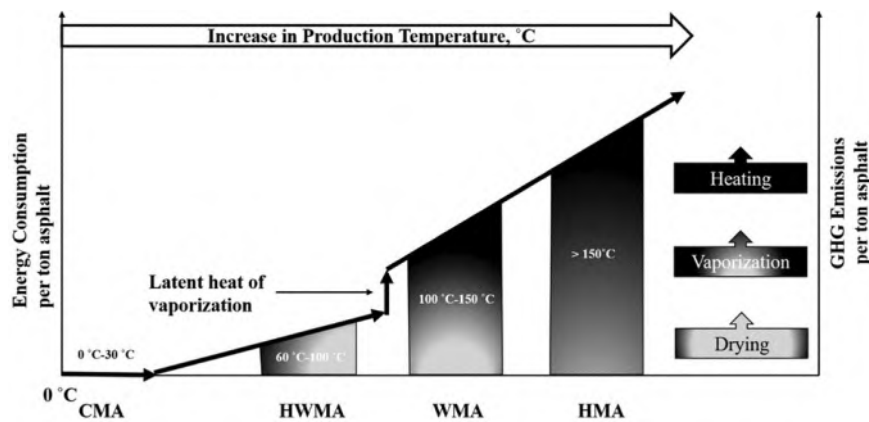


Fig. 6. Classification of asphalt mixtures [23].

Table 2
Components of asphalt mixture.

Component	Description [Ref.]
Aggregates (coarse & fine particles)	Provide strength and stability. Size, shape, and distribution impact mechanical properties [24].
Asphalt Binder (bitumen)	Binds the aggregates. Viscosity and elasticity affect performance under varying temperatures and loads [25].
Air Voids	Intentional spaces for slight movements and expansion. Amount and distribution impact durability and resistance to deformation [26].
Additives & Modifiers (polymers, fibers)	Enhance properties. Basalt fiber improves mechanical properties and crack resistance [25].

Table 3
Properties of asphalt mixture.

Property	Description	Ref.
Mechanical	- Stiffness, strength, and deformation resistance - Depends on the composition and interaction of components - High-temperature performance and rutting resistance are crucial	[27]
Thermal	- How the mixture responds to temperature changes - Includes thermal expansion and contraction that can lead to cracking	[28]
Viscoelastic	- Exhibits behavior dependent on loading rate and temperature - Crucial for understanding pavement performance under traffic	[29]
Chemical	- Chemical composition of asphalt binder affects physical properties - Includes types and proportions of asphaltenes, resins, and oils - Affects softening point and ductility	[30]

damage in Asphalt mixture. [4] have classified these theories as Mechanical interlocking, chemical reaction, molecular orientation, or surface energy theories. Since these theories were discussed in 1988, we tried to shed light on some recent developments made regarding these theories as of 2024.

5.1.1. Mechanical interlocking

This theory was proposed in 1925 by McBain and Hopkin. According to them, mechanical joints are possible only with porous materials. This theory assumes the absence of chemical interaction between asphalt and aggregate [4]. This theory is not preferred by researchers due to that very reason [44]. Surface texture i.e. shape, and size of the aggregates is one of the main factors that affect adhesion [20,45]. The three independent properties that characterize the geometry of aggregates are

form (shape), roundness (angularity), and surface texture, as identified by [46]. Open-graded aggregate has more air voids and larger pores, making it less resistant to stripping (increased creep strain) [47]. In contrast, fine particles fill voids and improve adhesion between the asphalt and larger aggregate particles, reducing water ingress [47,48]. Similarly, other available aggregates such as granite, limestone, dolomite, chert gravel, and quartz gravel were tested for surface area and porosity, and their stripping propensity was tested in the study conducted by [49].

5.1.2. Chemical reaction

According to [50], the chemical reaction theory is based on the idea that acidic and basic components of both asphalt and aggregate react to form water-insoluble compounds that resist stripping. Moisture damage is also influenced by pavement-mixture morphology and external environmental factors [51,52]. Aggregates like granite and limestone show varying resistance to moisture damage, with the aggregate's mineralogical and chemical properties affecting the bond strength with asphalt [53]. Also, oxidative aging of asphalt mixtures improves moisture resistance by altering the chemical composition at the asphalt-aggregate interface [54,55]. This theory is further explained by different researchers as shown in Table 5.

5.1.3. Surface energy theories

Surface energy is defined as the external work needed to create a new unit surface area of material in vacuum conditions. Surface free energy (SFE) method is used to forecast the moisture damage in asphalt pavement. It helps assess this bond by measuring the energy needed to break the asphalt-aggregate connection or create a crack within the asphalt itself (cohesion) [63,64]. Higher total SFE and positive adhesion-free energy indicate a strong bond and are less susceptible to water damage [65]. Later on, different research was done using the SFE method to determine moisture susceptibility. Some of the research has been summarized in Table 6.

Fowkes suggested that the total SFE of substances comprises two main components. These are scattered forces, such as nonpolar forces within a molecule like Lifshitz-van der Waals interactions, and forces related to specific interactions, such as hydrogen bonding. Fowkes stated that the total SFE or surface tension is a linear combination of these interactions [71]. The equations used to calculate SFE have been discussed by [72].

5.2. Contributing mechanisms in asphalt damage

Five well-known mechanisms can lead to a loss in adhesion and/or cohesion. These mechanisms are shortly described below.

Table 4

Comparison between HMA, WMA, and CMA.

Aspect	HMA	WMA	CMA
Production Temperature	High (150–180°C) [31]	Moderate (100–140°C) [32,33]	Ambient (0–40°C) [34]
Key Advantages	Superior durability and strength; suitable for high-traffic roads [31]	Reduced emissions (20–30 %) and energy consumption (30–40 %); extended paving season [32]	Low energy use; environmentally friendly [34]
Key Limitations	High energy consumption; thermal cracking in cold climates [31]	Potential moisture susceptibility; requires anti-stripping agents [33]	Long curing time; limited to low-traffic roads [34]
Climate Suitability	Best in warm climates; adjustments needed for cold climates [31]	Ideal for moderate to cold climates; extended paving window [32,33]	Performs poorly in extreme cold due to curing delays [34]
Traffic Suitability	Suitable for high-traffic and heavy-load roads [31]	Effective for moderate to high traffic with proper additives [32]	Limited to low-traffic roads and maintenance tasks [34]
Environmental Impact	High emissions; energy-intensive [31]	Reduced emissions and energy use; supports sustainability [32]	Lowest environmental impact; no heating required [34]
Economic Considerations	High production costs; durable long-term performance [31]	Cost savings on energy; higher initial costs for additives [32]	Low cost; frequent maintenance required [34]

Table 5

Chemical reaction in asphalt mixture.

Mechanism	Description	Ref.
Chemical bonding	Acidic and basic components of asphalt and aggregate react to form water-insoluble compounds, resisting stripping.	[56]
Selective chemical reactions	Specific interactions between asphalt and aggregate species occur.	[57]
Chemisorption	Asphalt functionalities bond chemically to aggregate surfaces.	[57]
Chemical sorption	Formation of strong bonds between asphalt and aggregate to minimize stripping.	[58]
Polarity attraction	Oppositely charged components in asphalt and aggregate attract each other.	[59]
Non-uniform charge distribution	Both asphalt and aggregate have areas with positive and negative charges, influencing attraction.	[60]
Environmental influence	Charge distribution on aggregates is affected by surrounding conditions.	[50]
Basic nitrogen adhesion	Pyridine compounds in asphalt strongly adhere to aggregate surfaces.	[50]
Carboxylic acid behavior	These asphalt components adhere to dry aggregates but detach easily in water (depends on acid type).	[50]
Monovalent cation salt detachment	Salts like sodium or potassium from carboxylic acids act as surfactants, detaching under water and traffic.	[61]
Divalent cation salt resistance	Salts like calcium from hydrated lime are more resistant to water compared to monovalent salts.	[50, 61,62]
Asphalt aging and moisture damage	Aged asphalts may become more susceptible to moisture due to highly acidic materials formed during oxidation.	[50]
Sodium salt formation and detergents	Asphalt acids converted to sodium salts (by some aggregates) can act as detergents, promoting stripping.	[50]
Calcium salt deactivation	Calcium salts of detergents are less moisture-sensitive and can be deactivated with lime.	[50]
Avoiding moisture-susceptible salts	Asphalts with free acids or monovalent salts should be avoided to prevent stripping.	[50]

5.2.1. Detachment

Bitumen-aggregate detachment, the microscopic separation of the bitumen or mastic film from the aggregate surface, is primarily driven by a thin water layer acting as a wedge [73]. This weakening effect is explained by the thermodynamics mechanism. Water's presence disrupts the bitumen-aggregate interface by lowering the surface energy between the two materials [73]. This stems from water's lower surface tension compared to bitumen, resulting in weaker dispersion forces, particularly when the aggregate surface exhibits high polarity and readily attracts water molecules [74]. In essence, water disrupts the attractive forces holding the bitumen film onto the aggregate, leading to its detachment.

Table 6

Use of SFE in the latest research.

Asphalt Type	Additive	Key Findings	Moisture Considered	Ref.
WMA	Dry-mixing	Effective adhesion work proposed to assess moisture damage	Yes	[66]
Conventional	WMA additive, Anti-stripping agent, Reclaimed Asphalt Pavement (RAP)	SFE promising for characterizing adhesion, WMA additive and anti-stripping agent improved moisture resistance	Yes	[67]
Epoxy Resin modified	Epoxy Resin	Epoxy resin increased adhesion and wetting in siliceous aggregates, adhesion only in limestone	Yes	[68]
WMA	Cecabase	Cecabase improved adhesion for both aggregates, and limestone less susceptible to moisture damage	Yes	[69]
Nanosize Hydrated Lime (NHL) modified	N/A	Adhesion improved with NHL	Yes	[70]

5.2.2. Displacement

Displacement stripping is a distress mechanism that occurs in asphalt pavements when water enters the asphalt film through a pre-existing weakness, causing the binder to detach from the aggregate surface [75]. These weaknesses may result from incomplete coating during mix production, film rupture due to traffic loads, or trapped air around dusty aggregates [76,77]. In addition to physical factors, chemical interactions also play a role. Changes in water pH can modify the types of polar groups adsorbed on the aggregate surface, creating opposing electrical charges between the aggregate and bitumen, which attract more water and ultimately lead to separation [73,77]. In summary, displacement stripping takes advantage of existing weaknesses in the asphalt film, allowing water to penetrate and disrupt the bond between the binder and aggregate.

5.2.3. Spontaneous emulsification

Spontaneous emulsification, a process that contributes to the separation of bitumen from aggregate, involves the creation of an inverted emulsion within the asphalt film [8]. This inverted emulsion is formed

by water droplets dispersed throughout the bitumen, leading to a breakdown of the binder's cohesive properties and resulting in separation from the aggregate surface (adhesive failure). This chemical process is influenced by various factors, including the type and viscosity of the bitumen, the duration of wet conditions, and the presence of specific additives [6]. For example, clays can act as emulsifiers, speeding up the breakdown of the binder by water [53]. Essentially, spontaneous emulsification weakens the asphalt binder internally through water interaction, ultimately causing it to detach from the aggregate.

5.2.4. Pore pressure

Pore pressure stripping occurs when water becomes trapped within the asphalt mix, leading to pressure buildup from traffic loads and freeze-thaw cycles. This pressure weakens the bond between the bitumen and aggregate. The susceptibility to pore pressure damage depends on the air void content of the asphalt. If the air void content is below 8 %, the voids are disconnected, making it hard for water to enter and saturate them. Conversely, if the air void content exceeds 10 %, the voids become interconnected, allowing water to flow out under traffic load and reducing the risk of pressure buildup. However, the critical zone lies within the 8–10 % air void range, where water can enter the voids but struggles to escape due to the lack of interconnected pathways. As traffic loads are applied, pressure builds up within these trapped pockets of water, forcing it through the asphalt structure. This pressurized water weakens the interface between the binder and aggregate, initiating micro-cracks and ultimately leading to stripping [78,79]. Fig. 7 shows how moisture damage occurs in asphalt mixtures when water infiltration generates pore pressure under traffic loads, causing the asphalt binder film to rupture and lose adhesion to the aggregate.

5.2.5. Hydraulic scouring

Hydraulic scouring, a process that contributes to bitumen-aggregate stripping, involves the removal of asphalt binder through the use of pressurized water forced into the pavement [79]. This pressurized water comes from water trapped between the tire and pavement surface as vehicles move. When the tire rolls, it pushes this water into the pavement, subjecting it to compression and tension cycles that lead to stripping [78]. The damaging effects of hydraulic scouring occur at both large and small scales [78]. Additionally, the presence of clay and silt particles in the water can worsen the problem. These particles act as abrasives, speeding up the stripping process by wearing down the weakened binder-aggregate interface [81]. Essentially, hydraulic scouring uses the force of moving traffic to remove the asphalt binder from the aggregate, causing pavement deterioration. Fig. 8 shows hydraulic scouring, where vehicle tires push and pull water through interconnected voids, creating pore pressure that stresses and weakens the binder film, leading to pavement damage.

Fig. 9 illustrates how moisture damage in asphalt pavements can be driven by either pore pressure-induced microcracking or diffusion-induced chemical deterioration, ultimately leading to the same outcome of mastic detachment and pavement failure.

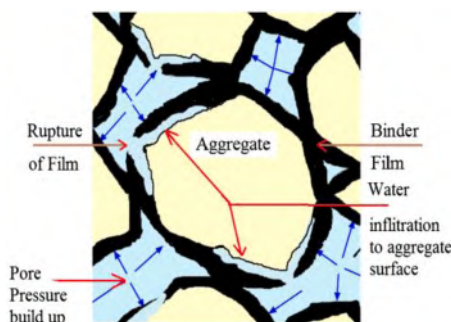


Fig. 7. Moisture damage due to pore pressure [80].

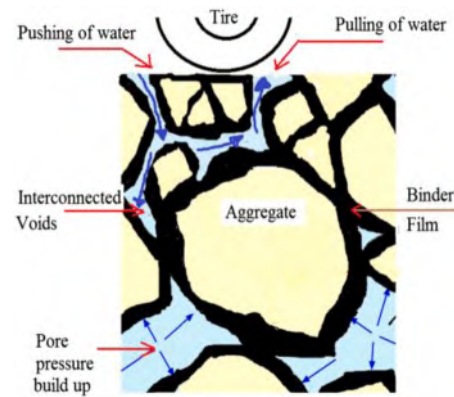


Fig. 8. Moisture damage due to hydraulic scouring [80].

Some other mechanisms have been discussed in the literature which are briefly summarized in Table 7.

5.2.6. Interactions of water damage mechanisms

The mechanisms of water damage in asphalt pavements—detachment, displacement, spontaneous emulsification, pore pressure, and hydraulic scouring—interact under specific environmental and structural conditions, collectively contributing to accelerated pavement deterioration. Detachment, caused by water acting as a wedge at the bitumen-aggregate interface, is prevalent in high-polarity aggregates exposed to wet conditions. Displacement occurs when water infiltrates pre-existing weaknesses in the asphalt film, often exacerbated by dusty aggregates or pH fluctuations that destabilize the bond between bitumen and aggregate. Pore pressure, resulting from water trapped within the critical air void range (8–10 %), intensifies under traffic loads or freeze-thaw cycles, generating micro-cracks that promote further water ingress. Hydraulic scouring, driven by pressurized water under vehicle tires, removes binder at both the micro- and macro-scale, especially in traffic-heavy areas. Spontaneous emulsification destabilizes the binder internally by forming water-induced emulsions, often accelerated in wet conditions or by the presence of clays acting as emulsifiers. Environmental factors, including freeze-thaw cycles and pH variability, further amplify these mechanisms, creating interdependent effects that degrade pavement performance. Understanding the specific conditions under which these mechanisms occur and their interactions is crucial for developing comprehensive mitigation strategies to enhance asphalt pavement durability.

6. Laboratory tests for evaluation of moisture damage

6.1. Qualitative and quantitative tests

Over the past six decades, understanding the performance of asphalt mixture in the presence of water has been a focal point for asphalt technologists and state highway agencies [84]. This pursuit has led to extensive research efforts aimed at developing reliable laboratory test protocols capable of predicting the behavior of asphalt pavements in the field, particularly concerning moisture sensitivity. According to [84], the tests that have been developed to evaluate moisture sensitivity can be divided into two main categories: qualitative and quantitative which are summarized in below Table 8.

Different standard procedures have been developed to assess the susceptibility of asphalt mixtures to moisture damage. Table 9 briefly gives an overview of those standards. Detailed description of these tests can be found on the paper by [85].

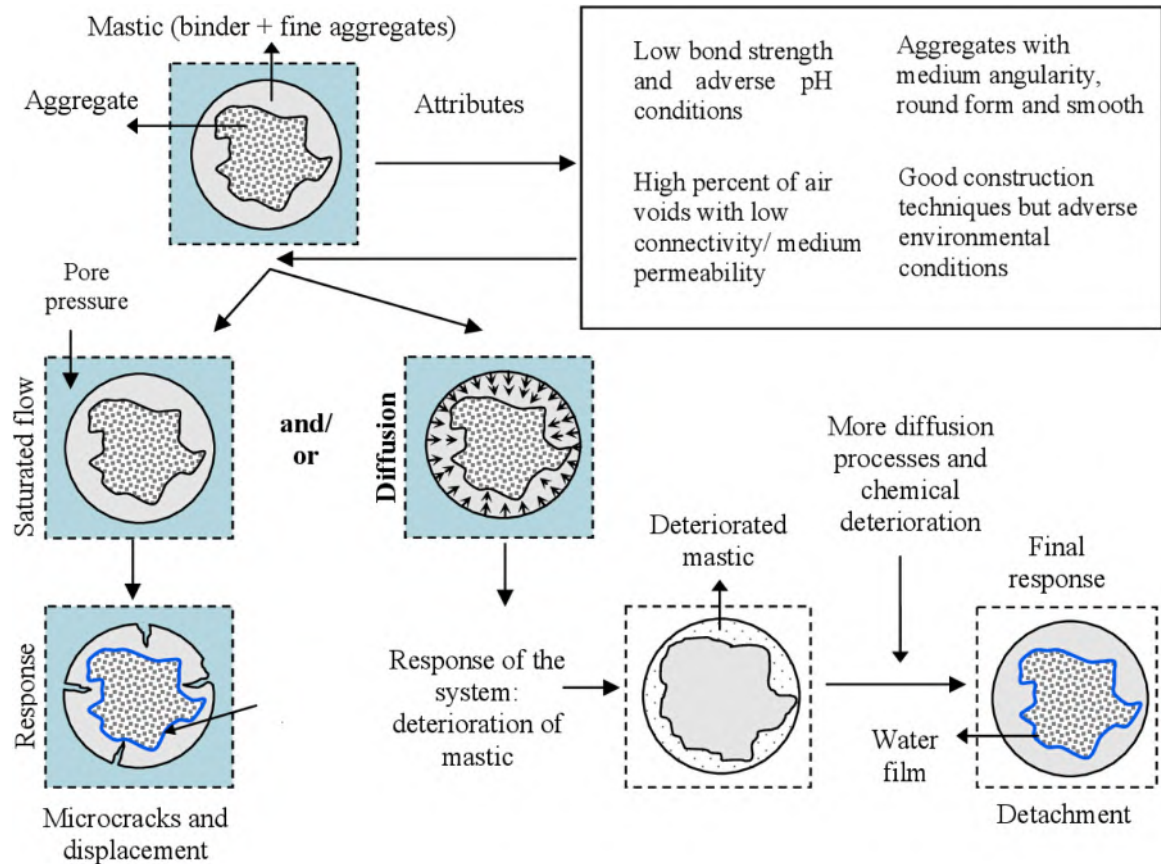


Fig. 9. Two potential mechanisms for moisture damage share the same attributes: a) pore pressure leading to microcracks and displacement, and b) diffusion causing detachment [82].

Table 7
Contributing mechanisms to moisture damage in asphalt concrete mixes [83].

Contributing Factor	Mechanism	Contributing factor	Mechanism
Cohesion	Loss of molecular cohesion in the binder	Spontaneous emulsification	Water penetration into the binder
Film Rupture	Breakdown of the asphalt film to the edges of the aggregate	Water penetration into the binder	Water penetration into the binder
Adhesion	Loss of adhesion in aggregate-asphalt interface	PH Instability	PH fluctuation at the interfacial zone
Pore Pressure	Effect of saturated voids in the mix	Displacement	Water ingress through a crack in the asphalt and displacing the binder
Environmental Effects	Abrupt changes in climate	Hydraulic Scouring	Action of tires on the saturated surface, osmosis effect, salt

6.2. Historical development

Since the 1920s, researchers have developed various methods to study moisture sensitivity in asphalt mixtures. Early tests included boiling and soaking (1937) and the Immersion-Compression Test (1940s). In the 1960s, simulating field conditions became important, while the 1970s introduced cyclic pore pressure application and vacuum saturation conditioning. The 1980s and 1990s saw standardized tests like AASHTO T283 and advanced methods such as the Hamburg wheel-

Table 8
Tests to evaluate moisture sensitivity (adopted from [84]).

Test Category	Test Name	Description
Qualitative	Boiling Water Test	Simulates hot water effect by submerging samples in boiling water
	Freeze-Thaw Pedestal Test	Assesses resistance to freezing and thawing by subjecting samples to cycles
	Quick Bottle Test	Rapid method using a bottle to observe moisture effects
	Rolling Bottle Method	Simulates moisture and mechanical stress by placing samples in a rolling bottle apparatus
Quantitative	Immersion-Compression Test	Measures material strength before and after immersion and compression
	Indirect Tensile Test	Evaluates tensile strength indirectly, often before and after moisture exposure
	Marshall Immersion Test (Asphalt Mixes)	Assesses moisture resistance of asphalt mixes by immersing samples in water
	Double Punch Method	Measures material strength by subjecting it to repeated mechanical stresses
	Resilient Modulus Tests	Determines the modulus of resilience before and after conditioning

tracking device (HWTd). Today, research continues with rapid centrifugation and surface energy studies. More historical developments are described in Table 10.

Other researchers have adopted the tests mentioned in the above tables in their research. Some of the findings from the literature have been summarized in Table 11.

Table 9
Standards for testing moisture-induced damage in asphalt.

Standard Name	Description
AASHTO T 283: Standard Method for Resistance of Compacted Bituminous Mixture to Moisture-Induced Damage [86]	Most widely used, modified Lottman Test for moisture-induced damage susceptibility
ASTM D 4867: Standard Test Method for Effect of Moisture on Asphalt Concrete Paving Mixtures [87]	Alternative to AASHTO T 283, evaluates moisture susceptibility
AASHTO TP 34: Determining Moisture Sensitivity of Compacted Bituminous Mixtures Subjected to Hot and Cold Climate Conditions	Simulates real-world conditions, promising but less common
Moisture Induced Stress Tester (MiST)	Emerging method, assesses adhesive and cohesive failure
EN 12697-43: Bituminous mixtures - Test methods - Part 43: Resistance to water attack by the boiling test	European standard, boiling test for quick evaluation of binder adhesion
AASHTO T 182: Standard Method of Test for Water Resistance of Compacted Bituminous Mixture Specimens	Basic screening method, visual inspection for stripping
AASHTO T 324: Standard Method of Test for Hamburg Wheel Tracking Device	Evaluates water rutting resistance, not solely for moisture damage

Table 10
Historical development of laboratory tests over time (adopted from [84]).

Year	Event	Author
1920s	Early work on moisture sensitivity	Nicholson, Riedel, Weber, McLeod, Hubbard, Powers, Winterkorn, Saville, Axon, Krchma, Nevitt, Loomis, Hveem
1937	Boiling and Soaking Tests	Saville, Axon
1940s	Immersion-Compression Test	ASTM
1950s	Surface energy of asphalt and aggregate	Thelen
1956	Sonic test for evaluation of stripping resistance	Andersland, Goetz
1960s	Importance of simulating field conditions	Johnson
1960s	Importance of simulating field conditions	Schmidt, Graf
1970s	Cyclic pore pressure application	Jimenez
1970s	Vacuum saturation followed by freezing and hot water bath conditioning	Lottman
1980s	AASHTO T283	Lottman, Root, Tunnicliff
1980s	Texas Freeze-Thaw Pedestal Test, Texas Boiling Test	Kennedy, Roberts, Anagnos, Lee
1980s	Bonding energy of the asphalt-aggregate system	Ensley
1980s	Evaluation of moisture damage test methods	Gharaybeh
1990s	Environmental conditioning system (ECS)	Al-Swailmi, Terrel
1990s	HWTD	Aschenbrener, Currier
2000s	Rapid centrifugation method	Western Research Institute (WRI)
2000s	Surface energy	Researchers
2000s	NCHRP Project 9-34	NCHRP

6.3. Comparison of testing methods

Table 12 highlights the trade-offs between simplicity and accuracy in moisture susceptibility tests. While rapid tests like the Texas Boiling Test are useful for preliminary assessments, more advanced methods such as HWTD and MIST are essential for high-stakes projects where field performance and durability are critical. These comparisons underscore the need for adopting complementary methods to achieve a balance between efficiency and reliability in evaluating moisture damage.

Table 11
Use of test methods in research.

Test	Purpose	Key Findings	Ref.
Indirect Tensile Asphalt Cracking Test (IDT)	Assess moisture susceptibility in asphalt mixtures	Increased oxidation and higher conditioning temperature (CIT) significantly reduced the TSR%. Moisture conditioning increased the CT Index, while oxidation decreased it.	[88, 89]
Bitumen Bond Strength (BBS) Test	Gauge the impact of moisture on the asphalt-aggregate bond strength	Bond strength is highly influenced by modification type and moisture exposure duration. Polymer modification enhanced both adhesion and cohesion.	[90]
Pull-Off Tensile Test	Quantify resistance to moisture damage	Fillers like bagasse and coconut peat can be as effective as traditional mineral fillers in preventing moisture damage.	[91]
Accelerated Pavement Testing (APT) with MMLS3	Evaluate moisture stability of specific asphalt mixtures (e.g., hard sandstone)	Hard sandstone asphalt mixtures exhibited good moisture damage resistance.	[92]
Loaded Wheel Tracking Test	Assess the susceptibility of asphalt mixtures to moisture damage	The conditioning protocols were effective in identifying moisture-sensitive mixtures.	[93]
Dynamic Modulus Tests	Characterize the viscoelastic behavior of porous asphalt mixtures	Lower air void content led to a higher dynamic modulus. Moisture conditioning increased the modulus, particularly at lower temperatures.	[94]
Asphalt Pavement Analyzer (APA) Test under Water	Estimate moisture resistance and rutting potential	While all mixtures performed adequately, APA testing did not detect a clear difference in moisture damage susceptibility due to WMA additives.	[95]

6.4. Emerging tools and technologies

Parallel to laboratory tests, image analysis methods have emerged as key tools for detecting moisture-induced damage early. Advanced thermal imaging, combined with deep learning algorithms, allows for accurately identifying moisture damage by analyzing thermal variations across the pavement surface [95,101]. This method, often paired with RGB image data, improves the detection of subsurface moisture-related issues like stripping. GPR also plays a critical role as a non-destructive evaluation tool, effectively identifying changes in dielectric properties that indicate moisture presence within pavement layers [102]. Additionally, digital image processing techniques, including crack segmentation and machine learning, automate the identification of surface defects, reducing the time and subjectivity associated with manual inspections [103]. These advancements enable proactive maintenance strategies, extending pavement lifespan and optimizing maintenance investments by providing timely data on moisture damage.

Table 13 summarizes emerging tools and technologies for detecting and managing pavement damage. Each tool is described based on its application, efficiency, and technical details, providing insights into their capabilities and advantages. The table highlights advancements in areas such as IoT integration, machine learning, digital twins, and remote sensing for effective pavement management and maintenance.

Adopting advanced pavement technologies presents significant cost and training challenges. Tools like IoT-based Pavement Management

Table 12

Comparison of common testing methods for moisture damage in asphalt mixtures.

Method	Pros	Cons	Applicability	Ref.
AASHTO T-283	- Simple, standardized, and widely adopted - Cost-effective and easy to implement.	- Limited ability to replicate dynamic field conditions - Poor correlation with extreme conditions.	Suitable for initial screening in moderate climates.	[96, 97]
HWTD	- Simulates both moisture-induced stripping and rutting - Strong correlation with field performance.	- Expensive equipment; lengthy testing time - Limited accessibility for routine testing.	Ideal for high-traffic areas and extreme weather conditions.	[97, 98]
MIST	- Realistically replicates field conditions - Faster than most other tests (results in under 3 hours).	- Requires specialized equipment - Not widely adopted yet in industry practice.	Best for high-stress environments and critical infrastructure projects.	[96, 99]
Superpave Indirect Tensile Test	- Measures multiple properties, including tensile strength and resilient modulus - Effective for advanced analysis.	Time-intensive and requires specialized preparation.	Used for research and high-stakes projects analyzing fracture mechanics and durability.	[99]
Texas Boiling Test	- Cost-effective, rapid, and straightforward - Requires minimal equipment and expertise.	- Highly subjective and prone to operator variability - Poor correlation with long-term performance.	Suitable for quick preliminary assessments of aggregate-asphalt adhesion.	[100]

Systems and Digital Twin Technology require substantial initial investments in sensors, hardware, and specialized software, such as SCADA systems, Autodesk Tandem, or Bentley Systems. These tools improve decision-making and reduce lifecycle costs by up to 15 % but necessitate recurring expenses for data integration and maintenance [104,108]. Similarly, advanced analysis methods like Deep Learning and Infrared Fusion and RetinaNet Damage Detection achieve high accuracy in crack and damage detection using frameworks like TensorFlow and PyTorch, but demand specialized training for operators to handle AI-driven systems effectively [109].

Additional tools such as Infrared Thermography (e.g., FLIR Tools) for subsurface anomaly detection, Computer Vision Systems (e.g., OpenCV) for automated crack classification, and Real-Time Road Surface Detection using frameworks like YOLO and MobileNet enable efficient monitoring but require adaptation to existing workflows. Remote sensing technologies using ArcGIS or UAV platforms like Pix4D further enhance large-scale assessments but involve significant costs and expertise.

To overcome these barriers, targeted training programs, cost-sharing models, and leveraging open-source software (e.g., ImageJ, scikit-learn) can facilitate adaptation. Addressing these challenges will ensure the successful implementation of innovative pavement management strategies in real-world applications.

6.5. Advancements and limitations

Several methods have been developed to evaluate the susceptibility of loose mixtures to stripping, offering advantages in terms of simplicity, speed, and ease of use [84]. However, these tests may not fully consider factors such as pore pressure, traffic action, and the mechanical properties of the mixture itself [84]. The most commonly used method, the Static Immersion Test (AASHTO T 182), involves submerging the mixture in water for a specified period [84]. Variations include the Dynamic Immersion test, which includes an agitation step for accelerated evaluation [84], and the Boiling Water Test (ASTM D3625), a quick procedure that boils the mixture for 10 minutes [117]. While the Boiling Water Test shows potential in identifying moisture mixtures [94], concerns exist regarding its representation of field conditions [118]. Other methods, such as the Rolling Bottle Test (BS EN 12697–2012) and the California Test 302, aim to integrate elements of rolling traffic and ease of use, respectively [84,119]. On the other hand, the Methylene Blue test is not a direct stripping test but rather a measure of potentially harmful clays and organic matter in the fine aggregate, which can contribute to stripping [120].

There are several other test methods developed based on calculation. The Net Adsorption Test (NAT) is used to determine the affinity between bitumen and aggregate as well as water sensitivity by measuring bitumen adsorption and desorption [121]. The Ultrasonic Energy method assesses the bond strength between bitumen and aggregate using sound waves, showing sensitivity to bitumen type, aggregate type, and the presence of additives [122]. The SFE method analyzes the surface energy of bitumen and aggregate to predict stripping potential. This method takes into account factors such as bitumen type, aggregate type, and aging, and utilizes various techniques like the Wilhelmy plate method [123,124]. The Pneumatic Adhesion Tensile Testing Instrument (PATI) provides a fast and cost-effective way to measure the adhesive strength between bitumen and aggregate using a pull-off test. This method allows for various aggregate types and conditioning options [125]. Lastly, Atomic Force Microscopy (AFM) employs a high-resolution approach to measure forces at the nanoscale, enabling the assessment of adhesive/cohesive forces and surface roughness. AFM has the potential to distinguish between different anti-stripping agents [126,127].

While substantial progress has been made in understanding and evaluating moisture susceptibility in asphalt mixtures, critical gaps remain. For instance, current methods such as AASHTO T-283 fail to replicate extreme environmental conditions like freeze-thaw cycles and heavy rainfall, reducing their applicability across diverse climates [96, 97]. Additionally, there is a lack of research on sustainable materials, such as bio-based and recycled binders, under standard test conditions [98,128]. Addressing these gaps could lead to more comprehensive and inclusive evaluation practices, bridging the discrepancy between laboratory tests and field performance [96,129].

To address the limitations of existing evaluation methods, innovative approaches have been proposed. Combining tests, such as the HWTD and MIST, can provide a more comprehensive assessment of static and dynamic moisture-induced stresses, enhancing predictive accuracy [130]. Modifications to AASHTO T-283, including pressure-based conditioning, can better simulate field conditions and improve relevance in extreme environments [96]. The integration of artificial intelligence for data analysis offers opportunities for more accurate predictions and optimized designs [97]. Furthermore, incorporating bio-based materials and nanomaterials under tailored protocols supports sustainability while ensuring material resilience [98,128].

7. Mitigation measures

Approximately 87 % of transportation agencies in the US attempt to address this issue by specifying moisture susceptibility tests and minimum standards. Additionally, 82 % of the agencies recommend the use

Table 13
Emerging tools and technologies for pavement damage detection and management.

Tool Name	Application	Efficiency/Features	Technical Data/Notes
IoT-based Pavement Management System	Real-time data collection and management of pavement conditions	Improves decision-making with real-time insights; integrates big data	Applicable to urban and highway pavements; integrates IoT sensors [104].
Computer Vision System	Low-cost automated road crack detection and classification	Accurate detection with minimal hardware cost; uses Hough Transform	Applicable to maintenance in rural and urban areas; accuracy 85–90 % [105].
RetinaNet Damage Detection System	Deep learning-based assessment of asphalt damage	High precision with RetinaNet architecture; large dataset support	Supports regional and national-level analysis; accuracy 92 % [106].
Infrared Thermography	Detection of subsurface debonding and thermal anomalies	Non-destructive; identifies flaws not visible to the naked eye	Effective for areas with heavy loads and thermal variability [107].
Digital Twin Technology	Lifecycle analysis and predictive maintenance	Combines real-time sensing with analytics; supports sustainability	Optimal for smart cities; reduces lifecycle costs by 15 % [108].
Deep Learning and Infrared Fusion	Crack severity classification using fused imaging	Enhanced accuracy in severity grading; uses infrared data fusion	Enhances thermal crack grading by 30 %; fusion accuracy 93 % [109].
DCNNs for Pavement Analysis	Automated pavement distress identification and analysis	Supports large datasets; efficient with convolutional neural networks	Widely used in urban infrastructure; efficiency 85 % [110].
High-Efficiency Digital Twin Modeling	High-fidelity pavement modeling for lifecycle predictions	Accurate predictions for maintenance using advanced simulations	Supports maintenance planning; 98 % accuracy in predictions [111].
Image-Based Crack Detection	Analysis of surface cracks using image processing	Provides robust crack detection via image contrast techniques	Best for localized maintenance tasks; detection accuracy 88 % [112].
Smart Pavement Decision Support System	Integrating smart sensors with decision support systems	Combines IoT data with AI for predictive maintenance	Efficient for long-term pavement management plans; ROI improvement 20 % [113].
Real-Time Road Surface Detection	Detection of road cracks and potholes in real-time	Improves maintenance response; utilizes lightweight frameworks	Scalable to urban networks; identifies 95 % of potholes [114].
Remote Sensing for Pavement Assessment	Remote sensing of pavement conditions and deterioration	Large-scale assessment using satellite and UAV data	Efficient for national networks; UAV integration improves coverage [115].
Moisture Susceptibility Machine Learning	Predicting HMA moisture susceptibility using ML models	Highly accurate classification with k-NN and PCA; uses MIST data	Applicable to new mix designs; k-NN accuracy 90 %; PCA reduces noise [116].

of antistripping additives to prevent damage caused by this phenomenon [131]. Using additives may delay these mechanisms but cannot prevent them totally [44]. The mitigation measures from different perspectives are briefly presented below.

7.1. Asphalt binder perspective

Effective mitigation strategies focus on improving the moisture resistance of the binder to enhance the overall structural integrity of asphalt mixtures. Approaches such as polymer modification, incorporation of anti-stripping agents, and the selection of appropriate binder grades have been shown to significantly enhance the binder's resistance to moisture-induced distresses, thereby extending the service life of pavements and reducing maintenance requirements. Table 14 below presents a summary of various modifications to asphalt binders aimed at enhancing moisture resistance in asphalt mixtures.

7.2. Mix design perspective

Moisture can weaken the bond between asphalt binder and aggregates, leading to issues such as stripping, reduced structural integrity, and increased maintenance needs. Effective mitigation strategies also include optimizing aggregate gradation which enhances the mechanical properties and moisture resistance of the pavement. Table 15 below highlights various studies focused on improving moisture resistance in asphalt mixtures through different approaches, such as aggregate gradation, cold-recycling techniques, production temperature adjustments, and the use of WMA with additives.

7.3. Asphalt binder and mix design synergy

The synergistic integration of advanced binder modifications and optimized mix designs is pivotal for mitigating water damage in asphalt pavements. Binder modifications such as bio-based additives, thermoplastic polyurethane (TPU), and organic montmorillonite (OMMT) enhance resistance to moisture damage and improve mechanical properties. For example, bio-modifiers derived from swine manure or bio-oil improve adhesion at the binder-aggregate interface by occupying active sites, thereby reducing moisture susceptibility [142]. Similarly, the

combination of TPU and OMMT demonstrates significant improvements in high- and low-temperature properties, elastic behavior, and moisture resistance when applied in optimized proportions [143]. Furthermore, combinations of polyphosphoric acid (PPA) and nano-silica demonstrate significant synergistic effects by enhancing binder stiffness and reducing moisture sensitivity. However, the presence of wax content in the binder can antagonize these effects by blocking active sites, highlighting the importance of precise formulation in binder design [144]. Technologies such as nanohybrid interfacial layers combining graphene oxide and halloysite nanotubes have also been shown to enhance binder-aggregate interaction, creating a robust three-dimensional matrix that improves adhesion and mechanical stability [145].

Optimized mix designs complement these binder advancements by ensuring mechanical performance and durability under various environmental conditions. Dense-graded asphalt mixtures with bio-binders, such as those derived from paper industry by-products, achieve comparable performance to conventional binders while offering improved sustainability [146]. These mixtures provide strong resistance to permanent deformation and cyclic loading while maintaining adequate water sensitivity. Moreover, the inclusion of modifiers like bio-oils and polymers in the mix design improves rutting resistance and stiffness, though careful calibration is required to balance performance across temperature extremes [147].

The synergy between binder improvements and optimized mix designs enables enhanced mitigation outcomes by leveraging their complementary effects. For instance, bio-binders in dense-graded mixes not only promote environmental benefits but also ensure mechanical stability, while TPU and OMMT-modified binders enhance adhesion and resistance to moisture-induced damage. Together, these strategies improve pavement durability and effectively address water-related challenges [143,145]. Future research should refine these approaches, particularly by exploring the interaction of bio-binders and nano-materials in sustainable pavement technologies.

7.4. Role of proper drainage

Proper drainage plays a crucial role in mitigating moisture-induced damage in asphalt mixtures by preventing water from infiltrating and accumulating within the pavement structure. Effective drainage systems

Table 14
Mitigation measures from asphalt binder perspective.

Modification Type	Findings	Supporting Data	Ref.
Carbon Nanotube (CNT) Modified Asphalt Binder	Improved moisture resistance due to enhanced adhesion between binder and aggregates. AFM results showed better adhesion forces with reduced moisture-induced weakening.	Tested with AASHTO T-283 method. AFM adhesion forces improved significantly.	[132]
WMA with Zycotherm and Sasobit	Zycotherm-modified mixtures exhibited higher resistance to moisture damage compared to Sasobit and conventional HMA. TSR and resilient modulus ratios were improved.	Resilient modulus ratio and TSR values higher for Zycotherm mixtures.	[133]
Nano-Silica (NS) Modified Asphalt Binder	2 % NS addition increased TSR value to 91 %, surpassing the AASHTO T-283 threshold of 80 %, showing enhanced moisture resistance and durability.	TSR value of 91 % for 2 % NS-modified binder compared to 74 % for conventional binders.	[134]
Styrene-butadiene-styrene (SBS) Modified Asphalt Binder	Higher SBS content (8–10 %) improved moisture resistance by increasing the energy required for debonding between binder and aggregates. Debonding energy improved by about 30 %.	30 % improvement in work of debonding with 8–10 % SBS content.	[135]
Toner-Modified Asphalt Binder	Toner-modified binder demonstrated improved viscosity and shear strength, enhancing moisture resistance. At 8 % toner content, significant improvements were noted.	8 % toner content improved binder viscosity and shear strength.	[136]

help direct surface water away from the pavement and manage sub-surface water, reducing the risks of moisture-related issues such as stripping, raveling, and loss of adhesion between the asphalt binder and aggregates. Table 16 below provides an overview of various aspects of drainage systems in asphalt pavements, highlighting methods like surface and subsurface drainage, their benefits, and the consequences of poor drainage.

Effective drainage is critical for mitigating moisture damage in coastal roadways. Structural measures, such as seawalls and elevated roadways, serve as barriers against storm surges and help reduce flooding. Additionally, nature-based solutions, including salt marshes, mangroves, and dune restoration, provide protection by attenuating wave energy and minimizing erosion [154]. Effective drainage system design integrates considerations of topographical and climatic factors, ensuring optimal water management. For instance, road slopes significantly influence drainage performance, with steeper gradients facilitating efficient runoff flow, while flatter terrains demand high-capacity infiltration systems to mitigate pooling and water-related damage [155, 156]. Climatic factors, such as rainfall intensity and frequency, further dictate the hydraulic capacity of drainage systems, requiring parameters like void ratios (20–30 %) and sufficient trench depths to handle peak runoff conditions [157,158].

Subgrade moisture content and seasonal variations play a critical role in influencing pavement durability, emphasizing the need for

Table 15
Mitigation measures from mix design perspective.

Study Focus	Findings	Supporting Data	Ref.
Impact of Aggregate Gradation on Moisture Susceptibility	Coarse-graded mixtures improved moisture resistance by enhancing internal stability. Higher TSR values indicated better performance.	ITS and TSR tests showed higher TSR values for coarse-graded mixtures.	[137]
Cold-Recycled Asphalt Mixtures with Controlled Gradation	Fine-graded CIR mixtures with high RAP binder content showed improved coating and moisture resistance. Proper gradation control achieved better performance.	Moisture-conditioned stability tests showed better stability for fine-graded RAP mixes.	[138]
Effect of Production Temperature and Aggregate Moisture Content in Foamed WMA	Using dry aggregates and reducing production temperature by 30 °F resulted in comparable moisture resistance to HMA. Moist aggregates led to higher moisture susceptibility.	Modified Lottman Test (AASHTO T-283) results supported the moisture resistance improvement.	[139]
Aggregate Blending for Achieving Optimal Voids in Mineral Aggregate (VMA)	Balanced aggregate gradation maintained optimal VMA, which enhanced moisture resistance by minimizing moisture intrusion.	Analysis demonstrated improved VMA and moisture resistance with balanced aggregate proportions.	[140]
WMA with Additives and Aggregate Gradation Control	Incorporating 20 % RAP with a well-graded aggregate blend in WMA achieved satisfactory moisture resistance, meeting Superpave® TSR thresholds.	TSR values met the Superpave® minimum threshold of 0.8 for moisture-conditioned samples.	[141]

tailored drainage solutions that maintain low moisture levels and prevent saturation-induced damage [159]. In cold climates, specific data on frost depth and water removal times underscore the importance of incorporating frost-resistant designs, which prevent frost heave by effectively removing free water from pavement layers [160]. In areas with high groundwater tables or frequent flooding, subsurface drainage systems such as French drains and permeable bases are particularly effective in reducing water ingress and improving structural resilience.

Advancements in drainage technologies have introduced systems such as permeable pavement systems (PPS), which incorporate porous materials to reduce surface runoff, enhance pollutant retention, and promote groundwater recharge. These systems have been shown to achieve pollutant reductions of up to 50 % for turbidity and phosphorus levels, demonstrating their dual functionality in water management and environmental sustainability [161]. Additionally, automated drainage assessment technologies, such as image acquisition and processing systems, have revolutionized the evaluation and optimization of drainage performance, enabling real-time identification of deficiencies and supporting data-driven maintenance strategies [162].

By integrating these advanced systems and carefully selecting design parameters based on environmental conditions, drainage systems can effectively mitigate water damage, enhance pavement longevity, and contribute to sustainable urban infrastructure development. This comprehensive approach highlights the necessity of aligning drainage design with specific topographical and climatic contexts while leveraging innovations to address emerging challenges in water management.

Table 16
Role of proper drainage in mitigating moisture-induced damage.

Aspect	Description	Supporting Data	Ref.
Surface Drainage	Ensures water is directed off the surface through slopes (1–2 %) and catch basins, reducing pooling and hydroplaning risks. A slope of at least 2 % is standard for effective water flow.	Standard slope recommendation for surface drainage is 2 % to prevent pooling, reducing repair costs due to water damage.	[148,149]
Subsurface Drainage	Involves using perforated pipes, permeable base layers, and geotextiles. Effective subsurface drainage can prevent water pooling under the surface, which can extend pavement life by 10–15 years.	Subsurface drainage systems can extend pavement life by up to 15 years through proper water management, potentially saving on reconstruction costs.	[148–150]
Prevention of Water Accumulation	Properly graded surfaces prevent standing water, reducing freeze-thaw damage in colder climates, which can decrease pavement deterioration rates by 30 %.	Freeze-thaw cycle reduction can decrease deterioration rates by 30 %, reducing repair needs and lowering costs of resurfacing by up to 20 %.	[149,151]
Impact of Poor Drainage	Water infiltration can weaken subgrade layers, causing structural instability. Poor drainage is a leading cause of potholes, increasing maintenance costs by up to 50 %.	Inadequate drainage can increase pothole repair costs by 50 % due to water-related structural failures, impacting annual maintenance budgets.	[151,152]
Innovative Drainage Solutions	Porous asphalt reduces runoff by allowing water to percolate through the surface, decreasing stormwater system loads by 60 %. Geotextiles prevent fine soil migration, ensuring stable subgrade layers.	Porous asphalt and geotextiles can reduce stormwater runoff load by 60 %, lowering costs associated with urban flood management systems.	[149,150]
Long-term Benefits of Proper Drainage	Reduces maintenance costs by 20–30 %, extends pavement lifespan by 5–10 years, and minimizes freeze-thaw cycle damage, preserving structural integrity.	Proper drainage reduces repair costs by 20–30 % over time, with investments in drainage systems showing a return through reduced maintenance expenses.	[151,153]

7.5. Advanced technologies

Recent research into mitigating moisture-induced damage in asphalt mixtures highlights both material modifications and advanced assessment methods. The use of nanomaterials such as NS and nano-titanium dioxide (NT) has shown significant improvements in enhancing the moisture resistance of asphalt binders, leading to better performance in terms of tensile strength and durability [163,164]. Incorporating these nanomaterials into asphalt mixtures has proven to increase TSR and Index of Retained Strength (IRS), crucial metrics for evaluating moisture resistance. Additionally, traditional additives like hydrated lime remain effective in improving the adhesion between asphalt and aggregates,

while the adoption of WMA technologies offers benefits by reducing compaction temperatures, thus enhancing moisture resistance [164, 165]. These advancements are complemented by testing techniques such as HWTD, which provides insights into the long-term moisture susceptibility of modified mixtures [165].

Table 17 shows advanced technologies developed to mitigate moisture-induced damage in asphalt pavements, addressing challenges like stripping, cracking, and rutting. These innovations include nano-materials such as nano-silica and carbon nanotubes, which enhance tensile strength and moisture resistance, and geosynthetics, which improve drainage and subgrade stability in high-moisture areas. Additionally, self-healing asphalt technologies and electrically conductive asphalt concrete (ECAC) provide long-term solutions by enabling autonomous crack repair and multifunctionality, such as snow melting. These technologies improve binder-aggregate adhesion, structural integrity, and proactive damage prevention, offering sustainable and durable solutions for asphalt pavement performance.

7.6. Role of construction practices

Effective construction practices are critical in minimizing moisture-induced damage in asphalt mixtures, with a primary focus on aggregate drying, quality control, and compaction. Proper drying of aggregates before mixing ensures effective coating with the asphalt binder, which is essential to prevent moisture from displacing the binder and causing stripping [173]. Quality control measures, such as the AASHTO T 283 test, allow for the evaluation of moisture susceptibility in asphalt mixtures before construction, facilitating necessary adjustments to the mix design [8]. Additionally, achieving appropriate compaction during construction reduces air voids in the mixture, thereby limiting potential moisture infiltration, which is crucial for maintaining the long-term integrity of the pavement [173].

Table 18 below serves as a summary of the key mitigation strategies discussed earlier, providing a concise overview of the various approaches to reducing moisture-induced damage in asphalt mixtures.

Table 19 below provides a summary of various additives and fillers used to improve the moisture resistance of asphalt mixtures, building on the detailed discussion provided earlier. Each additive is evaluated for its impact on moisture susceptibility, stability, and overall performance of the asphalt.

8. Implications for climate resilience and sustainable infrastructure

Climate change poses significant challenges to pavement infrastructure through intensified precipitation, increased freeze-thaw cycles, and rising temperatures. Heavy rainfall and extreme weather events lead to subgrade saturation, reducing stiffness and accelerating rutting and cracking, while freeze-thaw cycles increase thermal cracking and moisture ingress, particularly in colder regions like Northern Canada [182,183]. Urbanization compounds these issues as impervious surfaces hinder water infiltration, with inadequate drainage systems amplifying moisture-related failures during extreme rainfall, as evidenced on major highways like Ontario's 401 [182,184].

The economic implications are substantial, with the U.S. projecting an additional \$13.6 billion in pavement maintenance costs by 2040 due to climate-induced deterioration [185]. While adaptation strategies such as upgrading asphalt binders, increasing pavement thickness, and implementing advanced drainage systems have shown promise, the industry is increasingly turning to innovative solutions for long-term resilience [182,186].

Low-carbon technologies in asphalt pavement construction have emerged as effective solutions for reducing GHG emissions while enhancing sustainability [187]. Biochar, a carbon-neutral material derived from biomass waste, improves asphalt binder properties with an optimal particle size of < 75 µm, reducing CO₂ emissions by up to 10 %

Table 17
Advanced technologies for moisture mitigation in asphalt pavements.

Technology	Efficiency	Applicability	Other Technical Details	Ref.
Nano-Silica Modification	Improves moisture resistance of asphalt mixtures, enhances tensile strength by 60 % (ITS).	Suitable for heavy and medium traffic conditions.	Optimal content: 4 %; reduces moisture sensitivity and improves bitumen-aggregate adhesion.	[166]
Carbon Nanotubes (CNTs)	Enhances tensile strength and reduces rutting; significant moisture damage resistance.	Widely used for high-performance pavements.	Optimum CNT content: 0.75 %–2.25 %; increases viscosity, improving durability but reducing workability.	[167]
Nano Zinc Oxide (ZnO)	Improves moisture resistance indices like Marshall Stability and ITS by 40 %–60 %.	Ideal for regions prone to water-related damages.	Enhances aggregate coating and improves fracture energy; effective at 0.6 % content.	[168]
Nano-Hydrated Lime	Increases TSR by 60 % and resilience modulus (IRMr) by 56 %.	Effective for HMA in wet climates.	Produced using ball milling; distributes homogeneously in bitumen; resists freeze-thaw cycles.	[169]
Extrinsic Self-Healing	Thermal or rejuvenator-based healing improves crack repair efficiency by 30 %–50 %.	Applicable for low-carbon infrastructure projects.	Combines phase field and molecular diffusion theories for efficient crack repair.	[170]
ECAC	Provides multifunctional properties like snow melting and self-healing; reduces maintenance costs by 20 %.	Smart city projects and cold climates.	Uses carbonaceous and metallic-based materials; challenges include uniform additive dispersion.	[171]
Nano-Bentonite	Enhances moisture resistance and adhesion in bitumen by increasing surface energy.	Effective in low and moderate traffic areas.	Works well with hydrated lime to enhance moisture resistance; optimal content is ~4 %.	[169]
Nano-Reduced Graphene Oxide (RGO)	Improves ITS and fracture energy of SMA mixtures; reduces moisture susceptibility.	High-performance SMA pavements.	Increases viscosity and stiffness; more effective in specific mix designs compared to ZnO.	[168]
Geosynthetics	Improves drainage and reduces water infiltration, enhancing subgrade stability.	Effective for high moisture areas and flood-prone regions.	Works as a barrier and reinforcer; enhances aggregate interlock and prevents subgrade pumping.	[172]

Table 18
Strategies to reduce the effect of moisture on pavement (adopted from [31]).

Strategy	Description
Binder & Gradation	• Reduce air voids with proper gradation. • Use optimal binder content. • Select binder with low moisture diffusion coefficient.
Adhesion Promoters	• Add anti-stripping agents (especially amine-based), hydrated lime, or specific polymers.
Additives	• Anti-stripping agents: Improve adhesion, effectiveness may vary. • Hydrated lime: Reacts with bitumen to enhance adhesion and moisture resistance. • Polymers (SBS, EVA): Strengthen bitumen-aggregate bond, effectiveness depends on the type.
Nanomaterials (Emerging)	• Explore nanomaterials like nano-clay to improve adhesion and moisture resistance.
Construction Practices	• Minimize moisture exposure during construction. • Ensure proper compaction.
Drainage	• Maintain proper surface and subsurface drainage. • Control water table levels.

[188]. Similarly, carbon fiber powder (CFP) enhances microwave self-healing and low-temperature performance, with a 25 % replacement of limestone powder demonstrating superior durability and reducing lifecycle emissions by 32 % during maintenance [189].

The transition to net-zero infrastructure represents a fundamental shift in pavement engineering. Carbon capture technologies, including chemical absorption and CO₂ separation, have demonstrated the potential to abate emissions by 60–95 % in asphalt and concrete mixtures [190]. Energy-efficient practices, such as transitioning to renewable energy for asphalt production plants and deploying electric construction equipment, can reduce operational emissions by up to 95 %, while renewable energy sources for mixing plants achieve reductions of 36–90 % in carbon emissions [188].

Production processes have also seen significant innovations. WMA technologies, particularly LEADCAP, enable production at lower temperatures, reducing CO₂ emissions by 15–40 % while maintaining performance equivalent to HMA [191]. A Chinese case study revealed that adopting WMA technologies and transitioning to clean energy sources could reduce energy consumption by 20 % and emissions by 15–25 % [192].

The integration of recycled materials has proven crucial for achieving net-zero infrastructure goals. RAP, bio-based additives, and

WMA can reduce embodied carbon by up to 36 % [188,190]. Lifecycle assessment studies indicate that material production accounts for 70 % of road infrastructure’s carbon footprint, with secondary materials potentially reducing emissions by 12–42 % over a 40-year lifecycle [193].

Supporting sustainable development goals (SDGs), particularly SDG 9 and SDG 12, the industry has embraced innovative materials like marble dust, red mud, and fly ash. These alternatives can lower carbon emissions by up to 42 % over a pavement lifecycle while improving performance characteristics [194,195]. Modular bituminous blocks incorporating RAP and marble dust represent a significant advancement in sustainable maintenance practices, offering enhanced durability and installation efficiency.

Case studies worldwide demonstrate the potential of comprehensive approaches to sustainability. In Münster, Germany, the use of RAP and alternative materials could reduce CO₂ emissions by 133,608 tons and save 3668 GWh of energy over 100 years [195]. The UK has achieved emission reductions of up to 12 % through low-carbon alternatives, even under conservative grid decarbonization scenarios [193]. These successes underscore the importance of integrated approaches combining material innovations, technological advancements, and operational interventions in achieving sustainable infrastructure development.

9. Future works

Advancing the understanding and mitigation of moisture damage in asphalt mixtures requires targeted research in several critical areas. Developing predictive models that integrate SFE parameters with mechanical properties presents a promising avenue for accurately assessing moisture susceptibility. With advancement in AI, incorporating machine learning techniques into these models can enhance their predictive capabilities by leveraging diverse datasets that capture a range of asphalt mixture types and environmental conditions. This approach can bridge the gap between laboratory results and field performance, providing a more reliable basis for designing moisture-resistant pavements.

Refinement of evaluation techniques is equally essential, particularly through the integration of advanced image analysis methods. Leveraging AI to quantify moisture damage can enhance precision and repeatability, enabling rapid assessment of asphalt mixture performance. Future studies should aim to standardize AI-driven analysis protocols, ensuring their widespread applicability in asphalt mixture development and quality control. Nanomaterials and sustainable

Table 19
Feasibility of a variety of additives/fillers.

Additive/Filler	Description	Key Findings	Ref.
NS	Asphalt binder mixed with 2 %, 4 %, or 6 % NS by weight.	Improves pavement performance, stability, and volumetric properties, and reduces moisture susceptibility.	[174]
Ceramic Fiber	Added in 0.75 %, 1.5 %, or 2.25 % proportions to the asphalt mixture.	Improves Marshall properties and moisture resistance (optimal content: 1.5 %).	[175]
Bagasse & Coconut Peat	Ground into fine particles (passing sieve no. 200) to replace mineral fillers.	Effective in preventing moisture damage.	[90]
Natural Rubber Additive (NRA)	Used to modify high-viscosity asphalt mixtures (HVAM).	Superior resistance to temperature and moisture damage.	[176]
Re-refined Acidic Sludge (RAS) & Amorphous Poly Alpha Olefin (APAO)	Virgin bitumen (60/70) modified with 10 %, 20 %, or 30 % RAS and APAO polymer.	Improves moisture damage resistance, rutting resistance, and resilient modulus.	[177]
Polyethylene Terephthalate	Used with granite and limestone aggregates and neat 60/70 bitumen.	Improves moisture damage resistance.	[178]
Cecabase Warm Mix Additive	Used with PG–64 and PG–76 binders.	Improves work of adhesion and spreadability over limestone compared to granite.	[69]
Hydrated Lime & Nano Hydrated Lime	Used as mineral fillers.	Enhances moisture damage resistance.	[179]
Ordinary Portland Cement (OPC)	Used as a mineral filler with RAP at 15 %, 30 %, or 45 % and 4 % styrene butadiene styrene (SBS) polymer.	Improves performance and reduces moisture damage when used with RAP and SBS.	[180]
Cementitious Fillers	Fly ash, cement kiln dust, and three types of hydrated lime (varying fineness) were used.	Effective in reducing moisture susceptibility, with finer hydrated lime being more resistant.	[181]
Limestone Powder, Cement, Slaked Lime, & Brake Pad Powder	A Pen 70 base asphalt was modified with these fillers.	Slaked lime is the most effective, followed by brake pad powder, cement, and limestone powder.	[180]

additives offer substantial potential for improving moisture resistance. Materials such as nano-silica, nano-hydrated lime, and graphene oxide have demonstrated significant benefits in laboratory settings, but their long-term field performance remains underexplored. Future research should focus on evaluating these materials under diverse traffic and environmental conditions to establish their durability and cost-effectiveness. Parallel efforts should investigate bio-based additives and recycled materials to align moisture resistance strategies with sustainability objectives, thereby supporting the development of environmentally responsible infrastructure.

The design of climate-resilient asphalt mixtures is another critical research priority. Extreme weather conditions, such as heavy rainfall, flooding, and freeze-thaw cycles, exacerbate moisture-related damage and require innovative solutions. Studies should explore the integration of advanced drainage systems, hydrophobic materials, and temperature-stable additives to enhance pavement performance under these challenging conditions. Advancing testing methodologies to better replicate real-world conditions is vital. Hybrid approaches that combine established methods like AASHTO T-283 with advanced technologies such as

the MIST and HWTD can offer a more comprehensive evaluation of moisture susceptibility. Incorporating field simulations into these tests will improve their relevance to actual pavement performance, particularly in extreme climates or high-traffic environments.

The application of digital twin technology in pavement management represents a transformative opportunity for real-time monitoring and predictive maintenance. By integrating IoT sensors, AI analytics, and lifecycle modeling, digital twins can provide actionable insights into moisture-related damage and optimize maintenance interventions, enhancing pavement durability and extending service life. Field-validated case studies are essential to bridge the gap between research and practice. Long-term studies that document the performance of proposed mitigation strategies and materials across various geographies and traffic conditions can provide invaluable data for industry adoption. Furthermore, future research should adopt a holistic approach to sustainability, examining the synergy between low-carbon materials, recycled content, and moisture-resistant technologies. Lifecycle assessments that consider environmental benefits, cost efficiency, and long-term performance will guide the development of sustainable best practices.

Addressing these research priorities will drive significant advancements in understanding and mitigating moisture damage in asphalt mixtures. These efforts will not only improve pavement performance but also contribute to the development of durable, climate-resilient, and sustainable infrastructure.

10. Conclusions

Moisture-induced damage in asphalt mixtures continues to pose a significant challenge in pavement engineering, impacting the durability and performance of road infrastructure. This thorough review has emphasized the intricate nature of moisture-induced damage mechanisms and the different methods used to assess and mitigate this issue. The examination of various asphalt mixture types, such as HMA, WMA, and CMA, highlights their differing susceptibilities to moisture damage, underscoring the necessity for customized solutions based on mixture characteristics and environmental factors.

The assessment of current laboratory test methods illustrates the advancements made in evaluating moisture susceptibility as well as the limitations of existing techniques. While several standardized tests are available, there is a clear need for more dependable and field-representative methods that can accurately forecast long-term moisture damage performance. Advanced technologies, such as surface energy evaluations, thermal imaging, and machine learning, offer significant potential for refining these methodologies and bridging the gap between laboratory assessments and field conditions.

The review of mitigation strategies, particularly the use of different additives and fillers, presents encouraging findings in enhancing moisture resistance. Nanomaterials, polymers, and cementitious materials have shown potential in improving the bond between asphalt and aggregates, thus reducing susceptibility to moisture. Emerging technologies like ECAC, IoT-integrated pavement monitoring systems, and self-healing materials further extend the possibilities for proactive damage mitigation and durability improvements. However, the long-term effectiveness, scalability, and sustainability of these solutions require further exploration.

Emerging technologies are paving the way for innovative approaches to address moisture damage. Digital twin technology, coupled with IoT-based real-time monitoring systems, enables predictive maintenance and enhanced management of pavement conditions. Machine learning and artificial intelligence algorithms are revolutionizing moisture damage assessment by enabling rapid, accurate predictions of pavement performance and facilitating data-driven decision-making. Furthermore, the integration of advanced materials such as graphene oxide, nano-silica, and bio-based additives enhances mechanical properties while contributing to sustainable infrastructure development.

Future research should focus on the field validation of these emerging technologies and materials, the development of more precise predictive models based on surface energy parameters, and the refinement of image analysis techniques for evaluating moisture damage. Additionally, interdisciplinary approaches that integrate climate-resilient design, advanced drainage systems, and sustainable construction practices will be critical for addressing evolving environmental challenges.

In conclusion, addressing moisture damage in asphalt mixtures necessitates a multifaceted approach that combines advanced material science, improved testing methodologies, and the adoption of cutting-edge technologies. By embracing emerging innovations alongside traditional practices, the industry can ensure the long-term performance and resilience of asphalt pavements, fostering sustainable, cost-effective, and durable infrastructure solutions.

CRedit authorship contribution statement

Neupane Pratik: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Wu Shenghua:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Enhancing moisture resistance in hot mix asphalt: evaluating the effectiveness of amine-free anti-stripping agents

Bayan Abdullah Taha¹ · Suad Kamiran Rasheed¹ · Zaid Hazim Al-Saffar² · Hasanian Radhi Radeef³ · Anmar Dulaimi^{4,5}

Received: 9 May 2025 / Revised: 18 June 2025 / Accepted: 20 June 2025 / Published online: 2 July 2025
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Abstract

This research investigates the influence of moisture-induced degradation in hot mix asphalt (HMA), a critical issue that affects the performance and durability of asphalt pavements. To address this problem, the study examines the effect of adding a 0.3% amine-free anti-stripping agent based on the weight of the asphalt binder. The Marshall method was employed to prepare modified and unmodified samples with air voids of 7 ± 0.5 , and a vacuum device was used with varying saturation times (5, 10, and 15 min) to simulate moisture exposure. Saturation degree, Indirect Tensile Strength (ITS), and Tensile Strength Ratio (TSR) were utilised to evaluate performance. The results revealed that the modified mixes exhibited lower moisture content and greater moisture resistance compared to the unmodified mixes. In comparison to the unmodified mix, the modified mix demonstrated improved Marshall stability at 40 min, 24 h, and 48 h, a lower First Durability Index (FDI) and Second Durability Index (SDI), and a higher Retained Strength Index (RSI). Regression analysis indicated strong correlations, with R^2 values confirming the effectiveness of the additive. Overall, the study underscores that the amine-free anti-stripping agent enhances tensile strength, moisture resistance, and stability, particularly under short-term moisture exposure. These findings illuminate the potential of such additives to enhance the durability and performance of asphalt pavements.

Keywords Amine-free anti-stripping agents · Hot mix asphalt · Moisture sensitivity · Saturation

✉ Anmar Dulaimi
anmar.f@uokerbala.edu.iq; a.f.dulaimi@ljmu.ac.uk

Bayan Abdullah Taha
bayan.taha@dpu.edu.krd

Suad Kamiran Rasheed
Suad.rasheed@dpu.edu.krd

Zaid Hazim Al-Saffar
zaid.alsaffar@ntu.edu.iq

Hasanian Radhi Radeef
hasanainr.aldulaimi@uokufa.edu.iq

¹ Highway and Bridge Departments, Technical College of Engineering, Duhok Polytechnic University, Duhok 42001, Iraq

² Department of Building and Construction Techniques Engineering, Technical Engineering College/Mosul, Northern Technical University, Mosul 41002, Iraq

³ Department of Civil Engineering, College of Engineering, University of Kufa, Kufa, Najaf, Iraq

⁴ College of Engineering, University of Kerbala, University of Warith Al-Anbiyaa, Karbala 56001, Iraq

⁵ School of Civil Engineering and Built Environment, Liverpool John Moores University, Liverpool L3 2ET, UK

1 Introduction

Transportation infrastructure, particularly asphalt pavements, faces ongoing challenges due to traffic loading and environmental fluctuations. Over time, these pavements are susceptible to various forms of deterioration, including permanent deformation, cracking, and moisture-induced damage [1]. Among these issues, moisture damage, which results from the accumulation of moisture in asphalt pavements, stands out as a significant concern. This causes the aggregate and asphalt binder to separate, a phenomenon known as the “stripping phenomenon.” This phenomenon frequently results in the asphalt pavement layer cracking, shoving, and ravelling [1–3]. Ultimately, this affects the pavement’s performance and service life [4]. Several important factors are critical in influencing the stripping process of asphalt mixes, including the gradation and chemical composition of the aggregate, the type and amount of binder, the air void (Av) content of the mix, the type of modifying agent, and the amount/type of asphalt mix delamination agent [2]. A comprehensive understanding of these mechanisms is necessary

to mitigate the detrimental effects of moisture damage. The cohesive and adhesive forces that enable adherence between the binder and the aggregate are crucial in maintaining the integrity of pavements [5, 6].

The degree of adhesion between the aggregates and binder determines how much adhesion failure occurs within the mixture. In addition, a critical factor in influencing adhesion strength is the quality of the components of the asphalt mixture [7, 8]. Previous research has focused on reducing the bond between the aggregate and the binder to investigate the causes of moisture deterioration. These studies aim to determine how to enhance and improve this bond to reduce the effects of moisture [9].

There are several reasons why asphalt pavements suffer from early water damage. Apart from the significant air void ratio at the pavement site, the problem is closely related to other intrinsic factors. These include the hydrophilic and oil-repellent properties of the aggregate, as well as the emulsification of the asphalt layer [10]. This is particularly notable since the aggregate is hydrophilic, which makes it easier for water molecules to displace asphalt at the aggregate–asphalt contact. As a result of this displacement by vehicle loading, the adhesion between the aggregate surface and asphalt is destroyed [9–11]. In addition, various methods have been developed to evaluate the asphalt binder's ability to adhere to aggregates. As an illustration, techniques such as the water immersion tests, Marshall Stability (retained stability index), and tensile strength ratio (TSR) [12–14]. Improving the adhesion between asphalt and aggregate is the main approach to increasing the resistance of asphalt mixtures to water damage [15]. The traditional approach involves adding an anti-stripping agent to the asphalt [15–18]. Anti-stripping agents help the asphalt and aggregate physically adsorb or chemically react, creating a strong asphalt–aggregate interaction interface [19]. This notably improves the water damage resistance of asphalt pavements [10, 20]. To mitigate the negative effects of moisture, various anti-stripping agents have been extensively used to strengthen the bonding between the aggregate

and asphalt binder. These materials include lime, Portland cement, and liquid anti-stripping agents. Portland cement, Lime, liquid anti-stripping agents [19], passive adhesion promoters (PAPs) and active adhesion promoters (AAPs), amines, poly-amines, and amido-amine adhesion promoters (AMAPs) [21], amine-based anti-stripping agents [22], Zycotherm, and Wetbond-S [23], sulfur-based antistripping agents [24], epoxy-based anti-stripping agents [25], terpolymer-based additives [26], nano-engineered antistripping polymer modifiers [8], warm mix asphalt additives [27], liquid antistripping (amino-functional silanes) [8, 28], emulsified asphalt [29], warm mix asphalt (WMA) technologies [30], nano-engineered anti-stripping agents [31].

Moisture damage is a major concern in hot mix asphalt (HMA) because water infiltration and weakens the bonding between the aggregates and asphalt binder, leading to stripping, cracking, and reduced pavement life. To mitigate these effects, anti-stripping agents are often added to asphalt mixtures. This study aims to investigate the effectiveness of an amine-free anti-stripping agent in improving the moisture resistance of HMA. Understanding the agent's effect on tensile strength and moisture susceptibility is critical for enhancing pavement performance and reducing maintenance costs.

2 Experimental methodology

2.1 Materials used in the study

The experimental work employed in this study utilized an asphalt binder with a penetration grade ranging from 40 to 50, which is a standard specification in Iraq. This particular binder was obtained from the LANAZ refinery, which is strategically located about 365 km north of Baghdad, the country's capital. The properties of this asphalt binder are listed in Table 1. The quality of aggregate is important, as poor-quality aggregate may lead to weak bonds with the asphalt binder, making the mixture more susceptible to moisture damage. In this study, crushed gravel and sand were obtained from an asphalt mixture manufacturing factory in Zakho City, Iraq, where the source was from the Khabur River area. The main physical properties of the aggregate materials used in this study are shown in Table 2. Finally, the anti-stripping agent was the Sricote agent developed by SRIPATH Innovation LTD. Its key characteristics are shown in Table 3. The Sricote component gives asphalt mixtures excellent adhesion and fuel resistance. It is a special blend of silanes that was developed to alleviate concerns about worker health and safety when using amino compounds in asphalt binders and mixtures as an anti-stripping agent.

Table 1 Properties of asphalt binder

Properties	Unit	Value	Test Condition	ASTM Designation No.
Penetration	0.1 mm	42.6	25 °C, 100 gm, 5 s	D-5
Ductility	cm	160	25 °C, 50 mm/min	D-113
Specific gravity	----	1.03	25 °C	D-70
Softening Point	°C	53	Ring & ball	D-36
Retained Penetration	%	64	25 °C, 100 gm, 5 s	D-5
Residue Ductility	cm	87	25 °C, 50 mm/min	D-113
Rotational Viscosity	cp.	575	135°C	D4402
	cp.	146	165 °C	

Table 2 Properties of the used aggregate

Properties	Coarse Aggregate	Fine Aggregate	ASTM Standard
Bulk-specific gravity	2.666	2.658	ASTM C127 & 128
Apparent-specific gravity	2.739	2.723	ASTM C127 & 128
Los Angeles abrasion	23.6%	-----	ASTM C131
Water absorption	0.72%	1.64	ASTM C128&128
Soundness) Magnesium sulfate	3.1%	-----	ASTM C88
Deleterious materials	0.92	-----	ASTM C142
Crushed percentage	95%	-----	ASTM D5821

Table 3 Sricote data

Properties	Value
Specific gravity (g/cm ³)	1.07
pH	7
Viscosity @ 24 °C, cpc	75–125
Flashpoint COC, (°C)	> 150
Refractive index	1.4
Boiling point (°C)	290

2.2 Sample preparation

The study compares HMA samples with and without the 0.3% amine-free additive. The preparation of modified and unmodified Hot Mix Asphalt (HMA) samples using the Marshall method begins with aggregates being heated to 150–160 °C to remove moisture. The heated binder, with an optimum asphalt content of 4.7% by weight of the total mix, is then mixed with the aggregates in a mechanical mixer to ensure proper coating. For the modified samples, 0.3% (by weight of binder) amine-free anti-stripping agent was added to the hot asphalt binder. To ensure the homogeneity of the resulting mixture, a mixer operating at a temperature of 135 °C and a speed of up to 500 rpm was used to facilitate mixing for 10 to 15 min. The modified binder is mixed with the hot aggregate using the same method as the unmodified mixture. Once the mixing process is complete, the modified and unmodified mixtures are compacted using a Marshall compactor with 37 blows on each side to simulate field conditions, ensuring an air void content of 7% ± 0.5%. The compaction level of 37 blows was selected based on previous research and laboratory trials to target an air void content of approximately 7% ± 0.5%. However, actual air void content may vary slightly depending on the binder type and mix properties [32]. The air void content achieved at 37 strokes was within the target range for all tested mixes, regardless of binder type.

2.3 FTIR test description

In order to better understand the chemical structure of the amine-free anti-stripping agent, Fourier Transform Infrared Spectroscopy (FTIR) testing was conducted. Analyses were conducted using the attenuated total reflectance (ATR) mode, which allows direct sample evaluation without requiring extensive preparation. With a resolution of 4 cm⁻¹ and a total of 10 scans per sample, the spectrum was recorded over a range of 4000 to 500 cm⁻¹. These settings ensured reliable detection of functional groups associated with the additive, particularly to verify the absence of amine-related peaks while identifying other relevant functional groups that could enhance asphalt-aggregate adhesion.

2.4 Durability test

This methodology combines experimental testing with statistical analysis to comprehensively evaluate the performance of HMA and the effectiveness of anti-stripping agents under different humidity conditions. The primary test performed is AASHTO T283 (Modified Lotman Test), which evaluates the resistance of asphalt mixtures to moisture. The modified and unmodified specimens were divided into two groups, using three replicates per condition, and the average values were used for analysis. One group was kept unconditioned as a control, while the other was conditioned with moisture. For the conditioned samples, a vacuum saturation process (13 to 67 kPa) was applied. The samples were subjected to three different saturation durations—5, 10, and 15 min—to simulate varying exposure durations to moisture. The degree of saturation is calculated using the following equation:

$$S = \frac{100 J}{V_a}$$

Where S is a saturation degree, J is the volume of absorbed water, V_a is the volume of air voids.

After saturation, the conditioned samples are immersed in a water bath for 24 h at 60 °C. Then all samples are tested for indirect tensile strength (ITS) using a compressive load. The ITS is calculated using the following equation:

$$ITS = \frac{2000p}{\pi dt}$$

Where ITS is indirect tensile strength (kPa), p is maximum load before failure (N), and d and t are sample diameter (mm) and thickness (mm), respectively.

ITS test results are used to calculate the Tensile Strength Ratio (TSR), which quantifies the moisture susceptibility of

the asphalt mixture. (TSR) calculated using the following equation:

$$\text{TSR} = \left(\frac{\text{ITS conditioned}}{\text{ITS unconditioned}} \right) \times 100$$

The Marshall immersion test evaluates the moisture sensitivity and durability of asphalt mixtures and is an extension of the Marshall test conducted in compliance with ASTM D1075. In this test, the mixture was exposed to hot water at 60 °C for 40 min, 24 h, and 48 h to evaluate its resistance to the effects of water and temperature. Three sets of modified and unmodified samples, each consisting of six specimens, were prepared. The first set was immersed in a water bath at 60±1 °C for 40 min. (considered the control), and the other sets were conditioned for 24 h and 48 h at 60±1 °C. All sets were then tested at a constant compression rate of 50±5 mm/min until failure. The loss of strength or stability, which is typically expressed as an index, is the primary basis for evaluating asphalt pavement durability. Pavement durability is evaluated using three indices: RSI, FDI, and SDI [33]. A lower RSI indicated lower pavement durability. By contrast, a higher durability index indicates a greater loss of strength or stability, suggesting low pavement durability [34].

$$\text{RSI}\% = \frac{S_i}{S_o} \times 100$$

Where S_i is conditioned specimens or stability after immersion time (t_i) and S_o is unconditioned specimens or stability before immersion.

$$\text{FDI} = \sum_{i=0}^{n-1} \frac{S_i - S_{i+1}}{t_{i+1} - t_i}$$

$$\text{SDI} = \frac{1}{t_n} \sum_{i=0}^{n-1} A_i = \frac{i}{2t_n} \sum_{i=0}^{n-1} (S_i - S_{i+1}) \times [2t_n - (t_{i+1} - t_i)]$$

Where S_i and S_{i+1} are the RSI% following immersion times t_i and t_{i+1} , respectively, t_n is the total immersion time, and A_i is the square of the lost strength i immersion time.

2.5 Analysis of variance (ANOVA)

An analysis of variance (ANOVA) (two-way ANOVA) was conducted using Microsoft Excel at a 95% confidence level to examine potential differences between the control and modified mixtures across various saturation times. The key variables considered in this analysis were the saturation degree and the tensile strength ratio. The outcomes of the ANOVA provide insight into the effect of the modifiers on the measured properties. A statistically significant difference is indicated when the F-value exceeds the critical F-value and the p-value is below the significance threshold (typically 0.05). On the other hand, if the F-value is lower than the critical value and the p-value exceeds the significance level, it suggests that no meaningful variation exists between the groups.

3 Results and discussion

3.1 FTIR results and discussion

The FTIR spectrum of the amine-free anti-stripping agent was plotted using TQ Analyst™, a free software program for infrared data processing. The FTIR spectrum for the amine-free anti-stripping agent is presented in Fig. 1. In this analysis, we found that there were no peaks associated with amine functional groups. In particular, there was no N–H stretching vibrations in the range 3300–3500 cm⁻¹, nor any N–H bending vibrations between 1600 and 1650 cm⁻¹. This confirms that the tested additive is free of amine compounds,

Fig. 1 FTIR Spectrum of Amine-Free Anti-Stripping

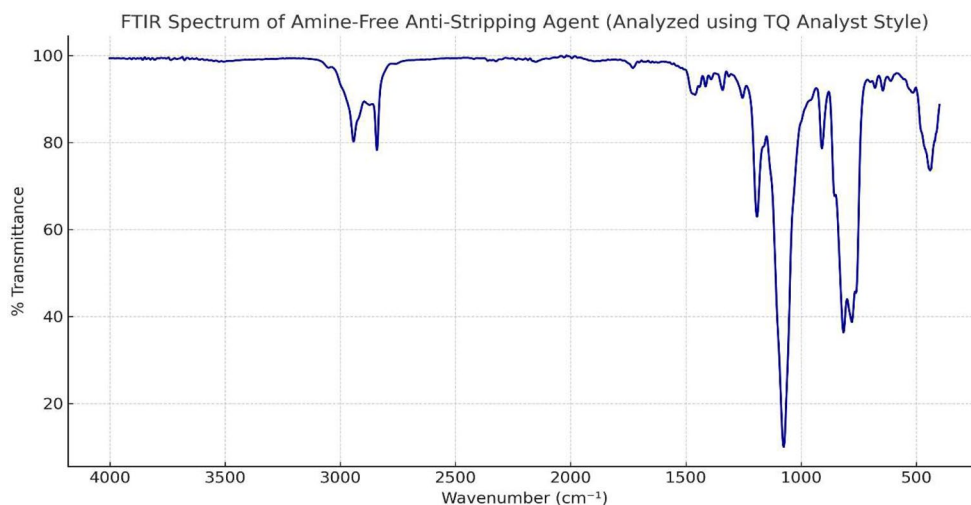


Table 4 Saturation degree result

Saturation Time (min)	Saturation Degree (%)	
	Unmodified	Modified
5	89.5	87.3
10	93.7	87.3
15	97.2	92.6

directly supporting the objective of this study to use amine-free modifiers in HMA.

There was a notable peak at 2941.97 cm^{-1} and 2841.41 cm^{-1} , which corresponds to the C–H stretching in aliphatic hydrocarbons, which is present in a multitude of surfactant-based anti-stripping agents. Further absorptions at 1254.07 cm^{-1} and 1190.87 cm^{-1} indicate that C–O bonds are present, most likely from esters or ethers. As a result, these functional groups improve asphalt-aggregate adhesion by promoting hydrophobicity [35]. The presence of these groups suggests the additive may provide improved resistance to moisture-induced stripping by enhancing binder-aggregate bonding through physical and chemical interactions.

Moreover, some spectral features in the range of $1000\text{--}1100\text{ cm}^{-1}$ may indicate the presence of siloxane (Si–O–Si) groups. These functional groups are characteristic of silane compounds, which are known to chemically bond with hydroxyl groups on siliceous aggregate surfaces. This interaction enhances adhesion through covalent bonding while also contributing to water resistance due to their hydrophobic nature.

The combined presence of aliphatic chains, oxygen-containing groups, and possible silane functionalities supports the role of the additive in improving binder–aggregate adhesion and resistance to moisture damage. These results highlight the additive's effectiveness in enhancing pavement durability without relying on traditional amine-based chemistries.

3.2 Saturation degree

The degree of saturation indicates the extent to which moisture has penetrated the air spaces in asphalt mixture samples. The saturation degree obtained for the modified and unmodified samples after 5, 10, and 15 min of vacuum saturation is listed in Table 4. In this study, the specimens were vacuumed under pressure ranging from 13 to 67 kPa, following the procedure outlined in AASHTO T283. This

process ensures that the target saturation level (usually 70–80%, as recommended by AASHTO T283) is achieved before subsequent testing.

An unmodified asphalt mixture reveals a steady increase in saturation degree with longer exposure to vacuum saturation. After only 5 min, the mixture reached an 89.5% saturation degree, which exceeds the optimum range of 70–80% for AASHTO T283 (Modified Lottman Test).

The saturation level increases with increasing saturation time; after 5 min, it reaches 89.5%, and after 15 min, it peaks at 97.6%. The rapid rise in the saturation degree of an unmodified mixture indicates that it is not sufficiently water resistant and therefore more susceptible to moisture damage. In addition, the results show that the mixture without a free amine stripping agent permits quick water absorption, which may have a detrimental effect on the long-term performance and durability of the pavement, specifically in locations subject to frequent freeze-thaw cycles or high rainfall.

On the contrary, the asphalt mixture containing 0.3% amine-free anti-stripping agent shows a distinctive behavior with respect to water absorption. After 5 min, the saturation of the mixtures containing 0.3% amine-free anti-stripping agent is 87.3%, which is less than the unmodified mixtures simultaneously in terms of saturation. Remarkably, after 10 min, the saturation degree stays at 87.3%, suggesting that the anti-stripping agent reduces the rate at which water penetrates the asphalt mixture. The saturation degree rises to 92.6% after 15 min, which is still less than the unmodified mixture, which reached 97.6% at the same time. This indicates that the amine-free anti-stripping agent enhances the resistance of the asphalt binder to water penetration, possibly by improving the adhesion between the aggregate and the binder. The unmodified mix showed a high saturation degree, indicating increased susceptibility to moisture damage and potential early pavement failure. Finally, the saturation degree data show how well the amine-free anti-stripping agent works to decrease water infiltration into asphalt mixtures. As a result, adding anti-stripping agents to asphalt mixtures is a valuable method for increasing pavement durability and moisture resistance.

Additionally, the ANOVA analysis in Table 5 outlines the sources of variation between the control and modified mixtures under various conditioning times. The results show that the p-value for the main effect of mixture type is less than 0.05, and the corresponding F-value exceeds the

Table 5 Analysis of variance (ANOVA) for the results of saturation degree under different condition times (5, 10, 15 min)

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Control and Modified Mixture	87.65108	1	87.65108	10.70065	0.006689	4.747225
Saturation Time (min)	132.6412	2	66.32059	8.096573	0.005946	3.885294
Interaction	13.32092	2	6.660459	0.813124	0.466476	3.885294

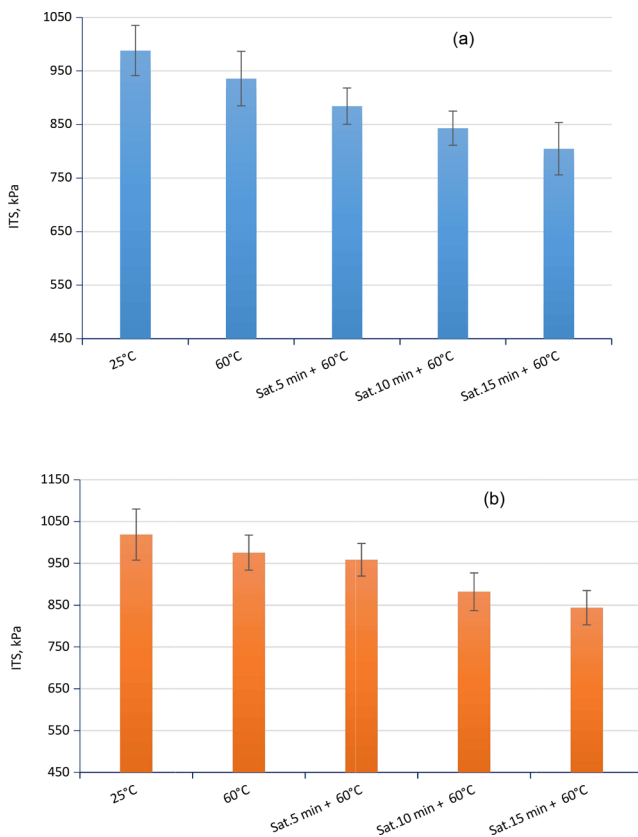


Fig. 2 Indirect Tensile Strength (ITS) Results, (a) for unmodified mixes, (b) for modified mixes

critical F-value, indicating a statistically significant effect of the amine-free anti-stripping agent on the asphalt mixtures. Moreover, the p-value related to the test conditions (e.g., saturation time (min)) is also below 0.05, suggesting that this factor significantly influences the test outcomes. However, the interaction effect between mixture type and test condition shows a p-value greater than 0.05, meaning the interaction is not statistically significant. This implies that the influence of the mixture type remains relatively consistent regardless of the conditioning time, and there is no notable variation resulting from the interaction between these two factors.

3.3 Indirect tensile strength and moisture sensitivity

Figure 2 shows indirect tensile strength (ITS) results for asphalt mixture samples, modified and unmodified, under different conditions with a particular moisture susceptibility and thermal conditioning. Both modified and unmodified mixtures exhibit their maximum ITS values in the initial unconditioned state at 25 °C. Despite the thermal conditioning with increased saturation time, the modified mixture shows relatively higher ITS values than the

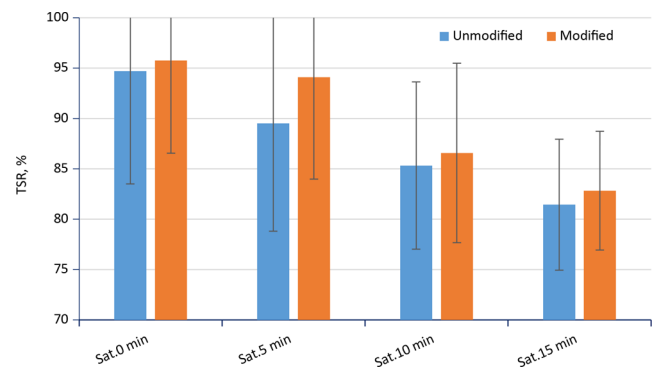


Fig. 3 TSR results

unmodified mixture, highlighting that the amine-free anti-stripping agent in the modified mixture is able to withstand the impact of thermal conditions by improving the binder-aggregate adhesion and overall cohesion within the mix better than the unmodified mixture. Asphalt binder adhesion to aggregates is primarily improved by amine-free anti-stripping agents due to chemical and physical interactions. As opposed to traditional amine-based additives, where amine groups bond with aggregates, amine-free alternatives often use organic silane compounds, polymers, or surfactants to enhance moisture resistance. The durability and resistance of a mixture to water damage are mostly determined by the tensile strength ratio (TSR).

Figure 3 provides important information about the moisture sensitivity of modified and unmodified asphalt mixtures under various saturation levels. With the introduction of saturation and heating conditions, both mixtures show a decrease in TSR values. Nonetheless, the rate of decline is noticeably different, indicating the impact of the anti-stripping agent in the modified mixture. The unmodified mixture exhibits a significant decrease in TSR value with increasing saturation time, reflecting its susceptibility to water-induced stripping. Although the unmodified mix exhibited a TSR value of approximately 95%, indicating acceptable moisture resistance according to standard specifications, the laboratory tests are conducted under controlled and conditioned environments, which do not fully replicate real-world field conditions where the pavement is exposed to repeated traffic loading, temperature variations, and long-term moisture exposure [36]. Even if the initial TSR is high, the rate of degradation under repeated saturation and harsh environmental conditions remains a concern. Our results showed that the modified mix exhibited slower TSR reduction with increasing saturation time, indicating stronger bond retention and improved resistance to long-term moisture damage. Therefore, even a TSR of 95% may not be sufficient to guarantee long-term performance over years of service, especially under harsher field conditions. The modification is intended

Table 6 Analysis of variance (ANOVA) for the results of TSR under different saturation times (5, 10, 15 min)

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Control and Modified Mixture	0.024885	1	0.024885	25.72192	0.000274	4.747225
Saturation Time (min)	0.011214	2	0.005607	5.795404	0.017323	3.885294
Interaction	0.001933	2	0.000966	0.99876	0.396991	3.885294

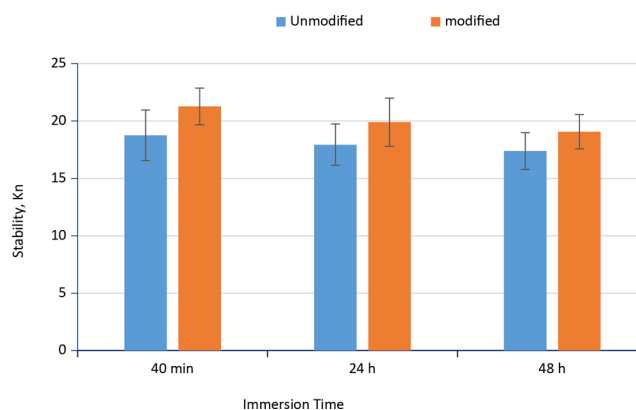
to enhance the durability, aging resistance, and long-term performance of the asphalt mixture.

In contrast, a modified mixture shows a steadier reduction, indicating improved adhesion of the binder to the aggregate and reduced potential for stripping. A higher TSR value of the modified mixture suggests it is more resilient to withstand moisture-related damage, resulting in improved performance and lifetime. In addition, the results showed that the anti-stripping agent is particularly adequate in areas with poor drainage or heavy rainfall, where sensitivity to moisture is a major concern. Thus, by mitigating the effect of moisture damage, the anti-stripping agent creates a more durable and cost-effective pavement, ensuring reliable performance over time.

Furthermore, ANOVA was also performed to examine the significance of the results for each mixture type (control and modified) and saturation time (5, 10, and 15 min) under the TSR, along with their possible interaction. The findings are presented in Table 6. The results indicate a statistically significant difference in TSR between the control and modified mixtures, as evidenced by a p-value of 0.0003. This confirms that the modification had a clear and measurable effect on the TSR. Furthermore, the influence of saturation time was consistent across all levels, suggesting that the modification maintained its effect regardless of the saturation time. The absence of significant interaction between the two factors indicates that the impact of the mixture type does not vary with changes in saturation time, which simplifies the interpretation of the results.

3.4 Marshall immersion

Marshall stability is one of the main criteria used to evaluate the strength and durability of asphalt mixtures. The Marshall stability of the unmodified asphalt mixture at 40 min, 24 h, and 48 h was 18.753 kN, 17.931 kN, and 17.388 kN, which shows a slight reduction in stability over time, with a decline of about 7.3% from the initial value at 40 min to 48 h. The stability values reported in this study are considered reasonable and accurate, assuming that standard laboratory preparation, conditioning, and testing procedures were rigorously followed. The relatively high stability observed can be attributed to several contributing factors, including the use of high-quality aggregates, improved binder–aggregate adhesion, and effective compaction, which collectively

**Fig. 4** Marshall Stability Result

provided sufficient structural integrity even at an air void content of 7%. Additionally, the controlled laboratory environment may have mitigated the extent of moisture-related damage compared to more severe field conditions, thereby helping to preserve the stability performance during immersion testing. However, as shown in Fig. 4, the modified asphalt mixture shows higher stability values across all periods: the decrease in stability over time is also pronounced which is less pronounced compared to the unmodified mixture. This indicates that the modified mixture has better resistance to environmental and loading conditions.

Retained Strength Index (RSI) is a fundamental parameter that assesses the capability of an asphalt mixture to retain its stability over time. The unmodified mix retained 90.5% of its strength, while the modified mix retained 93.6% at 24 h. In contrast, at 48 h, the unmodified mix dropped to 82.1%, while the modified mix maintained 89.6%, as shown in Fig. 5. The improved strength retention (RSI) indicates that the modification improves the asphalt's resistance to ageing and environmental factors such as moisture degradation. The smaller decrease in the RSI over time for the modified mix emphasizes its superior long-term performance compared to the unmodified mix.

The durability index, as shown in Fig. 6, indicates that FDI and SDI decrease over time for both modified and unmodified. The FDI for the unmodified mix is 4.6% and the SDI is 2.8%, while the modified mix shows significantly higher durability values, with an FDI of 7.6% and an SDI of 4.6%. The increase in FDI and SDI indicates that the modified binder effectively improves the ability of the mixture to withstand environmental stresses and repeated

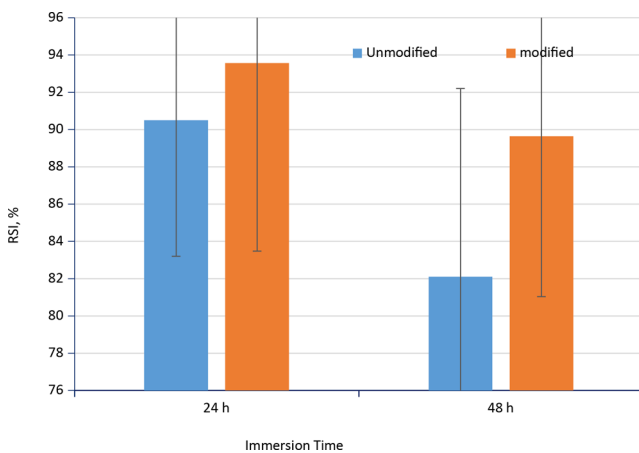


Fig. 5 Retained Strength Index (RSI) for sample immersed for 24 and 48 h

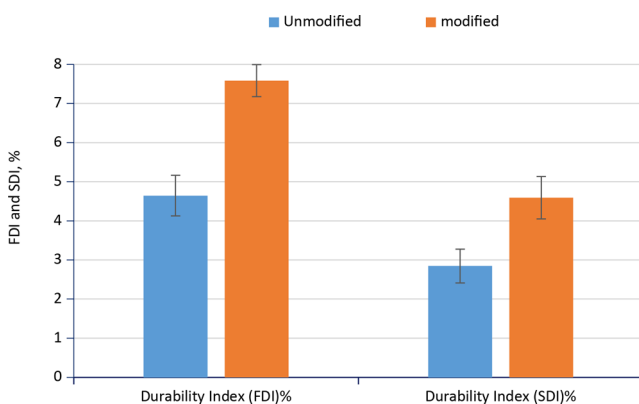


Fig. 6 FDI and SDI results of samples immersed for 48 h

loading cycles without premature deformation or cracking. The effectiveness of amine-free anti-stripping agents is due to the strong chemical bonds they form in asphalt mixtures by changing the chemical and mechanical binder properties. This is supported by FTIR analysis, which indicates enhanced hydrophobic characteristics. As a result of these interactions, asphalt mixtures are more cost-effective, environmentally friendly, and long-lasting.

3.5 Analysis of TSR% and ITS with saturation levels

To comprehend how moisture affects the structural integrity of materials, it is crucial to investigate the relationship between Indirect Tensile Strength (ITS), Tensile Strength Ratio (TSR%), and Saturation Levels. As illustrated in Fig. 7 the reliability of the observed trend lines has been quantitatively assessed through the coefficient of determination (R^2), which measures the proportion of variance in the dependent variables (ITS and TSR%) explained by the saturation level, both ITS and TSR% exhibit a clear inverse relationship with increasing saturation levels. indicating

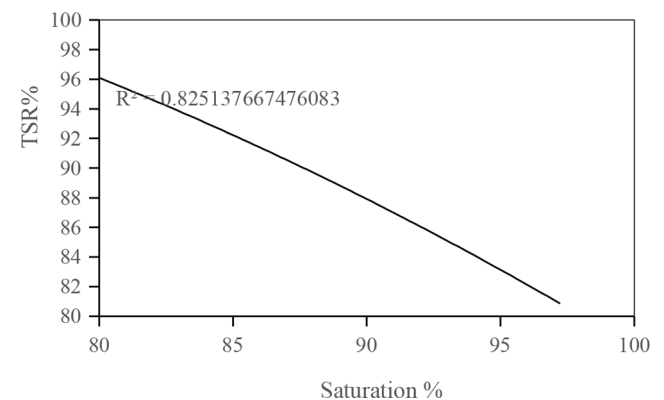
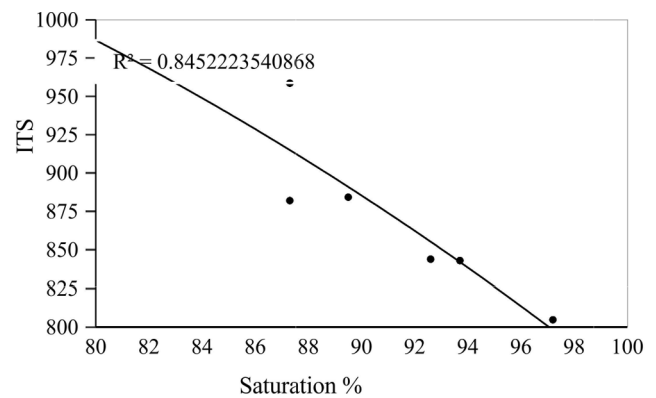


Fig. 7 Comparative Analysis ITS, TSR%, with Saturation

that moisture exposure adversely affects tensile properties. Although both show a high correlation with the coefficient of determination (R^2 of 0.8452), the regression model between TSR and saturation is slightly lower with R^2 (0.8251). While TSR% concentrates on the material's relative strength retention under wet conditions, ITS provides an absolute measure of tensile strength. This suggests that ITS provides a more direct and predictable measure of how moisture affects material performance. The information derived from this analysis is essential for improving the design and maintenance of infrastructure, especially in areas that are frequently or continuously exposed to water.

4 Conclusion

The study aims to evaluate a 0.3% amine-free anti-stripping agent in improving the mechanical performance and moisture resistance of hot mix asphalt. The following conclusions could be made in light of the different test results:

1. According to the study, the saturation values of modified HMA mixtures containing 0.3% amine-free anti-stripping agent were consistently lower than those of unmodified mixtures.

2. With extended saturation times (10 and 15 min), the modified mixtures exhibited a decrease in ITS and TSR%, despite their results remaining better than unmodified mixtures. This trend draws attention to the limitations of the additive when exposed to moisture for long periods.

3. The two-way ANOVA results demonstrate that the modified mixtures significantly improved the saturation and TSR compared to the control, with no evidence of interaction effects.

4. The durability of HMA mixtures was significantly increased by the use of an amine-free anti-stripping agent, which is essential to reduce maintenance costs and extend the life of the infrastructure.

5. Despite the benefits of the additive, as saturation times increased, both modified and unmodified mixes decreased in ITS and TSR. This suggests that even with the improvements, HMA remains susceptible to prolonged moisture exposure, requiring further research to develop stronger solutions.

6. The modified mixture not only shows higher initial stability but also retains its strength effectively over time compared to the unmodified mixture.

4.1 Limitations and future work

High-quality aggregates with low water absorption (0.72%) were used in this study to reduce aggregate property variability and concentrate on assessing how binder modification affects moisture susceptibility. Nonetheless, it is acknowledged that aggregates with higher water absorption (> 1.5–2%) are frequently utilized and pose more difficulties with moisture damage in many real-world scenarios. In order to provide a more thorough evaluation of the modified mixtures' long-term moisture resistance under more demanding field conditions, future research is advised to assess their performance using aggregates with higher water absorption.

Acknowledgements The authors express sincere gratitude for the support received from Northern Technical University in Iraq.

Author contributions B.A.T.: Writing—review and editing, Formal analysis, Visualization, Validation, Data curation. S.K.R.: Writing—review and editing, Visualization, Validation, Formal analysis. Z.H.A.S.: Writing—review and editing, Resources, Investigation, Visualization, Formal analysis. H.R.R.: Visualization, Writing—review and editing A.D.: Supervision, Resources, Investigation, Funding acquisition, Writing—review and editing. All authors have read and agreed to the published version of the manuscript.

Funding No funding was received for conducting this study.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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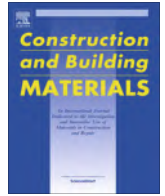
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Investigation of aggregate moisture content variation and its impact on pavement performance of WMA



Bin Yang^a, Huining Xu^{a,b,*}, Peiyan Zhou^a, Yiqiu Tan^{a,c}

^a School of Transportation Science and Engineering, Harbin Institute of Technology, Harbin 150090, China

^b Polar Academy, Harbin Institute of Technology, Harbin 150090, China

^c State Key Laboratory of Urban Water Resource and Environment, Harbin Institute of Technology, Harbin 150090, China

HIGHLIGHTS

- Analyze the effect of various factors on variations in aggregate moisture content.
- Explore the asphalt-aggregate interaction under different moisture contents.
- Investigate the effect of moisture content on the pavement performance of WMA.

ARTICLE INFO

Article history:

Received 2 August 2019

Received in revised form 8 April 2020

Accepted 25 April 2020

Available online 4 May 2020

Keywords:

Warm mix asphalt

Aggregate moisture content

Asphalt-aggregate interaction

Pavement performance

ABSTRACT

For warm mix asphalt (WMA), aggregates cannot be completely dried due to the low heating temperature, which further adversely affects the pavement performance. This study aimed to (1) analyze the effect of particle size, heating temperature and heating time on the variation in the moisture content in aggregates, (2) compare the interaction ability between warm mix asphalt binders and aggregates with different moisture contents from a rheological perspective and (3) to study the effect of aggregate moisture contents on the pavement performance of WMA. Results show that the quadratic function can be used to express the decline in moisture content over time. The decline rate of moisture content increases with the decrease in particle size. The moisture content of fine aggregate dominates that of the whole aggregate system. The decline rate of moisture content exhibits strong temperature sensitivity in the heating temperature ranging from 110 °C to 150 °C. The sensitivities of different evaluation parameters of the asphalt-aggregate interaction were compared by sensitivity analysis. With the decrease in the moisture content of aggregates, the pavement performance of WMA can be improved. These findings provide understanding on the variation in aggregate moisture content during the heating process, and prove that reducing aggregate moisture content can improve the pavement performance of WMA.

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1. Introduction

Warm mix asphalt (WMA) is an environmental-friendly asphalt mixture. This technology can significantly reduce the construction temperature, cut down the energy consumption and decrease the short-term aging of mixtures [1]. However, aggregates usually cannot be completely dried at the construction site due to the low heating temperature. Thus, the residual moisture will affect the adhesion of asphalt binders to aggregates, which poses a potential danger to the long-term performance of roads [2–4]. Therefore,

understanding the variation in aggregate moisture content is the key to improve the water stability and crack resistance of WMA.

Many efforts on the asphalt-aggregate interaction in WMA have been exerted for quite a few decades. Scholars [5–8] have found that moisture remains on the aggregate surface due to the lower heating temperature, and this condition leads to the poor bonding properties between warm mix asphalt binders and aggregates. As a result, durability of WMA becomes poor. Surface free energy theory has often been adopted by researchers to evaluate the effect of aggregate moisture on the bond strength between asphalt binders and aggregates. On the basis of this theory, Yang et al. [9] found that the adhesion of warm mix asphalt binders to aggregates with the moisture content of 10% decreases up to 50% compared with dry aggregates. From the perspective of micromechanics, some researchers [10,11] have used bitumen bond strength (BBS)

* Corresponding author at: School of Transportation Science and Engineering, Harbin Institute of Technology, Harbin 150090, China.

E-mail address: xuhn@hit.edu.cn (H. Xu).

and indirect tensile strength (ITS) tests to evaluate the bond between asphalt binders and aggregates. Moraes et al. [11] reported that moisture exposure time and modification type can highly influence the bond between asphalt binders and aggregates. With the help of advanced microscale equipment, the adhesion mechanism of asphalt binders to aggregates can be understood well [15]. Abdalla et al. [12] used atomic force microscopy (AFM) to investigate the influence of moisture on the adhesive and cohesive bonds between asphalt binders and aggregates. After comparing four kinds of cantilever tip materials, they found that the adhesion/cohesion energy of warm mix asphalt binders is similar or better than that of hot mix asphalt binders. Bairgi et al. [13] found that the frictional resistance of foamed asphalt binder decreases during the foaming process, which promotes the formation of a uniform coating on the aggregate surface in the mixing process and enhances the slide ability among aggregates in the molding process at low temperature.

Researchers have proposed many methods and suggestions to solve the problem of poor bond between asphalt binders and aggregates in WMA. Yang et al. [14] studied an efficient silicone based warm mix additive, which can significantly enhance the wettability and adhesion of asphalt binders to aggregates. Leng et al. [15] found that directly adding wax based additives into the rubberized asphalt (RA) will lead to high void ratio of asphalt mixtures and low wax content in the liquid phase of WMA. Thus they recommended adding warm mix additives into the RA after the mixing process of rubber powder and asphalt binders. Sangeftidi et al. [16] suggested that using antistripping agents or hydrophilic aggregates can reduce the adhesion failure in WMA.

In conclusion, scholars have conducted studies on the bond between warm mix asphalt binders and aggregates from microtopography to macromechanics [17–20]. However, the properties of the aggregate, which is a vital component that takes up nearly 95% of the weight of asphalt mixtures, can directly affect the pavement performance of WMA. Few studies [21,22] have paid attention to the variations in the aggregate moisture content and its effects on the pavement performance of WMA. Therefore, in the present study, the variation in aggregate moisture content was

investigated by controlling the heating temperature, heating time and particle size. Then, the effect of aggregate moisture content on the rheological properties of asphalt mortar was analyzed. Thereafter, the sensitivity of the evaluation parameter to moisture content was determined by sensitivity analysis. Finally, the effect of aggregate moisture content on the pavement performances of WMA was studied. The flowchart of the experimental plan is presented in Fig. 1.

2. Materials and Methods

2.1. Materials

The virgin asphalt binder with a penetration grade of 80/100 was prepared for warm mix asphalt binders. Two kinds of warm mix additives were used to prepare for WMA samples. The physical properties of the additives are listed in Table 1. The styrene-butadiene-styrene (SBS) triblock copolymer modified asphalt binder with a penetration grade of 60/80 was prepared for hot mix asphalt (HMA) samples. The coarse and fine aggregates were derived from basalt, and their bulk specific gravities were 2.738 and 2.760.

The aggregate gradation used in this study was AC-16, which is presented in Table 2. The asphalt mixtures were prepared according to the Marshall design method. The optimum asphalt content and the target void ratio of the WMA were 4.6% and 4.0%.

2.2. Sample preparations and experimental methods

2.2.1. Measurement of aggregate moisture content

In order to comprehensively investigate the variation of the aggregate moisture content, the aggregate underwent saturated water treatment according to the Test Methods of Aggregate for Highway Engineering (JTG E42-2005) in China, so the initial moisture content of aggregate of each particle size was at the same level. The specific operation is presented as follows. Firstly, all aggregates were saturated with water at normal temperature

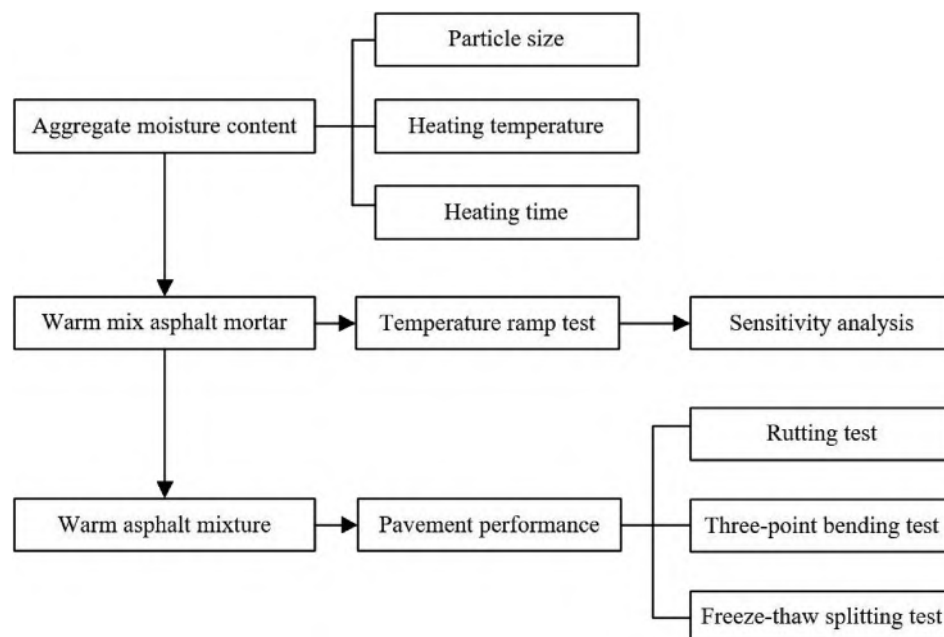


Fig. 1. Flowchart of the experimental design.

Table 1
Physical properties of warm mix additives.

	A wax based additive	A surfactant based additive
Type	Organic additive	Chemical additive
Origin	FT-wax	Chemical surfactant
Physical state	Milky white granule	Yellow-brown viscous liquid
Viscosity (mPa·s)	12 (135 °C)	15–30 (25 °C)
Density (g/cm ³)	0.94	1.05
Dosage	3% by weight of the asphalt binder	3% by weight of the asphalt binder

and under normal pressure for 24 h. Subsequently, the surface water of the coarse aggregate (4.75–16 mm) was wiped off with dry wipes, while that of the fine aggregate (0.075–2.36 mm) was filtered with the filter paper. Finally, the aggregates were placed in an oven, and the moisture content was measured per half an hour.

The moisture content at a specific time can be calculated by Eq. (1).

$$\omega = \frac{m(t_i) - m_0}{m_0} \times 100\% \quad (1)$$

where ω is the aggregate moisture content; $m(t_i)$ is the weight of the aggregate at t_i , m_0 is the weight when the aggregate is completely dried, t_i is the length of time after the saturated aggregate is placed in the oven.

2.2.2. Preparation and test method for warm mix asphalt mortar

The asphalt mortar was mixed by warm mix asphalt binders and 0.075 mm aggregates. The mass ratio of asphalt binders to aggregates was 2:1. The heating temperature of aggregates and the mixing temperature of asphalt mortars were 140 °C. In order to study the rheological properties of asphalt mortars with different aggregate moisture contents, the heating time of aggregates

was 1, 2, 3.5 h. The corresponding aggregate moisture contents were 1.5%, 0.4% and 0.0%.

However, foaming would occur when asphalt binders and aggregates were mixed due to the residual moisture in aggregates and the air among aggregates. This condition would affect the accuracy of the asphalt mortar rheological test results. Therefore, the Brookfield viscometer was used to mix and prepare the asphalt mortar. Meanwhile, the variation of asphalt mortar's viscosity was recorded. A spindle No. 27 and a specimen volume of 15.5 ml were used for this test at 135 °C. The mixing time was determined by the absence of surface foaming phenomenon to ensure the same mixing condition of different asphalt mortars. Under the different heating time of aggregates, the viscosity variation curves of two kinds of asphalt mortar over the stirring time are shown in Fig. 2.

Fig. 2 suggests that, when the aggregate moisture contents were 1.5%, 0.4% and 0.0%, the viscosity of two kinds of asphalt mortars tended to be stable after they were stirred for nearly 8, 5 and 3 min, respectively. Thus, when the heating time of aggregate was 1, 2, and 3.5 h, the optimal mixing time of asphalt mortars would be 8, 5, and 3 min, respectively.

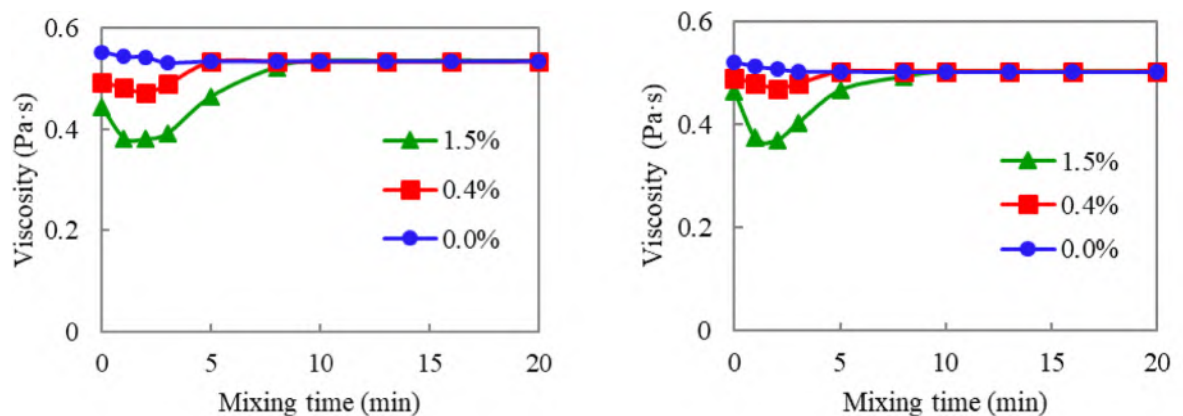
After warm mix asphalt mortars were prepared, the dynamic shear rheology test would be conducted to investigate the effect of moisture on the asphalt-aggregate interaction. The parallel plate with a diameter of 8 mm was adopted, and the gap between the upper and lower plates was set to 2 mm. The loading frequency was 1.59 Hz and the temperature range was controlled by the temperature ramp test. Complex shear modulus (G^*), complex viscosity coefficient (η^*), phase angle (δ), rutting factor ($G^*/\sin\delta$) were used to evaluate the effect of moisture content on the asphalt-aggregate interaction.

2.2.3. Preparation and test method for asphalt mixture

The molding temperature of WMA was determined by the equivalent volume method [23–25]. According to this method, when the aggregate gradation, aggregate type, and compaction

Table 2
Aggregate gradation.

Gradations (mm)	Percentage passing (%) through the following sieve size (mm)									
	19	16	13.2	9.5	4.75	2.36	1.18	0.6	0.3	0.15
AC-16	100	98	88	67	45	31	22	17	12	9



a) Asphalt mortar containing waxy additives

b) Asphalt mortar containing surfactant additives

Fig. 2. Temporal variation in the viscosity of asphalt mortar with different moisture contents.

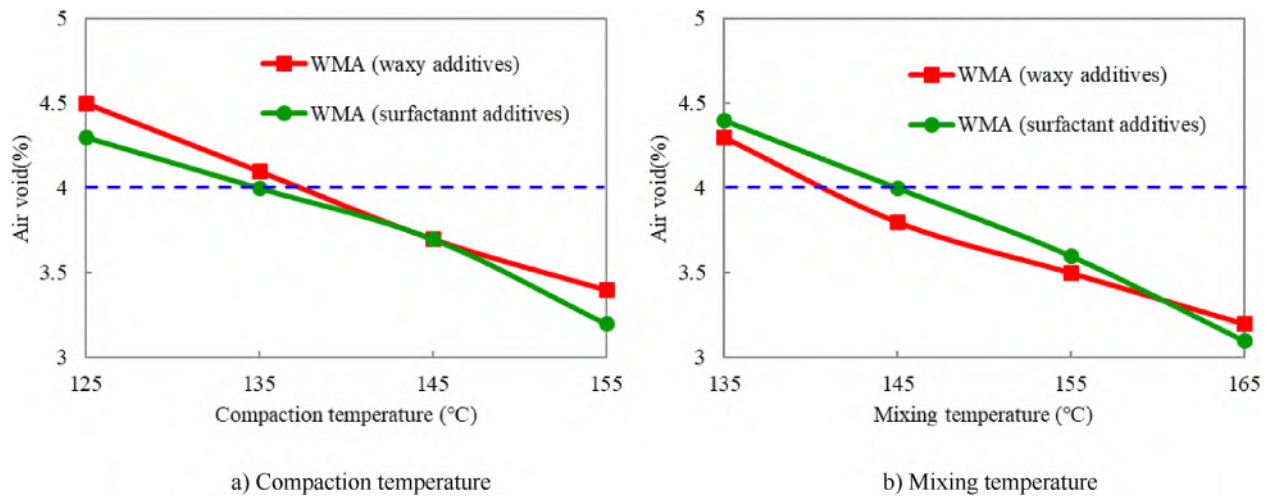


Fig. 3. Variation in void ratio of WMA under different molding temperatures.

Table 3
Optimal molding temperature of the WMA.

Asphalt mixtures	HMA	WMA	
	SBS	waxy additives	surfactant additive
Mixing temperature (°C)	175	140	145
Compaction temperature (°C)	165	135	135

method of asphalt mixtures are same, only the type of asphalt binder changes, so different kinds of asphalt mixtures will have a same void ratio by adjusting molding temperatures. The void ratio of the HMA determined by the Marshall test was 4.0%. The variation in the void ratio of WMA under different molding temperatures were shown in Fig. 3. So the mixing and compaction temperature of the WMA were determined by the same void ratio as the HMA. The optimal molding temperatures of WMA are listed in Table 3.

In this study, according to the Test Methods of Asphalt and Asphalt Mixtures for Highway Engineering (JTG E20-2011) in China [26], the pavement performance of WMA with different aggregate moisture contents have been compared and analyzed. The pavement performance mainly contains the rutting resistance at high temperature, cracking resistance at low temperature and the water stability after the freeze-thaw test. So the corresponding test parameters and test procedures are listed in Table 4.

Table 4
Primary information about the pavement performance.

Test name	Pavement performance	Test parameter	Test procedure
Rutting test	Rutting resistance at high temperature	Dynamic stability (time/mm)	The slab specimen (300 × 300 × 50 mm) was loaded by the round trip of experimental wheel under the loading frequency of 42 times/min, the specified temperature of 60 °C and the specified pressure of 0.7 MPa. Calculate the dynamic stability, that is, the average loading times to produce 1 mm deformation from 45 min to 60 min.
Three-point bending test	Cracking resistance at low temperature	Failure strain (με)	The trabecular specimen (250 × 30 × 35 mm) was used to carry out the three-point bending test under the loading condition of 50 mm/min at the specified temperature of −10 °C. Record and calculate the maximum failure strain when the specimen breaks.
Freeze-thaw splitting test	Water stability	Tensile strength ratio (TSR) (%)	The cylindrical specimen (φ101.6 × 63.5 mm) was used in this test. Firstly, half specimens were frozen at −18 °C for 16 h, and then placed in a 60 °C water bath for 24 h. While the other half specimens do not need carry out the freeze-thaw test. Then, after placing all specimens in a 25 °C water bath for 2 h, they were carried out the splitting test under the loading condition 50 mm/min at normal temperature. Record the indirect-tensile strength and calculate the TSR value.

3. Results and discussion

3.1. Evaluation of aggregate moisture content in the heating process

3.1.1. Temporal variation of aggregate moisture content in the heating process

In Section 3.1, all aggregates are saturated with water as described in Section 2.2.1, to ensure consistency of the initial aggregate moisture content of each particle size. Fig. 4 shows the temporal variation of aggregate moisture content when the aggregate heating temperature is 150 °C.

Fig. 4 indicates that, as the heating time increased, the moisture contents of aggregates with different particle sizes non-uniformly decreased. Instead of a linear equation, the quadratic function could be used to express the decline in moisture content over time. For example, the moisture content of the aggregate with a particle size of 0.6 mm decreased by 8.27%, 5.34%, and 0.40% in 0–1, 1–2, and 2–3 h, respectively. This result suggested that the aggregate moisture content rapidly decreased in the initial stage and then gradually decreased. Besides, the initial moisture contents of aggregates with various particle sizes were different. For instance, the initial moisture contents of aggregates with particle sizes of 0.6, 1.18 and 2.36 mm were 14.01%, 7.82% and 2.01%, respectively. The moisture was more likely to remain under the action of capillary forces due to quite a few gaps between fine aggregates. In other words, the initial moisture

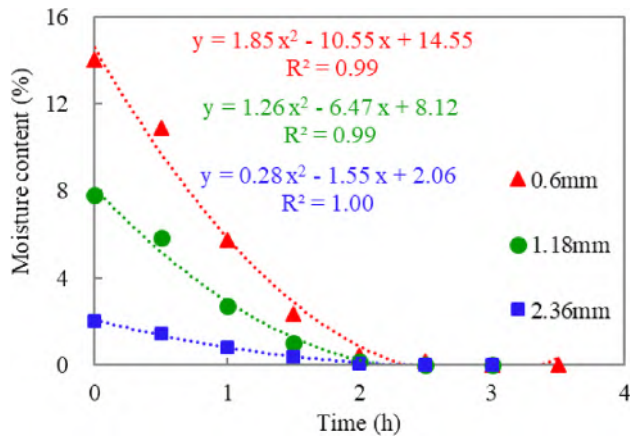


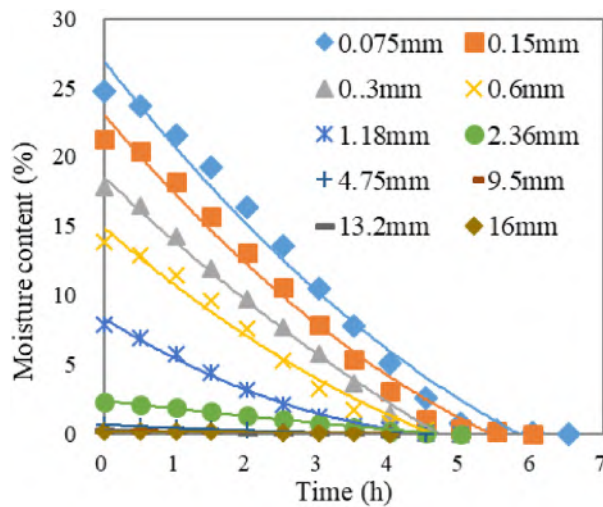
Fig. 4. Temporal variation of aggregate moisture content in the heating process.

content of aggregate quickly became smaller when the particle size turns coarser.

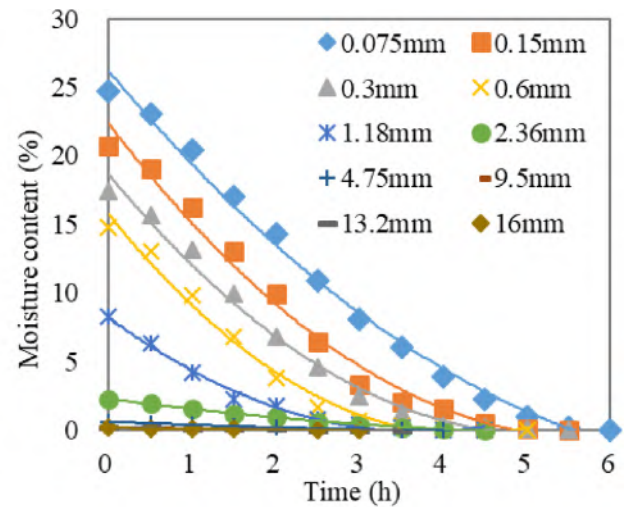
3.1.2. Effect of particle size on the decline rate of aggregate moisture content

The temporal variation curves of moisture content of aggregates with various particle sizes at four different heating temperatures are plotted in Fig. 5. To clearly reflect the difference between different particle sizes, the curve fitting was performed on each variation curve of moisture content to obtain their decline rate, as shown in Fig. 5. The basic form of curve fitting equation was the quadratic function.

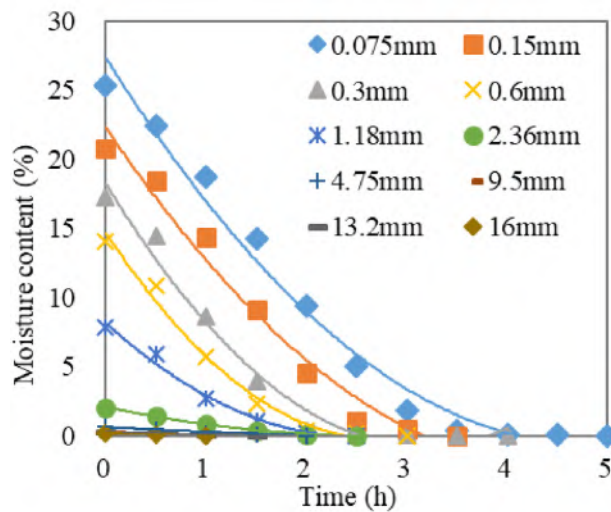
Fig. 5 suggests that regardless of the heating temperature, the decline rates of moisture content with various particle sizes were very different. For instance, when the heating temperature was 150 °C, the moisture content of aggregates with particle size of 0.075, 2.36 and 16 mm decreased by 6.67%, 0.56% and 0.16% in 0–1 h, respectively. The reason was that the fine aggregate had a



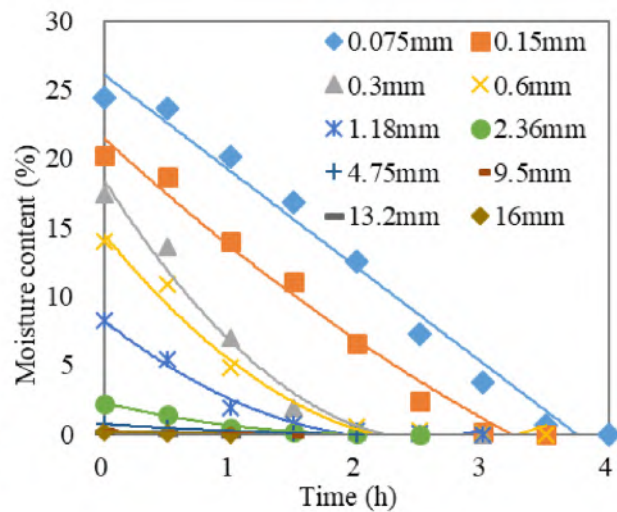
a) 110 °C



b) 130 °C



c) 150 °C



d) 170 °C

Fig. 5. Temporal variation of aggregate moisture content with different particle sizes.

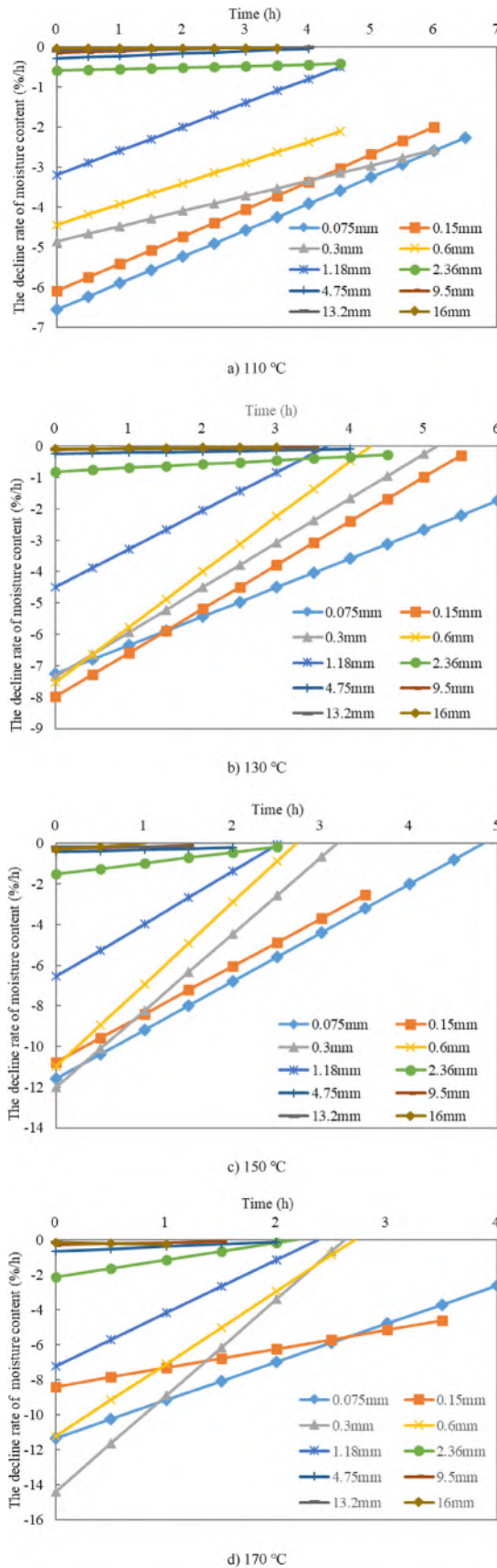


Fig. 6. Temporal variation in the decline rate of aggregate moisture content with different particle sizes.

larger specific surface area than the coarse aggregate, which suggested that the moisture in fine aggregates easily evaporated. Thus, a small aggregate particle size indicated a fast decrease in moisture content.

The slope of the curve in Fig. 5 represents the decline rate of moisture content. The experiments were tested at the same heating temperature, and the differences in the decline rate of moisture content of various aggregate particle sizes were evident. The decline rates of moisture content over time at four different heating temperatures are plotted in Fig. 6.

Fig. 6 suggested that the linear function could be used to express the decline rate of the moisture content over time. The slope of the line in Fig. 6 indicates the variation rate of moisture loss rate. Regardless of the heating temperature, the variation rates of moisture loss rate with various particle sizes were very different. For instance, when the heating temperature was 110 °C, the variation rates of moisture loss rate with the aggregate particle sizes of 16, 2.36, 1.18 and 0.075 mm was 0.01%/h², 0.04%/h², 0.6%/h² and 0.66%/h², respectively. This finding indicated that a large the aggregate particle size corresponded to a small variation rate of moisture loss rate. Thus, the coarse aggregate had a relative stable decline rate of moisture content over time.

Besides, Fig. 6 indicates that the line of fine aggregate was below the line of coarse aggregate. For instance, on the one hand, when the heating temperature was 110 °C, the decline rates of initial moisture content with the aggregate particle sizes of 16, 2.36, 1.18 and 0.075 mm were 0.03%/h, 0.59%/h, 3.19%/h and 6.55%/h, respectively. Meanwhile, when the aggregate was completely dried, the decline rates of moisture content with the aggregate particle sizes of 16, 2.36, 1.18 and 0.075 mm were 0.03%/h, 0.41%/h, 0.49%/h and 2.26%/h, respectively. This phenomenon again indicated that a fine aggregate particle size corresponded to a high decline rate of moisture content. Accordingly, the moisture content of fine aggregate primarily determined the moisture content of the whole aggregate system.

3.1.3. Effect of heating temperature on the decline rate of aggregate moisture content

It is well-known that the heating temperature can significantly affect the decline rate of aggregate moisture content. At the construction site, the heating temperature of aggregates for producing WMA and HMA usually set to range from 110 °C to 170 °C [27,28]. So in order to cover all the heating temperature of aggregates, the temporal variation of aggregate moisture content at the heating temperature of 110 °C, 130 °C, 150 °C, and 170 °C were tested and plotted in Fig. 7.

As shown in Fig. 7, the difference between the decline rates of aggregate moisture content under different heating temperatures was evident. For example, the aggregate with a particle size of 0.3 mm took 2, 2.5, 4 and 4.5 h from saturation to dryness at the heating temperatures of 170 °C, 150 °C, 130 °C and 110 °C, respectively. This phenomenon implied that as the heating temperature rose, the decline rate of aggregate moisture content gradually increased.

Interestingly, Fig. 7 indicates that regardless of the aggregate particle size, the decline rate of moisture content evidently changed under different heating temperatures when the heating temperature was in 110 °C–150 °C, especially in 130 °C–150 °C. This result suggested that the aggregate moisture content had high temperature sensitivity in this temperature range. However, when the heating temperature was 170 °C, the moisture content variation curve did not change considerably compared with that at

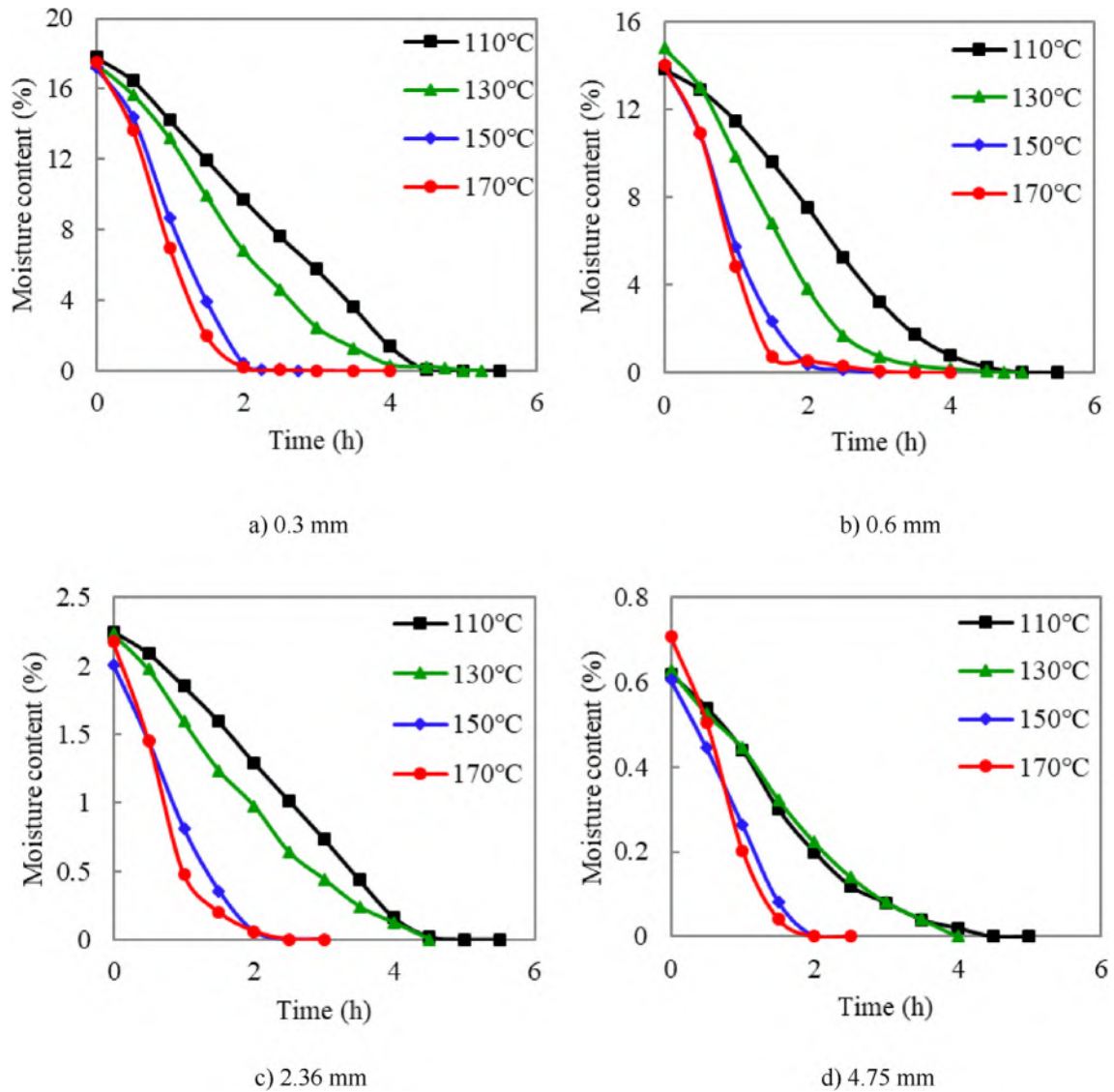


Fig. 7. Temporal variation of aggregate moisture content under different heating temperature.

150 °C. This result suggested that the decline rate of moisture content was kept smooth in the range of 150 °C–170 °C. For instance, when the particle size of aggregate was 0.3 mm, the aggregate took only 2 h to completely dry at 170 °C, which was 0.5 h faster than that at 150 °C. However, the aggregate took 4 h to completely dry at 130 °C, which was 1.5 h slower than that at 150 °C. Thus, for the WMA, the low heating temperature would cause a certain amount of water to remain in aggregates. And if the heating time was short, a large amount of water will also remain in aggregates. This condition would pose a hidden danger to the pavement performance.

3.2. Effect of moisture existence on asphalt-aggregate interaction

Fine aggregates can absorb more moisture than coarse aggregates because of their large specific surface area. Several studies [29–31] have indicated that the residual water in aggregates will affect the bond between asphalt binders and aggregates and even lead to the stripping, raveling and permanent deformation of asphalt mixtures. This section presents the effect of moisture in aggregates on the rheological properties of asphalt mortars.

3.2.1. Effect of moisture in aggregate on the rheological behavior of asphalt mortar

Two kinds of warm mixed asphalt mortars were prepared in accordance with Section 2.2.2. The heating temperature was 140 °C. The aggregate moisture contents after heating for 1, 2, and 3.5 h were 1.5%, 0.4% and 0.0%, respectively. G^* , η^* , δ , $G^*/\sin\delta$ were used to evaluate the rheological behavior of asphalt mortar. The results are shown in Figs. 8–11. The red, blue, and green curves represent moisture contents of 1.5%, 0.4% and 0.0%, respectively.

Figs. 8–10 suggest that, as the moisture content decreased, the G^* , $G^*/\sin\delta$ and η^* of the asphalt mortar increased at a relatively high temperature (> 50 °C). The higher values of G^* , $G^*/\sin\delta$ and η^* indicated a better rutting resistance of pavement at high temperature. Therefore, the existence of water could impair the resistance of asphalt mixtures to deformation. Fig. 11 also indicates that δ increased as aggregate moisture content increased in the temperature range of 10 °C–70 °C. The δ means the ratio of the viscous component to the elastic component. A small δ corresponded to high elastic components in the asphalt mortar, which suggested that the elastic recovery ability of asphalt mortar was strong. Thus,

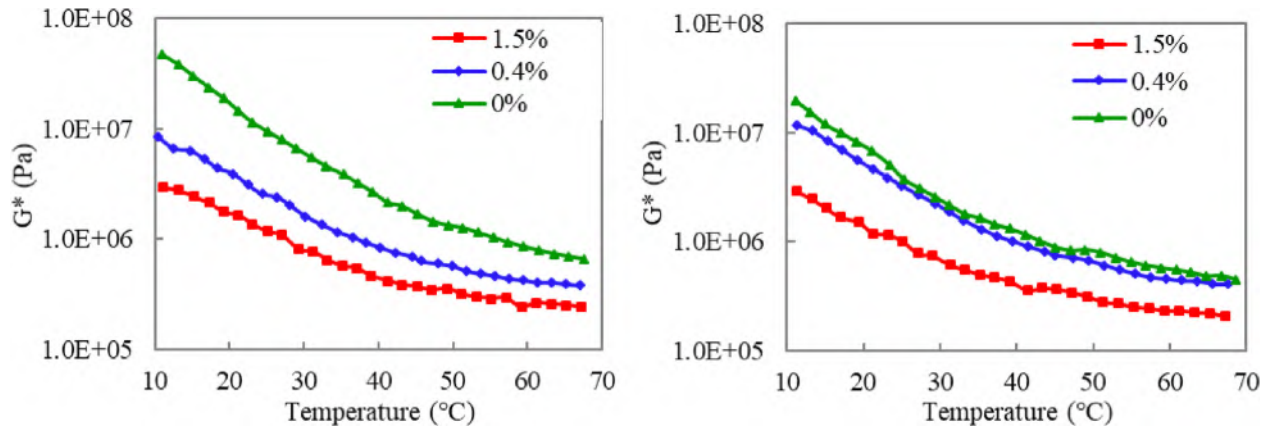
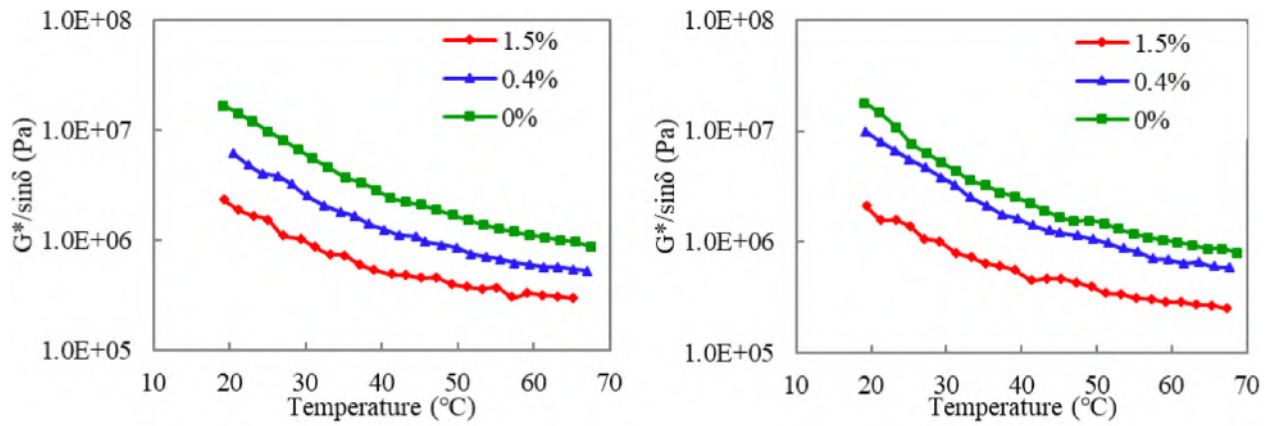


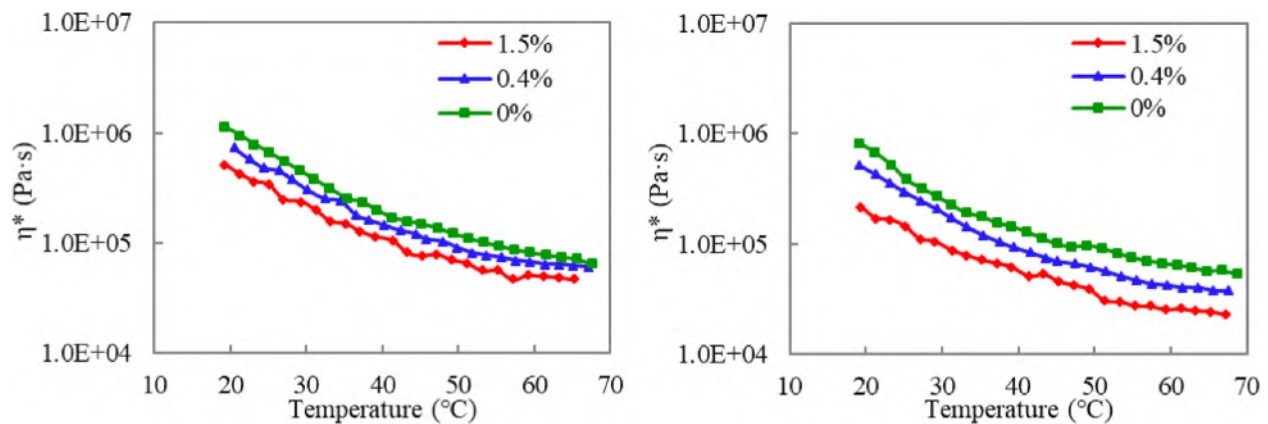
Fig. 8. Variation in G^* of asphalt mortar under different moisture contents.



a) Asphalt mortar containing waxy additives

b) Asphalt mortar containing surfactant additives

Fig. 9. Variation in $G^*/\sin\delta$ of asphalt mortar under different moisture contents.



a) Asphalt mortar containing waxy additives

b) Asphalt mortar containing surfactant additives

Fig. 10. Variation in η^* of asphalt mortar under different moisture contents.

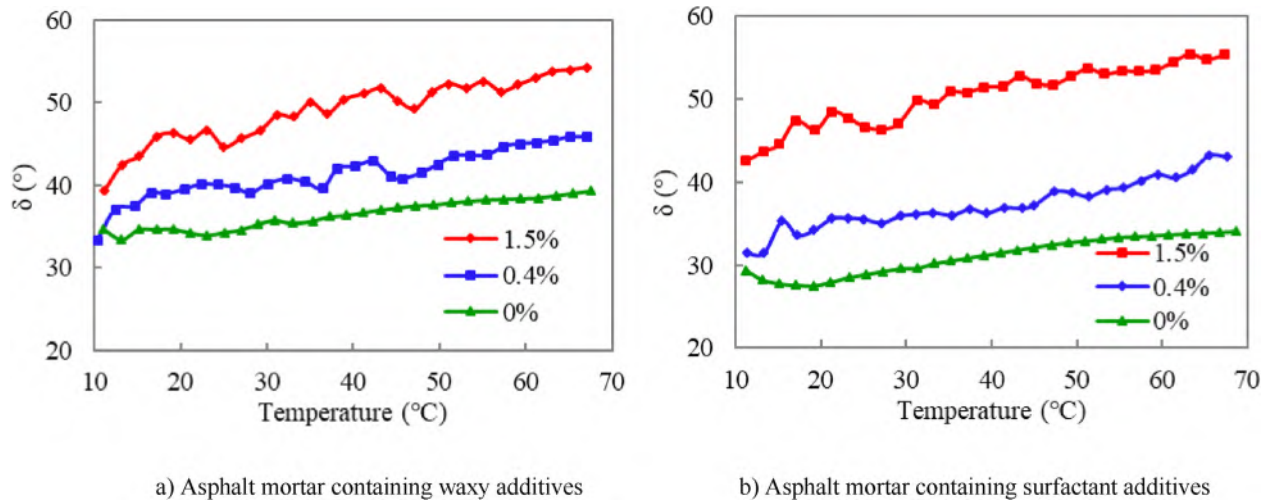


Fig. 11. Variation in δ of asphalt mortar under different moisture contents.

the residual moisture in the aggregate would affect the interfacial bond between warm mix asphalt binders and aggregates.

The Spearman correlation method was used to quantify the effect of moisture content and other factors on the rheological ability of WMA. The results are listed in Table 5.

Table 5 shows that under the evaluation of the significance level of 0.01, the absolute values of correlation coefficient of moisture content, test temperature, and warm mix additive type for four different evaluation parameters ranged from 0.60 to 0.91, from 0.33 to 0.37, and from 0.11 to 0.14, respectively. Thus, compared with test temperature and warm mix additive type, moisture content significantly affected the asphalt-aggregate interaction. For G^* , the correlation coefficients of moisture content, test temperature and warm mix additive type were -0.907 , -0.338 , and 0.132 , respectively. Evidently, the correlation between G^* and moisture content was much higher than that between G^* and the two other factors. Therefore, the aggregate moisture content significantly affected the δ , G^* , $G^*/\sin\delta$ and η^* . In other words, the aggregate moisture content significantly affected the asphalt-aggregate interaction.

3.2.2. Sensitivity analysis of evaluation parameters

The analysis in the previous section showed that, although the four evaluation parameters above could characterize the asphalt-aggregate interaction, they had different sensitivities to the variation of interaction ability caused by changing aggregate moisture contents. The sensitivity analysis method would be used to compare the sensitivity to the variation in aggregate moisture content for different evaluation parameters, as reported in this section.

Sensitivity refers to the ability of a parameter u_i to respond to the variation of a parameter x_j . Assuming that the parameter x_j

ranges from x_1 to x_2 , and parameter u_i ranges from u_1 to u_2 . Thus, the sensitivity is defined as a Eq. (2).

$$\text{sen}\left(\frac{u_i}{x_j}\right) = \frac{(u_2 - u_1)}{(x_2 - x_1)} \quad (2)$$

In this section, the aggregate moisture content can be regarded as parameter x_j . The evaluation parameters, namely δ , G^* , $G^*/\sin\delta$ and η^* , will be regarded as parameter u_i . In the case where the range of x_j is certain, given that different parameters have different dimensions, the sensitivity can be obtained according to the relative variation range of u_i . Accordingly, the sensitivity can be defined as a Eq. (3).

$$\text{sen}\left(\frac{u_i}{x_j}\right) = \left| \frac{u_2 - u_1}{u_1} \right| \times 100\% \quad (3)$$

A high sensitivity indicates a great effect of parameter x_j on parameter u_i . That is, when the variation range of x_j is certain, greater variation of u_i corresponds to a high sensitivity.

The WMA is easily damaged at high temperature when it has moisture [32,33]. Thus, the evaluation parameters of two kinds of asphalt mortars with different aggregate moisture contents at 60 °C were measured, and the results are listed in Table 6. Table 7 calculates and summarizes the sensitivity of each evaluation parameter.

Table 7 suggests that G^* and η^* had the highest sensitivity and similarity, $G^*/\sin\delta$ followed, and δ was the least. A higher sensitivity indicated a great effect of moisture content on an evaluation parameter. For example, when the aggregate moisture content of the asphalt mortar containing waxy additives decreased from 1.5% to 0.4%, the sensitivities of G^* , δ , $G^*/\sin\delta$ and η^* were 77.18, 13.71, 65.25, and 78.95, respectively. Therefore, the variations of aggregate moisture content could cause significant variations on

Table 5
Results of Spearman correlation analysis.

Factor	Statistical variable	δ	G^*	$G^*/\sin\delta$	η^*
Moisture content	Correlation coefficient r	0.611**	-0.907^{**}	-0.91^{**}	-0.909^{**}
	P-value (two-tailed test)	<0.001	<0.001	<0.001	<0.001
Test temperature	Correlation coefficient r	0.361**	-0.338^{**}	-0.332^{**}	-0.34^{**}
	P-value (two-tailed test)	0.004	<0.001	<0.001	<0.001
Type of warm mix additive	Correlation coefficient r	0.112**	0.132**	0.133**	0.133**
	P-value (two-tailed test)	0.004	<0.001	<0.001	<0.001

Note: “**” represents the significance level of 0.01, respectively.

Table 6

Comparison of evaluation parameters of asphalt-aggregate interaction.

Type of asphalt mortar	Aggregate moisture content	G^* (Pa)	δ (rad)	$G^*/\sin\delta$ (Pa)	η^* (Pa·s)
Asphalt mortar containing wax based additive	1.5%	2.41E + 05	52.14	3.05E + 05	1.52E + 04
	0.4%	4.27E + 05	44.99	5.04E + 05	2.72E + 04
	0%	7.15E + 05	38.40	8.19E + 05	4.49E + 04
Asphalt mortar containing surfactant based additive	1.5%	2.32E + 05	53.53	2.89E + 05	1.48E + 04
	0.4%	4.58E + 05	40.92	5.19E + 05	2.91E + 04
	0%	7.49E + 05	33.64	8.10E + 05	4.76E + 04

Table 7

Comparison of the sensitivity of each evaluation parameter.

Type of asphalt mortar	Aggregate moisture content	G^* (Pa)	δ (rad)	$G^*/\sin\delta$ (Pa)	η^* (Pa·s)
Asphalt mortar containing wax based additive	1.5%	–	–	–	–
	0.4%	77.18	13.71	65.25	78.95
	0%	67.45	14.65	62.50	65.07
Asphalt mortar containing surfactant based additive	1.5%	–	–	–	–
	0.4%	97.41	23.56	79.58	96.62
	0%	63.54	17.79	56.07	63.57

G^* and η^* . Thus, G^* and η^* could sensitively characterize the interaction ability between aggregates and asphalts.

3.3. Effect of aggregate moisture content on the pavement performance

Two kinds of WMA were prepared at the optimal mixing and compaction temperatures in accordance with Table 3. The aggregate heating temperatures of the WMA containing waxy and surfactant additives were 140 °C and 145 °C, respectively. The oven heating method was used to obtain different aggregate moisture contents by controlling the heating time. The pavement performances of two kinds of WMA prepared under different aggregate moisture contents are listed in Table 8. HMA was also considered to compare the differences in pavement performances between WMA and HMA. Its test results are listed in Table 9. Fig. 12 plots the high-/low-temperature performance and water stability of asphalt mixtures under different aggregate moisture contents.

Fig. 12 suggests that the performance of WMA showed the same growth trend regardless of high-/low-temperature performance or water stability. Take the WMA containing surfactant additives as an example, the moisture contents of the aggregate were 1.5%, 0.4%, and 0.0% after heating for 1, 2, and 3.5 h, respectively. Correspondingly, the dynamic stabilities were 7949, 10,941, and 11,347 time/min. This phenomenon indicated that, as the moisture content of the aggregate decreased, the performance of WMA would be improved.

Comparing the properties of WMA and HMA, it was found that, when the aggregate moisture content decreased to 0.4% or 0%, the rutting resistance and water stability of WMA were all better than

Table 9

Pavement performance of HMA.

	Dynamic stability (time/mm)	Failure strain ($\mu\epsilon$)	TSR (%)
HMA	7736	5731	81.2

those of HMA. However, the cracking resistance of WMA was worse than that of HMA even when the aggregate moisture content was 0%. Fig. 12 (b) shows that, when the aggregate moisture content was 0%, the failure strains of WMA containing surfactant additives and waxy additives were 19.28% and 35.58% lower than the failure strain of HMA. This result suggested that the crack resistance at low-temperature of WMA was worse than that of HMA, but it still met the basic requirement of the Chinese standard JTG E20-2011 of 2800 $\mu\epsilon$. Therefore, when the aggregate was totally dried, the WMA could exhibit better properties than HMA in terms of high-temperature performance and water stability.

Comparing the test results of pavement performances, it was found that the low-temperature performance of the WMA containing surfactant additives was generally better than that of the WMA containing waxy additives. Meanwhile, the high-temperature performance and water stability of the WMA containing surfactant additives were similar to those of the WMA containing waxy additives. Take the cracking resistance at low temperature as an example, when the aggregate heating time was 1, 2, and 3.5 h, the failure strain of WMA containing surfactant additives was 30.46%, 25.89%, and 25.29% higher than that of WMA containing waxy additives, respectively. Therefore, compared with wax based additives, surfactant based additives could improve the cracking resistance of WMA at low temperature.

Table 8

Pavement performance of WMA.

Type of WMA	Heating time	1 h	2 h	3.5 h
Containing wax based additives	Aggregate moisture content (%)	1.5	0.4	0
	Dynamic stability (time/mm)	7949	10,941	11,347
	Failure strain ($\mu\epsilon$)	2620	3541	3692
	TSR (%)	75	80	81.3
	Aggregate moisture content (%)	1.2	0.3	0
Containing surfactant based additives	Dynamic stability (time/mm)	7237	12,799	13,421
	Failure strain ($\mu\epsilon$)	3418	4458	4626
	TSR (%)	77.4	80.3	81.9

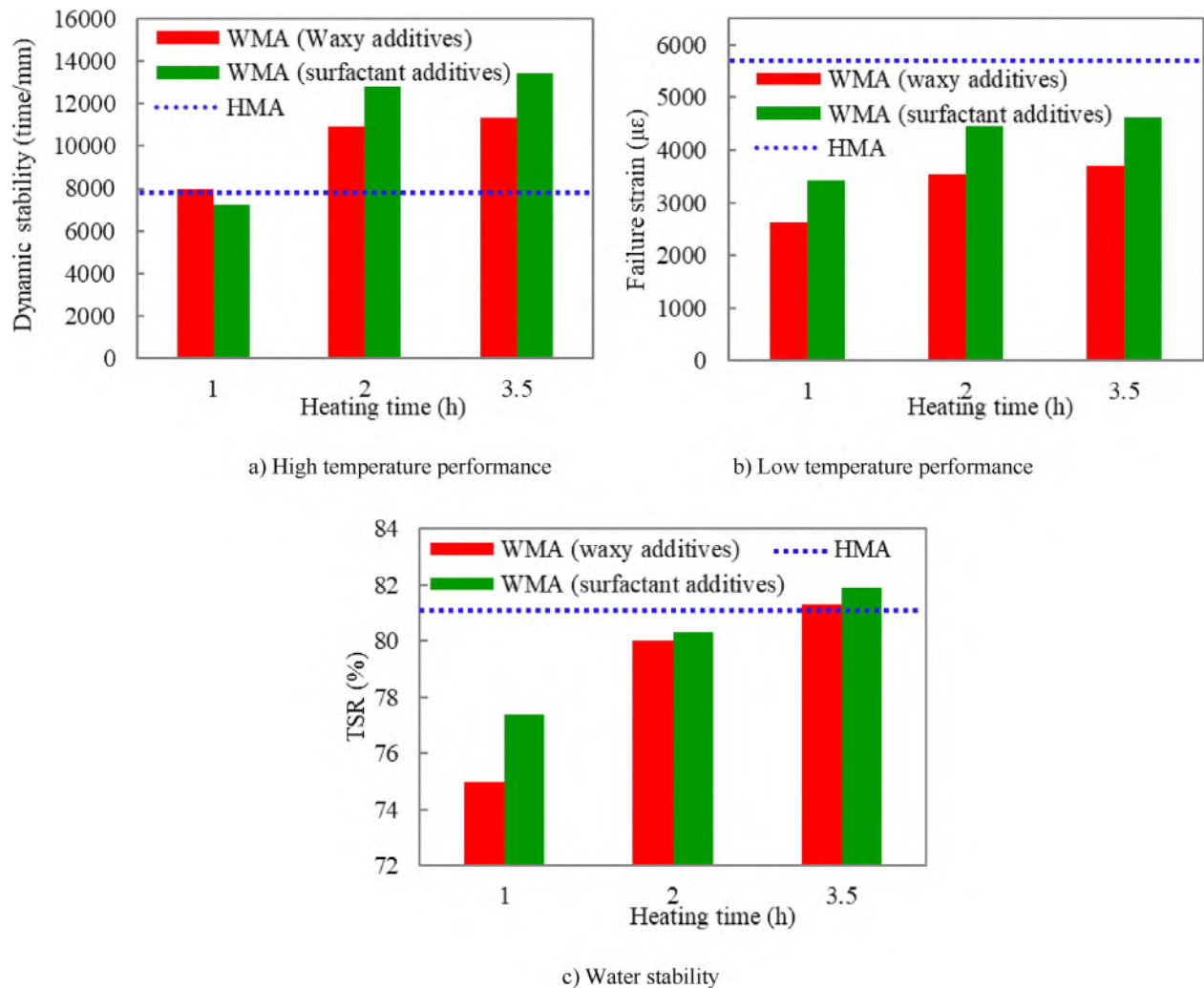


Fig. 12. The pavement performance of asphalt mixtures with different aggregate moisture contents.

4. Conclusions

The effects of different factors, such as heating time, heating temperature and aggregate particle size, on the moisture content of the aggregate was quantitatively analyzed. The initial moisture content of the aggregate quickly rose as the particle size became finer due to the action of the capillary force. The quadratic function could be used to express the temporal decline of moisture content. The specific surface area of the aggregate increased with the decrease in the particle size, which indicated that a small particle size corresponded to a fast the decline rate of moisture content. When the particle size increased, the variation rate of moisture loss rate decreased, which indicated that the coarse aggregate had a relatively stable decline rate of the moisture content over time. Compared with the heating temperature in the range of 150 °C–170 °C, the decline rate of moisture content exhibited a higher temperature sensitivity in the range of 110 °C–150 °C.

Rheological performances of asphalt mortar were characterized. As the aggregate moisture content increased, the G^* , $G^*/\sin\delta$ and η^* of asphalt mortar were decreased, and its δ increased. Therefore, the moisture remaining on the aggregate surface impaired the asphalt-aggregate interaction. Compared with the test temperature and the type of warm mix additive, the moisture content could significantly affect the asphalt-aggregate interaction as obtained using the Spearman correlation method. G^* and η^* could

sensitively characterize the interaction ability between asphalt binders and aggregates through the sensitivity analysis method.

Tests on the pavement performance of WMA were performed under different aggregate moisture contents. As confirmed, reducing the aggregate moisture content could improve the performance of WMA. When the aggregate was completely dried, the high-temperature performance and water stability of WMA were better than those of the HMA, while the low-temperature performance of WMA was worse than that of the HMA. Compared with wax-based additives, surfactant-based additives could improve the low-temperature performance of WMA.

CRediT authorship contribution statement

Bin Yang: Investigation, Methodology, Writing - original draft, Writing - review & editing. **Huining Xu:** Conceptualization, Methodology, Writing - review & editing, Formal analysis, Resources. **Peiyan Zhou:** Investigation, Data curation, Validation. **Yiqiu Tan:** Conceptualization, Methodology, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Key Research and Development Program of the Ministry of Science and Technology (Grant No. 2016YFE0202400) and the National Natural Science Foundation of China (Grant No. 51678207). Thanks to the anonymous reviewers for many comments that have notably helped us improve the manuscript.

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OPEN Investigating asphalt aggregate bonding degradation through aging and temperature effects using multiscale testing methods

Lemin Liu¹, Tao Liu^{2✉}, Xiaohua Fan¹, Hui Wei², Shengxu Wang¹ & Liang Fan^{2✉}

Moisture damage in asphalt mixtures is a critical durability concern influenced by interfacial adhesion between asphalt and aggregates. This study systematically evaluates adhesion performance using digital image analysis and surface energy theory across four asphalt types (OR50, OR70, OR90, ORSBS) and five aggregates (limestone, basalt, granite, diabase, diorite). Key findings reveal that limestone exhibits superior adhesion stability (100% adhesion rate under aging) due to chemical bonding, while acidic aggregates (e.g., granite) show significant sensitivity to moisture and aging. A novel quantitative method combining water immersion tests and pixel-based stripping area analysis demonstrated high correlation ($R^2 > 0.9$) with surface energy-derived ER values, validating its reliability. Temperature and aging effects were further quantified, showing adhesion loss under thermal/oxidative aging, with SBS-modified asphalt outperforming base binders. These results provide a robust framework for material selection in pavement design, emphasizing aggregate alkalinity and polymer modification as key factors for moisture resistance.

Keywords Asphalt-aggregate adhesion, Digital image analysis, Surface energy theory, Moisture damage resistance, Quantitative adhesion assessment

The moisture damage of asphalt mixtures represents a complex phenomenon influenced by multiple factors including the properties of asphalt and aggregates, interfacial adhesion characteristics, void content in mineral aggregates, asphalt film thickness, and degree of asphalt saturation^{1,2}. Among these factors, the interfacial bonding performance between asphalt and aggregates plays a decisive role in determining the moisture stability of asphalt pavement, as it directly governs the resistance against water-induced damage at the critical asphalt-aggregate interface^{3–5}.

The adhesion between asphalt and aggregates is essentially a complex physicochemical phenomenon. Currently, four main theories are widely used to evaluate the adhesion mechanisms and methods between asphalt and aggregates: mechanical theory⁶, surface energy theory⁷, chemical reaction theory⁸, and molecular orientation theory⁹.

The boiling water test and water immersion test have become the most widely used methods for evaluating asphalt-aggregate adhesion due to their simple principles, straightforward operation, high repeatability, and easily interpretable results^{10,11}. However, these methods present challenges in precisely controlling test conditions - for instance, maintaining the slight boiling state in the boiling water test is difficult, and different observers may assess the stripped surface differently. These significant human factors make it impossible to quantitatively evaluate the adhesion between asphalt and aggregates.

The net adsorption method employs ultraviolet spectrophotometry for measurement¹². Its fundamental principle involves determining the concentration changes in the solution system before and after water addition. By measuring the corresponding absorbance variations, the initial adsorption capacity and net adsorption of asphalt onto aggregates can be calculated, ultimately yielding the asphalt-aggregate adhesion rate^{13,14}. In the pre-hydration phase, aggregates adsorb asphalt molecules from the toluene solution, reducing the system concentration. Following water introduction, the system transforms into an asphalt-water-toluene ternary system, where water molecules displace asphalt from aggregate surfaces, consequently increasing the solution concentration¹⁵. Although the net adsorption method involves complex procedures, it offers high precision with minimal human interference, enabling quantitative assessment of asphalt-aggregate bonding performance.

¹Shandong High-Speed Infrastructure Construction Co., Ltd., Jinan 250098, China. ²Shandong Transportation Research Institute, Jinan 250100, China. ✉email: liutao1133@163.com; fanliang218@sina.com

Properties	OR50	OR70	OR90	ORSBS
Softening point/°C	51.0	48.5	45.0	76.5
Penetration (25 °C)/0.1 mm	47	67	84	46
Ductility (10 °C)/cm	16	37	> 100	48.7
Dynamic viscosity (135 °C)/ Pa·s	0.66	0.56	0.48	1.87

Table 1. Basic properties of four kinds of asphalt.

Aggregate type	SiO ₂	CaO	MgO	Al ₂ O ₃	Fe ₂ O ₃
Limestone	15.5	27.7	18.78	2.54	1.59
Basalt	53.99	7.02	4.9	15.5	8.88
Granite	72.77	1.43	0.22	14.8	1.16
Diabase	49.95	8.5	5.76	13.84	8.87
Diorite	60.24	5.83	3.86	18.45	4.26

Table 2. Mineral composition of aggregates (%).

The contact angle method (based on surface energy theory) provides a microscopic approach to analyze the interfacial bonding performance between asphalt and aggregates¹⁶. Essentially an energy conversion theory, surface energy theory postulates that the bonding strength at the asphalt-aggregate interface depends on the asphalt's wettability on aggregate surfaces - complete wetting indicates strong bonding, while incomplete wetting suggests weak adhesion^{17–20}. Wettability here refers to asphalt's ability to coat aggregates, which is fundamentally governed by the physicochemical properties of both materials. In aqueous environments, water's superior diffusivity enables it to rapidly coat aggregate surfaces, displacing the asphalt film through competitive adsorption and consequently degrading the interfacial bonding performance^{21–24}.

Several researchers have advanced this methodology: Little²⁵, Hefer⁶, and Bhasin^{26,27} employed surface free energy theory to analyze moisture susceptibility in asphalt mixtures; Little and Bhasin²⁸ utilized the Wilhelmy plate method to measure contact angles between solids and liquids for surface energy calculations.

While boiling water and immersion tests are widely used, their subjective interpretation and inability to quantify adhesion limit reliability. Surface energy theory, though precise, requires complex instrumentation unsuitable for routine practice. This study bridges the gap by proposing a hybrid approach: digital image analysis for quantitative stripping area measurement, validated against surface energy theory. We further systematically evaluate temperature and aging effects across diverse asphalt-aggregate combinations, providing actionable insights for pavement durability.

To facilitate convenient and quantitative characterization of adhesion between different asphalt-aggregate combinations, this study employed four asphalt types and five aggregate varieties. The digital image processing technique was utilized to quantify the stripping rate of asphalt from aggregate surfaces after water immersion. The results were subsequently correlated with surface energy measurements to validate the reliability of this methodology. Furthermore, the approach was systematically verified for its applicability across varying temperature conditions and aging states.

This study proposes a novel hybrid approach combining digital image analysis and surface energy theory to quantitatively assess asphalt-aggregate adhesion. By using digital image processing to measure stripping areas, it addresses the limitations of traditional tests. The approach is further validated across different temperatures and aging conditions, offering a more reliable and practical method for evaluating the moisture stability of asphalt pavements.

Materials and methods

Raw materials

The study selected a total of four different types of asphalt (OR50, OR70, OR90, SBS) and five types of aggregates (limestone, basalt, granite, diabase, diorite) for adhesion analysis. The basic properties of the four asphalt types are shown in Table 1.

The mineral composition of the five types of aggregates is shown in Table 2.

To investigate the changes in adhesion performance between asphalt and aggregates after aging, both short-term and long-term aging treatments were conducted on the asphalt binders.

Short-Term Aging: The rolling thin film oven (RTFOT) method was employed for short-term aging. A 35 g asphalt sample was placed in a glass container and aged in the RTFOT at 163 °C. The oven rotated at 15 r/min with a heated air flow rate of 4000 mL/min for a duration of 85 min.

Long-Term Aging: The pressure aging vessel (PAV) was used for long-term aging. Approximately 50 g of the short-term aged asphalt was transferred into a PAV pan and aged at 105 °C under compressed air at 2.1 MPa ± 0.1 MPa for 20 h.

The aged samples were designated as follows:

Short-term aged samples: OR50 → TR50, OR70 → TR70, OR90 → TR90, ORSBS → TRSBS.

Long-term aged samples: OR50 → PR50, OR70 → PR70, OR90 → PR90, ORSBS → PRSBS.

Research methodology

Asphalt four-component analysis

Based on the *Test Methods for Bitumen and Bituminous Mixtures in Highway Engineering* (JTG E20-2011)²⁹ four-component separation standard, TLRC achieves component separation through differential migration distances in specific solvent systems, followed by chromatographic scanning for quantitative analysis of component contents. Based on the test results, the asphalt was fractionated into four components: saturates (S), aromatics (Ar), resins (R), and asphaltenes (At).

1. Preparation: Activate the chromatographic rods. Dissolve the sample in toluene to prepare a 20 mg/mL asphalt solution.
2. Sample Application: Spot the sample at the origin point using a micro-syringe.
3. Development Process: Immerse the rod holder in Developer #1 until reaching target migration height. Dry for 5 min using air-blowing device. Condition in humidity chamber for 10 min. Repeat the process with Developer #2 (including drying and conditioning steps). Final development with Developer #3, followed by 10-minute drying.
4. Scanning: Place dried rods in chromatographic analysis workstation. Perform scanning to obtain component separation profiles.
5. Data Analysis: Analyze chromatograms to determine separated component contents.
6. Four-component analysis was performed on the asphalt samples both before and after aging.

Water immersion test

The water immersion test was conducted in accordance with JTG E20-2011. The detailed procedure is as follows:

1. Preparation of Asphalt Binder: Heat the asphalt sample to the specified mixing temperature (180 °C).
2. Aggregate Preparation: Sieve the aggregates to obtain particles sized between 9.5 mm and 13.2 mm. Wash 200 g of the selected aggregates (preferably flat and regular-shaped) with clean water, dry them in a desiccator, and then divide them using the quartering method to obtain a 100 g sample. Place the 100 g sample in a mixing container and heat it in an oven at 185 °C for 1 h.
3. Asphalt-Aggregate Mixing: Weigh 5.5 ± 0.2 g of asphalt (accurate to 0.1 g) per 100 g of aggregates. Pour the weighed asphalt into the mixing container with the heated aggregates. Return the container to the oven and heat for an additional 15 min.
4. Mixing and Cooling: Remove the container from the oven and stir thoroughly with a glass rod for 1–1.5 min until the asphalt is uniformly coated on the aggregate surfaces. Transfer the mixture to a glass plate using a small spatula and allow it to cool at room temperature for 1 h before the water immersion test.
5. Water Immersion Test: Submerge the prepared sample in a constant-temperature water bath maintained at 80 ± 1 °C. Immediately remove any loosened or floating asphalt film from the water surface.
6. Evaluation of Adhesion: After 30 min, carefully remove the glass plate from the water bath and immerse it in pre-prepared cold water.

Observe the extent of asphalt film stripping, estimate the stripping ratio, and classify the adhesion performance according to the specified rating scale.

Digital image analysis of adhesive failure

The adhesion performance was evaluated using Photoshop-based pixel analysis to determine the stripped or adhered asphalt area on aggregate surfaces after water immersion. The procedure is as follows:

1. Sample Preparation:
Select five flat-surfaced aggregates for water immersion testing.
2. Image Acquisition:
Photograph the post-immersion aggregates under uniform, high-intensity lighting using a macro lens to ensure consistent clarity.
3. Image Processing and Thresholding:
Since the asphalt regions appeared as black areas in the images, we applied a color-based selection strategy rather than grayscale conversion. Specifically, the Color Range / Select Color tool in Photoshop was used to isolate the black asphalt regions.
(1) Thresholding was performed in the RGB channels, with the following settings: Black asphalt regions were identified by selecting pixels within the low RGB value range (R: 0–50, G: 0–50, B: 0–50).
The contrast between the dark asphalt and the lighter-colored aggregates was enhanced by adjusting the selection fuzziness between 25 and 35 to ensure precise isolation of asphalt pixels without including aggregate edges.
(2) Manual fine-tuning was performed when necessary to ensure that all asphalt areas were accurately captured, especially at aggregate boundaries.
4. Boundary Handling and Error Minimization:
To address irregular aggregate edges, a contour smoothing step was applied after pixel selection to reduce jagged boundaries. Additionally, a light Gaussian blur (radius = 2 px) was applied to reduce pixel-level noise before quantification.
5. Calculation: Compute the following ratios.
Stripping ratio (%) = (Stripped area pixels / Total surface pixels) $\times$ 100.
Adhesion ratio (%) = (Adhered asphalt pixels / Total surface pixels) $\times$ 100.
The image processing effect is shown in Fig. 1.

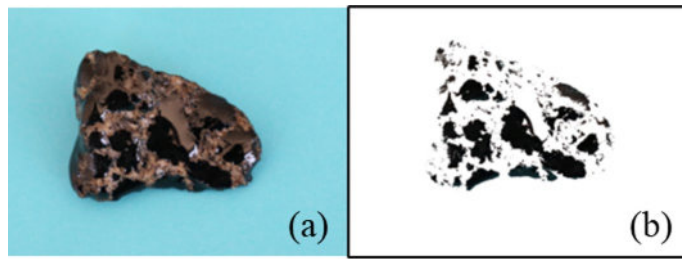


Fig. 1. Image processing effect. (a) Original image; (b) processed image (asphalt isolated).

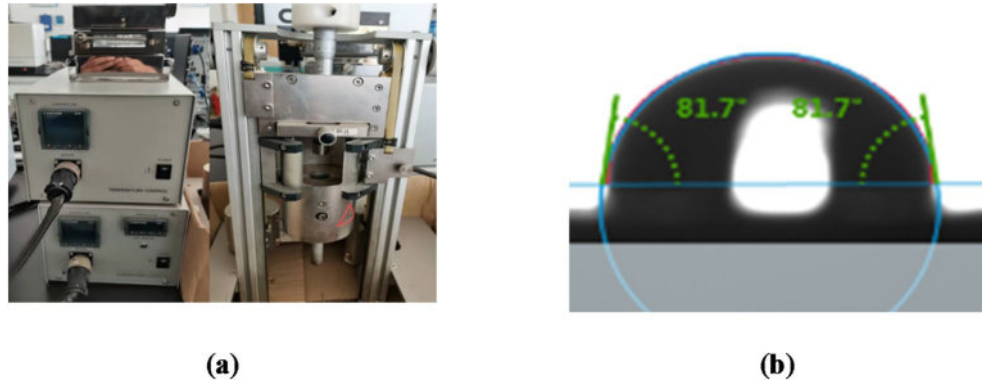


Fig. 2. The contact angle testing equipment and its schematic diagram: (a) contact angle testing equipment (with high-temperature goniometer); (b) schematic diagram of sessile drop method.

Surface energy measurements

To validate the rationality and stability of the quantitative evaluation method, the quantitative test results were compared and analyzed with the surface energy-based moisture susceptibility evaluation index *ER* (Eq. 1)³⁰.

$$ER = \left| \frac{W_{as}}{W_{aws}} \right| \quad (1)$$

where: *ER*: Moisture susceptibility evaluation index; W_{as} : Work of adhesion (mJ/m^2); W_{aws} : Work of debonding (mJ/m^2).

The calculation formulas for W_{as} and W_{aws} are given in Eqs. (2) and (3), respectively.

$$W_{as} = \gamma_a (1 + \cos\theta) \quad (2)$$

$$W_{aws} = \gamma_a (1 + \cos\theta) - \gamma_w \cos\theta_1 - \gamma_w \cos\theta_2 \quad (3)$$

In the equation, γ_a represents the surface energy of asphalt (mJ/m^2), γ_w denotes the surface energy of water (mJ/m^2), θ is the contact angle between asphalt and aggregate ($^\circ$), θ_1 indicates the contact angle between water and asphalt ($^\circ$), and θ_2 stands for the contact angle between water and aggregate ($^\circ$).

The contact angle measuring instrument (KRÜSS-DSA100) was selected as the testing equipment, employing the sessile drop method to measure the contact angles between water/bitumen and aggregates. The contact angle testing equipment and its schematic diagram are illustrated in Fig. 2.

The dynamic contact angle was quantified at 155 ± 1 °C using a high-temperature goniometer, replicating the actual asphalt-aggregate interfacial conditions during hot mixing. The contact angle changes dynamically over time. In this study, the contact angle at 10 s was chosen to calculate the surface energy, in order to avoid the influence of instantaneous fluctuations caused by the droplet's contact with the surface.

Results

Quantitative evaluation of asphalt-aggregate adhesion

The water immersion tests were conducted on all five aggregate types, with the quantitative results presented in Table 3.

As shown in Table 3, the visual adhesion grading obtained through the water immersion test exhibits relatively low precision, providing only approximate ranges. Moreover, this method is highly susceptible to subjective human judgment, resulting in limited accuracy and an inability to quantitatively evaluate the interfacial bonding performance between asphalt and aggregates.

Aggregate type	OR50	OR70	OR90	ORSBS
Limestone	5*	5	5	5
Basalt	3	2~3	4	5
Granite	2~3	2~3	2~3	4
Diabase	5	4	4	5
Diorite	4	3	3	5

Table 3. Adhesion grades between different aggregates and asphalt. *A higher numerical value indicates better adhesion performance.



Fig. 3. Test results of asphalt-aggregate samples after water immersion.

To address these limitations, the post-immersion aggregates were photographed, and the images were processed using Photoshop pixel analysis. This approach calculates the adhesion ratio, enabling a quantitative assessment of the asphalt-aggregate bonding characteristics. Representative images of the immersed aggregates are presented in Fig. 3.

The acquired images were analyzed using the Photoshop pixel area method, where the stripping ratio was calculated according to the methodology described in Sect. 2.1.2. The average results of ten experiments conducted under each experimental condition are recorded in Fig. 4.

Figure 4 demonstrates a clear adhesion performance ranking between the same asphalt grade and different aggregates, following the order of limestone > diabase > basalt > diorite > granite, which directly corresponds to their alkalinity levels from highest to lowest. The superior adhesion of limestone, composed primarily of CaCO_3 , stems from its strong chemical bonding with asphalt's acidic components, showing minimal variation across test points due to this effective chemisorption. In contrast, diabase and basalt, containing approximately 50% SiO_2 and classified as neutral aggregates, exhibit 20–30% lower adhesion than limestone because of their limited chemical reactivity. Granite, with about 70% SiO_2 content and acidic nature, displays the poorest adhesion performance as it lacks chemical affinity with asphalt and relies solely on mechanical interlocking. When examining different asphalt types with the same aggregate, the results reveal that the penetration grade of base asphalt does not significantly influence adhesion performance.

The successful application of digital image analysis through Photoshop pixel quantification objectively verifies these interfacial bonding characteristics, confirming the method's sensitivity to aggregate mineralogy, measurement repeatability, and superior reliability compared to traditional visual assessment techniques. These findings highlight the critical role of aggregate chemistry in pavement durability while demonstrating the effectiveness of quantitative image analysis for evaluating asphalt-aggregate compatibility.

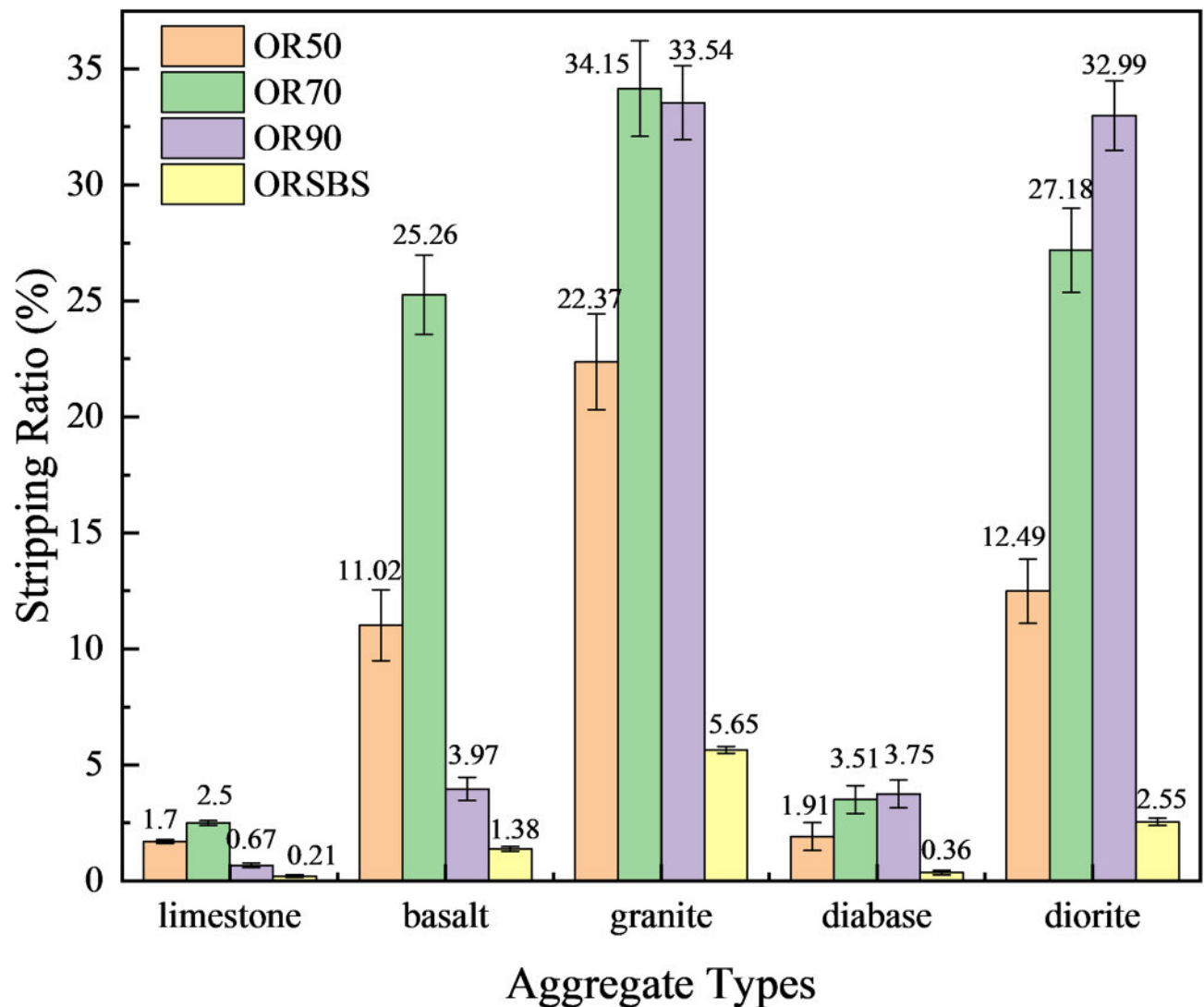


Fig. 4. Stripping rates of different asphalt-aggregate combinations.

Adhesion test results based on surface energy

Test results of contact angle between asphalt and aggregates

The spreading kinetics of distinct asphalt binders on aggregate surfaces exhibited significant variations, necessitating dynamic contact angle monitoring until stabilization. As evidenced in Fig. 5, the equilibrium contact angles were attained through distinct temporal patterns.

As observed in Fig. 4, the contact angles between all aggregates and asphalt gradually decreased over time before slowly stabilizing. The asymptotic contact angle behavior reflects thermodynamic equilibrium attainment, where: Limestone's higher final θ (20°–30°) indicates weaker wettability but stronger chemisorption. Granite's lower θ (5°–15°) manifests better wetting but poorer chemical bonding. This aligns with the stripping ratio ranking in Sect. 3.1.

The surface energy-based moisture susceptibility evaluation

Based on the results in Fig. 5, the work of adhesion (W_{as}) and the work of debonding (W_{aws}) were calculated using Eqs. 2 and 3, respectively. The computational results are presented in Figs. 6 and 7.

As shown in Fig. 6, the work of adhesion (W_{as}) varies significantly between different asphalt binders and the same aggregate type. Taking OR70 asphalt as an example, the adhesion performance follows the order: basalt > limestone > diabase > diorite > granite, which aligns with conventional understanding that alkaline aggregates exhibit better adhesion with asphalt. However, OR50 asphalt shows a completely different trend: limestone > diorite > granite > basalt > diabase.

This discrepancy suggests that using W_{as} alone to evaluate asphalt-aggregate adhesion has certain limitations. While OR70's results conform to the established rule (higher adhesion with alkaline aggregates), OR50's reverse trend indicates other factors (e.g., asphalt chemical composition, surface energy characteristics) may dominate the interfacial bonding behavior in some cases.

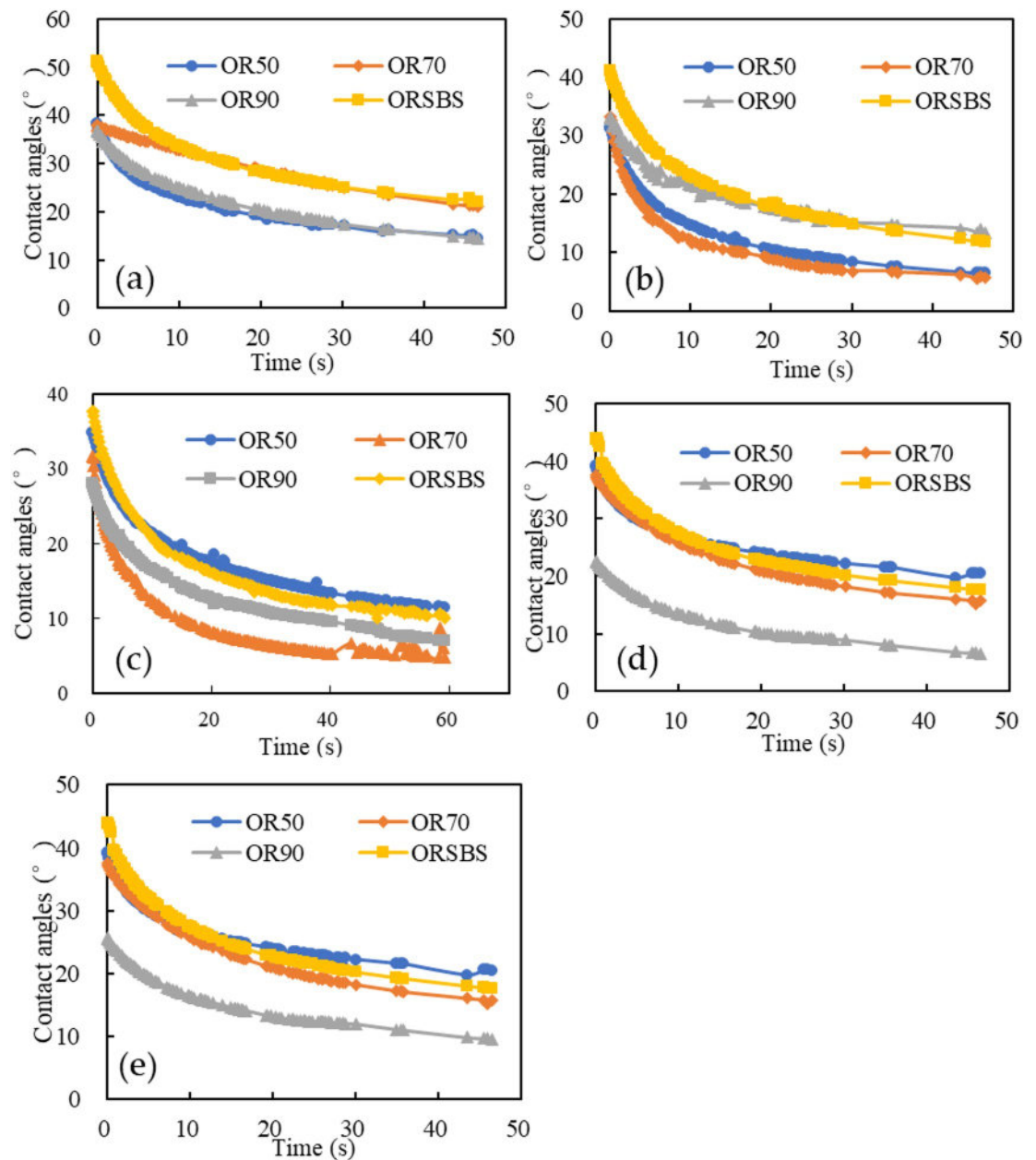


Fig. 5. Temporal evolution of contact angles for different asphalt-aggregate combinations (a) limestone; (b) basalt; (c) granite; (d) diabase; (e) diorite.

The results presented in Fig. 7 demonstrate a consistent hierarchy in the work of debonding (W_{aws}) across different asphalt-aggregate combinations, with limestone exhibiting the strongest moisture resistance, followed by diabase, basalt, diorite, and granite, confirming that alkaline aggregates maintain superior adhesion with asphalt in the presence of water while acidic aggregates perform poorest. This thermodynamic parameter W_{aws} , which specifically quantifies water's disruptive effect on the asphalt-aggregate interface, proves more reliable than dry adhesion measurements for evaluating moisture susceptibility, as evidenced by its strong correlation with both laboratory performance and real-world pavement behavior where limestone-based mixtures consistently outperform granite counterparts in wet conditions. The robust agreement between these surface energy measurements and practical engineering experience validates W_{aws} as a fundamentally sound indicator for predicting water damage potential in asphalt mixtures.

The energy ratio (ER), as a comprehensive moisture susceptibility indicator, was derived from the interfacial energy parameters (W_{as} and W_{aws}) in Figs. 5 and 6 through Eq. 1. Table 4 systematically presents the calculated ER values.

From the data in Table 4, it can be observed that basic aggregates generally maintain higher ER values, while acidic aggregates show lower ER values. The alkalinity/acidity of aggregates remains a key factor influencing their resistance to moisture damage.

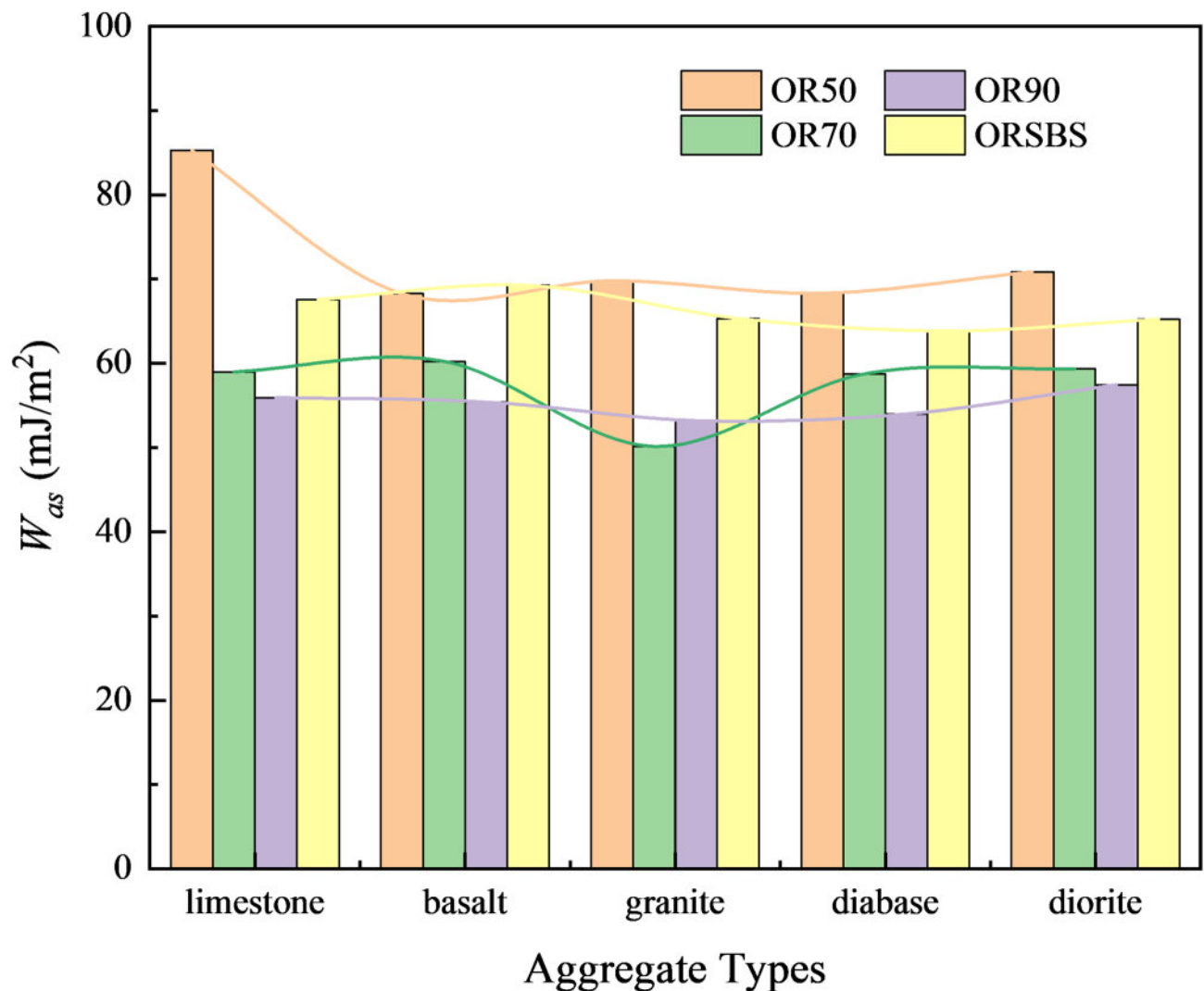


Fig. 6. Work of adhesion (W_{as}) under dry conditions.

Consistency analysis between quantitative analysis methods and ER value results

To validate the applicability and consistency of the quantitative results obtained from the water-immersion-based digital image analysis method, a correlation analysis was conducted between the stripping rate calculated in Sect. 3.1 and the moisture damage evaluation index (ER) from Sect. 3.2.2. The results are shown in Fig. 8.

The curve-fitting equations and variances for different asphalt binders in Fig. 8 are recorded in Table 5.

The statistical results in the table demonstrate a strong correlation between the stripping rate obtained through digital image analysis and the surface energy-based moisture damage evaluation index (ER). Notably, both OR70 and OR90 exhibit correlation coefficients exceeding 0.9. This indicates that the combined approach of water immersion testing and image processing technology provides an efficient method for evaluating the adhesion between asphalt and aggregates, with results demonstrating both stability and consistency.

Effect of temperature on adhesion performance

This section investigates the water immersion test at bath temperatures of 40 °C, 50 °C, 60 °C, 70 °C, and 80 °C. The stripping area at each temperature was quantified using pixel area analysis, with the resulting adhesion rates shown in Fig. 9.

Figure 9 shows the adhesion rates between various asphalt binders and aggregate types under different temperature conditions. The results demonstrate that the adhesion performance generally follows the order of limestone > diabase > basalt > diorite > granite. Limestone exhibits exceptional temperature stability, maintaining nearly 100% adhesion across all tested temperatures, while diabase, basalt, diorite and granite show significant temperature-dependent variations in adhesion performance.

The critical temperature at which stripping occurs differs among various asphalt-aggregate combinations. A clear trend emerges where adhesion rates progressively decrease with rising temperature. This phenomenon can be attributed to thermal expansion increasing intermolecular distances and consequently weakening interfacial forces. At elevated temperatures, the enhanced polarity of aggregates strengthens their water adsorption capacity,

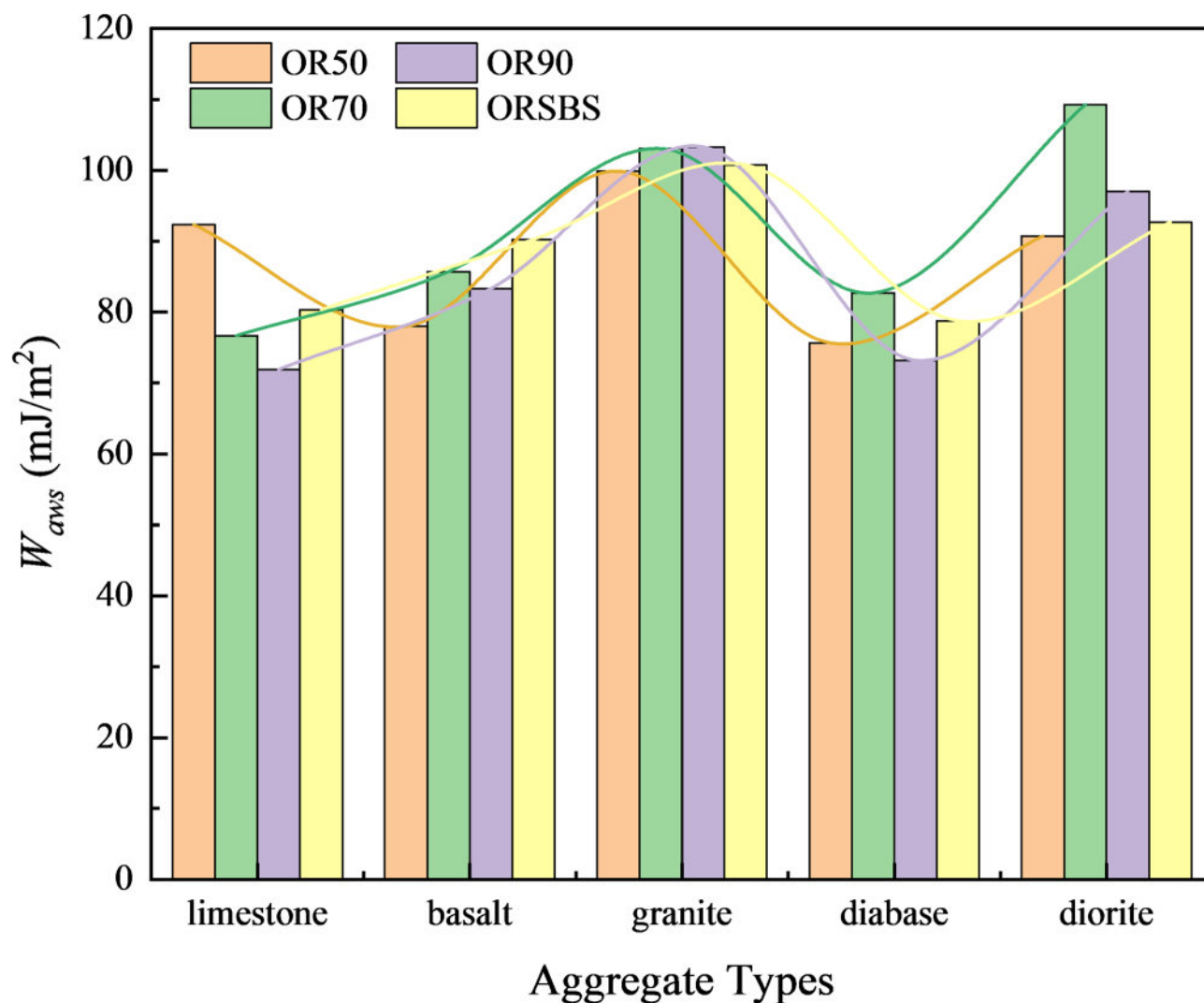


Fig. 7. Work of debonding (W_{aws}) after water exposure.

Aggregate type	OR50	OR70	OR90	ORSBS
Limestone	0.924	0.769	0.776	0.841
Basalt	0.876	0.703	0.665	0.767
Granite	0.698	0.486	0.515	0.648
Diabase	0.90	0.701	0.737	0.810
Diorite	0.781	0.543	0.592	0.703

Table 4. Energy ratio (ER) values for different asphalt-aggregate combinations.

leading to competitive displacement of asphalt films by water molecules. As temperature increases, this water-asphalt competition results in gradual film detachment from aggregate surfaces, manifested as reduced adhesion rates. Ultimately, these processes degrade the interfacial bonding performance between asphalt and aggregates, thereby diminishing the mixture's resistance to moisture damage.

The comprehensive data presented in these figures provides valuable insights into the temperature sensitivity of different asphalt-aggregate systems, particularly highlighting limestone's superior performance stability compared to other common aggregate types under thermal variations. These findings have important implications for pavement material selection in different climatic conditions.

The temperature at which stripping occurs varies for different asphalt-aggregate combinations. As shown in Fig. 9, limestone demonstrates superior adhesion with asphalt, requiring higher temperatures to induce stripping, whereas granite and basalt exhibit poorer adhesion performance with stripping occurring at relatively lower temperatures. Therefore, when employing the water immersion method to evaluate asphalt-aggregate adhesion,

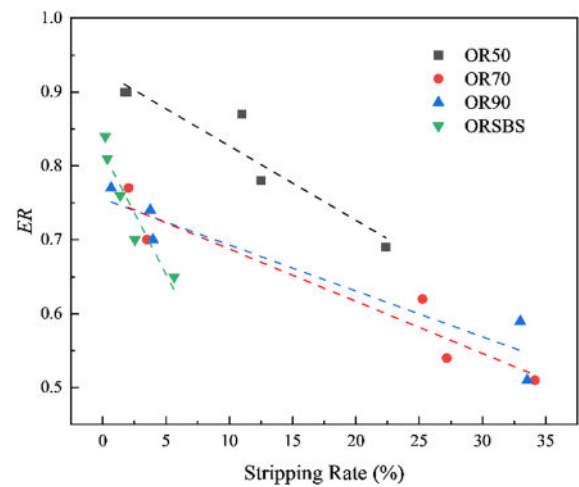


Fig. 8. Correlation analysis between stripping rate and ER.

Asphalt binders	Fitting equations	R ²
OR50	y = -0.0101x + 0.9276	0.891
OR70	y = -0.0071x + 0.7581	0.912
OR90	y = -0.00621x + 0.7550	0.910
ORSBS	y = -0.0331x + 0.8191	0.898

Table 5. Linear fitting equations and variances for different asphalt binders.

it is recommended to use different water bath temperatures for different aggregate types. For instance, a water immersion temperature of 80 °C may be considered for limestone, while 40 °C would be more appropriate for basalt and granite.

Effect of aging on adhesion performance

Changes in asphalt components before and after aging

The changes in component contents of various asphalts after aging are presented in Fig. 10, which clearly shows a progressive decrease in saturates, aromatics and resins contents coupled with a corresponding increase in asphaltenes content as the aging process intensifies. Among these components, saturates demonstrate the highest resistance to aging effects, exhibiting minimal content variations of less than 1% under different aging conditions. In contrast, asphaltenes display the most pronounced sensitivity to aging, with short-term aging resulting in an increase of at least 3% and long-term aging causing a more substantial growth ranging between 7% and 9%. These quantitative changes in component distribution provide direct evidence of the chemical transformations occurring during asphalt aging, particularly the oxidative conversion of lighter fractions into heavier asphaltenes compounds.

Changes in adhesion between asphalt and aggregates

The stripping areas at different aging levels were quantified using digital image analysis to determine adhesion rates, as shown in Fig. 11. Since the viscosity of asphalt undergoes significant changes after aging and is generally considered to affect the adhesion rate, the dynamic viscosity at 135 °C was selected as the abscissa variable to investigate their correlation in this study.

Figure 11 present the adhesion rates between base asphalt and limestone, diabase, and diorite aggregates at different aging levels. The results demonstrate a gradual but moderate decrease in asphalt-aggregate adhesion as the base asphalt undergoes aging.

The interfacial adhesion between asphalt and aggregates involves complex physicochemical interactions. While aging enhances mechanical interlocking forces, it simultaneously alters the asphalt’s composition and structure, disrupting its original equilibrium. These changes reduce the asphalt’s surface activity and wetting capability, ultimately leading to diminished adhesion performance with progressive aging.

The adhesion rates between base asphalt and basalt/granite aggregates under various aging conditions, revealing a distinctive pattern where adhesion initially increases before declining with prolonged aging. This behavior contrasts with other aggregate types and can be attributed to the unique physical adsorption mechanism governing the asphalt-basalt/granite interface. As the asphalt undergoes aging, its compositional and structural changes disrupt the original colloidal equilibrium, while simultaneously enhancing mechanical interlocking forces through increased asphaltene content (rising more than 3%). When exposed to moisture, water molecules initially displace the asphalt film, but the aged asphalt’s improved mechanical anchoring capability facilitates

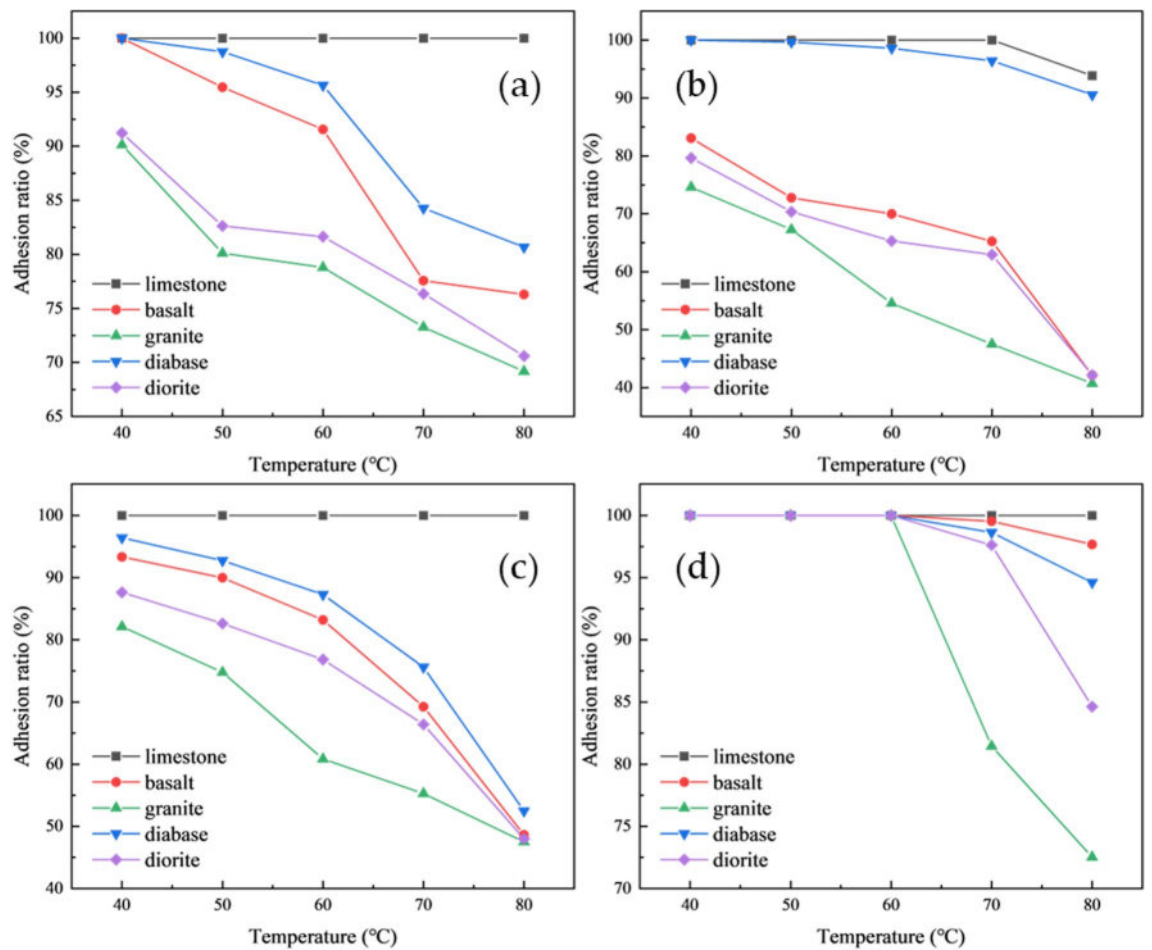


Fig. 9. Quantitative analysis of temperature-dependent adhesion performance using digital image processing (a) OR50; (b) OR70; (c) OR90; (d) ORSBS.

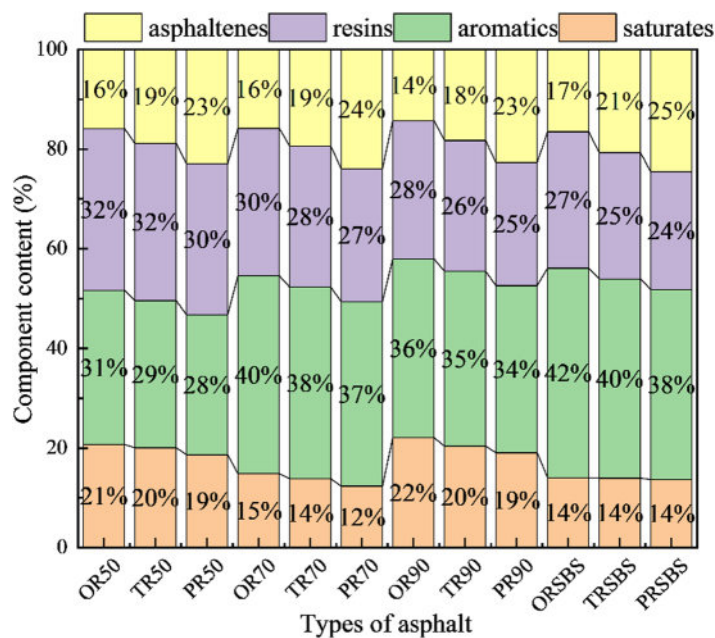


Fig. 10. Changes in four-component composition of aged asphalt.

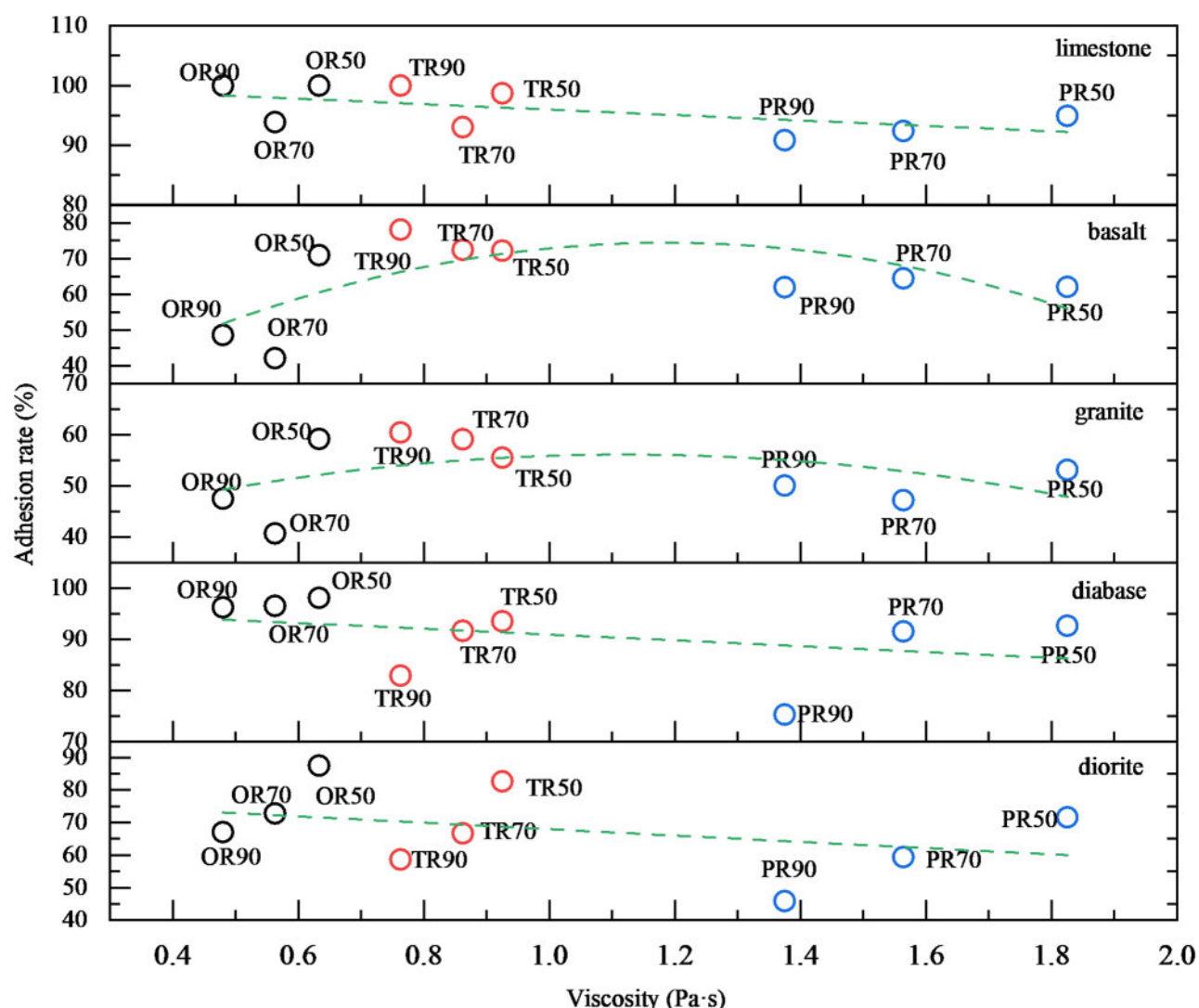


Fig. 11. Effect of base asphalt aging on adhesion rate.

re-adsorption onto the aggregate surface, creating additional binding sites and temporarily boosting adhesion performance. However, with continued aging, the progressive deterioration of asphalt's wetting properties and surface activity ultimately leads to the observed decline in adhesion, demonstrating the complex interplay between physical interlocking and chemical bonding in these systems.

Figure 12 specifically presents the relationship between viscosity and adhesion rate changes in SBS-modified asphalt after aging. Due to SBS's significantly higher viscosity compared to base asphalt (typically 2–3 times greater at 135 °C), its aging characteristics exhibit distinct patterns from unmodified binders. The plotted data reveals how the substantial viscosity increase during SBS aging - from approximately 1.8 Pa s (unaged) to over 2.8 Pa s (after PAV aging)—correlates with corresponding adhesion rate variations.

As shown in the Fig. 12, the SBS-modified asphalt maintains a constant 100% adhesion rate with limestone aggregates regardless of aging severity, indicating superior interfacial compatibility. This exceptional performance stems from the mechanical interlocking forces between aged SBS asphalt and limestone surpassing the chemical bonding effects, thereby preserving adhesion integrity. In contrast, the adhesion rates progressively decline for diabase, basalt, diorite, and granite aggregates with extended aging. This deterioration results from two synergistic mechanisms: firstly, aging degrades the polymer modifier's effectiveness, causing structural and compositional changes that reduce crosslinking between asphalt and SBS modifiers, consequently weakening cohesive strength; secondly, while aging enhances mechanical adhesion, it simultaneously diminishes asphalt's surface activity and wetting capability, ultimately impairing overall adhesion performance with these aggregate types.

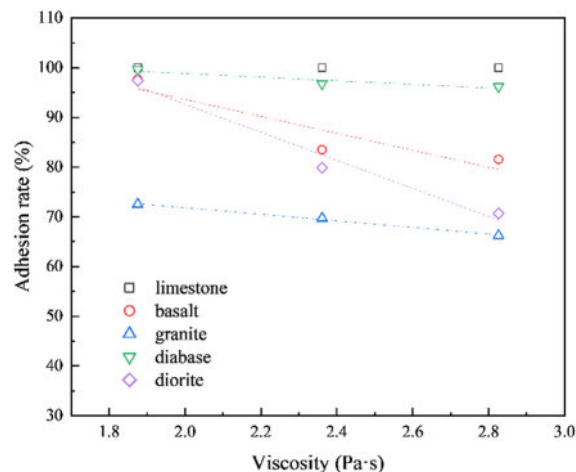


Fig. 12. Adhesion rate of SBS-modified asphalt at different aging levels.

Conclusions

In summary, this study provides a comprehensive evaluation of asphalt-aggregate adhesion mechanisms through advanced quantitative methods, revealing critical insights into moisture damage resistance under varying environmental and aging conditions. The main findings and recommendations are as follows:

Limestone, with its CaCO_3 -rich surface, demonstrates stable adhesion (100% under aging), while granite, due to its high SiO_2 content, shows poor performance mainly due to reliance on mechanical interlocking. SBS-modified asphalt improves adhesion with granite, highlighting the importance of polymer modification in enhancing interfacial bonding.

Aging reduces adhesion by approximately 10%, primarily due to increased asphaltenes and resin depletion, which degrade wettability. SBS modification mitigates these effects, improving performance and maintaining adhesion strength, especially with granite aggregates.

Adhesion decreases with temperature, with granite failing at 40 °C and limestone at 80 °C, suggesting that testing conditions should be tailored to specific aggregates. Alkaline aggregates and SBS-modified asphalt are recommended for areas with high moisture exposure.

Strong correlation ($R^2 > 0.9$) is observed between stripping areas and surface energy values, providing a reliable alternative to subjective visual assessments. This method should be incorporated into quality control for better accuracy in measuring asphalt-aggregate adhesion.

Author contributions.

Data availability

Data is contained within the article.

Received: 25 July 2025; Accepted: 29 October 2025

Published online: 27 November 2025

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Author contributions

Funding acquisition, Liang Fan; Methodology, Xiaohua Fan; Project administration, Lemin Liu; Resources, Shengxu Wang; Writing – original draft, Hui Wei; Writing – review & editing, Tao Liu.

Funding

This research was funded by Shandong Provincial Natural Science Foundation, grant number No. ZR2024QE425. Hainan Provincial Key R&D Program, grant number No. ZDYF2023GXJS147

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to T.L. or L.F.

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Full length article

Molecular dynamics investigation of interfacial adhesion between oxidised bitumen and mineral surfaces

Yangming Gao^a, Yuqing Zhang^{a,*}, Yang Yang^b, Junhui Zhang^c, Fan Gu^a^a Aston Institute of Materials Research, and Engineering Systems & Management Group, Aston University, Birmingham, B4 7ET, UK^b Bioenergy Research Group, European Bioenergy Research Institute, Aston University, Birmingham, B4 7ET, UK^c National Engineering Laboratory of Highway Maintenance Technology, Changsha University of Science & Technology, Changsha, China

ARTICLE INFO

Keywords:

Bitumen aging

Interfacial adhesion

Oxidised bitumen

Mineral surfaces, molecular dynamics

Water damage

ABSTRACT

The interfacial adhesion between oxidised bitumen and mineral surfaces at dry and wet conditions was investigated using molecular dynamics (MD) simulations. Molecular models were built for virgin and oxidised bitumen components including saturate, aromatic, resin and asphaltene. The bitumen models and four representative mineral substrates (namely quartz, calcite, albite and microcline) were employed to construct bitumen-mineral interface systems. These models were validated by the experimental results and MD simulations reported in the literature. The hardening mechanism of the aged bitumen was analysed by comparing the density, cohesive energy density and fraction of free volume between the virgin and oxidised bitumen. Work of adhesion was computed to quantify the adhesive bonding property of the bitumen-mineral interface systems for the virgin, lightly oxidised and heavily oxidised bitumen models under dry and wet conditions. Results show that the oxidised products (carbonyl and sulfoxide) strengthen the intermolecular bonding, resulting in molecular aggregation and physical hardening of the aged bitumen. When bitumen becomes oxidised at the dry condition, the interfacial adhesion of bitumen-acidic minerals (quartz) is dominated by van der Waals interaction which decreases due to the increased bitumen-quartz intermolecular distance caused by the aggregated bitumen molecules during aging. In comparison, the interfacial adhesion of bitumen-strong alkali minerals (albite and microcline) is dominated by electrostatic energy which increases due to higher polarity introduced by the oxidised products. For the bitumen-weak alkali mineral (calcite), the interfacial adhesion is attributed to both electrostatic energy and van der Waals energy, where compared to the virgin bitumen, the electrostatic energy becomes lower for the lightly-oxidised bitumen due to the increased bitumen-mineral distance but becomes higher for the heavily-oxidised bitumen due to higher polarity. At wet condition, water is the dominating factor that affects (weakens) the interfacial adhesion between the bitumen and the acidic minerals (quartz), and the oxidative aging of bitumen is the major factor that affects (strengthens) the interfacial adhesion between the bitumen and the strongly alkaline minerals (albite and microcline). For the weak alkali minerals such as calcite, both water and bitumen aging can significantly affect the interfacial adhesion.

1. Introduction

Bitumen is an important organic material for industrial applications, and it is the refinery residues from petroleum (crude oil). As an adhesive and thermoplastic material, bitumen has been widely used for road pavement construction [1] and water-proofing coatings. In addition, it can also be used as an electrically conductive material when mixed with graphite particles [2]. Bitumen's durability is significantly affected by oxidative aging where bitumen molecules react with atmospheric oxygen [3]. The oxidative aging can cause chemical changes of the bitumen including the formation of carbonyl and sulfoxide,

which have a considerable impact on the physical and mechanical properties of the bituminous materials, such as viscoelastic properties, cohesive and adhesive bond strengths, and fracture properties [4,5].

A number of experimental studies have been performed to understand the bitumen aging at the micro and macro dimensions in the pavement engineering applications. Rheological tests were developed to characterize and quantify the aging behaviour of the bitumen [6–9]. Recently, the aging effects of bitumen on the rheology [10,11], intrinsic healing [12], and surface energy and bonding property [13] were also evaluated using a variety of material characterisation technologies including Dynamic Shear Rheometer (DSR), Fourier Transform Infrared

* Corresponding author.

E-mail address: y.zhang10@aston.ac.uk (Y. Zhang).<https://doi.org/10.1016/j.apsusc.2019.02.121>

Received 4 July 2018; Received in revised form 6 February 2019; Accepted 13 February 2019

Available online 14 February 2019

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Spectroscopy (FTIR) and Atomic Force Microscopy (AFM) tests. It has been recognised that the oxidation causes the hardening of the bitumen, leading to a higher cracking susceptibility of the asphalt pavements [14]. While these experimental studies were commendable, they have not fully addressed the oxidative aging effects on the thermodynamic properties (such as intermolecular energies, interactions and cohesion) of bitumen at the atomistic and molecular scales. Aged bitumen with the introduction of oxygen atoms differs from virgin bitumen in terms of chemical composition that links closely to thermophysical behaviours of bitumen [15,16].

Molecular dynamics (MD) simulation is advantageous over the traditional laboratory-based methods in understanding the nanostructure and properties of the bitumen at the molecular scale [17]. Several studies using MD simulation have been carried out to investigate the physical property changes of the bitumen from the perspectives of chemical compositional or molecular structural changes due to the oxidative aging. Tarefder and Arisa [18] determined the changes in thermodynamic properties of asphaltene and resin in bitumen before and after oxidative aging based on the defined molecule structures [19,20]. Pan et al. [21] used the 3-component model bitumen system [22] to explore the molecular-scale processes of bitumen oxidation in the quantum chemistry (QC)-based environment and analysed the hardening mechanisms of bitumen's oxidation. Recently, Li and Greenfield [23,24] developed three 12-component model bitumen systems to represent the AAA-1, AAK-1 and AAM-1 bitumens used by the Strategic Highway Research Program (SHRP) and analysed the properties of the model bitumens using molecular simulation such as viscosity, relaxation time and dynamics. Pan and Tarefder [25] investigated the density, bulk modulus, viscosity and mechanical property changes of the 12-component AAA-1 model bitumen system before and after oxidative aging at room temperature (298 K). Xu and Wang [26] used the 12-component AAA-1 model bitumen system to analyse the intermolecular interactions of virgin and aged bitumen and to study the aging effects on self-healing and moisture damage of bitumen due to oxidation.

However, limited work has been focused on the effects of bitumen's oxidative aging on the adhesion of the bitumen-mineral interface at the molecular scale, particularly when the minerals are different and water are present. The authors' previous studies have found that the interfacial adhesion between the bitumen and aggregate played a critical role in determining the viscoelastic characteristics and cracking performance of the asphalt mixtures [27–31]. The interfacial interaction between bitumen and mineral surfaces has been investigated using atomic force microscopy (AFM) and X-ray photoelectron spectroscopy (XPS) to improve the interfacial adhesion performance of asphalt mixture [32,33]. MD simulations have been used to quantify the adhesion properties and debonding behaviours between a virgin (unaged) bitumen and four different minerals [34]. It was found that the bitumen-mineral interfacial adhesion is primarily attributed to van der Waals force for neutral minerals (quartz) and to electrostatic interaction for alkali minerals (calcite, albite and microcline). This interfacial adhesion strength was greater for the alkali minerals than that of the neutral minerals. The adhesion would be significantly reduced when water presents. The authors' previous study used the virgin bitumen, thus it is unknown how the oxidative aging of the bitumen will alter the aforementioned findings. Therefore, there is a need for understanding the changes of the bitumen-mineral adhesion at the molecular scale when the bitumen becomes oxidatively aged to different levels, which is investigated in the study.

The main objective of this study is to investigate the interfacial adhesion between oxidised bitumen and mineral surfaces using molecular dynamics (MD) simulation method. Molecular models of virgin and oxidised bitumen were constructed based on saturate, aromatic, resin and asphaltene (SARA) four fractions. The bitumen models together with four different mineral substrates (quartz, calcite, albite and microcline) were employed to build bitumen-mineral interface systems.

The molecular models were validated by the laboratory results from the experiments or simulations reported in the literature. The hardening mechanism of aged bitumen was analysed using the molecular models of the virgin and oxidised bitumen. The adhesive bond energy of the bitumen-mineral interface systems was quantified by the work of adhesion and the effects of oxidative aging on the bitumen-mineral adhesion were investigated at dry and wet conditions.

2. Modelling of virgin and oxidised bitumen components

Bitumen is a chemical mixture composed of a wide variety of hydrocarbons with the heteroatoms such as oxygen, sulphur and nitrogen atoms, which leads to the chemical complexity of the bitumen. Modern separation techniques divide bitumen into different fractions according to their similarity in terms of polar and molecular characteristics. Based on the concept of four fractions proposed by American Society for Testing and Materials (ASTM) D4124-09, a bitumen can be separated into saturates, aromatics, resins and asphaltenes (SARA) four components.

Asphaltenes act as the viscosity-building agents in bitumen and are the largest and most polar component fraction in the bitumen system. They consist of the condensed aromatic and naphthenic molecules with a trace amount of heteroatoms (O, S and N) [35]. Three different types of asphaltene (asphaltene-phenol, asphaltene-pyrrole and asphaltene-thiophene) are used to represent asphaltene fraction in the bitumen system. These asphaltene molecular structures modelled the laboratory measurements and calculations, thus being capable of representing the true asphaltene [36,37]. The molecular structures of the virgin asphaltene fraction are shown in Fig. 1(a). Resins (polar aromatics) are the second polar fraction in bitumen and act as the stabilizer for the asphaltene [38]. Heteroatoms such as oxygen, sulphur and nitrogen are also found in resin molecules composed of polar aromatic rings and long non-polar paraffinic groups. Fig. 2(a) shows the five different molecules (quinolinohopane, thioisorenieratane, benzobisbenzothio-phenene, pyridinohopane and trimethylbenzene-oxane) used to represent the virgin resin fraction. These molecular models were selected based on the analyses of petroleum found in sedimentary rock deposits in the geochemistry studies [39–43]. Aromatics (naphthene aromatics) are the softening component in bitumen and have the minimal polarity in the bitumen system. They are hydrocarbons structures with light molecular weights. Fig. 3(a) illustrates the molecular models of the virgin aromatic fraction, which are named as perhydrophenanthrene-naphthalene (PHPN) and dioctyl-cyclohexane-naphthalene (DOCHN), respectively. PHPN was proposed as a main aromatic compound [44] and DOCHN was shown as another common naphthene aromatic structure on the basis of the n-d-M method [45]. Saturates consist of the normal, branched and cyclic alkanes, which rarely change with time due to the lack of polar chemical functional groups. Squalane and hopane shown in Fig. 4 are selected as representative saturate fraction, which is consistent with the findings from the direct separation and quantitative determination of n- and isoalkanes in neat SHRP (Strategic Highway Research Program) bitumen using urea adduction and high temperature gas chromatography (HTGC) by Netzel and Rovani [46]. The specific selections of these molecular models for asphaltene, resins, aromatics and saturates were reported and explained in detail by previous experimental and simulation studies [23].

Bitumen composition changes when it is exposed to atmospheric oxygen. New chemical functional groups are formed in bitumen, which is responsible for the increase of the viscosity [3]. Extensive experimental studies have been conducted to investigate the mechanism of bitumen aging due to oxidation. It was found that there are two types of the readily oxidizable locations in bitumen. One is the carbon atom adjacent to an aromatic ring. The hydrogen atom attached to the benzylic carbon atom is displaced by oxygen to form a ketone. The other is sulphide that reacts with oxygen to form sulfoxide. The sulfoxides formed at sulphur atoms and ketones formed at benzylic carbon

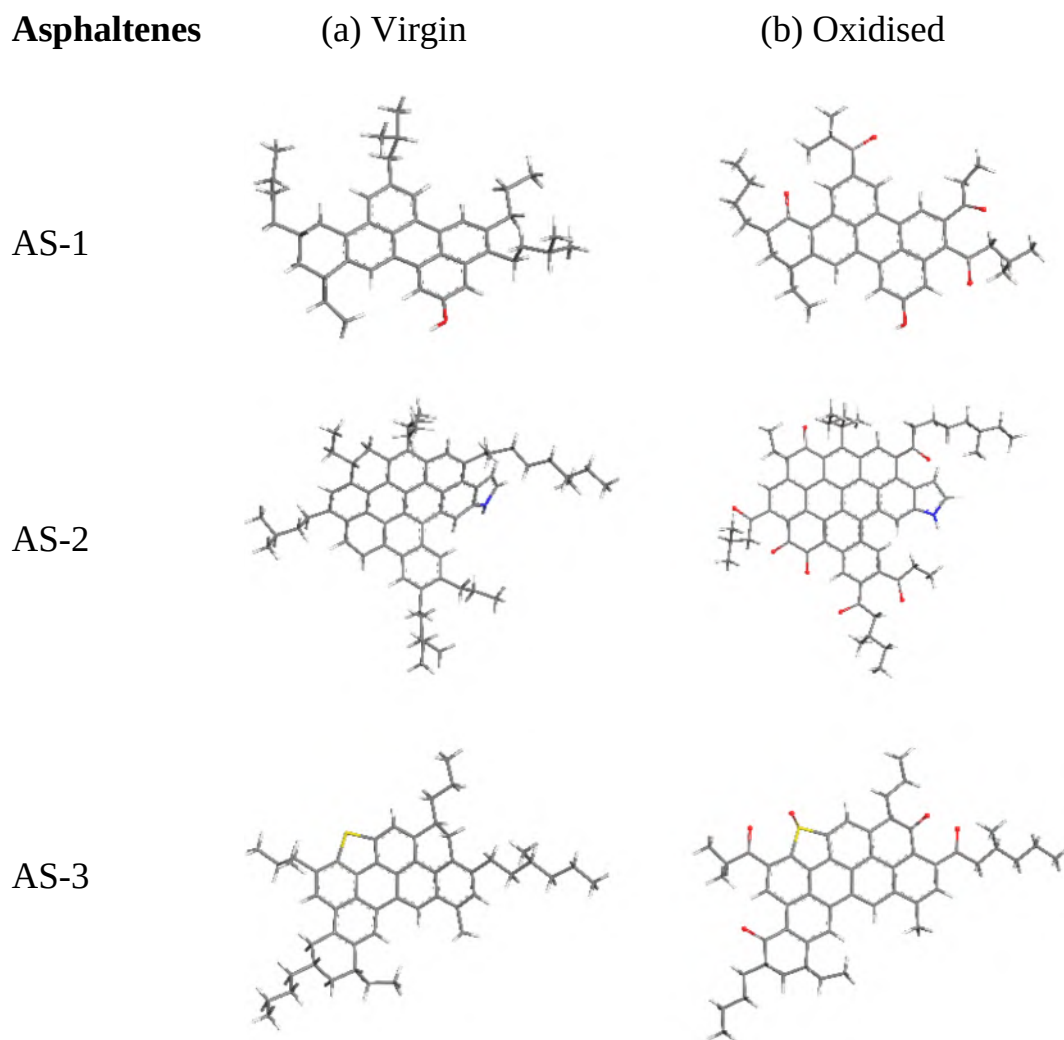


Fig. 1. Molecular structures of asphaltene fraction (AS-1, asphaltene-phenol; AS-2, asphaltene-pyrrole; AS-3, asphaltene-thiophene). (a) Virgin asphaltene molecules [23]. (b) Oxidised asphaltene molecules [25]. (Carbon atoms are grey, hydrogen atoms are white, oxygen atoms are red, sulphur atoms are yellow, and blue is nitrogen).

atoms have been identified as the major oxidation products [15,47].

Asphaltene and resin contain the chemically functional groups of oxygen, sulphur and nitrogen atoms, which results in their high polarity. Therefore, they have strong interaction and are highly reactive with the electronegative oxygen. King and Corbett showed that aromatic was slightly oxidised and saturate were relatively inert to the reaction with oxygen due to the lack of sensitive functional groups [48]. Based on the oxidation mechanism discussed above, the ketones and sulfoxides were added to the possible oxidizable sites of the 12-component bitumen model to build the oxidised bitumen molecular models due to aging. Figs. 1(b)–3(b) show the molecular structures of the oxidised asphaltene, resin and aromatic fractions in the aged bitumen, respectively [25]. The saturate fraction has been assumed to remain unchanged during the oxidative aging. It is noted that the saturate volatilization is also one of the mechanisms causing aging in bitumen in addition to oxidation. The study of volatilization was not considered in this study due to the following reasons: (a) Volatilization mainly occurs during the mixing and compaction of asphalt mixtures at high temperatures such as 150 °C or even higher. This contributes to the short-term aging rather than the long-term aging. The research scope of this study was focused on the long-term aging mechanism involving the formation of ketone and sulfoxide functional groups due to oxidation in bitumen. (b) Volatilization is a time-dependent process. It is still a numerical challenge to study a time-dependent process such as

volatilization taking more than seconds in classical MD simulations. Thus the authors decided to focus the study on the long-term aging of bitumen due to oxidation, which is also the primary contributor for the bitumen's aging. The short-term aging mechanisms including volatilization will be investigated by the laboratory study in the future.

3. Molecular dynamics (MD) simulation method

3.1. Virgin and oxidised model bitumen systems

Based on the molecular models of the virgin and oxidised bitumen components presented in Section 2, the virgin and aged model bitumen systems were constructed in this section. The SHRP AAA-1 model bitumen system developed by Li and Greenfield [23] is chosen for this study, which has been demonstrated to have the similar or equivalent chemical and mechanical properties and performance to a real bitumen. The number of each type of the component molecules in the virgin and aged model bitumen systems can be seen in Table 1.

To build the virgin model bitumen system, all these molecules of the virgin bitumen were put into a cubic simulation box to form a bulk bitumen model. The initial density of the box was set as 0.1 g/cm³ to ensure the random distribution of the molecules. A geometry optimization process for 5000 iterations was first conducted to minimize the system energy. The system was then subjected to molecular dynamic

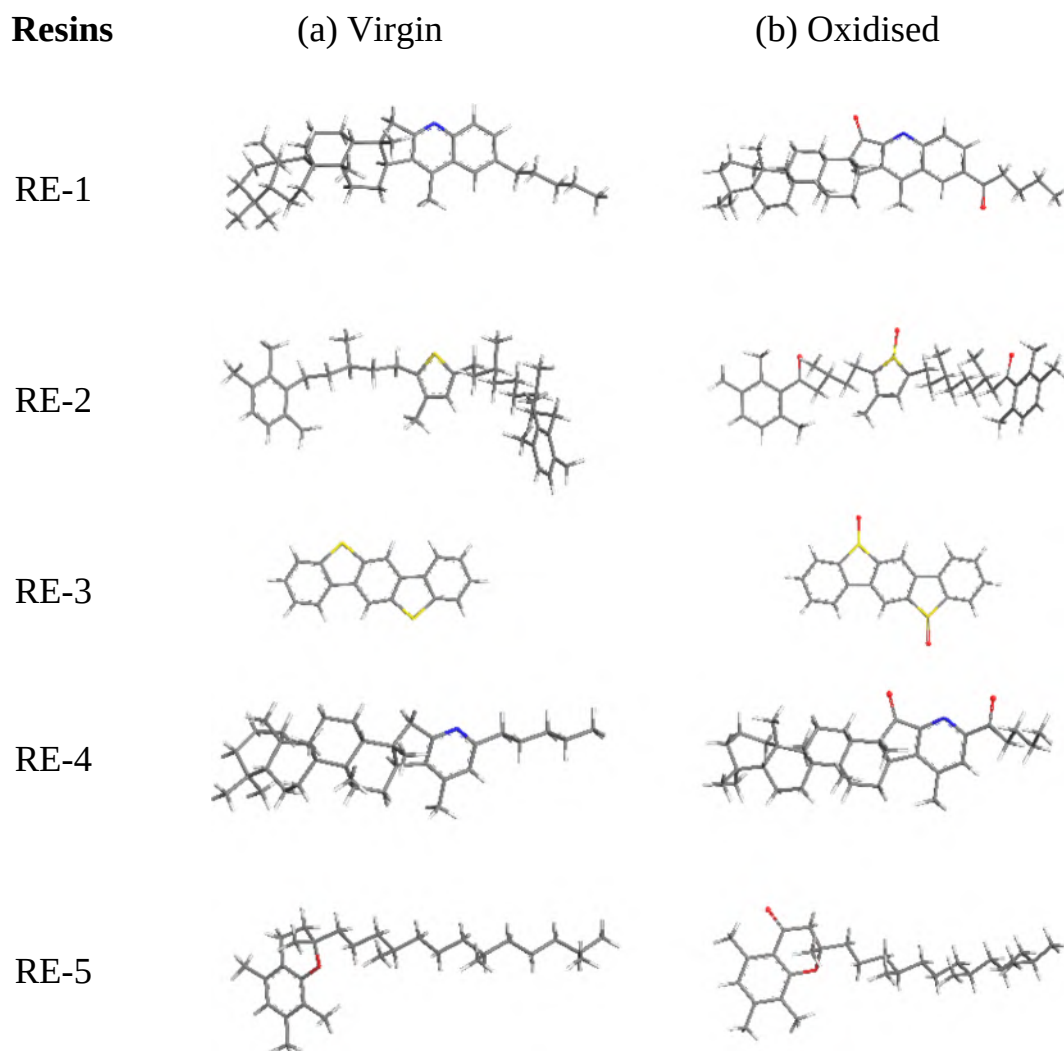


Fig. 2. Molecular structures of resin fraction (RE-1, quinolinohopane; RE-2, thioisorenieratane; RE-3, benzobisbenzothiophene; RE-4, pyridinohopane; RE-5, trimethylbenzene-oxane). (a) Virgin resin molecules [23]. (b) Oxidised resin molecules [25].

(MD) equilibration under the NPT ensemble (in which the number of particles, pressure, and temperature were controlled unchanged) for 100 ps with 1.0 fs time step to obtain a stable system volume. The MD simulation was performed at a temperature of 298 K and a pressure of

1.0 atm. After which, NVT ensemble (where the number of particles, system volume, and temperature were controlled unchanged) at 298 K was used to further equilibrate the system for 100 ps to generate an equilibrated model bitumen system with stable volume and energy

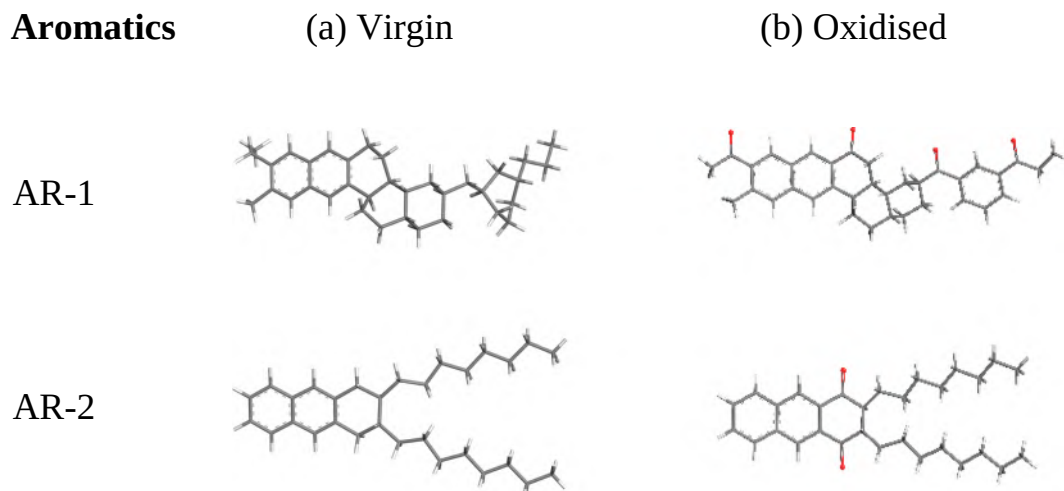


Fig. 3. Molecular structures of aromatic fraction (AR-1, PHPN; AR-2, DOCHN). (a) Virgin aromatic molecules [23]. (b) Oxidised aromatic molecules [25].

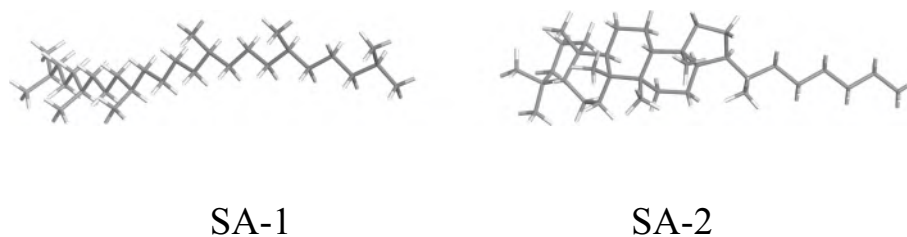


Fig. 4. Molecular structures of saturate fraction (SA-1, squalane; SA-2, hopane) [23].

fluctuation.

For the aged model bitumen systems, two oxidation levels, namely lightly-oxidised bitumen (LOB) and heavily-oxidised bitumen (HOB), are considered in this study by assigning a different number of the oxidised molecules in a model bitumen system, as shown in Table 1. The number of the oxidised molecules in the lightly-oxidised bitumen is half of that in the heavily-oxidised bitumen. This method has been employed to model the aging of bitumen at different oxidation levels in the previous studies [17,25]. The aged model bitumen systems were built following the same procedure as the virgin model bitumen system above. Fig. 5 shows the final structures of the model bitumen systems at different oxidation levels.

3.2. Construction of the bitumen-mineral interface systems

In order to construct the bitumen-mineral interface system models, the mineral substrate and bitumen layer models are first prepared following the method described in the authors' previous work [34].

Quartz, calcite, albite and microcline are selected to build the mineral substrate models. The four types of minerals commonly exist in the aggregates such as granite, limestone and basalt used for constructing asphalt concrete. To model the mineral substrate, the crystal unit cell was firstly cleaved along the Miller plane (100) to expose the corresponding surface and then this surface was extended to create an orthogonal mineral supercell with the periodic boundary condition in all three dimensions. The basis for selecting Miller plane (100) is built

upon an evaluation the effects of different Miller planes on the interfacial adhesion simulation results. Three Miller planes (100), (010) and (001) of a mineral (using quartz as an example) were used and the results obtained from the three Miller planes showed a minor difference, indicating that the use of Miller plane would have little effect on the qualitative analysis of the interfacial adhesion. Since the Miller plane (100) makes the interface model with the minimum lattice mismatch for the four minerals, the Miller plane (100) was therefore selected in the study. The details of four mineral substrates are shown in Table 2.

The bitumen layer was built with the same width and length as the mineral substrate. The layer model was subjected to a geometry optimization followed by an MD simulation of 100 ps with NVT ensemble to equilibrate the system.

Once the developed bitumen layer and mineral substrate models were obtained, the bitumen-mineral interface system models could be constructed. The bitumen layer was firstly combined with the mineral substrate and then a vacuum layer of 30 Å was placed over the top surface of the bitumen layer in order to avoid interaction across the mirror image in the z-direction. For the wet interface model, a layer containing 200 water molecules was added at the bitumen-mineral interface to form a bitumen-water-mineral interface system. The dry and wet interface systems for the heavily-oxidised bitumen with albite substrate are illustrated in Figs. 6(a) and (b). After a geometry optimization process, the interface systems were subjected to dynamics equilibration with NVT ensemble for 100 ps to reach the stable state. An

Table 1

Molecular compositions of the SHRP AAA-1 model bitumen system at different oxidation levels used for MD simulations.

Label	Molecules in model	Molecular formula	Number of molecules			
			VB ^a	LOB ^b	HOB ^c	
SA-1	Saturate	Squalane	C ₃₀ H ₆₂	4	4	4
SA-2		Hopane	C ₂₉ H ₅₀	4	4	4
AR-1	Aromatic (naphthene aromatic)	PHPN	C ₃₅ H ₄₄	11	6	0
OAR-1		Oxidised PHPN	C ₃₅ H ₃₆ O ₄	0	5	11
AR-2		DOCHN	C ₃₀ H ₄₆	13	7	0
OAR-2		Oxidised DOCHN	C ₃₀ H ₄₂ O ₂	0	6	13
RE-1	Resin (polar aromatic)	Quinolinhopane	C ₄₀ H ₅₉ N	4	2	0
ORE-1		Oxidised Quinolinhopane	C ₄₀ H ₅₅ NO ₂	0	2	4
RE-2		Thioisorenieratane	C ₄₀ H ₆₀ S	4	2	0
ORE-2		Oxidised Thioisorenieratane	C ₄₀ H ₅₆ SO ₃	0	2	4
RE-3		Benzobisbenzothiophene	C ₁₈ H ₁₀ S ₂	15	8	0
ORE-3		Oxidised Benzobisbenzothiophene	C ₁₈ H ₁₀ S ₂ O ₂	0	7	15
RE-4		Pyridinohopane	C ₃₆ H ₅₇ N	4	2	0
ORE-4		Oxidised Pyridinohopane	C ₃₆ H ₅₃ NO ₂	0	2	4
RE-5		Trimethylbenzene-oxane	C ₂₉ H ₅₀ O	5	3	0
ORE-5		Oxidised Trimethylbenzene-oxane	C ₂₉ H ₄₈ O ₂	0	2	5
AS-1	Asphaltene	Asphaltene-phenol	C ₄₂ H ₅₀ O	3	2	0
OAS-1		Oxidised Asphaltene-phenol	C ₄₂ H ₄₆ O ₅	0	1	3
AS-2		Asphaltene-pyrrole	C ₆₆ H ₈₁ N	2	1	0
OAS-2		Oxidised Asphaltene-pyrrole	C ₆₆ H ₆₇ NO ₇	0	1	2
AS-3		Asphaltene-thiophene	C ₅₁ H ₆₂ S	3	2	0
OAS-3		Oxidised Asphaltene-thiophene	C ₅₁ H ₅₄ SO ₅	0	1	3

^a VB, virgin bitumen. The number of each molecule in VB is based on the Li and Greenfield's work [23].

^b LOB, the lightly-oxidised bitumen.

^c HOB, the heavily-oxidised bitumen.

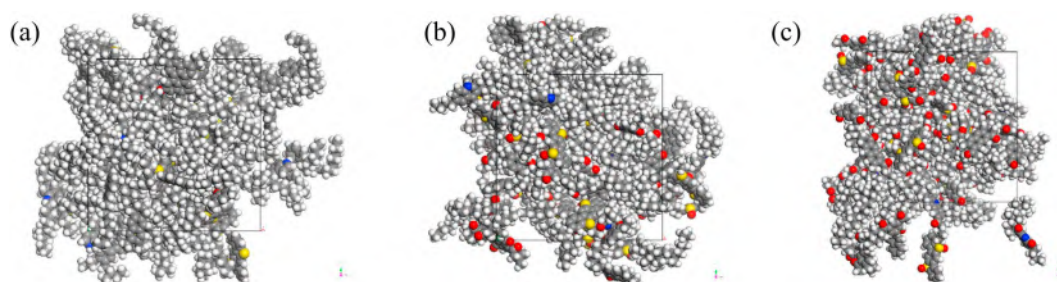


Fig. 5. Model bitumen systems built for MD simulations. (a) Virgin model bitumen system. (b) Lightly-oxidised model bitumen system. (c) Heavily-oxidised model bitumen system.

Table 2

Details of four mineral substrates used for MD simulations.

Minerals	Chemical formula	Lattice parameters of unit cell	Supercell size (Å × Å × Å)
Quartz	SiO ₂	a = b = 4.910 Å, c = 5.402 Å α = β = 90°, γ = 120°	39.28 × 37.81 × 20.36
Calcite	CaCO ₃	a = b = 4.990 Å, c = 17.061 Å α = β = 90°, γ = 120°	34.93 × 34.12 × 23.65
Albite	NaAlSi ₃ O ₈	a = 8.115 Å, b = 12.762 Å, c = 7.158 Å α = 94.218°, β = 116.803°, γ = 87.707°	38.29 × 35.79 × 22.10
Microcline	KAlSi ₃ O ₈	a = 8.573 Å, b = 12.962 Å, c = 7.219 Å α = 90.567°, β = 115.917°, γ = 87.750°	38.89 × 36.10 × 23.89

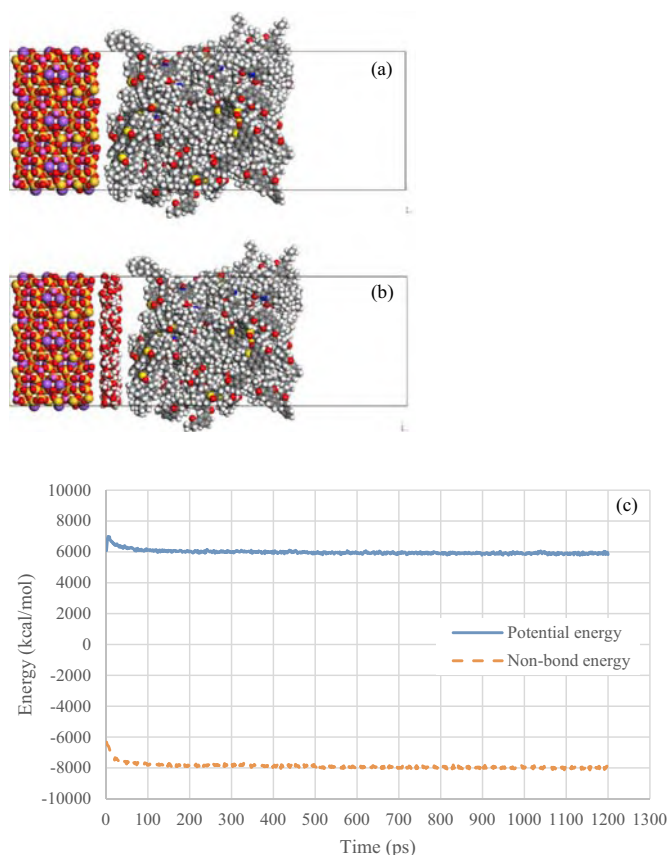


Fig. 6. Interface systems between a heavily-oxidised bitumen and an albite substrate. (a) The heavily-oxidised bitumen-albite model. (b) The heavily-oxidised bitumen-water-albite model. (c) The convergence of energies during the simulation for the virgin bitumen-albite model.

additional 400 ps simulation under the NVT ensemble was performed on all bitumen-mineral interface systems for the subsequent trajectory analysis and calculations. In this study, the simulation time was

demonstrated to be sufficiently long. To support our assertion, a MD study with a longer simulation time (1.2 ns) was performed, in which the bitumen/albite interface model was used as an example. As shown in Fig. 6(c), a stable energy system (including potential energy and non-bond energy) has been reached after the first 100 ps. It was also found that the work of adhesion (1328.73 mJ/m²) obtained from 1.2 ns simulation coincides with the result (1257.25 mJ/m²) using 500 ps as the simulation time. Therefore, the simulation time used is reasonable considering the balance between the accuracy of simulation results and the computational time consumed.

3.3. Force field and simulation parameters

In MD simulations, the interatomic interactions are characterised by force field that defines the interatomic potential. The Condensed-phase Optimized Molecular Potentials for Atomistic Simulation Studies (COMPASS) force field is adopted in this study, which is an ab initio force field and is able to predict dynamics of organics, inorganics and their interface systems. The COMPASS force field has a broad coverage in covalent molecules including most common organics (polymers), more importantly, it has an extended coverage for the inorganic materials (metals, metal oxides, and metal halides using various non-covalent models). The combination of force field parameters for organics and inorganics makes the COMPASS force field possible to study the interfacial interaction in the mixed systems. The COMPASS force field parameters have been validated using various calculation methods [49] and it has been successfully applied for the study of the bitumen-mineral interfacial systems [34,50,51].

The potential energy (E) in the COMPASS force field is comprised of valence (or bonded) interactions (E_{val}) and non-bonded interactions (E_{nb}), as expressed in Eq. (1) [49].

$$E = E_{\text{val}} + E_{\text{nb}} \quad (1)$$

The valence interactions (E_{val}) are contributed by covalent bonds between atoms, while the non-bonded interactions (E_{nb}) include Coulomb electrostatic (E_{Q}) and van der Waals (E_{vdW}) terms, as shown in Eq. (2) [52].

$$E_{\text{nb}} = E_{\text{Q}} + E_{\text{vdW}} \quad (2)$$

The electrostatic interactions describe the Coulombic forces

between two charged atoms. The electrostatic energy (E_Q) is calculated using the Coulombic function:

$$E_Q = \sum_{ij} \frac{q_i q_j}{r_{ij}} \quad (3)$$

where r_{ij} is the distance between atoms i and j , and q_i and q_j are the atomic charges, respectively.

The van der Waals interactions describe the van der Waals forces between two atoms, which are the sum of the intra and inter molecular attractive or repulsive interactions. The van der Waals energy (E_{vdW}) is calculated using the Lennard-Jones (LJ) 9–6 potential

$$E_{vdW} = \sum_{ij} \epsilon_{ij} \left[2 \left(\frac{r_{ij}^0}{r_{ij}} \right)^9 - 3 \left(\frac{r_{ij}^0}{r_{ij}} \right)^6 \right] \quad (4)$$

where r_{ij} is the distance between atoms i and j , r_{ij}^0 and ϵ_{ij} denote the LJ-9-6 parameters (constant) for the ij atom pair, respectively.

In this study, all these simulations were performed at 298 K with a time step of 1 fs. Ewald summation method was applied for the electrostatic interactions, and atom-based summation with a cut-off distance of 15.5 Å was used for the van der Waals interactions. Nose-Hoover-Langevin (NHL) thermostat and Andersen barostat were applied to control the temperature and pressure, respectively. These simulation parameters were extensively tested and successfully applied for MD simulation studies of bituminous materials [26,34].

4. Characterisation of the bitumen-mineral interfacial adhesion

To quantify the adhesion between bitumen and mineral, the adhesive bond energy of the bitumen-mineral interface systems is characterised by the work of adhesion. It is defined as the energy required separating a unit area of the bitumen-mineral interface into a bitumen surface and a mineral surface. The work of adhesion (W_{BM}) is expressed as [34,50].

$$W_{BM} = \Delta E_{BM}/A \quad (5)$$

$$E_{BM} = E_B + E_M - E_{BM} \quad (6)$$

where ΔE_{BM} is the interaction energy between bitumen and mineral in the bitumen-mineral system; E_B , E_M , and E_{BM} represent the potential energies of the bitumen layer, mineral substrate, and bitumen-mineral system at a thermodynamic equilibrium, respectively; and A is the interface contact area between bitumen and mineral.

Note that these parameters in Eqs. (5) and (6) are calculated based on the bitumen-mineral models and the bitumen-water-mineral models at the equilibrium state in order to determine the work of adhesion between bitumen and mineral at dry and wet conditions, respectively.

5. Results and discussion

5.1. Model validation and hardening mechanism of aged bitumen

To validate the molecular simulation method and analyse the hardening mechanisms of bitumen due to oxidative aging at the molecular level, the thermophysical properties of virgin and aged bitumen were calculated including density, cohesive energy density (CED), and fraction of free volume (FFV).

The density of the bulk bitumen model at different oxidation levels was calculated at the temperature of 298 K (25 °C) based on MD simulations. Fig. 7 shows the bitumen density changes due to oxidative aging. The bitumen density became higher after oxidative aging, where the density was 1.016 g/cm³ for the lightly-oxidised bitumen and 1.048 g/cm³ for the heavily-oxidised bitumen and the density was 0.981 g/cm³ for the virgin bitumen. This trend is in good agreement with Pan and Tarefder's results [25] as well as the experimental results for the virgin and aged PG 58–28 bitumen [53]. It can also be seen from

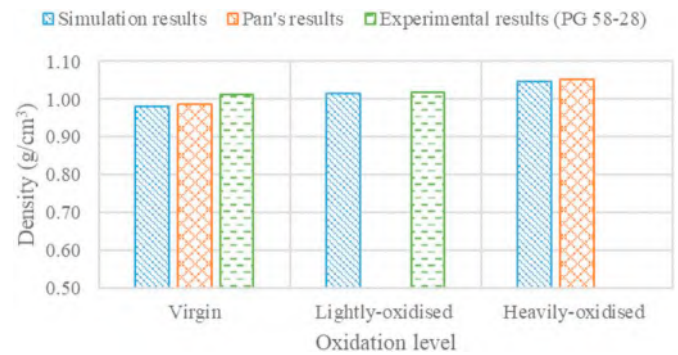


Fig. 7. Density of bulk bitumen model at different oxidation levels. (Note Pan's results [25] for lightly-oxidised bitumen and the experimental results [53] for heavily-oxidised bitumen are not available.)

Fig. 7 that the density change for simulation results before and after oxidative aging was greater than that of the experimental results. This is because the loss of volatile components due to the volatilization during the aging process was not considered in the simulation.

Cohesive energy density (CED) is used to calculate the cohesive energy per unit volume to evaluate the intermolecular bonding strength of a bulk bitumen model. The cohesive energy is attributed to the non-bonded interactions including van der Waals and electrostatic interactions. Fig. 8 shows the calculated CED for the bulk bitumen model at different oxidation levels including the van der Waals and electrostatic components. It was found that the CED value increased with the increasing oxidation level and the majority of the CED resulted from Van der Waals interaction. This indicates that the intermolecular bonding in oxidised bitumen is stronger than that in the virgin bitumen. This is mainly due to that, when the oxygen atoms are introduced to bitumen, both the molecular weight and the polarity of the oxidised bitumen became higher than that of the virgin bitumen, which results in a greater van der Waals and electrostatic intermolecular interactions in the bulk bitumen model.

The fraction of free volume (FFV) is used as a measure of the volume that has not been occupied by the molecules and to determine the percentage of volume available in a bulk bitumen model. Fig. 9 shows the calculated FFV of the bulk bitumen model at different oxidation levels. The FFV value for the oxidised bitumen (approximately 37.5% for both the lightly-oxidised bitumen and the heavily-oxidised bitumen) was lower than that of virgin bitumen (38.2%). This means that the oxidised bitumen is agglomerate further than the virgin bitumen due to a higher molecular weight and polarity of the oxidised bitumen system.

The lower FFV for the oxidised bitumen is consistent with the higher

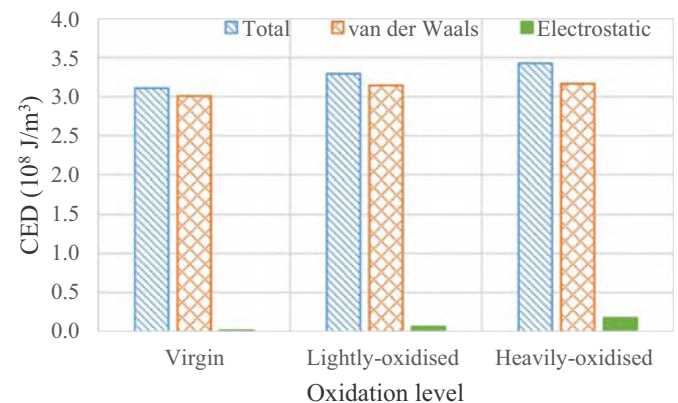


Fig. 8. Cohesive energy density (CED) of bulk bitumen model at different oxidation levels including two components, namely van der Waals and electrostatic intermolecular interactions.

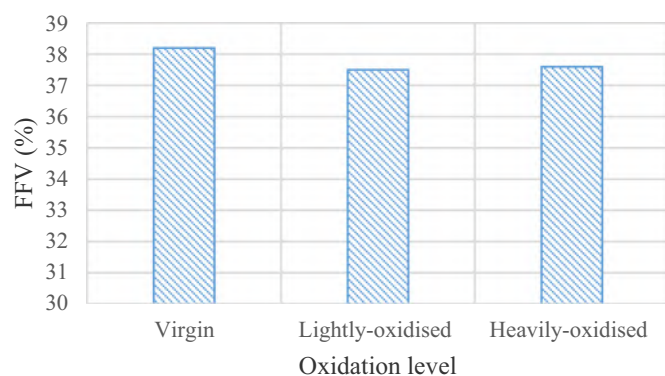


Fig. 9. Fraction of free volume (FFV) of the bulk bitumen model at different oxidation levels.

density and CED, which demonstrates a stronger intermolecular bonding in the oxidised bitumen system that leads to a further shrinkage of the bitumen system when oxygen is present in the material. The analysis of the thermophysical properties of the virgin and aged bitumen from the molecular perspective explains the physical hardening happened during the oxidative aging which was recognised through a number of experimental studies at the macroscopic level [14]. This also demonstrates the molecular simulation method is suitable for describing the aging behaviours of the bitumen due to oxidation and can be further applied for the investigation of the aged bitumen-mineral interface behaviours.

5.2. Molecular structure of bitumen on the mineral surfaces

Bitumen is regarded as a colloidal structure where asphaltene is

dispersed in the maltene composed of resins, aromatics and saturates [38,54]. Fig. 10 illustrates the nanostructures of bitumen at the heavily-oxidised bitumen-quartz interface and the heavily-oxidised bitumen-water-quartz interface obtained from the last frame of simulation trajectories, respectively. It can be observed that within the interfacial systems, asphaltene molecules (in blue colour) were dispersed in the maltene medium. Asphaltenes tended to form the nano-aggregation structures without a continuous network. This type of structure is also referred to bitumen sol-gel structure, which has been reported in prior experimental and simulation studies [38,54].

To characterize the molecular arrangement and mobility of bitumen on the mineral surfaces, the radial distribution function (RDF) and mean squared displacement (MSD) of the asphaltenes were analysed, respectively. RDF describes the probability of finding a particle at a radial distance r away from a reference particle in a system of particles. In this study, the intermolecular distance was calculated based on the asphaltene molecule's centre of mass. The trajectory obtained from the last 200 ps of MD simulations under NVT ensemble was applied for RDF analysis. Fig. 11 shows the RDFs of asphaltene-asphaltene pair in the bitumen-microcline interfacial models at different oxidation levels. Radial distances corresponding to the first peak value of RDF curves for asphaltene-asphaltene pair in bulk bitumen model and four interfacial models at different oxidation levels were summarised in Table 3. In bulk bitumen models, the first peak value of RDF for the virgin bitumen occurred around 10–11 Å, while the distance became smaller for the heavily-oxidised bitumen (8–9 Å) and the lightly-oxidised bitumen (4–5 Å). This indicates that the oxidative aging strengthened the bitumen's aggregation due to the higher molecular polarity, which is consistent with the finding in Section 5.1. In the interfacial models, it can be found from Table 3 that the mineral surfaces had a significant influence on the nano-aggregation structure of the bitumen, especially under the oxidation conditions. The aggregation of the bitumen on the

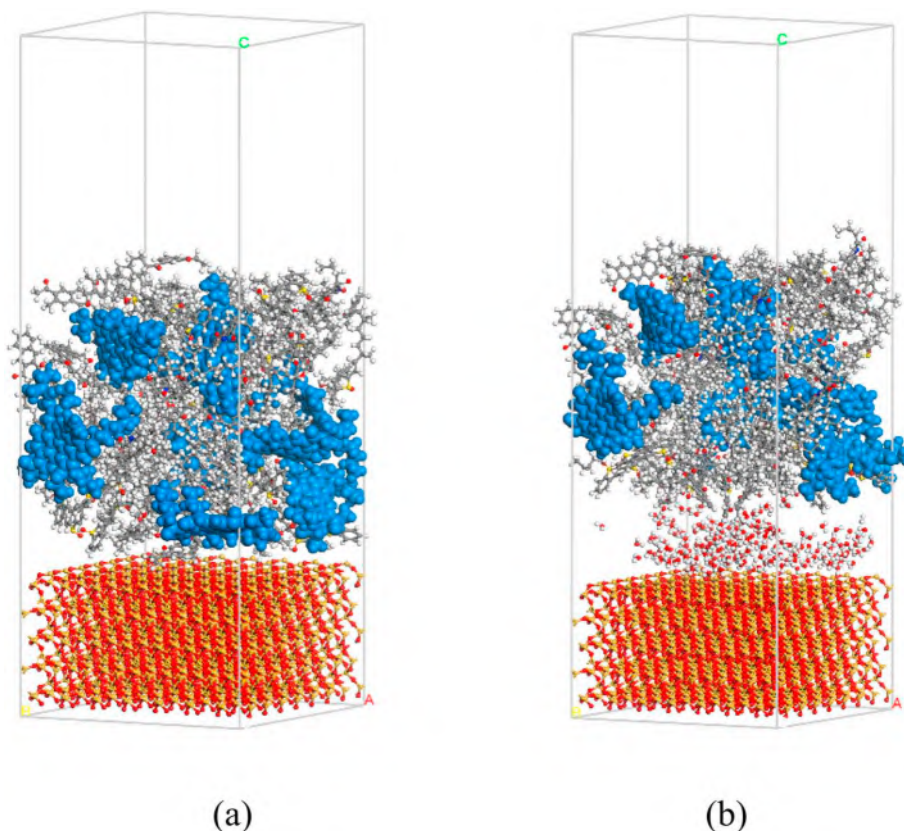


Fig. 10. Nanostructures of bitumen at (a) heavily-oxidised bitumen-quartz interface and (b) heavily-oxidised bitumen-water-quartz interface obtained from MD simulations. The asphaltene molecules (in blue colour) were dispersed in the maltene medium composed of resins, aromatics and saturates.

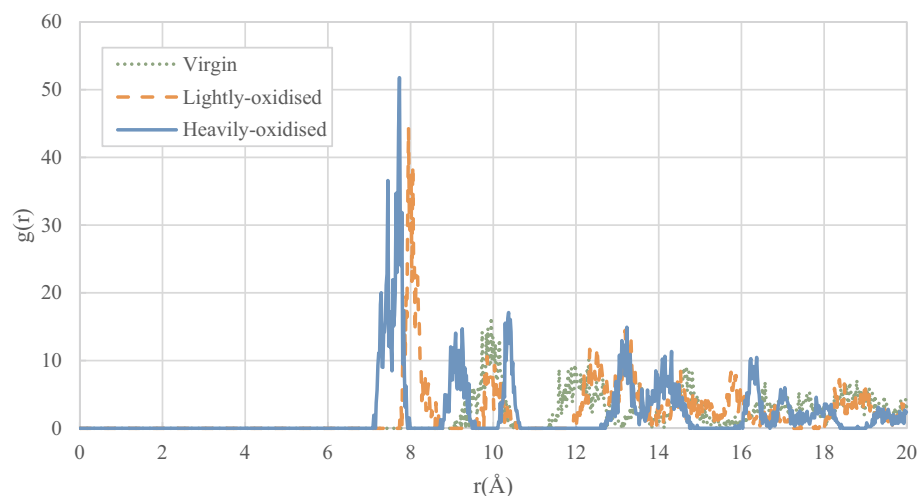


Fig. 11. RDFs of asphaltene-asphaltene pair in bitumen-microcline interfacial model at different oxidation levels.

Table 3

First peak positions of RDF curves for asphaltene-asphaltene pair in bulk bitumen model and four interfacial models at different oxidation levels.

Models	Virgin	Lightly-oxidised	Heavily-oxidised
Bulk bitumen model	10–11 Å	4–5 Å	8–9 Å
Bitumen-quartz model	9–10 Å	6–7 Å	10 Å
Bitumen-calcite model	9–10 Å	9 Å	11–12 Å
Bitumen-albite model	10–11 Å	11–12 Å	8 Å
Bitumen-microcline model	10 Å	8 Å	7–8 Å

quartz surface was consistent with that of the bulk bitumen because of the very weak interaction between quartz and bitumen. The aggregation of the oxidised bitumen on the calcite surface was weakened since calcite attracted a part of the oxidised bitumen from the bulk bitumen due to the slightly strong electrostatic interaction. While for albite and microcline the very strong electrostatic interaction with bitumen has significantly strengthened the aggregation of the oxidised bitumen through attracting all the bitumen molecules to their surfaces, demonstrated by the reduced first peak position when the bitumen became oxidised.

MSD is used to study the molecular mobility of bitumen on the mineral surfaces over time that is associated with the interaction between bitumen and minerals. In this study, MSD of bitumen was measured based on the asphaltene molecule's centre of mass. A higher slope of MSD curves means a greater mobility of molecules. Fig. 12 illustrates the MSDs of asphaltenes in bitumen-microcline interfacial model at different oxidation levels. Table 4 shows the slopes of MSD curves for

Table 4

Slopes of MSD curves for asphaltenes in bulk bitumen model and four interfacial models at different oxidation levels.

Models	Virgin	Lightly-oxidised	Heavily-oxidised
Bulk bitumen model	0.0052	0.0013	0.0050
Bitumen-quartz model	0.0124	0.0014	0.0092
Bitumen-calcite model	0.0049	0.0139	0.0057
Bitumen-albite model	0.0034	0.0031	0.0006
Bitumen-microcline model	0.0025	0.0020	0.0014

asphaltenes in bulk bitumen model and four interfacial models at different oxidation levels. It can be seen that in all these models except for the bitumen-calcite model, the mobility of the oxidised bitumen was smaller than that of the virgin bitumen, which means that the oxidative aging slowed down the mobility of the bitumen molecules. This is because the oxidised bitumen had an enhanced aggregation due to oxidation as well as a larger molecular weight with the introduction of the oxygen atoms. It is noted that the lightly-oxidised bitumen on the calcite surface had a greater molecular mobility since calcite caused the weak aggregation of the lightly-oxidised bitumen that has been shown in Table 3. It was also found from Table 4 that the bitumen molecules on different mineral surfaces exhibit distinctive molecular mobility. The mobility of the bitumen on the quartz and calcite surfaces was faster than that on the albite and microcline surfaces as the strong attraction between bitumen and alkaline minerals (albite and microcline) constrained the mobility of the bitumen molecules.

5.3. Effect of oxidative aging on work of adhesion at dry condition

To quantify the adhesion between the bitumen (at different oxidation levels) and the different minerals at the dry condition, the work of adhesion of the bitumen-mineral interface systems was calculated based on the MD simulation results. Figs. 13 and 15–17 illustrate the work of adhesion for the bitumen-quartz, bitumen-calcite, bitumen-albite and bitumen-microcline interface models at the dry condition and at different oxidative aging levels of the bitumen. It's noted that, based on Eqs. (1) and (2), the total energy of a molecular system equals the summation of valence energy and non-bonded energy, where the non-bonded energy is composed of van der Waals and electrostatic interaction energy. It was found that total energy equalled to the non-bond energy for all of the four bitumen-mineral interface systems, which indicates that the interfacial adhesion between bitumen and mineral purely results from the non-bond interaction and the valence energy does not contribute to the total energy of the bitumen-mineral system.

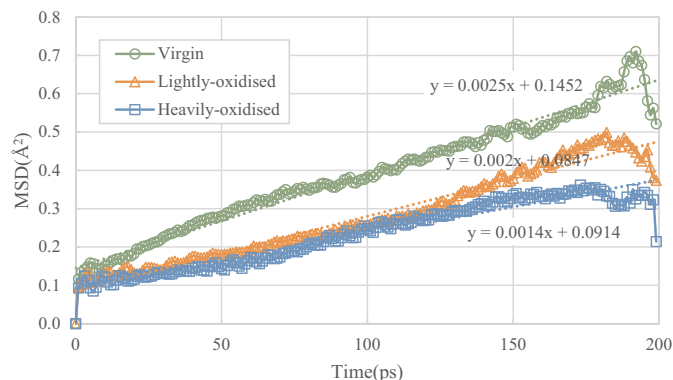


Fig. 12. MSDs of asphaltenes in bitumen-microcline interfacial model at different oxidation levels.

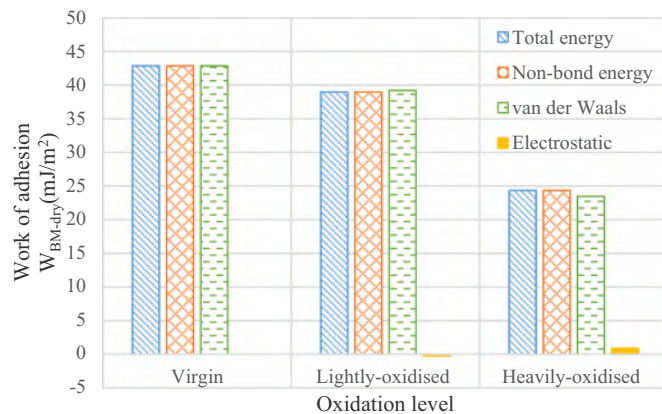


Fig. 13. Work of adhesion for bitumen-quartz interface model at the dry condition and different oxidation levels. (Note, based on Eqs. (1) and (2), total energy = valence energy + non-bonded energy, and non-bonded energy = van der Waals + electrostatic energy.)

This is consistent with the previous finding that the bitumen-mineral adhesion is attributed to non-bond interaction energy [34].

The work of adhesion of the bitumen-quartz model at the dry condition and at different oxidation levels of the bitumen is shown in Fig. 13. It was found that the work of adhesion ($W_{\text{BM-dry}}$) decreased with the increase of the oxidation level, which indicates that the bitumen aging weakens the interfacial adhesion between the bitumen and the quartz. This is in agreement with the experimental finding from Yi et al. [13] who found that the aging of bitumen weakened the bonding performance between bitumen and granite (in which quartz is the major component). The negative effect of aging is due to the decreased van der Waals energy after aging, where the van der Waals is the primary component of the adhesion between the bitumen and the quartz as the electrostatic energy contributes little to the total energy, as indicated in Fig. 13.

The decreased van der Waals energy is fundamentally caused by the increased distance between the bitumen and quartz molecules when the

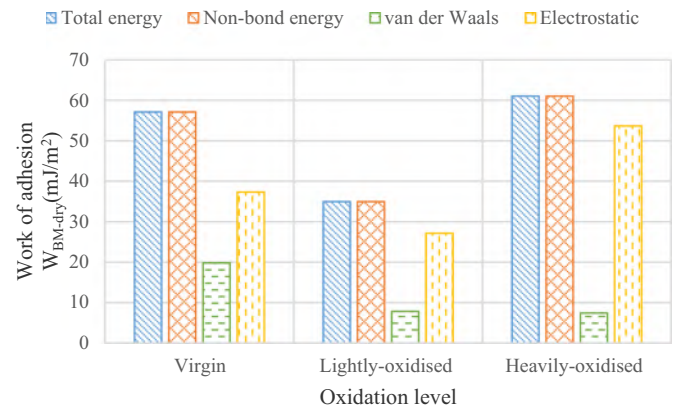


Fig. 15. Work of adhesion for bitumen-calcite interface model at dry condition and different oxidation levels. (Note, based on Eqs. (1) and (2), total energy = valence energy + non-bonded energy, and non-bonded energy = van der Waals + electrostatic energy.)

bitumen becomes aged. In Fig. 14(a), the centre-to-centre distance between the bitumen mass and the quartz matrix was computed. It was found that the centre-to-centre distance of the bitumen-quartz models increased from 30.65 Å (10^{-10} m) for the virgin bitumen to 31.15 Å and 34.34 Å for the lightly-oxidised bitumen and the heavily-oxidised bitumen, respectively. According to Eq. (4), a greater distance between atoms leads to a lower van der Waals energy. Thus the van der Waals energy between bitumen and quartz is reduced when the bitumen becomes more severely aged. Furthermore, the increased centre-to-centre distance between bitumen and quartz at a higher oxidation level is believed due to the stronger aggregation of the oxidised bitumen molecules, resulting from higher polarity and molecular weight when the oxygen atoms are introduced into the bitumen. In sum, the severer agglomeration of the oxidised bitumen caused by more polar carbonyl and sulfoxide functional groups leads to a greater distance between the mass centre of the bitumen and quartz, which decreases the van der Waals interaction and eventually weakens the work of adhesion of the

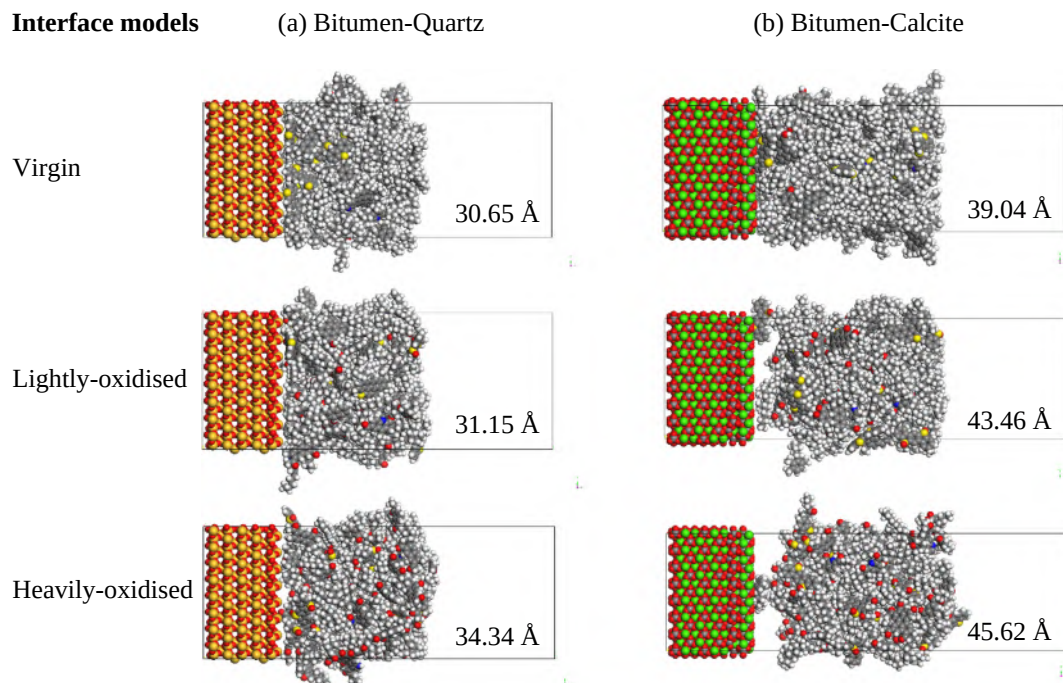


Fig. 14. Centre-to-centre distance (Å or 10^{-10} m) between bitumen mass and mineral matrix for (a) bitumen-quartz interface model and (b) bitumen-calcite interface model at different oxidation levels.

bitumen-quartz model at the dry condition.

Fig. 15 shows the work of adhesion of the bitumen-calcite interface model at the dry condition and at different oxidation levels. The work of adhesion ($W_{\text{BM-dry}}$) first decreased from 57.13 mJ/m² for the virgin interface model to 34.97 mJ/m² for the lightly-oxidised interface model but then increased to 61.08 mJ/m² for the heavily-oxidised interface model. This change results from the combined aging-induced changes of the van der Waals energy and the electrostatic energy that both contribute to the adhesion between the bitumen and the calcite, as shown in Fig. 15. Cucalon et al. [55] also found that, compared to the virgin bitumen, the long-term aged bitumen using a Pressure Aging Vessel test (PAV) has a higher adhesive bond energy with limestone (in which calcite is the main component). The results in Fig. 12 verifies the findings of Cucalon et al. [55] from the molecular modelling perspective.

The van der Waals energy is inversely proportional to the distance between atoms according to Eq. (4). It can be seen from Fig. 14(b) that the centre-to-centre distance between the bitumen mass and calcite matrix has increased from 39.04 Å for the virgin bitumen-calcite model to 43.46 Å for the lightly-oxidised bitumen-calcite model and 45.62 Å for the heavily-oxidised bitumen-calcite model, respectively. This leads to a constant decrease of the van der Waals energy with the increasing oxidation level in the bitumen-calcite interface system, which is consistent with the findings for the bitumen-quartz interface system as shown in Fig. 13.

However, the electrostatic energy is inversely proportional to the distance between the atoms and directly proportional to the atomic charges according to Eq. (3). When the bitumen becomes more severely aged, the greater bitumen-calcite distance (due to more aggregation of bitumen molecules) causes a lower electrostatic energy and the higher atomic charges (due to more oxygen atoms and higher polarity) results in a higher electrostatic energy. Fig. 15 shows that the electrostatic energy of the bitumen-calcite interface system decreased when the bitumen became lightly-oxidised and then increased when the bitumen was heavily-oxidised. Thus it may be concluded that the centre-to-centre distance is the dominating factor for determining the electrostatic energy of the lightly-oxidised bitumen-calcite interface system, whereas the atomic charge is the dominating factor for determining the electrostatic energy of the heavily-oxidised bitumen-calcite interface system.

Fig. 15 also indicates that the electrostatic energy is the primary contributor and the van der Waals energy is the secondary contributor for the non-bonded energy (that is the same as the total energy). Thus the change of the work of adhesion for the bitumen-calcite model is dominantly determined by and follows the same trend as the electrostatic energy, which decreases when bitumen is lightly-oxidised and then increases when the bitumen becomes heavily-oxidised.

The work of adhesion values of the bitumen-albite model and the bitumen-microcline model at the dry condition and at different oxidation levels are shown in Figs. 16 and 17, respectively. It was found that the work of adhesion ($W_{\text{BM-dry}}$) significantly increased with the increase of the oxidation level, which indicates that the oxidation of bitumen is able to strengthen the adhesion for the bitumen-albite interface and the bitumen-microcline interface. This is because of the considerable increase of the electrostatic energy that contributes to the primary interfacial adhesion in the bitumen-albite model and the bitumen-microcline model, as shown in Figs. 16 and 17. The strong electrostatic interaction for the oxidised bitumen-albite or bitumen-microcline interface system is caused by the very strong electrostatic attraction between the negatively charged oxidised bitumen due to the introduction of oxygen atoms during aging and the positively charged alkaline minerals (albite and microcline). It is noted that the van der Waals energy is a very minor contributor to the total energy as shown in Figs. 16 and 17. Thus the work of adhesion is primarily dominated by the electrostatic energy for the bitumen-albite or bitumen-microcline interface system.

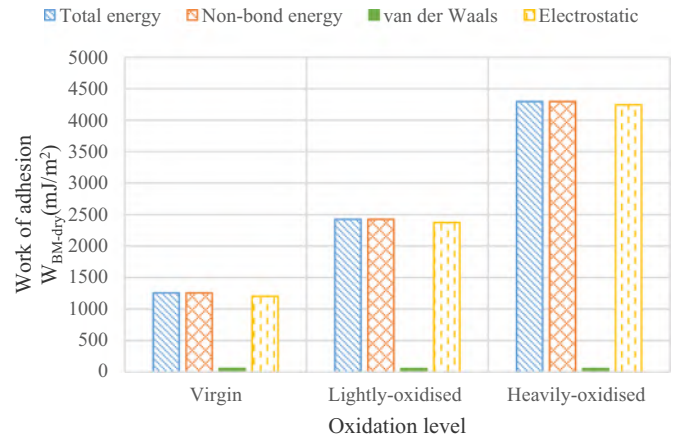


Fig. 16. Work of adhesion for bitumen-albite interface model at the dry condition and different oxidation levels. (Note, based on Eqs. (1) and (2), total energy = valence energy + non-bonded energy, and non-bonded energy = van der Waals + electrostatic energy.)

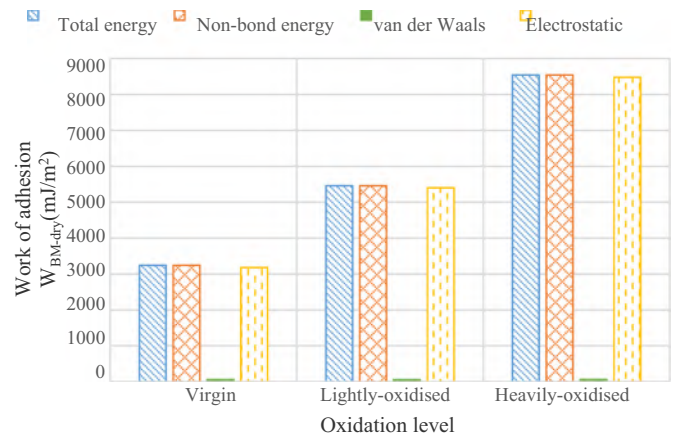


Fig. 17. Work of adhesion for bitumen-microcline interface model at the dry condition and different oxidation levels. (Note, based on Eqs. (1) and (2), total energy = valence energy + non-bonded energy, and non-bonded energy = van der Waals + electrostatic energy.)

As observed above, the variation of the work of adhesion for the four bitumen-mineral models at dry condition is significantly distinct when bitumen becomes more severely aged (decrease for the bitumen-quartz model, decrease and then increase for the bitumen-calcite model, and increase for the bitumen-albite model and the bitumen-microcline model). The difference results from the different chemical properties of the minerals. Quartz is an acidic mineral, having a rather weak or negligible electrostatic interaction with the acidic bitumen, while albite and microcline are the strongly alkaline minerals, having a considerably strong electrostatic interaction with the oxidised bitumen containing more polar molecules. Calcite presents weakly alkaline, having a relatively weak electrostatic interaction with the acidic bitumen. The previous experimental studies [56] and molecular simulation results [34] have shown that a mineral with an alkaline nature has a stronger electrostatic adhesion with the weakly acidic bitumen than an acidic mineral. Therefore, it can be concluded that the effect of bitumen oxidative aging on the bitumen-mineral interfacial adhesion strongly depends on the mineral types.

5.4. Effect of oxidative aging on work of adhesion at wet condition

To study the effect of water on the oxidised bitumen-mineral adhesion, the work of adhesion of the four bitumen-mineral interface

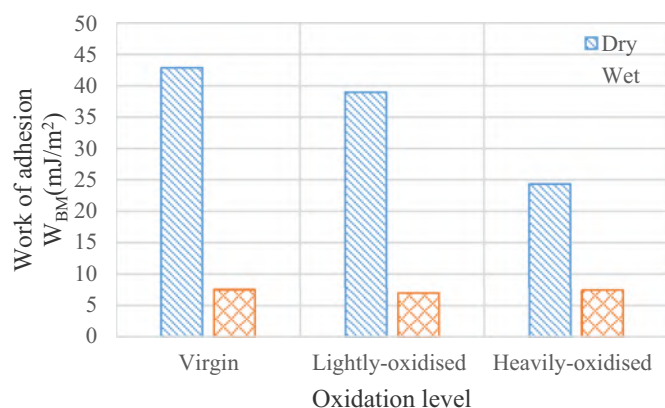


Fig. 18. Work of adhesion of bitumen-quartz model at dry and wet conditions and different oxidation levels.

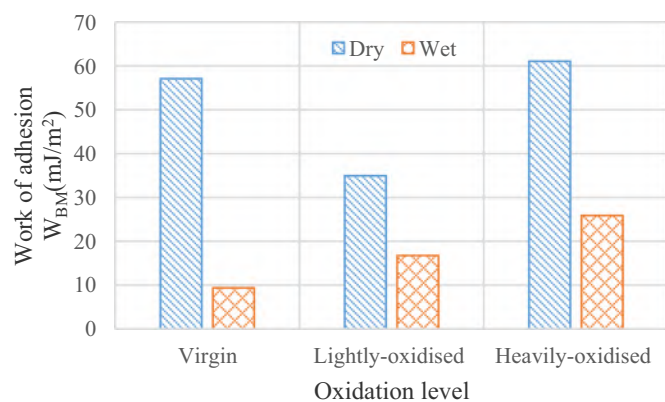


Fig. 19. Work of adhesion of the bitumen-calcite model at dry and wet conditions and different oxidation levels.

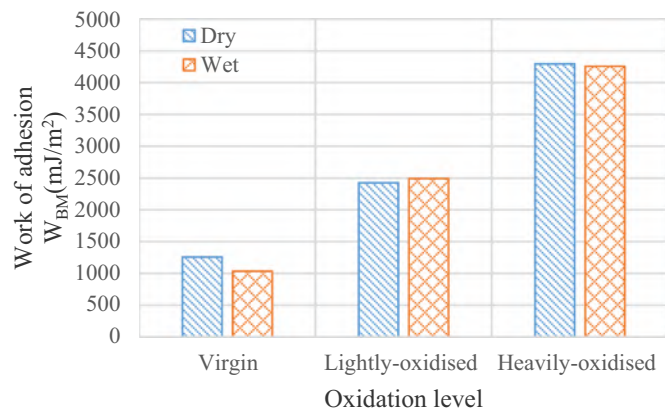


Fig. 20. Work of adhesion of the bitumen-albite model at dry and wet conditions and different oxidation levels.

systems at wet condition and different oxidation levels was also calculated based on the MD simulation results and was compared with the work of adhesion at dry condition, as shown in Figs. 18–21.

The work of adhesion of the bitumen-quartz model at dry and wet conditions with different oxidation levels is shown in Fig. 18. It can be seen that the work of adhesion at wet condition had little change with the oxidation level, which indicates that the oxidative aging has a very weak or none influence on the bitumen-quartz interfacial adhesion at the wet condition. This is because the already very low van der Waals interaction (serving as the major contributor to the bitumen-quartz adhesion) at wet condition cannot be further reduced by the stronger

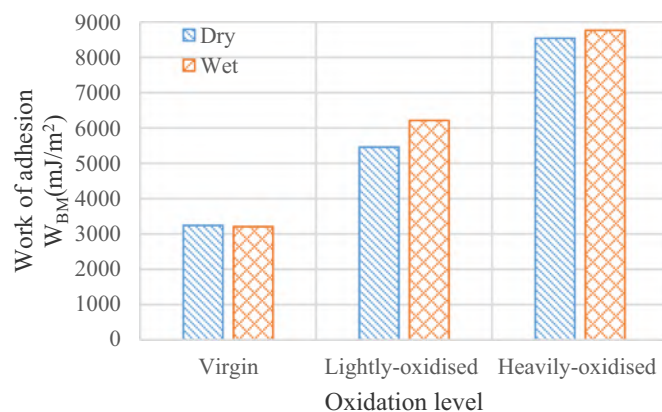


Fig. 21. Work of adhesion of the bitumen-microcline model at dry and wet conditions and different oxidation levels.

aggregation of the oxidised bitumen at the aging condition. Fig. 18 also shows that the work of adhesion was much smaller at the wet condition as compared to that at the dry condition. This is because the presence of water molecules has separated the bitumen from the quartz and increased the distance between the bitumen molecules and the quartz atoms, leading to the very weak van der Waals interaction according to Eq. (4).

Fig. 19 shows the work of adhesion of the bitumen-calcite model at dry and wet conditions with different oxidation levels. It was found that the work of adhesion for the oxidised bitumen-calcite model was slightly greater than that of the virgin model at the wet condition, which indicates that the oxidation of bitumen has made the bitumen-calcite adhesion stronger at the wet condition. This is because of the increase of the electrostatic interaction between bitumen and calcite. This explains the findings from Cucalon et al. [55] that the resistance of the interfacial bonding between bitumen and limestone to moisture damage improves when the bitumen becomes aged. Fig. 19 also shows that the work of adhesion at wet condition was always smaller than that at dry condition since the presence of water molecules makes the bitumen-calcite van der Waals interaction weaker. This means that water tends to weaken the adhesion between bitumen and calcite, which remains consistent with the bitumen-quartz model.

The work of adhesion values of the bitumen-albite model and the bitumen-microcline model at dry and wet conditions and different oxidation levels are shown in Figs. 20 and 21, respectively. It was found that the work of adhesion at both dry and wet conditions significantly increased with the increase of oxidation level, which indicates that the oxidative aging has been able to strengthen the interfacial adhesion for the bitumen-albite model and the bitumen-microcline model, regardless the effect of water. This results from the considerable increase of the electrostatic interaction which is the major contributor to the interfacial adhesion of the bitumen-albite model and the bitumen-microcline model. From Figs. 20 and 21, it can be seen that the work of adhesion at wet condition was very close to and consistent with that at dry condition. This differs from the case of the bitumen-quartz model and the bitumen-calcite model where the work of adhesion at wet condition was always weaker than that at the dry condition, as shown in Figs. 18 and 19. This difference is mainly caused by the much stronger electrostatic interaction at the bitumen-albite interface and the bitumen-microcline interface, resulting from the stronger attraction between the highly alkaline minerals (albite and microcline) and the oxidised bitumen with the high polarity due to oxidative aging.

5.5. Combined effect of oxidative aging and water on bitumen-mineral interfacial adhesion

To clearly describe the combined effects of oxidative aging and

water on the bitumen-mineral adhesion at wet condition, the sensitivity of adhesion to oxidative aging (S_O) and the sensitivity of adhesion to water (S_W) are proposed using the work of adhesion. The sensitivity of adhesion to oxidative aging (S_O) at wet condition is defined as the difference of the work of adhesion before and after oxidative aging divided by the work of adhesion before oxidative aging, as shown in Eq. (7). The sensitivity of adhesion to water (S_W) is calculated as the ratio of the change of the work of adhesion from dry to wet conditions as compared to the work of adhesion at dry condition, as expressed in Eq. (8). A higher absolute value of the sensitivity of adhesion (S_O or S_W) means a greater effect of oxidative aging or water on the bitumen-mineral adhesion. A positive (or negative) value of the sensitivity of adhesion (S) indicates the strengthening (or weakening) effect.

$$S_{O-H} = (W_{BM-H} - W_{BM-V}) / W_{BM-V} \quad \text{or} \quad S_{O-L} = (W_{BM-L} - W_{BM-V}) / W_{BM-V} \quad (7)$$

$S_W = (W_{BM-wet} - W_{BM-dry}) / W_{BM-dry}$ (8) where W_{BM-H} , W_{BM-L} and W_{BM-V} are the work of adhesion between bitumen and mineral at wet condition for heavily-oxidised, lightly-oxidised and virgin bitumens, respectively; and W_{BM-dry} and W_{BM-wet} are the work of adhesion between bitumen and mineral at dry and wet conditions, respectively.

Fig. 22 shows the sensitivity of adhesion to the oxidative aging (S_{O-H} and S_{O-L}) and water (S_W) for quartz (acidic), calcite (weak alkali) and albite (strong alkali) minerals. It can be seen that the sensitivity of the bitumen-mineral adhesion to the oxidative aging becomes higher and the absolute sensitivity of the bitumen-mineral adhesion to water becomes lower when the mineral changes from acidic to weak alkali and to strong alkali types.

This is fundamentally due to that, for the acidic minerals, van der Waals energy is the major contributor for bitumen-mineral adhesion which can be significantly reduced by water through increasing the distance between the bitumen and the acidic minerals. In this case, the electrostatic energy of the acidic mineral with the acidic bitumen is so weak that the increase of the polarity of the bitumen due to aging will not alter the overall adhesion between the bitumen and the acidic minerals. Therefore, the adhesion of the acidic minerals (such as quartz) with bitumen is more susceptible to the water but much less sensitive to the bitumen aging.

For the strong alkali minerals, the electrostatic energy is much higher than the van der Waals energy and becomes the dominating contributor to the bitumen-mineral adhesion. The electrostatic energy becomes even higher when the bitumen is aged due to higher polarity of the oxidised bitumen. However, the van der Waals is so minor for the adhesion between bitumen and the strong alkali minerals that the decrease of the van der Waals due to water cannot alter the adhesion significantly. Thus the adhesion of the strong alkali minerals (such as albite and microcline) with bitumen is more susceptible to bitumen

aging but less sensitive to water.

For the weak alkali minerals, both the van der Waals and the electrostatic energy contribute to the adhesion with bitumen. The van der Waals energy can be significantly reduced by water due to greater distance and the electrostatic energy can be dramatically increased by bitumen aging due to higher polarity. Thus the adhesion of the weak alkali mineral (such as calcite) with bitumen is susceptible to both bitumen aging and the water.

6. Summary and conclusions

A study has been conducted to discover the interfacial adhesion mechanism between oxidised bitumen and mineral surfaces at dry and wet conditions using molecular dynamics simulation method. Based on the results, the main conclusions are as follows:

- (1) The oxidised functional groups (i.e., carbonyl and sulfoxide) in bitumen due to oxidative aging strengthens the intermolecular bonding of the bitumen including van der Waals and electrostatic interactions, which results in the aggregation of bitumen molecules and physical hardening of the aged bitumen. This is revealed by the increase of density and cohesive energy density and the decrease of the fraction of free volume, when the bitumen becomes oxidative aging.
- (2) When bitumen becomes more severely aged at dry condition, the adhesive bond energy consistently decreases for bitumen-quartz (an acidic mineral) interface, decreases first and then increases for the bitumen-calcite (a weak alkali mineral) interface, and constantly increases for the bitumen-albite or microcline (strong alkali minerals) interfaces.
- (3) The interfacial adhesion of bitumen-acidic minerals (e.g., quartz) at dry condition is dominated by van der Waals interaction which decreases when bitumen becomes aged. This is due to the increased bitumen-quartz distance caused by the aggregated bitumen molecules. In comparison, the interfacial adhesion of bitumen-strong alkali minerals (e.g., albite and microcline) is dominated by electrostatic energy which increases during aging. This results from the higher polarity introduced by the oxidised products including carbonyl and sulfoxide in an aged bitumen.
- (4) The interfacial adhesion of bitumen-weak alkali mineral (calcite) at dry condition is attributed primarily to electrostatic energy and secondarily to van der Waals energy. The electrostatic energy between aged bitumen and the weak alkali mineral decreases due to the increased bitumen-mineral distance when the bitumen is lightly-oxidised and increases due to the higher bitumen polarity when the bitumen is heavily-oxidised.
- (5) For the bitumen-mineral interfacial adhesion at wet condition, water is the dominating factor that affects (weakens) the interfacial adhesion between the bitumen and the acidic minerals such as quartz. While the oxidative aging of bitumen is the major factor that affects (strengthens) the interfacial adhesion between the bitumen and the strongly alkaline minerals (i.e., albite and microcline). For the weak alkali minerals such as calcite, both water and bitumen aging can significantly affect the interfacial adhesion.

The research findings provide a fundamental understanding of the interfacial adhesion between oxidised bitumen and mineral surfaces at dry and wet conditions from the perspective of molecular modelling. The results can potentially be used for guiding the pairing and selections of the virgin and aged bitumen (such as reclaimed asphalt binder) with different minerals for optimal performance of asphalt mixtures. Further work will be focused on studying the effect of bitumens aging on the cohesive and interfacial adhesive debonding under various loading conditions for the asphalt materials.



Fig. 22. Sensitivity of adhesion to oxidative aging (S_{O-H} and S_{O-L}) and water (S_W) for quartz (acidic), calcite (weak alkali) and albite (strong alkali) minerals.

Acknowledgements

The authors acknowledge the financial support provided by Aston University via a PhD studentship and by European Union's Horizon 2020 programme via a Marie S. Curie Individual Fellowship project (Grant No. 749232-AMAM).

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Review

Adhesion in Bitumen/Aggregate System: Adhesion Mechanism and Test Methods

Katarzyna Pstrowska ^{1,*} , Volodymyr Gunka ² , Iurii Sidun ³, Yuriy Demchuk ² , Nataliya Vytrykush ⁴ , Marek Kułazyński ⁵ and Michael Bratychak ^{2,*}

¹ Department of Advanced Materials Technologies, Wrocław University of Science and Technology, 7/9 Gdańska Street, 50-344 Wrocław, Poland

² Institute of Chemistry and Chemical Technology, Lviv Polytechnic National University, 12 Bandera Street, 79013 Lviv, Ukraine

³ Institute of Building and Environmental Engineering, Lviv Polytechnic National University, 12 Bandera Street, 79013 Lviv, Ukraine

⁴ Viacheslav Chornovil Institute of Sustainable Development, Lviv Polytechnic National University, 12 Bandera Street, 79013 Lviv, Ukraine

⁵ Innovation and Implementation Company Ekomotor Ltd., 1A Wyścigowa Street, 53-011 Wrocław, Poland

* Correspondence: katarzyna.pstrowska@pwr.edu.pl (K.P.); mykhailo.m.bratychak@lpnu.ua (M.B.)

Abstract: A literature review of the five main theories describing the interaction mechanisms in the bitumen/aggregate system was conducted: theory of weak boundary layers, mechanical theory, electrostatic theory, chemical bonding theory, and thermodynamic theory (adsorption theory). The adhesion assessment methods in the bitumen/aggregate system are described, which can be divided into three main groups: determination of adhesion forces for bitumen with different materials, determination of bitumen resistance to the exfoliating action of water with different materials, and determination of adhesion as a fundamental value (contact angle measurements, interfacial fracture energy, adsorption capacity and others). It is proposed to evaluate the quality of adhesive interaction in the bitumen/aggregate system in two stages. The authors recommend using the adhesion determination methods for these two stages from the second group of methods the determination of bitumen resistance to the exfoliating action of water with different materials. In the first stage, the adhesion in the bitumen/aggregate system is determined by an accelerated technique in which the used bitumen binder and mineral material are considered as test materials. After the first stage, there are positive results in the second tests on compacted mixtures (indirect tensile strength test, Modified Lottman indirect tension test, immersion-compression test, and Hamburg wheel tracking test).

Keywords: adhesion; bitumen; aggregate; moisture damage; laboratory testing



Citation: Pstrowska, K.; Gunka, V.; Sidun, I.; Demchuk, Y.; Vytrykush, N.; Kułazyński, M.; Bratychak, M. Adhesion in Bitumen/Aggregate System: Adhesion Mechanism and Test Methods. *Coatings* **2022**, *12*, 1934. <https://doi.org/10.3390/coatings12121934>

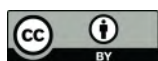
Academic Editor: Valeria Vignali

Received: 9 November 2022

Accepted: 3 December 2022

Published: 8 December 2022

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1. Introduction

Asphalt pavements include two main ingredients: bitumen and aggregates (crushed stone). Bitumen performs the adhesive function that binds mineral material particles to form an asphalt concrete coating. Being non polar, bitumen has high water resistance properties. It is known that low-paraffinic oil is the most suitable for the bitumen production, but, due to the shortage of this type oil, almost any oil residues are used. This leads to the bitumen materials' low quality and, as a result, to the asphalt concrete pavement's low quality.

In general, asphalt concrete appears as a substantially water resistance material, but water can penetrate into the pores in different ways: impregnation (surface water), under the action of capillary forces (water from the bottom of the road base rises to the asphalt concrete), and water vapor (air moisture can penetrate into the asphalt concrete pores and condense). Furthermore, transport wheel pressure accelerates the water penetration into the asphalt concrete pores [1]. Therefore, both passive and active adhesion of bituminous binder to the asphalt concrete must be ensured.

Active adhesion is the ability of the binder to completely surround the particles of the aggregate and adhere to it during the mixing of the components of the asphalt concrete mixture (Figure 1). This type of adhesion arises due to the mutual attraction of positively charged surface-active substances of bitumen molecules to negatively charged aggregate molecules. This type of interaction makes it possible to displace water located at the interface of phases and ensures maximum bitumen coverage of the surface of the mineral material. In case of loss of active adhesion, this can lead to increased movement of moisture in asphalt concrete layers [2,3].

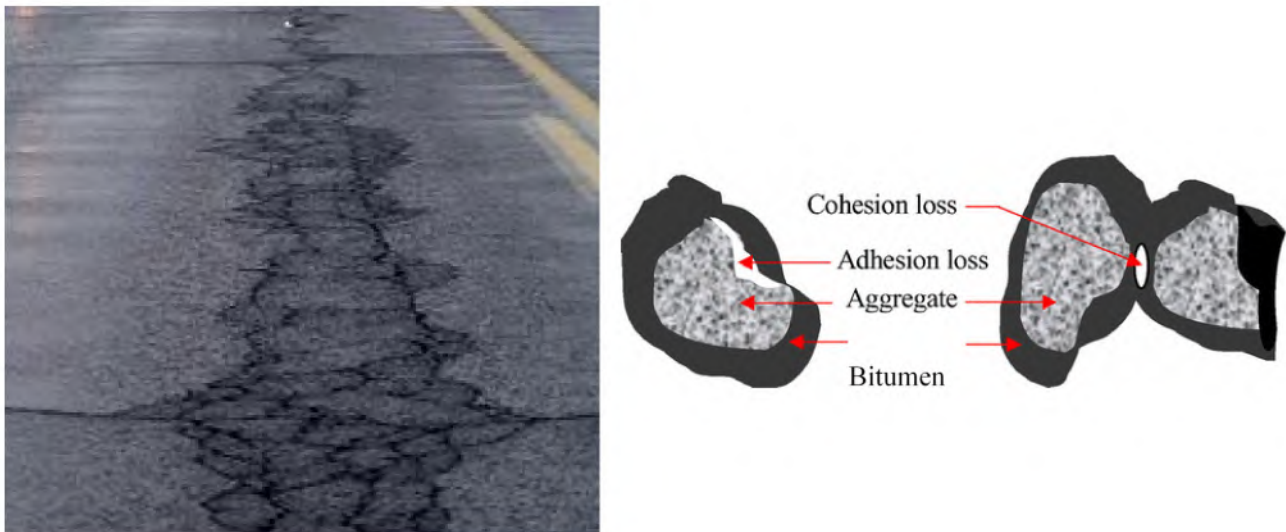


Figure 1. Fatigue cracks caused by low-adhesion in bitumen.

In contrast to active adhesion, passive adhesion occurs as a result of external forces, for example, due to increased pressure in the pores and can be described as the ability to resist the penetration of water into the asphalt mixture. Passive adhesion indicates the ability of bitumen to bind to the surface of the mineral material throughout the further service life of the asphalt concrete without the risk of destruction of these bonds under the action of vehicle wheels or water [4]. Loss or insufficiency of passive adhesion can cause premature appearance of cracks, potholes and ruts [5].

Poor adhesion between bitumen and aggregate leads to the binder stripping in the presence of water, which ultimately leads to the pothole formations (Figure 1). Moisture damage in asphalt concrete pavements is considered as primary cause of distresses in the asphalt pavement layers. The resistance of bitumen to stripping is determined by its adhesion to the aggregate surface: it must be not only high, but also stable over time, which is one of the conditions for the road surface durability. The situation is exacerbated by the traffic and car numbers constant increase on the roads. Therefore, to obtain high quality coatings, a key factor is high adhesion ensure between bitumen and the road pavement mineral components.

This work is devoted to fundamental knowledge related to adhesion in the Bitumen/Aggregate system, namely there is a need to explain and understand how bitumen adheres to the aggregate surface, show the most well-known tests that numerically evaluate the adhesion between bitumen and aggregate, and group them by common features.

The following section highlights the theories which explain the adhesion phenomena between bitumen and aggregate.

2. Adhesion Mechanism in Bitumen/Aggregate System

Researches in this area were conducted for more than 100 years, but the question about adhesion mechanism between bitumen and aggregate remains debatable. There were proposed many theories to explain the adhesion mechanism between bituminous and

aggregate. The main theories explaining the adhesion mechanism in the bitumen/aggregate system are presented in the Table 1 and Figure 2.

Table 1. Theories explaining the adhesion mechanism in the bitumen/aggregate system.

Theory	Interaction	References
Weak Boundary Layer	layering	Bikerman, 1961 [6] and other [7–11]
Mechanical interlocking	interlock	McBain and Hopkins, 1925 [12] and other [6–11]
Electrostatic	potential	Deryaguin and other, 1948 [7] and other [6,8–11]
Chemical bonding	covalent and ionic bonds	[6–11,13–16]
Thermodynamic (Surface free energy)	surface free energy	[6–11]

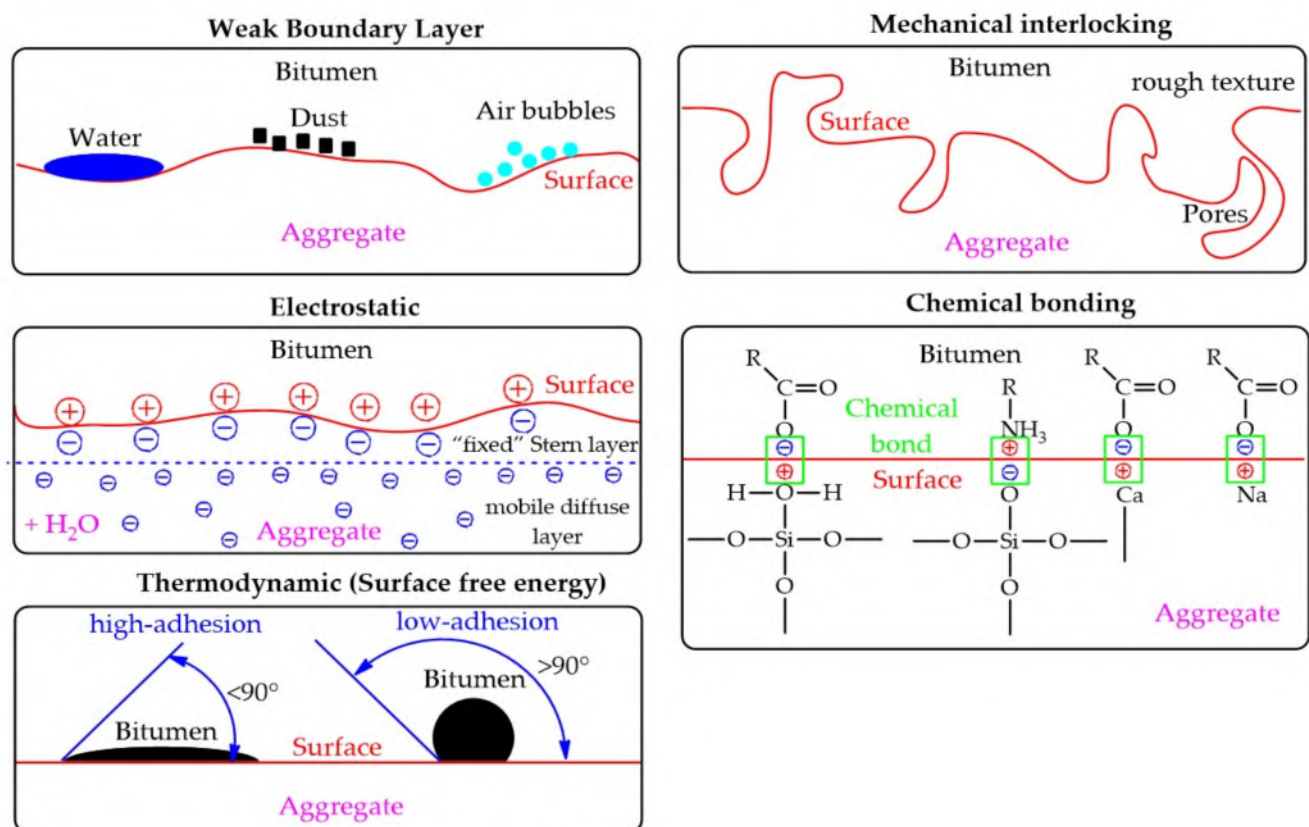


Figure 2. Representation of theories explaining the adhesion mechanism in the bitumen/aggregate system.

Let us briefly describe each of the theories.

2.1. Theory of Weak Boundary Layers

According to the weak boundary layer theory, the breakdown of adhesive bonds occurs either in the adhesive or in the substrate due to the presence of an interfacial region with low cohesive strength. While there is debate as to how common these layers are, they do exist and cannot be ignored. However, certain types of boundary layers, according to some researchers, are important factors for good adhesion. Common causes of weak boundary layers are low molecular weight (and therefore low cohesive strength), and surface contaminants such as organics and water. Dusty substrates can prevent effective wetting and close contact with the adhesive, and dust on aggregates is able to trap air, mixing with bitumen, and thus weaken the bitumen/aggregate bond (Figure 2).

2.2. Theory of Mechanical Interlocking

Adhesion occurs through the binder material penetration into the aggregate surface irregularities (pores and cavities). Adhesion occurs due to the “mechanical connection” between the binder and aggregate (Figure 2). Rough surfaces provide greater adhesion compared to a smooth surface. It should also be noted that rough surfaces are not the only reason for better adhesion. This may also be due to other factors (clean surfaces, highly reactive surface formation, or a contact area increase). The surface texture of an aggregate also affects its coat-ability, or wettability, in that a smoother surface coats easier than a rough surface. In addition, wettability, including pore filling, also depends on the viscosity of the bitumen and the chemical composition of the surface of both aggregate and bitumen. While many researchers claim that aggregate surface texture is the primary factor affecting adhesion, others report that chemical and electrochemical effects dominate. This theory simply explains the mechanical adhesive bond without taking into account the chemical interaction that can occur between the bitumen and the surface of the aggregate, so this theory has not found wide support among scientists.

2.3. Electrostatic Theory

Electrostatic theory is based on the presence of free charges in any material, where the opposite charges attract each other (Figure 2). The presence of free charges in the materials causes an electrochemical potential difference between two contacting materials and, thus, establish an electrical double layer. Most surfaces are charged in the presence of water. This is due to the high dielectric permeability of water, which makes it a good solvent for ions. When the bitumen/aggregate surface system is exposed to water, the mineral structure of the aggregate easily collapses and ionizes, and thus the adhesive structure of the bitumen film on the aggregate can be broken and ionized. Two layers are formed (“fixed” Stern layer and mobile diffuse layer), together they form an electric double layer. The electric potential at the shear plane between the fixed and mobile layers is measurable, and called the zeta potential (ζ). The zeta potential value depends on two main factors: diffusion of external water to the bitumen/aggregate interface (the amount of moisture in the system) and the pH value of the water. The introduction of various substances can change the pH value of water into the bitumen/aggregate system (for example, mineral powders), which leads to a decrease in zeta potential. Researchers consider this approach to be of fundamental importance for quantifying adhesion in the presence of water, with knowledge of the pH effect. Determination of zeta potential can serve to quantify adhesion in the presence of water, especially taking into account the pH effect.

2.4. Chemical Bonding Theory

This theory is based on the formation chemical bonds in the system of bitumen and aggregate components. Although bitumen mainly consists non-polar and low-reactive hydrocarbons in relation to the aggregate, it also contains polar compounds that contain heteroatoms (O, S, N and others). The study of adsorption of bituminous model compounds containing specific functional groups is presented in the works [13–16]. Relative affinity of functional groups for aggregate surfaces and their relative displacement by water are presented in these studies.

Tests involving the adsorption affinity of the model compounds representing the various functional groups present in asphalt averaged over a series of aggregates (including granites, limestone, gravels, and greywacke) gave the following ranking [7]:

sulfoxide > carboxylic acid > nitrogen base > phenol > ketone > pyrrole > 4-ring aromatic > 2-ring aromatic

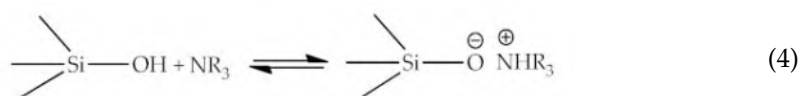
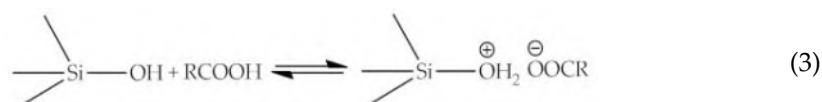
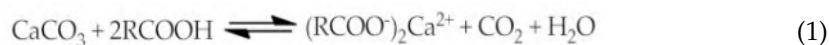
The order of desorption:

sulfoxide > carboxylic acid > pyrrole > ketone > nitrogen base > phenol

Functional groups that had the highest adsorption capacity for aggregates also had the highest sensitivity to water (sulfoxide and carboxylic acid), while nitrogen base and pheno-

lic functionalities had the lowest sensitivity. The adsorption of bituminous components and their subsequent desorption by water depends on both the bitumen and aggregate (adsorbent) composition.

As an example of the interaction between organic and inorganic parts can be the following reaction equations [7,11]:



2.5. Thermodynamic Theory (Surface Free Energy)

The physical and chemical adhesion between bitumen and aggregate can be seen as a thermodynamic phenomenon, which requires the removal of bitumen from the aggregate surface (bond failure), that is, the work or energy required to produce new units of area in a vacuum. The physical and chemical adhesion between bitumen and aggregate depends on the surface free energy (SFE) of the material. The higher surface free energy (adhesion work) indicates that the adhesion strength between the bitumen and aggregate interface is better. This theory, is the most widely used approach in adhesion science as indicated by most comprehensive references on this subject.

SFE is directly proportional to the contact angle (θ). The theory explains the adhesion phenomenon on base of wettability. The contact angle is the angle formed by a liquid with a solid surface when both materials are in contact. If the contact angle is $>90^\circ$, the wettability is bad, and if $<90^\circ$ then wettability is good (Figure 2). Thus, if the wetting decreases, it leads to the adhesion decrease.

The aggregate wetting remains an important condition for good adhesion. Wetting is necessary for contact between materials, and thus the basic physical and chemical forces establishment that are ultimately responsible for adhesion. Without wetting, it is unlikely to establish the surface texture favorable impact by mechanical adhesion.

If you follow this theory, which is the most accepted and proven by researchers, adhesion in the bitumen/aggregate system can be numerically determined as derivatives of surface free energy or contact angle.

3. Test Methods for Adhesions

The term adhesion is widely used but its direct measurement is less evident. Far easier to define the lack of adhesion directly correlated with failure or in situ damage (e.g., raveling). Adhesion is considered as an important aspect of durability.

Directly in the bitumen/aggregate system the adhesion determination is difficult for several reasons: excess in the positive temperatures range of adhesive strength over cohesive; polyminerality of stone materials; results significant dependence on the temperature, strain (deformation) rate, binder layer thickness, etc. Therefore, for practical purposes, in laboratories usually perform an indirect adhesion assessment for the bitumen resistance assess to the exfoliating action of water (as the most significant destructive factor). The adhesion assessment indirect methods have a long history of use and are the most numerous and widely used in production conditions.

Currently there are a large number of adhesion testing methods for bitumen—more than 150 [16–23], and there are no generally accepted test methods, but the most popular are those regulated by European standards (EN) and US standards (AASHTO and ASTM).

This section provides an overview of the main laboratory test methods for adhesion in the bitumen/aggregate system in the world.

Based on the processed literature data, these methods can be divided into three groups:

1. Determination of adhesion forces for bitumen with different materials;
2. Determination of bitumen resistance to the exfoliating action of water with different materials;
3. Determination of adhesion as a fundamental value (contact angle measurements, interfacial fracture energy, adsorption capacity and others).

3.1. Determination of Adhesion Forces for Bitumen with Different Materials

The main methods in which the base principle is mechanical separation include: a group of methods—Peel test, Bitumen Bond Strength Test, Vialit Plate Test.

Peel test. The main principle of this group methods is the force determination required for the separation of a certain area bituminous film from the surface under study (Figure 3). The advantages of peel test methods are testing and devices design simplicity, ability to determine the adhesive strength of any binder, assessment of temperature effects, strain rate, stripping action of water and aggressive water environments. A similar method is the Bitumen Bond Strength Test (BBS Test) (Figure 4), which is based on the force determination generated by the pneumatic system of the device required for separation working fragment glued with bitumen to the tested surface [19,24,25]. The disadvantage of these methods is that during the test there is both adhesive and cohesive separation of the bituminous film, and also there can be inhomogeneity of values over the entire sample area [19,26,27].

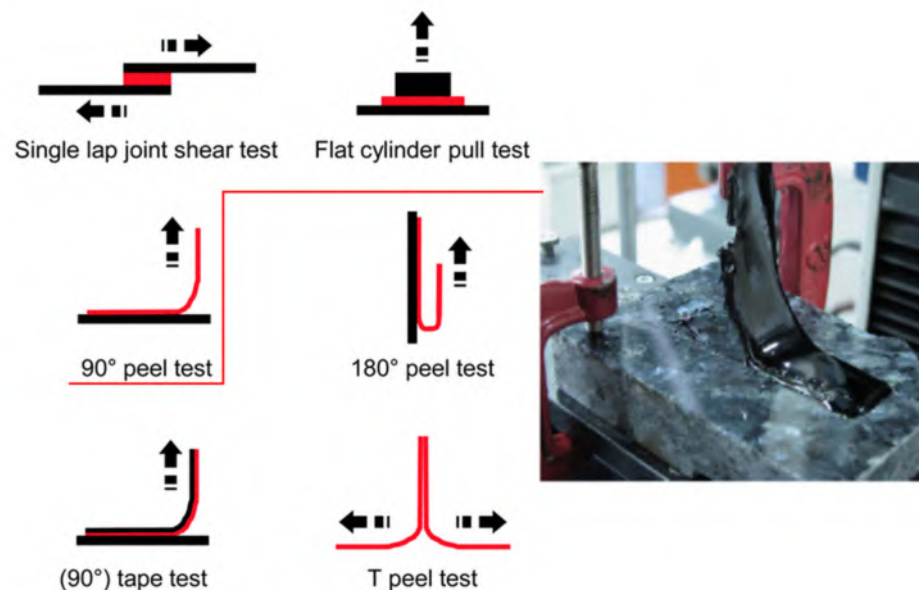


Figure 3. Possible schemes on peel test.

Vialit Plate Test [28,29] used in European countries is also one of the typical for the peeling methods. The essence of the method is that on a metal plate is applied a bitumen layer and then 100 or 50 grains of stone material are glued by the appropriate fractions. After that, the plate is fixed with the grains down and a weighing 500 g ball is thrown three times from a fixed height. After the third stroke, the sample on the plate is examined and the grains of crushed stone are counted as follows: the number of unstuck gravels, the surface of which is not covered with binder; the number unstuck gravels, the surface of which is partially covered with binder; the number of stuck gravels.



Figure 4. Tested surfaces after BBS Test, cohesive, adhesive and mixed separation (from left to right) Reprinted with permission from [26] Copyright Taylor & Francis 2010.

3.2. Determination of Bitumen Resistance to Exfoliating Action of Water with Different Materials

Different surfaces are used for testing: stone, metal and glass plates; grains of stone materials that are treated with bitumen and after cooling (or immediately after the treatment) are immersed in distilled water or in an aqueous solution with a given temperature and kept for a certain time (from several minutes to several days). After an aqueous medium keeping, the exfoliation of bituminous binders from the surface of stone materials is estimated in percentages or points, usually by visual method.

Characteristic advantages this group of methods are:

- Easy implementation;
- No additional special equipment;
- Short test time in most methods (when boiling in water the test time does not exceed 1.0–1.5 h);
- Due to the use of grains of stone material of the same size it is possible to assess the influence on the adhesion value the bitumen film thickness on the surface of the grains.

At the same time, this group of methods has the significant disadvantages, which include:

- Visual evaluation of results, leads to significant errors;
- Results dependence on the size and surface of stone materials grains;
- In some methods, low sensitivity to the presence of modifying additives;
- Inability to compare the results obtained by different methods.

These methods adhesion assessment (exfoliating action of water) of bitumen with a mixture of stone materials were used the first in production laboratories.

The simplest methods of this group include adhesion to the glass surface [30–40]. A portion of bitumen 0.35 g is weighed on a scale and placed on a glass plate (70 mm × 24 mm). The bitumen is distributed by an even layer along the contour marked on the plate. Its overflow outside the rectangular contour marked on the glass plate should be prevented. This plate is kept for 30 min at a temperature 80 °C higher than the bitumen softening point. After that, the plate is cooled by air for 30 min at room temperature and photographed. Then, the glass plate with bitumen is kept in water at 85 °C for 25 min. After holding the bitumen plate, cold water is added to the water bath to make its temperature lower on 10 °C than the bitumen softening point. The plate is removed from the water and photographed a second time. After that, on the basis taken before and after test photos and by use the applied program software the bitumen film retained area on the glass is determined in percent (Figure 5).



Figure 5. The glass plates after testing two bitumen samples for adhesion to the glass surface, 33 and 87%, respectively.

The main disadvantage of the above method is the use a glass surface, which has much worse adhesion to bitumen than the surface of stone materials. This disadvantage is absent in the method of determining bitumen adhesion to gravel [31,41–48]. Gravel grain (not less than 10 mm) is tied with a thread and heated in an oven to a temperature depending on the technology. After 1 h heating, it is immersed for 15 s in bitumen (at the temperature according to the technology), then removed and hung on a tripod to drain excess bitumen. The tests are carried out not earlier than 1 h after the treatment of gravel grains with bitumen. Gravel grain is immersed in a glass and kept in boiling water for 30 min. Then remove and immerse in a glass of cold distilled water for 1–3 min to cool and fix the bitumen film left on the gravel surface. The gravel grains surface is checked and evaluated on a 5-point scale, with an interval of 0.5 points (Figure 6).

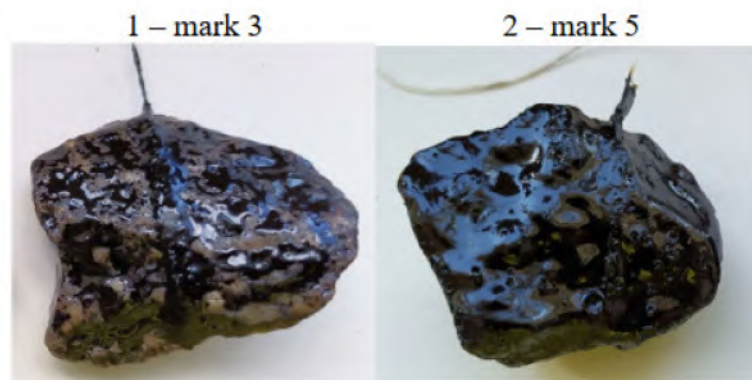


Figure 6. The gravel grains after testing two bitumen samples for adhesion to the gravel surface, 3 and 5 points, respectively.

A modification of this test, which is used in some laboratories in Ukraine, is that the stone material coated with bitumen is weighed before and after boiling, then according to the ratio of these masses the adhesion (in %) is calculated. That is, only the data interpretation is different, not in points, but in %. The main advantage of this method is a fairly high-test speed and it also allows to test the adhesion properties of bitumen on various stone materials.

Due to the detrimental effect of moisture damage, it is important to test the susceptibility of an asphalt mixture to moisture damage. The tests for asphalt mixes are divided into tests for loose mixtures and tests for compacted mixtures [6,19,23,49–51]. Despite the availability of tests for moisture susceptibility, none of them provides high correlation with field performance.

3.2.1. Tests on Loose Mixtures

Moisture-susceptibility tests on loose mixtures are carried out on particles with an asphalt coating in the presence of water. The main advantages of these tests are their simplicity and low cost compared to the compacted-specimen test costs. The use of simple equipment and procedures to perform experiments is another advantage.

Tests on Loose Mixtures can be divided into:

- Static
 - Static test [52];
 - Static immersion test [53];
 - Boiling water test or Texas boiling test [54,55];
 - Boiling water stripping test [52,56];
 - Chemical immersion test [57];
 - Surface reaction test [58]; many others.
- Dynamic
 - Rolling bottle test [41,52,59–62];
 - Film stripping test [63];
 - Dynamic immersion test [49,64];
 - many others.

Conditions the most common main methods that use tests on Loose Mixtures are presented in Table 2.

Table 2. Conditions of the most common methods that use Loose Mixtures treated with bitumen.

Parameter	Test					
	Static	Static Immersion	Texas Boiling	Boiling Water Stripping	Rolling Bottle	Film Stripping
Aggregate Coating						
Binder amount	4% ($\pm 0.5\%$)	5.5 g (5.2%)	2.3%–8%	30 g (2.0%)	16 g (3.0%)	3.6%
Aggregate (size)	150 particles (8/11.2 or 6/10 mm)	100 g (6.3/9.5 mm)	300 or 100 g (3/8 in)	1500 g (8/11.2 or 6/10 mm)	510 g for 3 sets (8/11.2 (5.6/8, 6.3/10) mm)	60 g (3/8 in)
Amount for testing						
Loose Mixtures	150 particles	100 g	300 or 100 g	200 g	150 g	—
Water	full cover	600 mL	500 mL	600 mL	500 mL *	—
Water conditioning						
Conditioning type	static	static	static	static	dynamic	dynamic
Water temperature	19 $\pm$ 1 °C	25 °C	boiling water	20 $\pm$ 5 °C	25 °C	
Conditioning time	48 $\pm$ 1 h	16–18 h	10 and 13 min	10 min	6 h and 24 h if necessary 48 h and 72 h	15 min
Assessment of the coating degree						
Validation of the result	number of not completely coated aggregates (visual)	visual estimation of the coating, less than or greater than 95%	visual estimation of the coating	visual estimation of the coating	visual estimation of the coating	validation of the result

* including mixtures.

Most static methods of analysis on loose mixtures differ mainly from each other in:

- Bitumen/aggregate ratio;
- Aggregate size;
- Methods of bitumen and aggregate preparation and mixing (for example, mixing Temperature, cooling time and other);
- Water conditioning (test temperature from room to boiling; it should also be noted that as the water temperature increases, the test time decreases significantly; adding different amounts of Na₂CO₃ to distilled water during chemical immersion test [57];

- Coating degree assessment (it's usually a visual estimation of the coating, but digital image processing is possible [61].

The difference between the various dynamic methods of analysis on loose mixtures is the same as static methods, the only difference from static methods is that the bitumen/aggregate system is mixed in water. There are different ways of mixing (Figure 7).

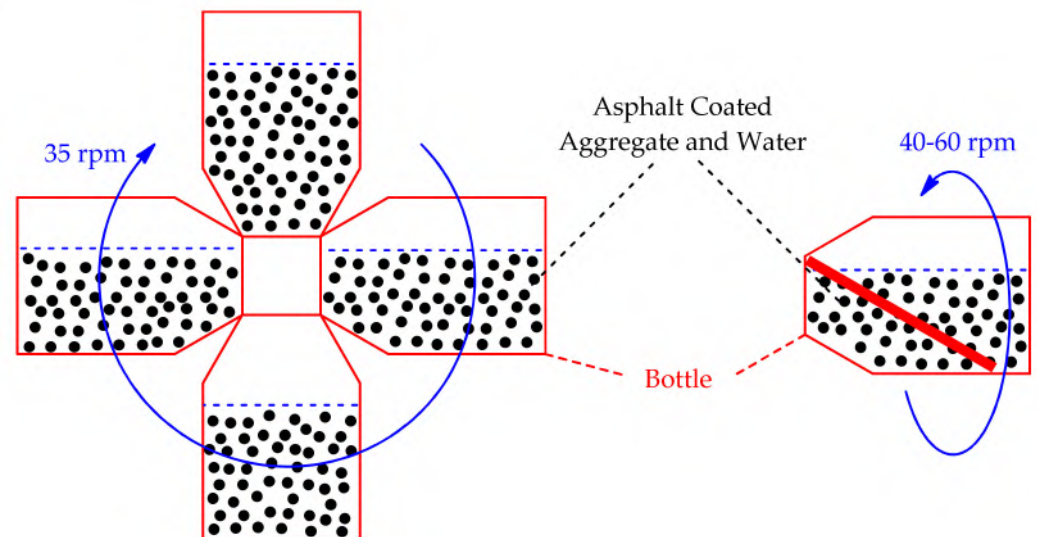


Figure 7. Rotating bitumen/aggregate mixture in a sealed bottle for film stripping test and rolling bottle test.

3.2.2. Tests on Loose Mixtures

Many researchers believe that the adhesion determination on loose mixtures is uninformative and does not reflect the real conditions of asphalt concrete in the coatings and prefer a group of test methods for compacted mixture. This group includes methods based on the characteristic changes of asphalt concrete samples (different shapes) before and after their deposition in water at different temperatures. A significant advantage of these methods is that they can be used to quantify the water impact on reducing the asphalt concrete adhesive properties. The disadvantages of these group methods are that they are used in different test conditions, which differ significantly from the actual operating conditions of asphalt pavements, and also have a relatively low results accuracy.

Mainly for this group of tests use compacted cylindrical mixtures:

- Indirect tensile strength test [65];
- Original Lottman indirect tension test [49,66];
- Modified Lottman indirect tension test [49,67,68];
- Tunncliffe–Root test [69,70];
- Compression strength test [71];
- Immersion-compression test [72,73];
- Marshall immersion test [74];
- Texas freeze-thaw pedestal test [49,75,76].

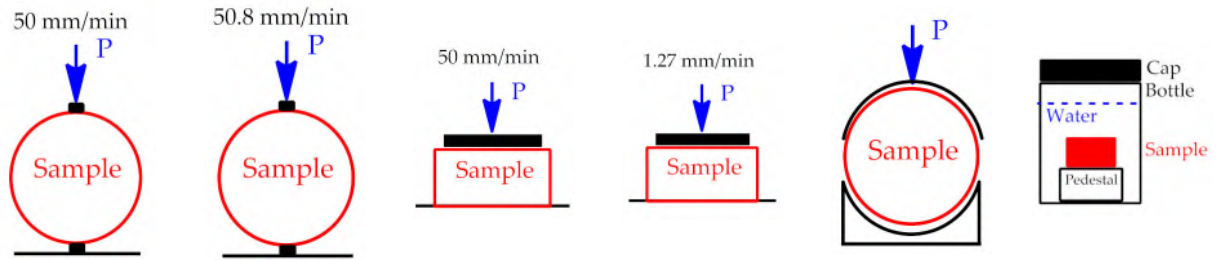
The Tunncliffe–Root test is comparable with modified Lottman indirect tension test. In both methods, the freeze cycle is optional. However, curing of the loose mixture in a 60 °C oven for 16 h is eliminated in the Tunncliffe–Root test procedure [23,49].

Therefore, most methods on compacted cylindrical mixtures differ mainly from each other in (Table 3):

- Size samples;
- Preparation the dry and wet samples;
- Determination of the different performance indicators for dry and wet samples (indirect tensile strength);

- Compressive strength;
- Marshall stability and other.

Table 3. Conditions the main most common methods on Compacted Cylindrical Mixtures.

Parameter	Test					
	Indirect Tensile Strength	Modified Lottman Indirect Tension	Compression Strength	Immersion-Compression	Marshall Immersion	Texas Freeze-Thaw Pedestal
Sample Size						
Diameter	80, 100, 120, 150 and 160 mm	63.5 and 150 mm	80, 100, 120, 150 and 160 mm	101.6 mm	100 mm (cylinder height 75 mm)	41.3 mm (cylinder height 19.5 mm)
Preparation of dry samples						
	20 °C (air)	25 °C, 2 h (in plastic film, water)	18 °C (air)	25 °C, 4 h (air)	25 °C, 4 h (air)	-
Preparation of wet samples						
Water saturation	(1) 30 min (vacuum)	(1) 10 min (water, vacuum)	(1) 60 min (vacuum)	(1) 60 °C, 24 h or 49 °C, 4 days (water)	(1) 60 °C, 24 h (water)	(1) 3 days (air)
	(2) 40 °C, 68–72 h (water)	(2) 18 °C, 16 h (in plastic film, freezer)	(2) 120 min (water, vacuum)	-	-	(2) freeze-thaw cycles bottle (−12 °C, 24 h (freezer); 49 °C, 9 h (water))
	-	(3) 60 °C, 24 h (in plastic film, water)	(3) 18 °C, 7 days (water)	-	-	-
Testing						
Cooling Temperature	2–4 h (water) 25 °C	2 h (in plastic film, water) 25 °C	- 18 °C	2 h (water) 25 °C	2 h (water) 25 °C	- -
Method	Indirect tensile strength (ITS; kPa)	Indirect tensile strength (ITS, kPa)	Compressive strength (CS; kPa)	Compressive strength (CS; kPa)	Marshall stability (MS; kN)	After each freeze-thaw cycle, the briquette surface is checked for cracks
						
Result						
Index	indirect tensile strength ratio (ITSR)	tensile strength ration (TSR)	index of retained strength (IRS)	retained compression strength (RCS)	retaining strength index (RSI)	the number of cycles required to induce cracking is a measure of water susceptibility (typically 10 freeze-thaw cycles)
Calculation	$\frac{ITS_{wet}}{ITS_{dry}} \times 100$ (%)	$\frac{ITS_{wet}}{ITS_{dry}} \times 100$ (%)	$\frac{CS_{wet}}{CS_{dry}} \times 100$ (%)	$\frac{CS_{wet}}{CS_{dry}} \times 100$ (%)	$\frac{MS_{wet}}{MS_{dry}} \times 100$ (%)	-

One of the relatively recent methods for assessing the asphalt concrete samples on long-term adhesive strength without water use is developed in the UK at the end of the last century Saturation Aging Tensile Stiffness (SATS) test [77]. This method allows to estimate the complex influence of moisture and aging binder on the strength of asphalt concrete samples. The test is performed on five asphalt concrete samples with diameter 100 mm and height 60 mm, which are kept in the test chamber at pressure 2.1 MPa and temperature 85 °C for 65 h and then another 24 h with a gradual decrease in pressure to atmospheric and temperature to 30 °C. After keeping samples, their modulus stiffness is determined and the durability indexes are calculated by the modulus ratio of initial and sustained samples. The SATS method is standardized in the countries of the European Union [78].

In addition to the cylindrical samples the tests are performed on prismatic samples of asphalt mixture, as well as rods drilled out from aggregate rock and stuck together with

bitumen film (Direct tensile strength ratio—DTSR). The unit axial direct tensile strength test was introduced for investigating adhesion strength properties in this study. Again, strength is determined before and after conditioning the test sample in a water bath according to [16,65].

The Hamburg Wheel-Tracking Device (HWTDD) measures combined effects of rutting and moisture damage and is gaining popularity because of fast and reliable testing of various hot mix asphalt (HMA) mixes [23,78–82]. Two HMA beam samples, each measuring 260 mm wide, 320 mm long and 40 mm thick. It measures the effects of rutting and moisture damage using a steel wheel that runs over compacted beams immersed in hot water (usually 50 °C). HMA sample is loaded for 20,000 passes or until a final strain of 20 mm appears (Figure 8).

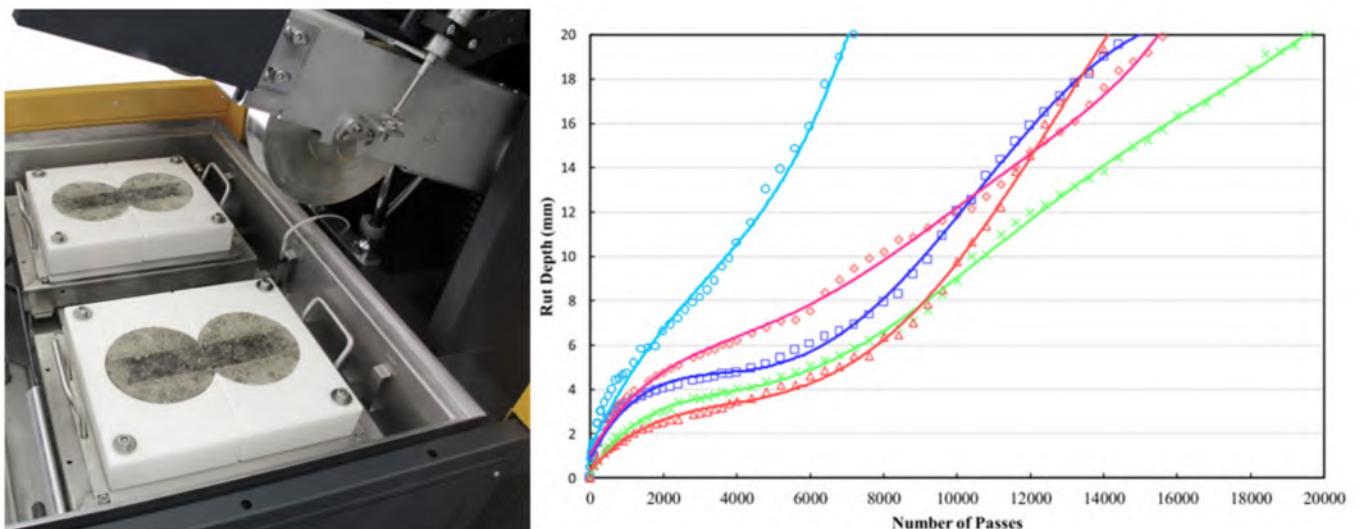


Figure 8. HWTDD rutting curves of asphalt mixtures. Reprinted with permission from [79] Copyright Elsevier 2019.

3.3. Definition of Adhesion as a Fundamental Value

The main problem of the previously presented tests is the dependence on visual observation to detect stripping, low reproducibility and results accuracy. This group of tests allows to get rid of these shortcomings.

As already mentioned, the aggregate high wetting is essential for good adhesion. Therefore, the measurement of its derivatives—contact angle and surface free energy is important for the adhesion characterization in the bitumen/aggregate system. Many methods are based on the determination of these indicator.

For the analysis of bitumen, the contact angle measurement can be used in different ways. Three different approaches were covered in this study [16]: bitumen drop on glass, Pendant drop and water drop on bitumen surface. Contact angle measurements were conducted on a DSA100 (KRÜSS Ltd., Hamburg, Germany), a goniometric setup including an automated sample dosing device. The sample is illuminated by a strong light source and filmed and recorded with a high-resolution camera with manual zoom and focus.

It is known that the adhesion of bitumen to aggregate can also be determined based on the calculation of SFE [83–85] and its polar and dispersive components [83,86], by measuring the contact angle of liquids with different hydrophilicity on bitumen binder films under ambient conditions. Having determined the contact angle of the liquid (water, ethylene glycol, and others) to the bitumen (in the solid state), you can calculate the SFE of bitumen. To calculate SFE of bitumen, using the dispersion force component and polarity component (hydrogen bonding and dipole-dipole interactions) into the equation (using

a modified Young-Good-Girifalco-Fowkes equation), which was adopted by Owens and Wendt [87,88]:

$$1 + \cos(\Theta) = 2 \cdot \sqrt{\gamma_S^d} \cdot \left(\frac{\sqrt{\gamma_L^d}}{\gamma_{LV}} \right) + 2 \cdot \sqrt{\gamma_S^h} \cdot \left(\frac{\sqrt{\gamma_L^h}}{\gamma_{LV}} \right) \quad (5)$$

where θ is the contact angle between the interface (Figure 9); γ_S^d and γ_L^d is SFE of dispersion force component for solid and liquid, respectively; γ_{LV} is SFE for liquid-vapor; γ_S^h and γ_L^h denotes the component of SFE due to hydrogen bonding and dipole-dipole interactions (polarity component) for solid and liquid, respectively.

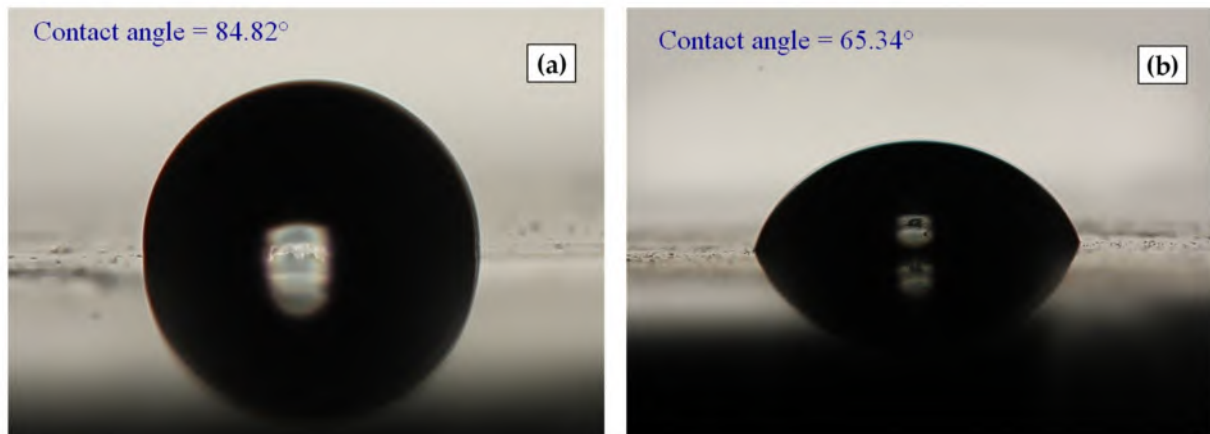


Figure 9. Photos of sessile drops (5 μ L) in contact with a solid surface: (a) water drop on the bitumen; (b) ethylene glycol on the bitumen 70/100.

The samples were illuminated by a strong light source and recorded with digital camera Canon EOS 1100D equipped with macro lens (Canon Inc., Tokyo, Japan). Drop images were analyzed for CA measurement by open-access software ImageJ with plugins such as Low Bond Axisymmetric Drop Shape Analysis (LB-ADSA) technique [89] and Half-Angle technique [90].

Equation (5) is written in a condensed form relative to the known components and terms containing the values of the components (dispersive and polar) for the free energy of the solid surface for both test liquids ($i = 1; 2$, where water is index 1 and ethylene glycol is index 2):

$$0.5 \cdot \gamma_{LV-i} \cdot (1 + \cos(\Theta_i)) = \sqrt{\gamma_S^d} \cdot \sqrt{\gamma_{LV-i}^d} + \sqrt{\gamma_S^h} \cdot \sqrt{\gamma_{LV-i}^h} \quad (6)$$

To form a system of two equations with two unknowns, the corresponding designations X, A, B, C were entered (the contributions from the dispersive and polar parts to the free energy of the solid surface):

$$X_1 = \sqrt{\gamma_S^d}; X_2 = \sqrt{\gamma_S^h}; A_i = \sqrt{\gamma_{LV-i}^d}; B_i = \sqrt{\gamma_{LV-i}^h}; C_i = 0.5 \cdot \gamma_{LV-i} \cdot (1 + \cos(\Theta_i))$$

The values of the components (dispersive and polar) for the surface tension of both test liquids were taken from the work [84] and are represented in Table 4.

Table 4. Values of the components (dispersive and polar) for the surface tension of both test liquids.

Liquid	γ_{LV-i}^d (mJ/m ²)	γ_{LV-i}^h (mJ/m ²)	γ_{LV-i}^{total} (mJ/m ²)
Water	21.80	51.00	72.80
Ethylene glycol	29.00	19.00	48.00

Solutions of the system of equations:

$$X_1 = \frac{B_1 \cdot C_2 - B_2 \cdot C_1}{A_2 \cdot B_1 - A_1 \cdot B_2}; X_2 = \frac{C_1 - A_1 \cdot X_1}{B_1}$$

The SFE calculation is summarized in Table 5.

Table 5. Contact angles and SFE of bitumen 70/100.

Contact Angle (°)		Surface Energy Components (mJ/m ²)		Total Surface Energy (mJ/m ²)
Water	Ethylene glycol	γ_s^d	γ_s^h	γ_s^{total}
84.82	65.34	14.90	9.21	24.11

Note: γ —surface energy (i.e., surface tension); subscript indices s pertains surface energy of solids; superscript indices d mean dispersion force component of SFE for solid; h mean polarity component (hydrogen bonding and dipole-dipole interactions) of SFE for solid.

The determination of contact angle and SFE for bitumen characterizes their adhesive properties, the lower the value of contact angle and SFE, the higher the adhesion and properties of bitumen.

Labib et al. [91] report use of electrophoreses to investigate the proton transfer surface properties of both aggregate and bitumen in the presence water. Previous studies in the field have shown consistency with the results obtained in this study, proving that the zeta potential test can be a viable and accurate method for predicting aggregate-bitumen adhesion. The zeta potential test has the potential to act as an alternative to conventional adhesion tests with substantially shorter turnaround times [92].

The surface reaction test allows a researcher to quantify the level of stripping on loose asphalt mixtures [58]. The reactivity of calcareous or siliceous aggregates and the reaction to the presence of highly toxic and corrosive acids is assessed by the surface reaction test. The gas released during the chemical reaction creates a pressure proportional to the aggregate surface area. The basis of this test is the assumption that different levels (intensity) of stripping lead to exposed aggregate surface parts [49,58].

These group tests includes Net Adsorption Test [13,23] and Modified Net Adsorption Test [23,93]. These methods are based on physic-chemical adsorption of bitumen from solution (in toluene) by the mineral aggregate surface. Adhesion is estimated by the amount of adsorbed bitumen from the solution. These methods are not widely used because they are quite difficult to implement.

Blister test (interfacial fracture energy (IFE)) allows to determine the binder film elasticity modulus and the interfacial fracture energy, which corresponds to the adhesive strength. The principle of the test is to break the interface bonding by pressurizing the interface between the adhesive and adherent. The amount of pressure and the deformation of the adhesive before and during the debonding period are used to calculate the IFE [22].

Wilhelmy plate method [21] is to determine the thin plate weight and the depth at which it will sink into the liquid. At touching a thin plate (a plate is usually made from glass, filter paper, foil or platinum), suspended on a thin metal wire or spring, on the liquid surface there is retraction of this plate into the liquid. At the same time, retraction is accompanied by spring elongation and weight gain. The retraction is due to the surface tension force acting on the thin plate perimeter, and the force of the plate retraction into the liquid is determined by the appropriate formula.

4. Conclusions and Recommendations

Due to the excellent adhesion in the bitumen/aggregate system, scientists and technologists can use these two different materials together. Therefore, laboratory determination of the adhesion value between different bitumen and stone materials used in road construction is very important.

First, a review of the various mechanisms of interaction in the bitumen/aggregate system was conducted. There are five main theories why the interaction between these materials occurs, but to some extent the theories contradict each other. None of the theories

is absolutely correct, because according to the authors, excellent adhesion depends on a number of factors listed in different theories.

The literature review analysis shows the absence of a single generally accepted adhesion evaluation method in the bitumen/aggregate system, which would allow the evaluation of the adhesion characteristics relatively quickly with high accuracy and reproducibility. A significant number of existing methods that evaluate the adhesion of bitumen, stone materials mixtures with bitumen, and asphalt mixtures differ in test conditions, and the obtained results are not comparable. At the same time, the number of adhesion evaluation methods increases constantly and the issue of assessing and predicting changes in the bitumen affinity to stone materials is relevant.

The article shows that the adhesion tests in the bitumen/aggregate system, according to the authors, can be divided into three groups: determination of adhesion forces for bitumen with different materials; determination of bitumen resistance to stripping action of water with different materials; and definition of adhesion as a fundamental value (contact angle, surface free energy, zeta potential, interfacial fracture energy, adsorption capacity and others). The most common tests to determine bitumen resistance to water stripping with different materials in turn are divided into: those that are conducted on loose bitumen/aggregate mixtures and those that are conducted on compacted mixes. The first evaluate the bitumen and aggregate compatibility and stripping potential, the second try to take into account various properties (as an example, tensile strength) of asphalt concrete on the water action. Many researchers believe that the determination of adhesion on loose mixtures is uninformative and does not reflect the asphalt concrete real conditions in the coatings and prefers a group of tests methods for compacted mixtures.

Therefore, today, there are no single common adhesion test in the bitumen/aggregate system that are regional. Thus, it is important to develop a single procedure for interpreting data in this area and the possibility of comparing them with each other. According to the authors, the test should consist of two successive stages: the first should be fast, which will confirm bitumen and aggregate compatibility (for example, “adhesion to gravel” test); the second should be the Compacted Mixtures test (indirect tensile strength test, Modified Lottman indirect tension test, immersion-compression test and Hamburg wheel tracking test).

Author Contributions: Conceptualization, V.G. and M.B.; methodology, I.S. and Y.D.; software, N.V.; validation, M.B., V.G., Y.D. and K.P.; formal analysis, M.B. and M.K.; investigation, V.G., I.S. and Y.D.; resources, Y.D. and N.V.; data curation, V.G. and Y.D.; writing—original draft preparation, V.G. and Y.D.; writing—review and editing, M.B., M.K. and K.P.; visualization, V.G.; supervision, M.B.; project administration, M.B. and V.G.; funding acquisition, M.B. and K.P. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Research Foundation of Ukraine, Kyiv (Grant No. 2020.02/0038) and statutory activity subsidy from the Ministry of Science and Higher Education Republic of Poland for the Faculty of Chemistry of Wrocław University of Science and Technology.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data supporting reported results are stored by V. Gunka.

Acknowledgments: The paper is a part of the research done within the project 2020.02/0038 (National Research Foundation of Ukraine). KP gratefully acknowledge statutory activity subsidy support (2022) from the Ministry of Science and Higher Education Republic of Poland for the Faculty of Chemistry (Department of Advanced Materials Technologies) of Wrocław University of Science and Technology.

Conflicts of Interest: The authors declare no conflict of interest.

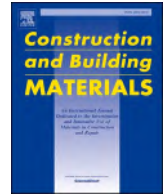
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Effect of two-phase mixing on moisture susceptibility of hot mix asphalt

Ajit Kumar, Sudhir Varma*

Department of Civil and Environmental Engineering, Indian Institute of Technology Patna, Bihta, Bihar 801103, India

ARTICLE INFO

Keywords:

Asphalt mix
Two-phase mixing
Moisture damage
Dynamic modulus
Tensile strength ratio
Mix design
Aggregate-bitumen bond strength

ABSTRACT

Moisture damage is one of the major concerns of transportation agencies. Improper coating of asphalt binder over aggregate surface provides easy access at the aggregate-binder interface for moisture induced adhesive failure. Due to sheer size, there is a large difference in specific surface area of coarse and fine aggregates. During conventional mixing process, this difference in area leads to non-homogeneous distribution of binder. Fines receive thicker binder thickness and more absorption compare to coarse aggregates. Also, filler particles inhibit the binder from penetrating into micro-pores of coarse aggregate and devoid binder the opportunity to penetrate its surface deeply. Aggregate-binder interface is significantly influenced by the sequence in which aggregate and bitumen are mixed, aggregate type and moisture conditioning. The current practice in industry is to mix the entire proportion of aggregates together with bitumen in one-phase. In this study, a two-phase mixing sequence is proposed in which initially half of the binder content is premixed with coarse aggregates, followed by adding fines and remaining binder in the second step. The paper presents a novel study to evaluate the effect of two phase mixing on viscoelastic properties of dense asphalt mix. Further, effect of the proposed two-phase mixing on moisture sensitivity has been investigated using mechanical performance test. For this, conditioned dynamic modulus test was performed after 0, 1, 3 and 6 freeze-thaw cycles and then compared with tensile strength ratio test. Furthermore, the influence of moisture damage on aggregate-bitumen interface was studied using bond strength test. Granitic aggregates had higher silica content and were more moisture susceptible as compared to basalt. The interfacial failure mode in both the aggregates were cohesive in unconditioned specimens, which on conditioning drifted from cohesive to adhesive. For all the aggregates, two-phase mixing was found to significantly improve moisture damage resistance of mixes compared to the conventional one-phase mixing.

1. Introduction

Moisture can cause loss of asphalt mix integrity, resulting in premature failure in field. It creates an additional economic burden of early maintenance and rehabilitation. Moisture damage in asphalt mix is generally referred as degradation in mechanical properties due to presence of moisture in its microstructure [1]. Moisture damage can result from loss of adhesion or loss of cohesion or from the combination of both [2]. Adhesive failure is caused by detachment of bitumen film from aggregate surface due to loss in bitumen-aggregate interface bond strength. Cohesive failure is caused due to loss of cohesion in the bitumen itself. Generally adhesive failure is expected if the bitumen film thickness is thin and cohesive failure if it is thick [1]. Adhesive failure is the most accepted failure mode emanating to stripping of bitumen [1,3]. Ingression of moisture into asphalt mix may occur in several manners such as, vapour diffusion, runoff from road surfaces, seepage from nearby areas and capillary action from the water table [1,4]. This

moisture may develop pore pressure inside asphalt mix due to traffic loading and expansion under freezing condition. Stresses in bitumen film due the pore pressure may cause microcracks [1], providing moisture an access to the aggregate-bitumen interface. Once the water reaches at the interface, it causes stripping. The damage occurs due to thermodynamic, chemical, physical, and mechanical processes [1] which results into weakening of bond between aggregate and bitumen [5].

The chemical bond formed between aggregate and bitumen depends upon their chemical composition [1,6–9]. Siliceous aggregates have a large concentration of hydroxyl groups, which have a stronger affinity for water and carboxylic acid. In general, carboxylic acids and sulfoxides present in bitumen are strongly absorbed by aggregates, but are also more susceptible to displacement in the presence of water [6,8,9]. The functional groups of bitumen that are highly adsorbed on the surface of the aggregate are also more likely to get dissolved by water [7]. Therefore, basic aggregates like those higher in carbonate content and

* Corresponding author.

E-mail address: svarma@iitp.ac.in (S. Varma).

<https://doi.org/10.1016/j.conbuildmat.2024.135275>

Received 24 July 2023; Received in revised form 6 January 2024; Accepted 29 January 2024

Available online 7 February 2024

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lower in siliceous content are easier to coat and more resistant to moisture damage. Functional groups in aggregate that are associated with aluminium, iron, magnesium, and calcium, have strong affinity for bitumen, whereas, sodium and potassium are associated with lower affinity [10]. Chemical composition may not have significant influence on moisture sensitivity of mixes if the bitumen binders have same viscosity [11]. Generally, high-grade bitumen mixed with clean and dry aggregates having rough surface texture perform better against moisture damage [2]. This is because, aggregates with rough surface texture and more pores and angularities are expected to have higher mechanical interlocking. The damage is aggravated when coating of bitumen over aggregate is not homogeneous and some of the surface is not coated [12–14]. Transportation of moisture and displacement of bitumen occurs as water is absorbed into aggregate through the uncoated aggregate surface [15,16]. Higher affinity towards water than bitumen results in the displacement of asphalt from the three-phase interface [1]. Bitumen film over fine aggregates is relatively thicker than that of coarse aggregates. This happens because of non-uniform distribution of bitumen between the coarse and fine aggregates owing to significant difference between their surface areas [17]. As coarse and fine aggregates are mixed together there is a tendency of smaller particles to move to the top which can result in an inhomogeneous mixture, furthermore, these filler particles may agglomerate and trap some of the bitumen forming dullness in the mix [13]. When the bitumen coating is very thin, the moisture may pass through diffusion, furthermore, the thin film at sharp corners and edges and rough textures is under higher stress and may tear [18]. When coarser aggregates are mixed with bitumen before the fine aggregates, it provides thicker coating around the coarse particles [12, 13,17]. The occurrence of improper coating is more in the conventional (one-phase) mixing, where coarse and fine aggregate are mixed with heated bitumen in a single step. During the mixing process, fines get trapped or attached inside the surface texture of coarse aggregate which prevents the full coating over the surface [12,14]. Furthermore, the coarse aggregates are not able to absorb bitumen in the micro-pores on their surface because of the hindrance from finer particles.

The effect of aggregate-binder physical interaction can be influenced by altering mixing process. Thus, adhesion at the interface depends on the mixing process including sequence, temperature and time [17]. Here the term “sequence” means the order in which different fractions of aggregate (coarse, fine, filler) and bitumen are introduced in the mixing process. Researchers have shown that certain mixing sequence can improve moisture resistance [12,17], workability [13,17], resilient modulus [13] and fatigue and rutting resistance [12,19]. Studies have shown that pre-coating of coarse aggregate with partial bitumen content before mixing fines and filler improves the mix performance [12,17,19]. This has been commonly referred as two-phase mixing in literature. Huang et al. (2006) used gasolite to pre-coat the coarse aggregates before mixing fines and bitumen. The three layer system was found to improve dynamic modulus, rutting, indirect tensile strength and moisture susceptibility of hot mix asphalt (HMA). In seal coats, Kandhal and Motter (1991) showed that pre-coating 90% or more can improve adhesion and retention of aggregates.

In the two-phase mixing process suggested by Kok and Kuloglu (2011), first the coarse aggregates are pre-coated with 20% of optimum binder content. Then the coated aggregate was left for 1 h at 10 °C below the mixing temperature. After conditioning, in the second-phase the pre-coated aggregate was mixed with fines and remaining bitumen. Although the mixes exhibited improvement in moisture damage resistance and fatigue and rutting performance, the approach is not easy for asphalt plants to execute due to the following: (i) conditioning period required a separate coated aggregate bin for conditioning, (ii) overall time and fuel efficiency of the production is drastically reduced, (iii) separate air dryer drum was proposed for producing pre-coated aggregates. Furthermore, the additional conditioning will induce aging of bitumen. The infiltration of bitumen inside the pores of aggregate surface mainly occurs in the initial period of mixing and its rate reduces as

time progresses [20]. Kiselev et al. (2021) studied two-phase mixing for porous asphalt. In the first step 20% modified bitumen was mixed with coarse aggregate. Then the fines and rest of the 80% bitumen were mixed. Fatigue and rutting performance were improved in the two-phase mixing. Liu et al. (2020) studied effect of two-phase mixing sequence on total adhesion between aggregate-bitumen using surface energy theory. The sequence included first mixing coarse and partial amount of fines with 100% bitumen followed by mixing the remaining fines and then finally mixing filler. It was suggested that the coarse and fine aggregates should be so adjusted that adhesion work in both the mixing phases should be approximately same, this was to ensure homogeneous distribution of bitumen between the aggregate size fractions. The two-phase process is more suitable for batch type asphalt plant in which coated aggregates can be directly introduced in plug mill. Viman et al. [21] implemented a three phase mixing process (KGO-III) in which first coarse aggregates were mixed with bitumen, then in the second step filler was mixed followed by mixing of fines in the last step. Five test sections on three different roads were prepared using KGO-III approach. The approach was found to decrease mixing temperature by 30 °C and bitumen requirement by 0.5%. Compared to conventional HMA, cores from KGO-III sections showed improvement in moisture resistance, stiffness and rut resistance. Since 1998, the method has been used in asphalt plants to produce wearing course for highways and secondary roads in Sweden [13]. Hesami et al. [13] found that the mixing sequence can significantly influence the mastic viscosity. From X-ray tomography it was observed that the sequential mixing produce more homogeneous mix, resulting in even air-void distribution along the depth.

Studies have observed that adding 20% bitumen achieves at least 90% coating in the first phase of mixing [12,19] and then temperature conditioning facilitates better binder absorption [12]. However, coating coarse aggregate with lower binder content may result in incomplete coating and lower binder absorption that would defeat the purpose of first phase of mixing. In Viman et al. [21] and Hesami et al. [13], 100% binder content is added in the first phase itself. Although, this will provide complete coating and higher binder absorption in coarse aggregates, it may result in drain down of bitumen and difficulty in mixing fines. In the next step, before adding fines filler was mixed which will control drain down and provide formation of better mastic to coat the fines. But then the process becomes a three phase mixing in which the three different fractions of aggregates have to be introduced in the plug mill at different time. Furthermore, complete separation of dust from stockpiles or in hot bins is not feasible. This would result in cumbersome and longer production time. Based on the literature, in the present study a two-phase mixing process is proposed in which a higher percentage of bitumen (50% of the total design binder content) is mixed with coarse aggregates in the first phase. After pre-coating the coarse aggregates, fines were mixed followed by mixing the remaining bitumen. The main goal of the study was to examine the effect of the proposed two-phase mixing on moisture damage resistance of dense graded bituminous mix.

Moisture damage is a complex phenomenon, which mainly depends on the combined effect of material characteristics, mixture design parameters, loading type and environmental factors [2,22]. Tensile strength ratio (TSR) obtained from Modified Lottman indirect tension test is mostly used to evaluate moisture susceptibility of asphalt mixes [2,23–25]. TSR is ratio of indirect tensile strength (ITS) of moisture conditioned to unconditioned specimens. Several studies have used TSR to evaluate effect of mixing sequence on moisture damage resistance [12,17,26]. Adverse environmental and traffic conditions can accelerate moisture damage in asphalt mix. Inducing multiple cycles of moisture damage can be used to study damage evolution and longer and aggressive field conditions [27,28]. But, being a destructive test, new specimens are required to obtain TSR at multiple moisture conditioning. Furthermore, TSR is not a characteristic property of asphalt mix and it cannot be used to predict mechanical behaviour of bituminous mixtures [29,30]. Also, in some cases, TSR alone may not be a reliable parameter [25,31] to distinguish moisture sensitivity of mixes. Instead of static

loading that is applied to obtain TSR, a test simulating field traffic conditions through dynamic loading would be a better indicator of moisture damage [32]. Solaimanian et al. (2007) studied different testing procedures to improve moisture susceptibility evaluation for HMA. The tests included simple performance tests (SPTs) developed in NCHRP projects 9–19 and 9–29, TSR and Hamburg wheel tracking. Retained dynamic modulus after moisture conditioning was found to provide the best moisture performance prediction. Flow number test do not successfully evaluate moisture sensitivity due to hardening of mix in environmental conditioning, effect of permanent deformation and high variability in the test [25,27]. Researchers have successfully used the loss in dynamic modulus due to moisture conditioning for evaluating moisture sensitivity [27,30–34]. Studies have found that dynamic modulus mastercurve monotonically shifts downwards as the number of freeze-thaw (FT) cycles increase. Addition of antistripping agents, polymers and hydrated lime can reduce the damage in dynamic modulus [33]. One of the major advantages of using dynamic modulus is that it's a non-destructive test, so it can be performed over same specimen before and after moisture conditioning [25,32]. Another important application of retained dynamic modulus could be to mechanistically access the impact on fatigue and rutting due to moisture damage [25]. Other than mechanical properties of asphalt mixes, aggregate-bitumen bond strength is a useful parameter to understand mode of moisture damage failure [35]. Factors such as moisture conditioning [35–38], bitumen and aggregate chemical composition [39,40], binder ageing [39,41], film thickness [38,42] aggregate texture and its porosity [43] influence the aggregate-bitumen bond strength.

As stated earlier, considering the benefits, a two-phase mixing sequence has been proposed in this paper. Effect of the proposed mixing sequence on moisture sensitivity of dense HMA has been investigated. In literature no work has been done to assess moisture sensitivity of dense graded two-phase mixes using viscoelastic characteristic property. For this, conditioned dynamic modulus test was performed after 0, 1, 3 and 6 FT cycles and then compared with tensile strength ratio test. Therefore, the main objectives of this study were to first devise a mixing scheme which will improve moisture damage resistance of mixes, second, to use viscoelastic and aggregate-bitumen interfacial properties to investigate moisture damage resistance of the proposed mixing scheme.

2. Materials and methodology

2.1. Materials

Mixes in this study have been laboratory produced using VG30 grade unmodified bitumen and aggregates from four different sources. The aggregates were used to prepare dense gradations for single and two-phase mixing. The one-phase mixes have been hereafter referred as Mix-1 C, Mix-2 C, Mix-3 C and Mix-4 C (where “C” indicates conventional one-phase mixing). The corresponding two-phase mixes have been, hereafter referred as Mix-1CC, Mix-2CC, Mix-3CC and Mix-4CC (where “CC” indicates that the mix has been prepared in two-phases). More details on the constituent materials are presented in the sections below.

2.1.1. Aggregate

The aggregates used in this study were obtained from four different states in India. The aggregate used in Mix-1 (C and CC), hereafter referred as A1, was basalt. The rest of the three aggregates, hereafter referred as A2, A3 and A4, were granite, and were used in C and CC combinations of Mix-2, Mix-3 and Mix-4, respectively. The aggregate gradation of Mix-1, Mix-2, Mix-3 and Mix-4 are shown in Fig. 1.

Maximum aggregate size of all the mixes was 25.0 mm. The aggregates were subjected to XRF test to obtain its chemical composition. The basic engineering properties and elemental composition of the aggregates are shown in Table 1 and Table 2, respectively. It can be seen from Table 1 that, the aggregates satisfied the MORTH [24] specification.

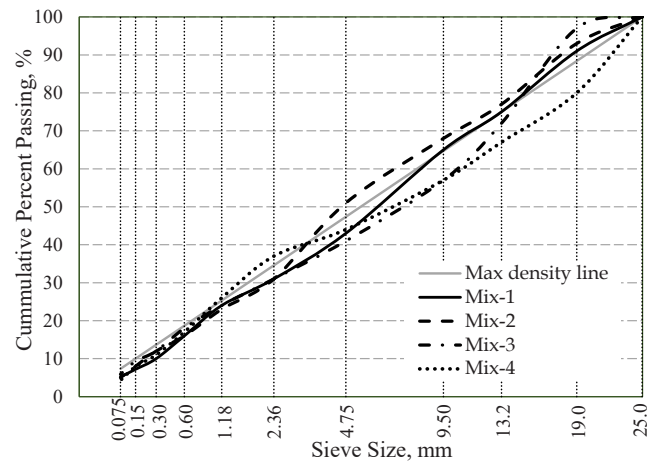


Fig. 1. Aggregate gradations used in the study.

Table 1

Basic properties of the studied aggregates.

Properties	Result				Limits	Test standard
Aggregate Type	A1	A2	A3	A4		
Aggregate impact value, (%)	9.2	24.3	12.3	10.8	< 30	BS 812-112
Specific gravity (Coarse)	2.790	2.740	2.687	2.676	-	ASTM-C127
Specific gravity (Fine)	2.746	2.628	2.640	2.627	-	ASTM-C128
Water absorption (Coarse)	1.18	1.29	0.91	0.74	Max 2%	ASTM-C127
Water absorption (Fine)	1.48	1.75	1.43	1.14	Max 2%	ASTM-C128
Crushing value, (%)	15.7	17.4	15.9	14.3	-	BS 812-110
Los Angeles abrasion, (%)	16.6	18.3	16.2	14.8	-	ASTM C 131

Table 2

Chemical composition of the studied aggregates.

Component	Content (% by mass)*			
Source	A1	A2	A3	A4
SiO ₂	50.1	56.7	59.4	65.4
Al ₂ O ₃	14.6	12.3	11.6	14.1
K ₂ O	0.6	2.5	2.3	3.0
CaO	10.4	8.6	4.6	3.5
Fe ₂ O ₃	11.4	9.1	8.7	4.7
Na ₂ O	5.6	5.5	5.8	5.3
MgO	4.8	3.4	5.8	2.7
TiO ₂	1.5	0.8	0.7	0.6
ZnO	0.6	0.2	0.6	-
SO ₃	0.4	0.4	0.2	0.2
MnO	-	0.2	0.1	0.1

* The constituents which is less than 0.1% are not presented in the table.

2.1.2. Bitumen

Bitumen used in this study was unmodified VG30 binder obtained from a refinery. VG30 grade bitumen is generally used in road works for traffic less than 30msa. The specific properties including its specific gravity, ductility and consistency properties were measured according to standard testing procedures. The basic properties of the bitumen are presented in Table 3.

The mixing and compaction temperatures were determined using rotational viscometer (ASTM D4402), corresponding to the viscosities 170 ± 20 cSt and 280 ± 30 cSt, respectively [23]. The mixing and

Table 3
Basic properties of the studied bitumen.

Properties	Result	Standard
Specific Gravity	1.01	ASTM-D70
Ductility at 25 °C, cm	100 +	ASTM-D113
Softening Point (°C)	48	ASTM-D36
Penetration at 25 °C (0.1 mm)	56	ASTM-D5
Flashing Point (°C)	> 220	ASTM-D92
Viscosity at 135 °C, Pas	0.546	ASTM-D4402

compaction temperatures were 160.9 ± 3 °C and 153.7 ± 2 °C, respectively.

2.2. Methodology

In this study, a two-phase mixing approach has been proposed in which the coarse aggregates are first mixed with bitumen followed by mixing of fine aggregates and the rest of the binder. Schematic of the mixing process is shown in Fig. 2. In the proposed approach, there is no conditioning time between the first phase and the second phase stages of mixing. Instead, in first phase stage of mixing, the percentage of bitumen added is 50% of the total design binder content which is more than the binder required in coating the coarse aggregates to achieve the final film thickness of the mix. This has been done with the hypothesis that, excess binder may result in better coating and higher binder absorption, eliminating the need for conditioning after the first phase of mixing. The two-phase mixes produced using the proposed methodology have been hereafter referred as CC. In an initially trial, when the entire 100% bitumen was added in the first phase itself, then the excess bitumen formed a thick coat on the mixer, resulting in difficulty mixing fines in

the next phase. Whereas, when smaller percentage of bitumen was used (20%) in the first phase of mixing, it was observed that thorough coating of aggregates could not be achieved. However, to examine the effectiveness of the proposed method, it was compared with the mixing proportion used by Kok and Kuloglu (2011) and Kiselev et al. (2021) for two-phase mixing. Kok and Kuloglu (2011) and Kiselev et al. (2021) used 20% bitumen for mixing with the coarse aggregates in the first phase. The remaining 80% bitumen was added in the second phase of mixing. However, the deviation in the process adopted is that no curing period was provided during the mixing. The mixes produced using Kok and Kuloglu (2011) and Kiselev et al. (2021) proportioning have been hereafter referred as CC20.

The current practice in industry is to mix the entire proportion of aggregates together with bitumen in one-phase. In this mixing process, the coarse aggregates may get deprived of proper coating due to (i) the hindrance of fines and dust at the surface of coarse aggregates, (ii) higher absorption of bitumen by fine and dust. Therefore, the two-phase mixing process is expected to produce mixes with higher moisture damage resistance. In this study effect of two-phase mixing on moisture damage resistance of dense graded asphalt mixes have been studied. For this, the effect of moisture conditioning on modulus and strength of one-phase and two-phase mixes have been compared. Being a characteristic property of asphalt mix, dynamic modulus has been measured to understand the effect on viscoelastic behaviour due to moisture damage. ITS has been measured to understand reduction in strength due to moisture damage. To study moisture damage evolution, the tests were conducted on specimens conditioned for 0, 1, 3 and 6 FT cycles.

2.2.1. Mix design

All the one-phase mixes in the study were prepared using Marshall mix design method, satisfying MORTH [24]. Marshall parameters of all the two-phase mixes were obtained from the tests at the design binder content of the one-phase mixes. Details of the mixes are shown in Table 4. The mixing sequences followed in the study to produce asphalt mixes is shown in Fig. 2. Compaction and testing of Marshall samples were as per ASTM D6926 and ASTM D6927, respectively. Since, the maximum size of aggregate in all the mixes was 25 mm, 4-inch diameter Marshall samples have been used in the mix design.

Uncoated aggregates in the first phase of mixing with partial bitumen content would restrict the full benefit of two phase mixing. At the same time, excessive bitumen on coarse aggregates would increase the risk of draindown during mixing and transportation [13,15]. To overcome these issues, in this study the percentage of bitumen required to achieve complete coating of coarse aggregates was obtained from trial and error as the proportion of bitumen for the first phase of mixing.

As discussed earlier, some of the two-phase mixing studies [12,19] in literature conduct curing of asphalt mix between the phases or split the mixing sequence in more than two aggregate addition phases [17,21]. These additional operations would complex asphalt plant production process compared to the conventional mixing. However, mixing in phases increases workability during mixing and reduces the effort required in coating aggregates compared to the conventional one-phase mixing [13]. In the present study aggregates have been mixed in two phases without any curing. Duration of mixing in one-phase and two-phase processes in laboratory are shown in Fig. 2. The duration of mixing was decided based on visual observation until thorough coating

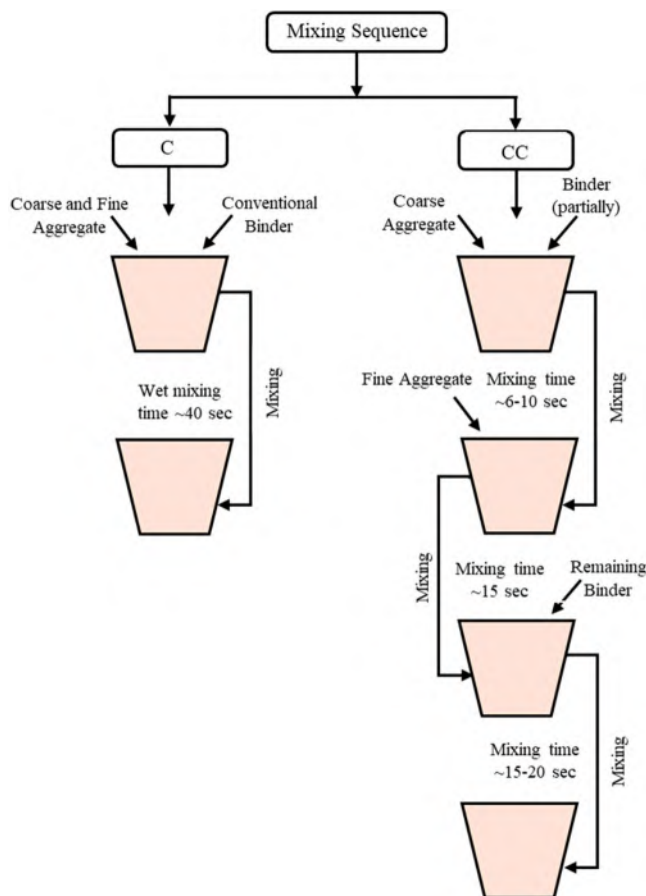


Fig. 2. Mixing sequence followed in the study.

Table 4
Marshall mixes used in the study.

One-phase mixes	Two-phase mixes	Aggregate gradation	% passing 4.75 mm sieve	Binder grade
Mix-1 C	Mix-1CC	A1 (Basalt)	43%	VG30
Mix-2 C	Mix-2CC	A2 (Granite)	51%	VG30
Mix-3 C	Mix-3CC	A3 (Granite)	41%	VG30
Mix-4 C	Mix-4CC	A4 (Granite)	44%	VG30

of aggregates was achieved. In one-phase mixing, the time for preparing mix was found approximately 40 s to achieve coating in a 20 litres capacity mixer with a single mixing paddle. In two-phase mixing, coating of aggregates was ensured at the end of all the mixing sequences. The approximate coating time in two phase mixing sequence were (i) 6–10 s in phase-1 for mixing coarse aggregates with half of the bitumen content (ii) 15 s for mixing fine aggregates and (iii) 15–20 s for mixing the remaining bitumen. Therefore, the total mixing time in two-phase was also approximately 40 s. Thus, it was observed that the mixing time in laboratory for the one-phase and two-phase mixing were nearly same. The swift coating of aggregates in first phase is due to smaller surface area of coarse aggregates and excess bitumen. The brief mixing time required for coating after adding the remaining bitumen could be because the aggregates at this stage already possessed a bitumen coating from the previous mixing step. Typical asphalt mixing time in a batch type plant is minimum 30 s for wet mixing and minimum 4 s of dry mixing preceding the addition of bitumen [44]. As only coarse aggregates are being mixed in first phase of the two-phase mixing, a separate dry mixing period may not be needed in the plant, except for the time required for the opening of weigh box gate and start of application of bitumen. Therefore, it is envisaged that the proposed mixing method would not affect the production efficiency of HMA plant.

2.2.2. Aggregate-bitumen interfacial bonding

To evaluate the effect of moisture on aggregate-bitumen adhesion, bond strength test has been performed on specimens prepared as per AASHTO T361. Metallic stubs with cavity thickness 0.8 mm were used. The test was performed at 25 °C for unconditioned and moisture conditioned specimens of all the four aggregates with the VG30 binder. Moisture conditioning was performed at 40 °C (recommended in AASHTO T361) and 60 °C [38] to study damage evolution and aggressive field conditions. Another reason for selecting 60 °C was that the conventional conditioning in TSR test (ASTM D4867) is also conducted at the same temperature. The conditioning time was taken as 24 h. After conditioning the specimens for 24 h, they were placed in water at the test temperature for 1 h before testing. Specimen preparation and the test setup are shown in Fig. 3.

2.2.3. Moisture damage evaluation of mixes

In this study, moisture sensitivity of bituminous mixes has been studied using TSR and dynamic modulus test at 1, 3 and 6 FT cycles. TSR test was conducted as per AASHTO T283 on specimens compacted to $7 \pm 0.5\%$ air void. TSR is basically the ratio of the indirect tensile strength with and without moisture conditioning. Deviation from the standard protocol was that the dry subset specimens were not kept in water bath. Instead, the specimens were placed in environmental chamber without any wrapping. Four specimens were tested each for unconditioned and moisture conditioned cases. TSR values after 1, 3 and 6 cycles of conditioning were obtained at 25 °C using Eq. 1 and Eq. 2.

$$ITS_i = \frac{2000P_{maxi}}{\pi td} \quad (1)$$

where, d : diameter of the specimen in mm, t : thickness of the specimen in mm, P_{maxi} : peak load after i^{th} freeze-thaw cycle, N , ITS_i : indirect tensile strength of sample after i^{th} freeze-thaw cycle, kPa

$$TSR_i = \frac{(\text{Indirect Tensile strength of conditioned specimen})_i}{(\text{Indirect Tensile strength of unconditioned specimen})_i} \quad (2)$$

where, the subscript i indicates freeze-thaw cycle number.

Specimens for dynamic modulus test were prepared as per AASHTO R83. The gyratory compacted samples had 150 mm diameter and at least 160 mm height, such that after coring and face cutting, specimens of 100 mm diameter, 150 ± 2.5 mm height and $7 \pm 0.5\%$ air void was obtained. Dynamic modulus is a characteristic property of asphalt mix which is used to define its linear viscoelastic behaviour and used as input in mechanistic empirical pavement design [30,45]. It is analytically described as the ratio of the peak amplitudes of cyclic stress and strain and the time lag between the cyclic stress-strain input-output is referred as the phase angle [46]. Therefore, the viscoelastic response of asphalt mix depends on the loading frequency as well as the test temperature. The test was conducted as per AASHTO T342 at 25, 10, 5, 1, 0.5, 0.1 Hz frequencies at each temperature – 10 °C, 4 °C, 21 °C, 37 °C, 45 °C.

Mastercurves of dynamic modulus and phase angles were developed at a reference temperature to be able to compare the effect of moisture conditioning at different temperatures and different frequencies in a single plot. For developing mastercurve, time-temperature superpositioning theory of thermo-rheological simple material was used. In this study, the mastercurves were generated at 21 °C reference temperature for reduced frequency f_R , computed using Eq. 3 to Eq. 6.

$$f_R = f \cdot a_T(T) \quad (3)$$

where, f is the test frequency

$$\log(a_T(T)) = a_1 \left(T^2 - T_{ref}^2 \right) + a_2 (T - T_{ref}) \quad (4)$$

where, T is test temperature, T_{ref} is reference temperature, 21 °C in this study, $a_T(T)$ is shift factor and a_1 and a_2 are shift factor fitting coefficients.

$$\log(|E^*|) = b_1 + \frac{b_2}{1 + e^{(-b_3 - b_4 \log(f_R))}} \quad (5)$$

where b_1 , b_2 , b_3 and b_4 are sigmoid coefficients for dynamic modulus.

$$\varphi = \frac{lab^2}{(f_R - c)^2 + b^2} \quad (6)$$

where a , b and c are phase angle coefficients.

Researchers have demonstrated that dynamic modulus test can be used to predict moisture susceptibility of asphalt mixes [28,30,32,33].



Fig. 3. Bond strength test specimen and test setup.

In this paper, after testing dynamic modulus in dry condition, the same specimens were subjected to AASHTO T283 moisture conditioning process for 1, 3 and 6 FT cycles. Samples were allowed to dry after each FT cycle before conducting the test. Deviation from the standard protocol was that the dry subset specimens were not kept in water bath. Instead, the specimens were placed in environmental chamber without any wrapping. The ratio of damaged (moisture conditioned) and undamaged dynamic modulus is reported as dynamic modulus ratio (DMR) as shown in Eq. 7.

$$DMR(f_R)_i = \frac{|E^*(f_R)|_i}{|E^*(f_R)|_o} \quad (7)$$

where, $i = 1, 3, 6$ is FT cycle number, $DMR(f_R)_i$ is dynamic modulus ratio at reduce frequency f_R and FT cycle i . $|E^*(f_R)|_o$ is undamaged dynamic modulus and $|E^*(f_R)|_i$ is the damaged dynamic modulus at reduced frequency f_R and i^{th} FT cycle.

3. Results and discussion

3.1. Mix design

The one-phase mixes were designed using Marshall mix method using conventional mixing as per the Asphalt Institute [23] procedure. The design binder content was selected at 4% air void. The design binder content obtained in the one-phase mixes was used to produce corresponding two-phase mixes.

The following steps were followed in deciding the percentage of bitumen to be added in the first phase of two-phase mixing:

- (i) In step-1, the mixes were prepared with 20% of the total bitumen content in the first phase mixing followed by adding the rest 80% in the second phase mixing. Image processing tool in Matlab was used to perform 2D image analysis to determine percentage of uncoated aggregate surface after the first phase mixing. After the first phase mixing, 4032×3024 pixel image of the coated coarse aggregates were taken using the system shown in Fig. 4. Table 5 shows sieve size wise percent of aggregates uncoated after the first phase mixing. It can also be seen from Fig. 5 that in all the mixes the coarse aggregates were found to have more than 10% uncoated surface at the end of the first phase mixing. In general, it was observed that the percentage of uncoated aggregate surface was more in larger size aggregates.
- (ii) Next in step-2, the mixes were prepared with 50% of the total bitumen content in the first phase mixing followed by adding the rest 50% in the second phase mixing. It was observed that in all the mixes the coarse aggregates were found to have 100% coating at the end of the first phase of mixing. As 50% of the total bitumen content in the first phase mixing produced 100% coarse aggregate coating, it was used to produce CC mixes.

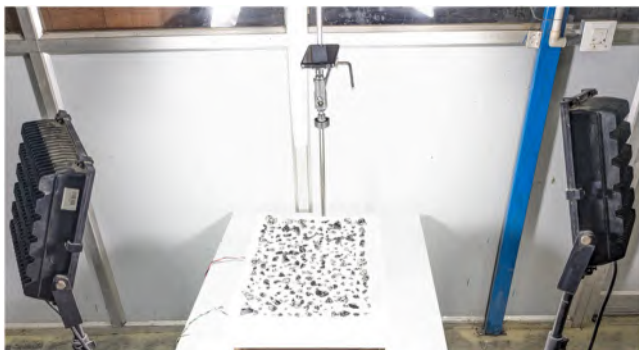


Fig. 4. System to take 2D images of coated aggregates.

Table 5

Percent uncoated coarse aggregate surface at the end of first phase mixing.

Aggregate Size (mm)	Mix-1CC20	Mix-2CC20	Mix-3CC20	Mix-4CC20
4.75-9.5	9.6%	11.5%	11.4%	9.9%
9.5-12.5	8.6%	11.7%	12.1%	8.6%
12.5-19	10.6%	11.2%	11.6%	9.7%
19-25	12.6%	17.0%	12.7%	11.4%

- (iii) Coating of coarse aggregate was also achieved with 40% of the total bitumen, but the time required for mixing increased for bitumen to coat the texture of larger size aggregates. Also, the time required to homogeneously mix fines with the precoated aggregates increased.

The Marshall test results of all the mixes at design binder content are shown in Table 6. All the results shown in the table are average of four verification test replicates. It can be seen from the table that the two-phase mixes had air void within $\pm 0.2\%$ of the corresponding one-phase mixes. The design binder content of Mix-4 is significantly lower compare to the other mixes. This was expected as Mix-4 was significantly coarser compare to the other mixes with 20% aggregate retained on the 19 mm sieve. It was observed that the binder absorption in two-phase mixes was 16% to 36% higher compare to the one-phase mixes. The higher binder absorption indicates a better coating of bitumen and stronger bond strength in the mixes. It can be seen that Marshall quotient (MQ) (ratio of stability and flow) was higher in two-phase mixes, which indicates that the two-phase mixing produces a stiffer mix as compared to the conventional mix. Higher MQ was due to around 5–15% increase in Marshall stability when the one-phase mixing processes was replaced by two-phase mixing. Further, it can be seen from the results that two-phase mixes had minor reduction in VFB% and insignificant change in VMA%.

The Marshall mixes were verified for Superpave mix design criteria at N_{design} and N_{max} gyratory compactive effort of 100 and 160 gyrations, respectively, corresponding to a traffic level of 3 to < 30 mESAL [23]. The mixes were within $4 \pm 0.5\%$ air void content at N_{design} and greater than 2% air void at N_{max} .

3.2. Bond strength test

Bond strength test was conducted to understand moisture damage and its mechanism in the aggregate-bitumen system. It can be seen from Fig. 6 that, for all the aggregates the pull-off tensile strength (POTS) was highest in unconditioned specimens followed by moisture conditioned specimens at 40°C and 60°C . After conditioning at 40°C , the POTS decreased by 16.9%, 35.5%, 23.6% and 48.8% for aggregate A1, A2, A3 and A4, respectively. Conditioning at 60°C resulted in even more reduction of POTS by 48.2%, 70.8%, 67.9%, 81.8% for aggregates A1, A2, A3 and A4, respectively. The reduction in strength is expected as change in failure mode sets in due the water penetrating the aggregate pores and weakening the interfacial bond [35]. The failure mechanism was defined as adhesive when more than 50% of the bitumen de-bonded from the aggregate surface. In case of unconditioned specimens, the failure was found to be cohesive. With moisture conditioning the failure mechanism was found to be drifting from cohesive to adhesive. The mode of failure for each aggregate can be seen in Fig. 7. Moraes et al. (2007) found similar observation for limestone, granite, and diabase aggregates. In aggregates with similar chemical properties, water absorption is a critical factor affecting aggregate-bitumen bond strength [38]. Failure in adhesion could have been due to the compounded effect of porosity and composition of the aggregates. For basalt (A1), the decrease in POTS was found to be lower compared to the granitic aggregates (A2, A3 and A4). This could be primarily due to the presence of higher silica content in the granitic aggregates which imparts acidic nature, resulting in more moisture susceptibility [1,47,48]. Also, based

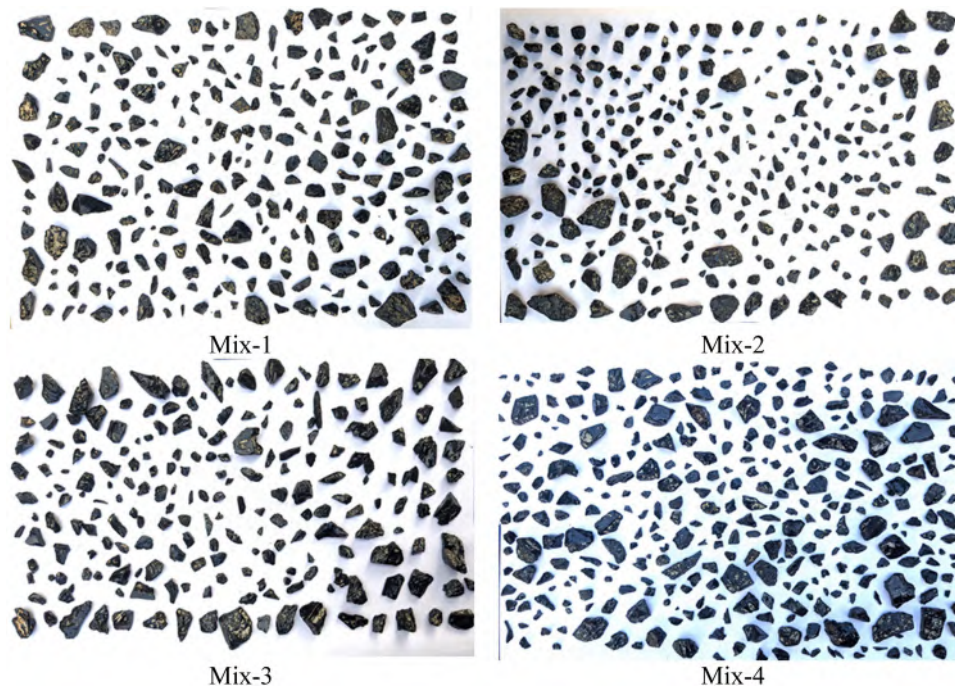


Fig. 5. Coarse aggregates at the end of first phase mixing using 20% of the total bitumen content.

Table 6
Marshall mix design results of one-phase (C) and two-phase (CC) mixes.

Parameters	Mix-1		Mix-2		Mix-3		Mix-4	
	C	CC	C	CC	C	CC	C	CC
Design bitumen content (%)	5.6	5.6	5.7	5.7	5.5	5.5	4.7	4.7
Max. theoretical specific gravity, G_{mm}	2.554	2.558	2.490	2.497	2.480	2.492	2.496	2.503
Air void (%)	4.1	4.2	4.1	4.3	3.9	4.0	4.1	4.3
VMA (%)	15.9	15.8	16.0	16.0	15.6	15.3	14.1	14.0
VFB (%)	74.0	73.5	74.5	73.2	74.7	73.6	70.8	69.0
Marshall Stability (kN)	12.9	13.5	14.4	15.5	12.3	14.2	12.2	14.0
Flow Value (mm)	3.30	3.22	3.27	3.37	3.33	3.38	3.14	3.34
Marshall Quotient (kN/mm)	3.90	4.21	4.41	4.61	3.70	4.21	3.89	4.20
Bitumen absorption (%)	0.48	0.55	0.70	0.81	0.59	0.80	0.52	0.64

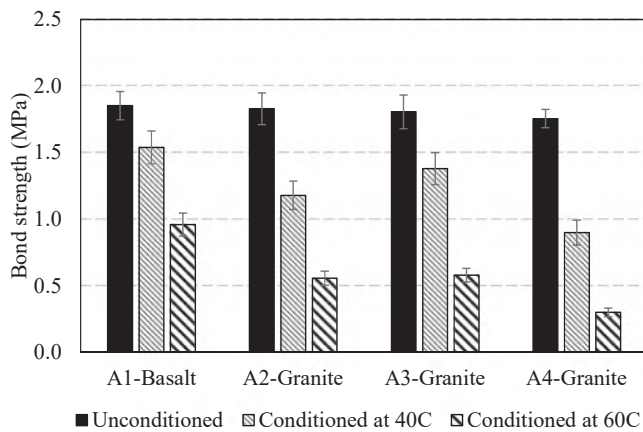


Fig. 6. Effect of moisture on aggregate-bitumen bond strength of different aggregate type.

on the ratio of chemical composition of Fe/Si and Ca/K, the basalt aggregate was less moisture susceptible compare to the granitic aggregates [48]. Among the three granitic aggregate (A2, A3 and A4), A4 was found to have greatest silica content and highest decrease in POTS after

conditioning. Although, A2 and A3 had similar silica content, the water absorption in A3 was relatively smaller compared to A2, which may have resulted in its better bond resistance.

3.3. Indirect tension test

From the indirect tension test data, the one-phase and two-phase mixes were compared for unconditioned ITS, conditioned ITS, TSR, toughness indices (TI), strain energy at failure (u_f) and fatigue index (f_i). The TSR results at different FT cycles were used to assess strength and TI, u_f and f_i were used to assess cracking resistance of the mixes before and after the moisture conditioning. The results were used to investigate effect of two-phase mixing on moisture susceptibility of the mixes.

3.3.1. Two-phase mixes with 20% bitumen in first phase

As shown in Fig. 8(a), TSR values at all the FT cycles were higher in the two-phase CC20 mixes compared to the corresponding one-phase mixes. The two-phase mixes had 4.0%–8.0% improvement in TSR over the one-phase mixes after FT cycle-1. This difference further increases with increase in number of FT cycles. The two-phase mixes had 2.2%–15.8% and 4.4%–26.9% improvement in TSR over the one-phase mixes after FT cycle-3 and FT cycle-6, respectively. This shows that although two-phase CC20 mixes could produce some improvement over one-phase mixes, but were significantly inferior compared to CC

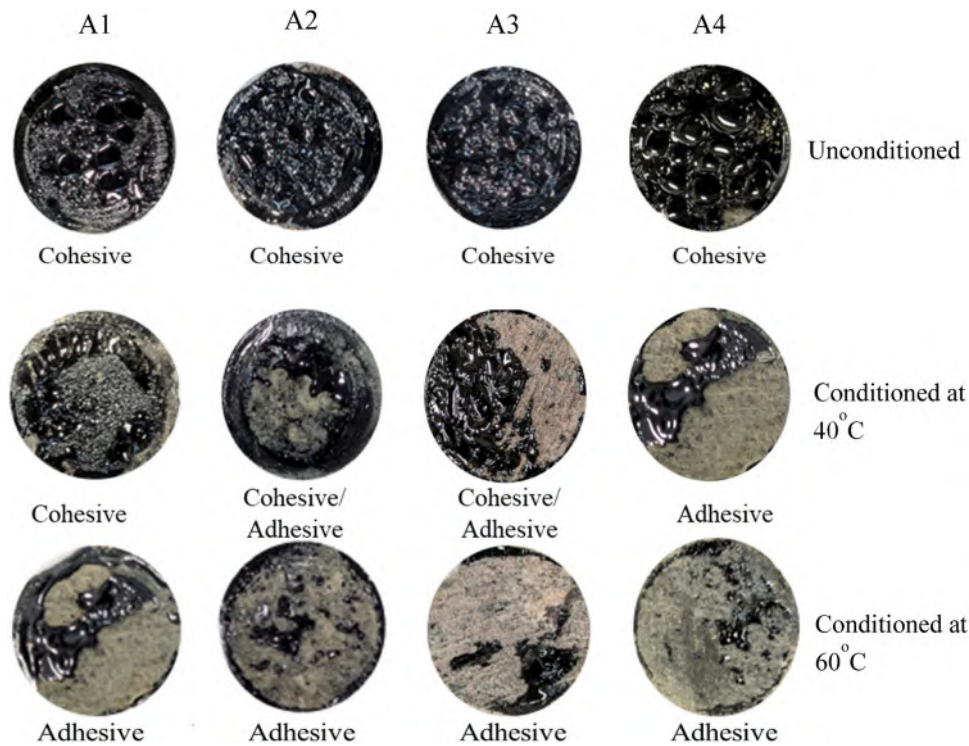


Fig. 7. Effect of moisture on aggregate-bitumen interface failure mode.

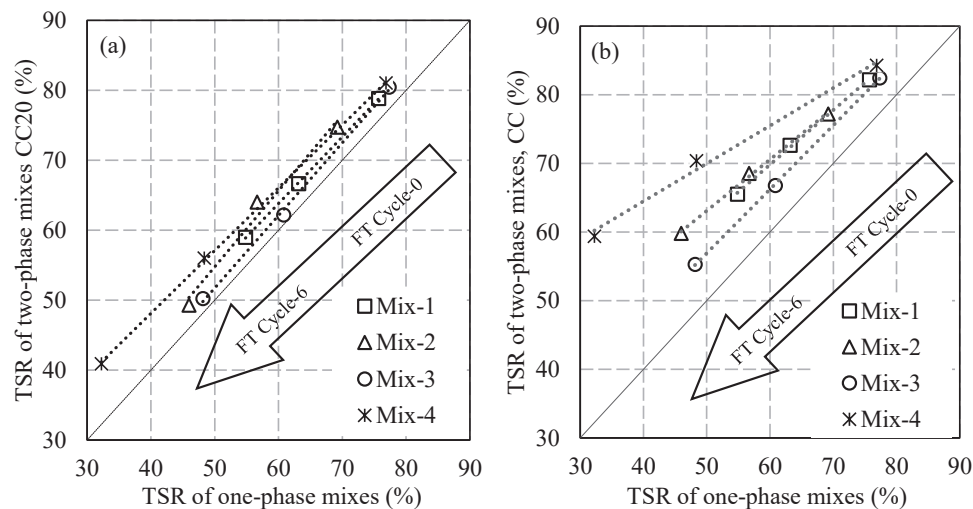


Fig. 8. TSR (%) after freeze-thaw cycle-1, 3 and 6 for two-phase mixes with (a) 20% bitumen and (b) 50% bitumen in the first phase.

mixes especially after freeze-thaw cycle-1. This can also be noticed from the TSR trend with freeze-thaw cycles (refer Fig. 8(a)) in which CC20 mixes showed nearly same inclination as one-phase mixes. It can be seen from Fig. 9 that both the conditioned and unconditioned average ITS values were higher in two-phase CC20 mixes compared to one-phase, but the difference was statistically insignificant in almost half of the results. The two-phase mixes had 0.5%–4.2%, 5.9%–8.5%, 4.0%–18.7% and 6.2%–30.1% increase in ITS values over the one-phase mixes after FT cycle-0, 1, 3 and 6, respectively.

3.3.2. Two-phase mixes with 50% bitumen in first phase

As shown in Fig. 8(b), TSR values at all the FT cycles were higher in the two-phase CC mixes compared to the corresponding one-phase mixes. The two-phase mixes had 6.6%–11.6% improvement in TSR

over the one-phase mixes after FT cycle-1. This difference further increases with increase in number of FT cycles. The two-phase mixes had 9.8%–45.5% and 14.9%–84.2% improvement in TSR over the one-phase mixes after FT cycle-3 and FT cycle-6, respectively. This shows that two-phase mixing produces a significantly more moisture resistant mix even in aggressive field conditions. The ITS results of one-phase and two-phase mixes are shown in Fig. 9. It can be seen from the figures that both the conditioned and unconditioned ITS values were higher in two-phase CC mixes. The two-phase mixes had 2.9%–8.8%, 11.2%–18.2%, 14.5%–51.4% and 19.8%–91.6% increase in ITS values over the one-phase mixes after FT cycle-0, 1, 3 and 6, respectively. This improvement in both unconditioned and conditioned specimens is due to better binder-aggregate interaction resulting in higher binder absorption and proper coating over the aggregate surface in two-phase mixing. Since, in

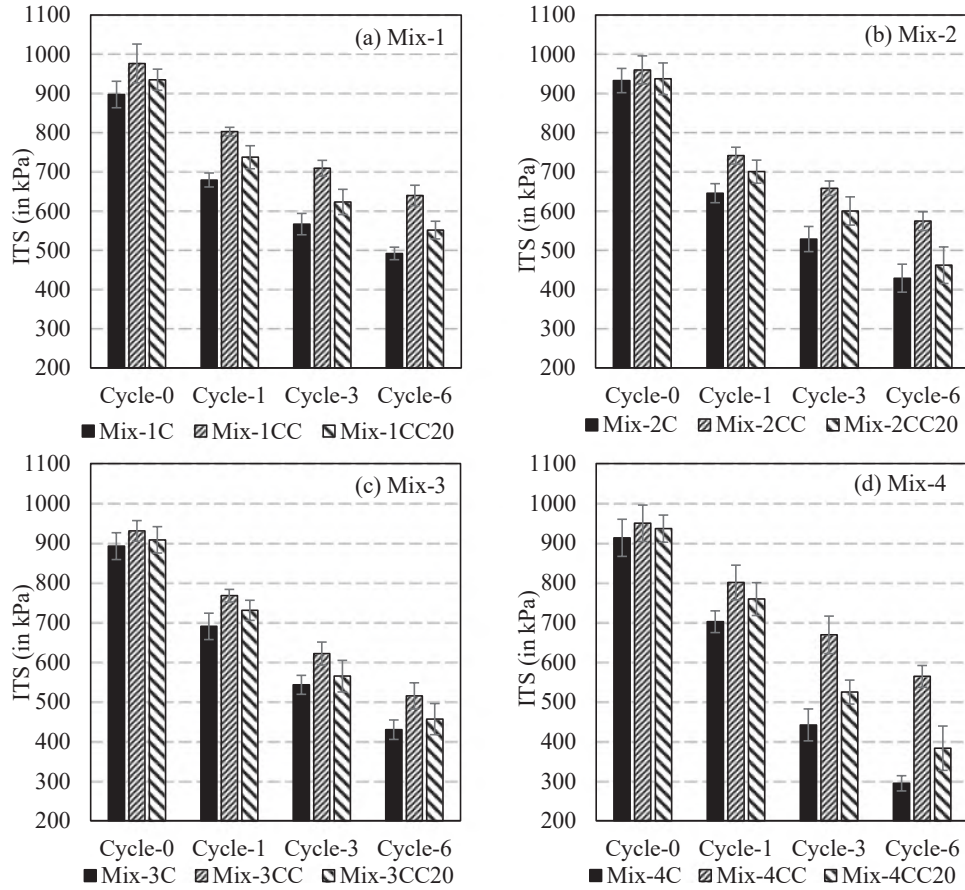


Fig. 9. Comparison of ITS of one-phase and two-phase mixes at different freeze-thaw cycles.

all the cases two-phase CC20 mixes showed inferior moisture damage resistance compared to CC mixes, hereafter they were discarded from further analysis.

The stress values from the ITS test were normalized by the peak indirect tensile strength to generate normalized stress-strain curve. A typical normalized stress-strain curve is shown in Fig. 10. It can be seen from the figure that, if the failure is perfectly plastic then the value of TI computed using Eq. 8 would be constant one and if the failure is ideally brittle then zero. Thus, the parameter estimates the flexibility of the mixes.

$$TI = \frac{A_e - A_p}{\epsilon - \epsilon_p} \quad (8)$$

where, A_e is the complete area forming under the normalized stress-strain curve till terminal strain (ϵ) and A_p is area under the normalized curve till the strain corresponding to peak stress (ϵ_p).

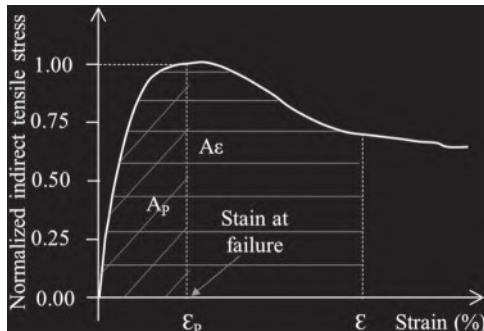


Fig. 10. Typical normalized stress-strain curve.

The TI values were calculated using Eq. 8 at a range of 4 to 8% terminal strain (ϵ). TI values for one-phase and two-phase mixes at FT cycle 0, 1, 3 and 6 are shown in Fig. 11 and Fig. 12, respectively. It can be seen from the figures that, for all the mixes TI decreased with increase in terminal strain and FT cycles. This was expected because, as the mixes were subjected to FT cycles it induced moisture damage, which resulted in deterioration in cracking resistance. However, in general the TI values were higher and the rate of reduction in TI was lower in two-phase mixes compared to one-phase mixes. As the moisture induced damage in the two-phase mixes were lower, it resulted in better cracking resistance compared to the one-phase mixes. Although TI is useful information, it struggles to sufficiently differentiate cracking resistance of the mixes. Barman et al. proposed fatigue index (f_i) as a more appropriate parameter, which can be used to better discriminate the cracking resistance of mixes [49]. The f_i values were calculated using Eq. 9. Strain energy at failure and fatigue index of all the mixes are shown in Fig. 13 and Fig. 14, respectively. From the figures it can be seen that u_f and f_i values of two-phase mixes were significantly higher compared to one-phase mixes.

$$f_i = -\frac{u_f}{\left(\frac{dT_I}{d\epsilon}\right)} \quad (9)$$

where, u_f is the strain energy at the failure, $\frac{dT_I}{d\epsilon}$ is the slope of TI vs ϵ curve.

3.4. Dynamic modulus

Damaged dynamic modulus and phase angle mastercurves have been developed after FT cycles 1, 3 and 6. Three replicates were tested for

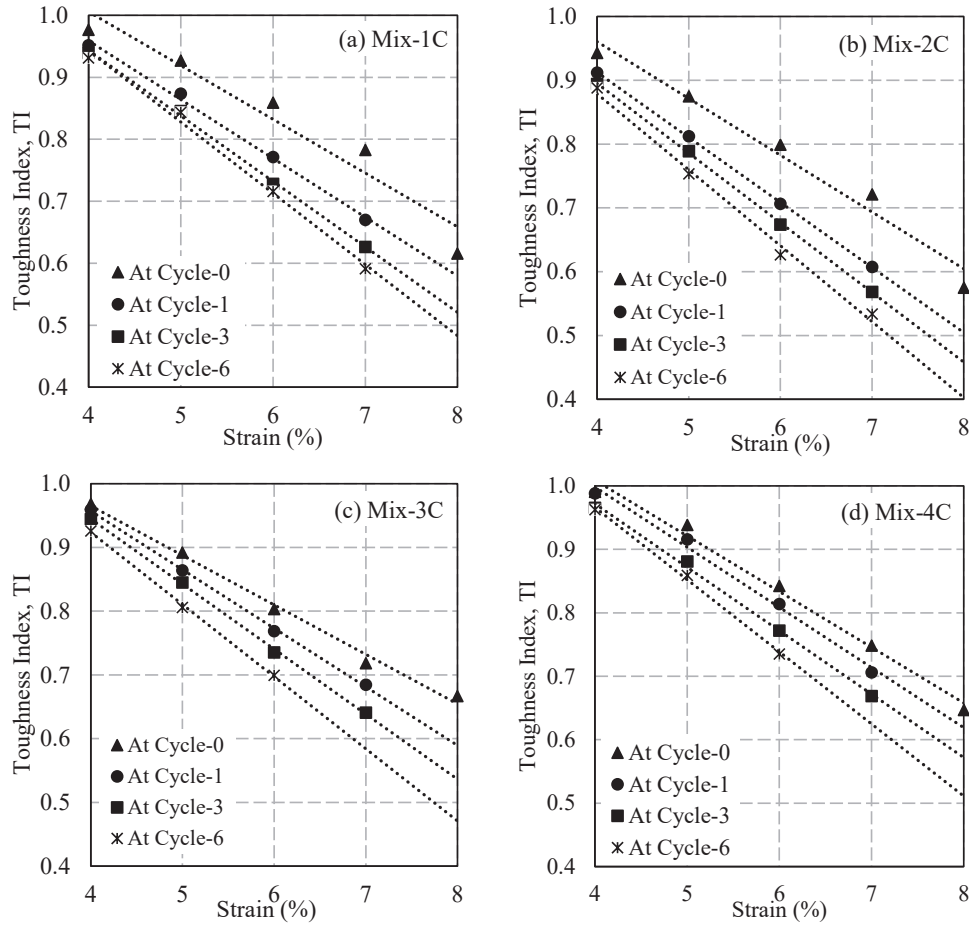


Fig. 11. TI at different terminal strains and freeze-thaw cycles for one-phase mixes: (a) Mix-1 C, (b) Mix-2 C, (c) Mix-3 C, (d) Mix-4 C.

each mix and each FT cycle. As dynamic modulus test is a non-destructive test, same specimens have been used to provide different conditioning cycles. The damaged mastercurves have been compared with undamaged mastercurve to understand evolution of damage. The fitted mastercurves at 21 °C reference temperature for one-phase and two-phase mixes of Mix-1, Mix-2, Mix-3 and Mix-4 are shown in Fig. 15 to Fig. 18, respectively. It can be seen from the figures that the dynamic modulus monotonically decreases with increase in FT cycles. Same trend has been also reported for one-phase hot mix asphalts by Corrales-Azofeifa et al. (2020) and Tan et al. (2020).

DMR, the ratio of damaged dynamic modulus, (after FT cycle(s)) and undamaged dynamic modulus was calculated at different reduced frequencies. Vargaas-Norbeck et al. (2016) and Tan et al. (2020) observed that, in general the reduction in dynamic modulus is greatest at high temperature and low frequency. Tan et al. (2020) studied SBS and basalt fiber mixes at FT cycles 1 to 21 at an interval of 3 cycles. In the first 6 cycles, reduction in dynamic modulus was relatively insignificant between reduce frequency 10^{-3} to 10^3 . At higher FT cycles, reduction in modulus was significant at higher temperatures. It was found that reduction in modulus significantly decreased and then slightly increased with increase in reduced frequency. Lee et al. (2017) measured dynamic modulus before and after moisture conditioning at 25, 10, 5, 1, 0.5, and 0.1 hz frequencies at temperature -10 °C, 5 °C, 20 °C, 40 °C, 54.4 °C. DMR values were found decreasing and then increasing with increase in temperature. All the DMR values calculated from the measured data (at each cycle, each temperature -10 , 4 , 21 , 37 , 45 °C and each frequency 25, 10, 5, 1, 0.5, 0.1 hz, i.e. 960 DMR values) was less than 1 except for: Mix-2CC DMR = 1.02 at cycle-1 45 °C and 0.1 hz, Mix-3 C DMR = 1.01 at cycle-1 4 °C 25hz, and Mix 4 C DMR = 1.02 at cycle-1 45 °C and 0.5hz.

As no specific trend was observed in DMR and temperature, an aggregate approach was considered in which average DMR over a range of frequencies was computed. In the mastercurves, modulus at 1 hz reduced frequency represents measured value for material at reference temperature and 1 hz loading frequency. In the mastercurve, modulus values at higher reduced frequencies are in fact measured values at test temperatures lower than the reference temperature. Similarly, modulus values at lower reduced frequencies are measured values at test temperatures higher than the reference temperature. Three parameters shown in Eq. 10 to Eq. 12 were proposed to compute average DMR.

$$DMR_{pre}^i = \frac{1}{n} \sum_{f_R=10^{-4}}^{10^0} \frac{|E^*(f_R)|^i}{|E^*(f_R)|^0} \quad (10)$$

where, $|E^*(f_R)|^i$ is the dynamic modulus after FT cycle i and reduce frequency f_R , $|E^*(f_R)|^0$ is the undamaged dynamic modulus at reduce frequency f_R and n is the number of discretization of the frequency range.

$$DMR_{post}^i = \frac{1}{n} \sum_{f_R=10^0}^{10^4} \frac{|E^*(f_R)|^i}{|E^*(f_R)|^0} \quad (11)$$

$$DMR_{tot}^i = \frac{1}{n} \sum_{f_R=10^{-4}}^{10^4} \frac{|E^*(f_R)|^i}{|E^*(f_R)|^0} \quad (12)$$

DMR_{pre}^i is the average of summation of DMR at reduce frequencies between 10^0 to 10^{-4} hz. Thus, DMR_{pre}^i represents average damage in modulus after FT cycle i at temperatures higher than the reference

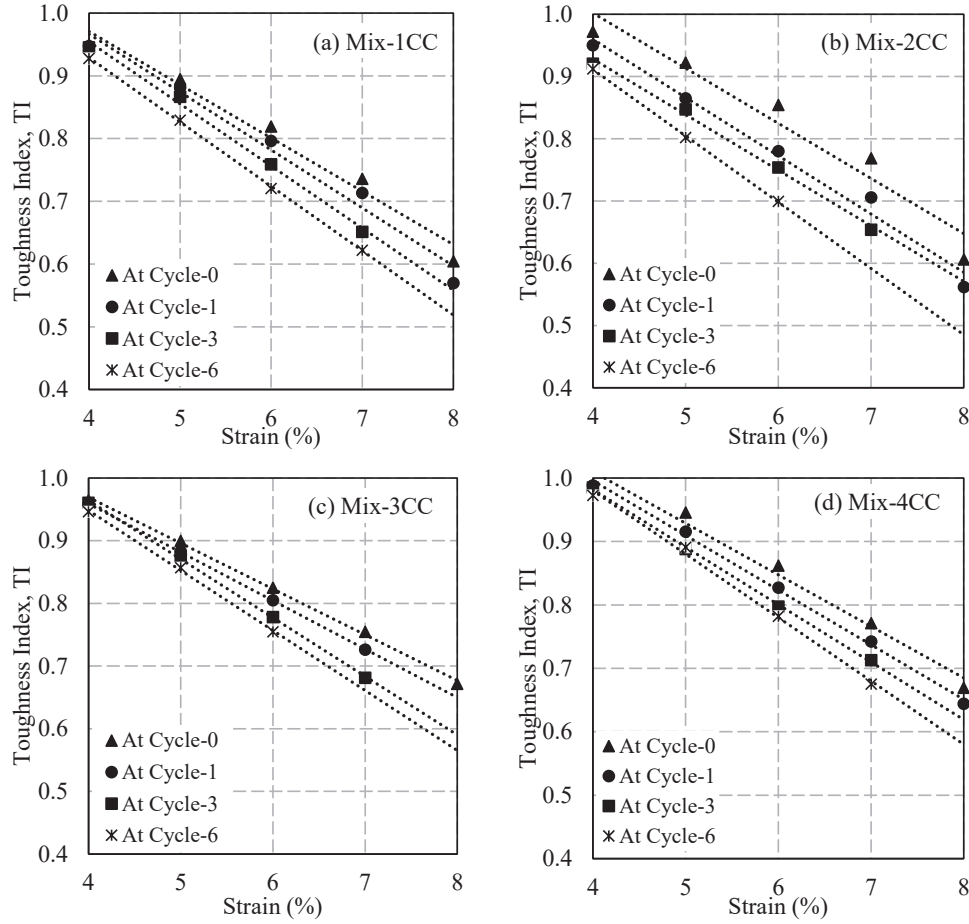


Fig. 12. TI at different terminal strains and freeze-thaw cycles for two-phase mixes: (a) Mix-1CC, (b) Mix-2CC, (c) Mix-3CC, (d) Mix-4CC.

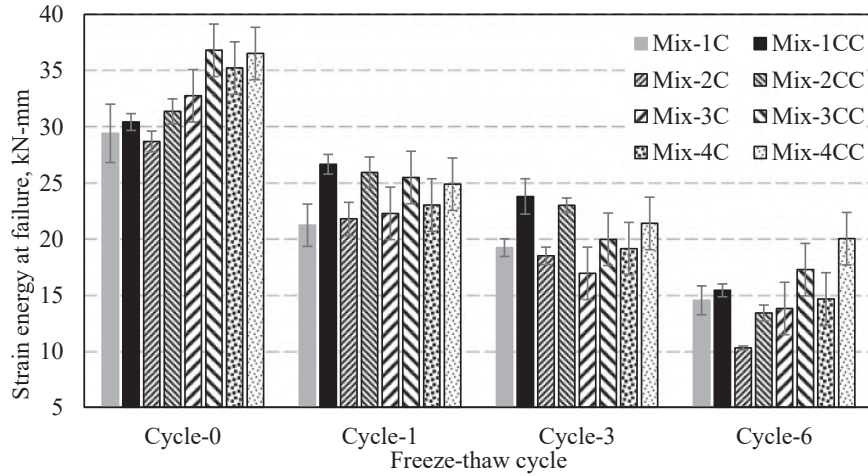


Fig. 13. Strain energy at failure at different freeze-thaw cycles.

temperature. DMR_{post}^i is the average of summation of DMR at reduce frequencies between 10^4 to 10^0 hz. Thus, DMR_{post}^i represents average damage in modulus after FT cycle i at temperatures lower than the reference temperature. DMR_{tot}^i represents the average damage in modulus after FT cycle i over the entire mastercurve. The reference frequency range of 10^{-4} to 10^4 hz was chosen for defining the parameters based on the test temperatures. Based on time-temperature super-positioning, -10°C to 45°C corresponded to reduced frequency

range of around 10^{-4} to 10^4 hz at 21°C . Similarly, for computing damage in phase angle, parameters PAR_{pre}^i , PAR_{post}^i and PAR_{tot}^i were defined. This is similar to DMR defined by Brondani et al. [50] in which the ratio is defined between the area under damaged to undamaged dynamic modulus mastercurves. The advantage of the proposed average DMR definition is that it does not require any integration of complex fitted functions. The DMR and PAR values for all the mixes and FT cycles are shown in Table 7. It can be seen from Fig. 15 to Fig. 18 that, the effect of moisture damage on dynamic modulus is higher at higher frequencies

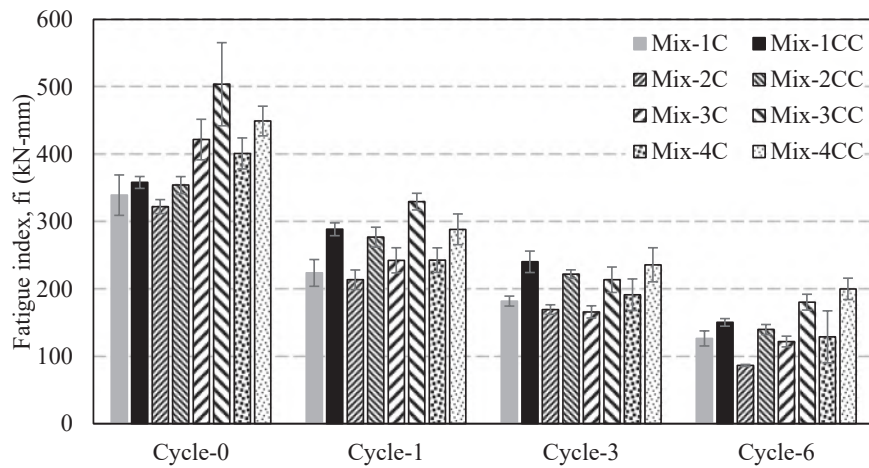


Fig. 14. Fatigue index at different freeze-thaw cycles.

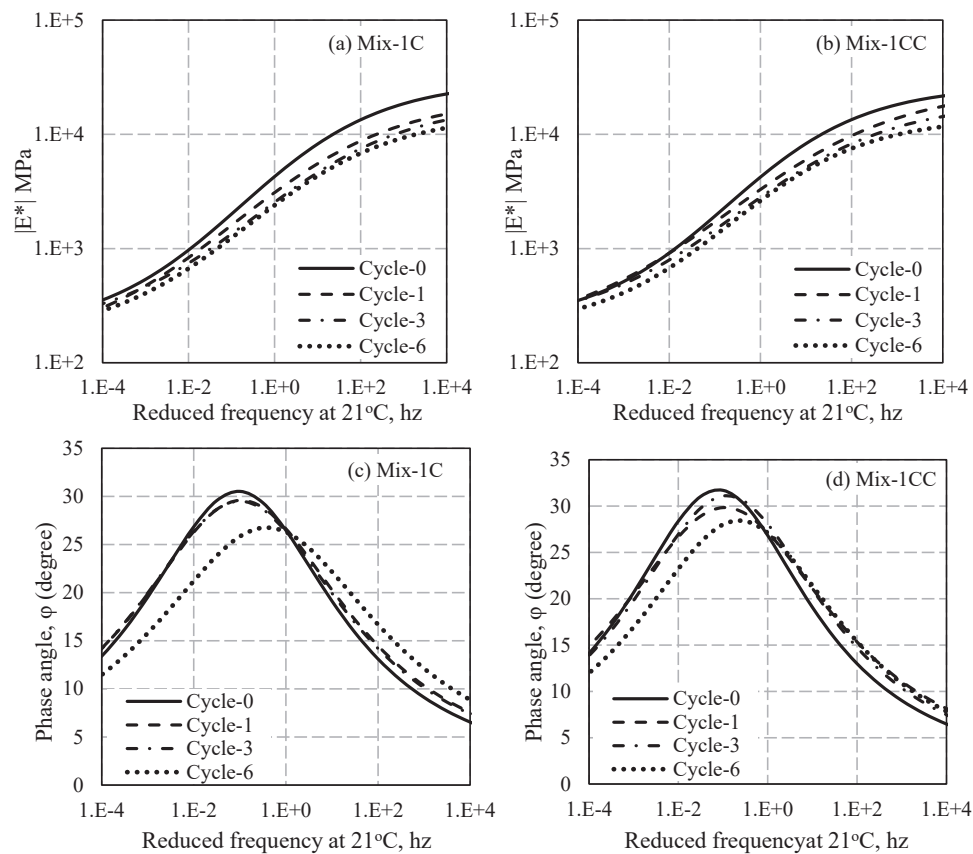


Fig. 15. Fitted mastercurves: (a) $|E^*|$ Mix-1 C, (b) $|E^*|$ Mix-1CC, (c) Phase angle Mix-1 C and (d) Phase angle Mix-1CC.

compared to lower frequencies. The same can be observed from Table 7, where the DMR_{post}^i for all the mixes are smaller than DMR_{pre}^i . Also, it can be seen from the figures that, drop in the dynamic modulus curves are smaller in two-phase mixes compare to one-phase mixes. The same can be observed from Table 7, where DMR_{pre}^i , DMR_{post}^i , and DMR_{tot}^i are higher in two-phase mixes compare to one-phase mixes at all the FT cycles. Thus, the DMR results show that two-phase mixing is more effective in resisting moisture damage compare to one-phase mixing. Fig. 15 to Fig. 18 indicate that the phase angle mastercurve translates towards higher frequencies with increase in FT cycles. Also, it can be seen that the peak phase angle drops with increase in FT cycles. During the conditioning process, specimen is moisture damaged through saturation and

freezing, however, it also gains some ageing from thawing at 60 °C. The results indicate that, at higher temperatures the effect of ageing dominates over the effect of damage, resulting in drop in phase angle, whereas, at lower temperatures, the effect of damage dominates over the effect of aging, resulting in higher phase angle. It can be seen from Table 7 that the average PAR_{post}^i and COV values for two-phase mixes were significantly lower compare to one-phase mixes. Similar results were reported by Vivar and Haddock for phase angle in conventional mixes [51]. In the study, dynamic modulus and phase angle were measured for unconditioned and conditioned samples at 20 °C and 40 °C for four asphalt mixes. At 20 °C, decrease in phase angle was reported with increase in frequency. At 40 °C, increase in phase angle was

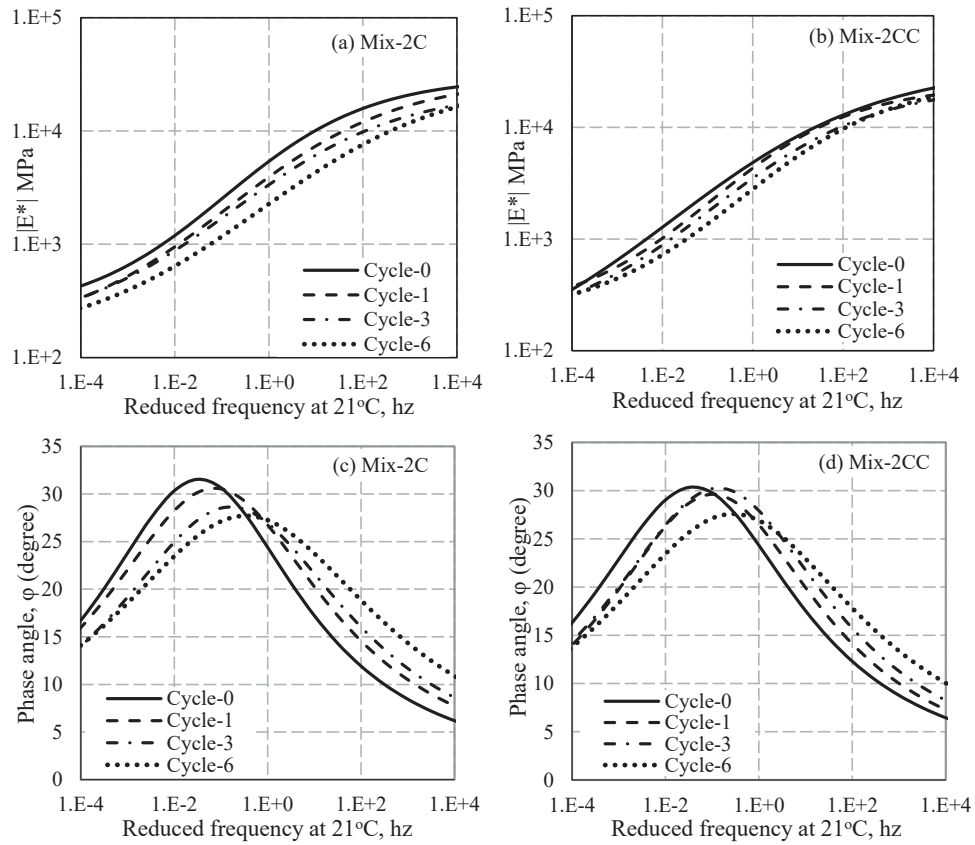


Fig. 16. Fitted mastercurves: (a) $|E^*|$ Mix-2 C, (b) $|E^*|$ Mix-2CC, (c) Phase angle Mix-2 C and (d) Phase angle Mix-2CC.

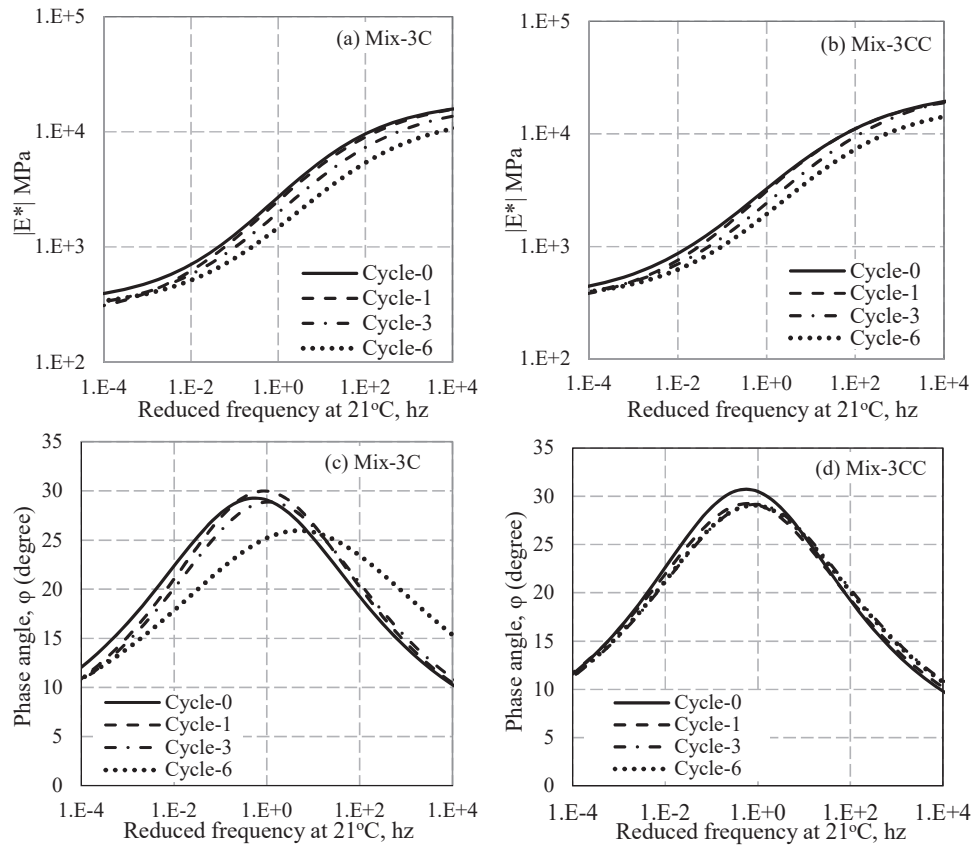


Fig. 17. Fitted mastercurves: (a) $|E^*|$ Mix-3 C, (b) $|E^*|$ Mix-3CC, (c) Phase angle Mix-3 C and (d) Phase angle Mix-3CC.

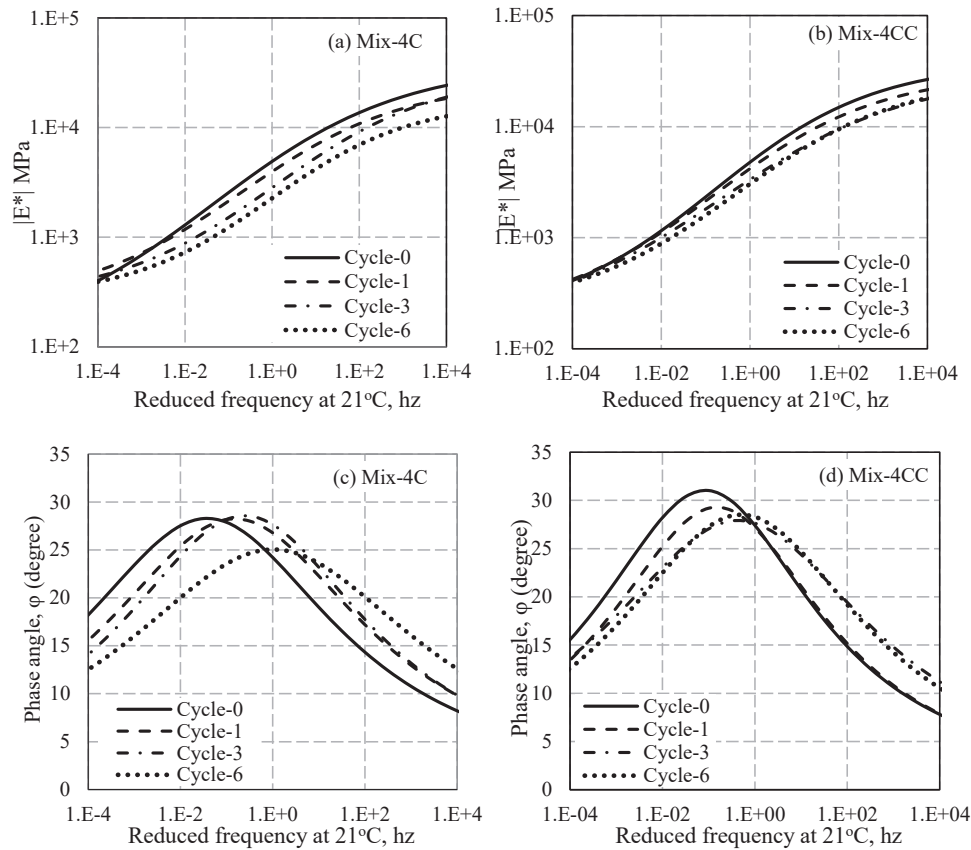


Fig. 18. Fitted mastercurves: (a) $|E^*|$ Mix-4 C, (b) $|E^*|$ Mix-4CC, (c) Phase angle Mix-4 C and (d) Phase angle Mix-4CC.

Table 7

DMR and PAR of one-phase and two-phase mixes.

Mix Type		Average of all one-phase mixes						Average of all two-phase mixes					
Reduce frequency, hz		Cycle-1		Cycle-3		Cycle-6		Cycle-1		Cycle-3		Cycle-6	
		DMR	PAR	DMR	PAR	DMR	PAR	DMR	PAR	DMR	PAR	DMR	PAR
10^4 to 10^0	Avg	0.80	1.12	0.67	1.16	0.62	1.25	0.87	1.07	0.73	1.16	0.65	1.20
	Stdev	0.11	0.08	0.09	0.12	0.13	0.19	0.10	0.09	0.11	0.12	0.07	0.16
10^{-4} to 10^0	Avg	0.86	0.99	0.76	0.93	0.69	0.89	0.92	0.96	0.82	0.95	0.72	0.90
	Stdev	0.08	0.04	0.04	0.05	0.12	0.04	0.04	0.05	0.05	0.06	0.06	0.04
10^{-4} to 10^4	Avg	0.83	1.06	0.72	1.04	0.66	1.07	0.90	1.01	0.78	1.06	0.69	1.05
	Stdev	0.08	0.03	0.06	0.05	0.12	0.09	0.03	0.06	0.04	0.05	0.03	0.06

reported with increase in frequency. In another study, Tan et al. (2020) found that, in general the phase angle increases with increasing FT cycles, which was attributed to the damage after moisture conditioning.

3.5. Comparison between $|E^*|$ and TSR for moisture susceptibility evaluation

Vargas-Nordbeck et al. (2016) measured DMR at all the temperatures and frequency as per AASHTO TP 79. They observed higher variability at certain temperatures and frequency and for comparison purpose results obtained at 21 °C and 10 hz was used. Nadkarni et al. (2009) determined DMR at 25 °C at 25, 10, 5, 1, 0.5, 0.1 hz for one FT cycle. They found that DMR results at each frequency for 25 °C were statistically insignificant and DMR and TSR were statistically same. Lee et al. (2017) observed that DMR results are similar to TSR results at 5 °C and 20 °C for 13 mm dense asphalt mix and 20 °C and 40 °C for 19 mm dense asphalt mix. All the above observations in literature are from one-phase conventional mixing.

In the present study, it was observed that the effect of FT cycles on

reduction in dynamic modulus vary with temperature and frequency. To study this, difference between the minimum and maximum DMR values were computed at each temperature and FT cycle using Eq. 13.

$$DMR_{mm} = \max[DMR \text{ at } f = 25, 10, 5, 1, 0.5, 0.1 \text{ hz}] - \min[DMR \text{ at } f = 25, 10, 5, 1, 0.5, 0.1 \text{ hz}] \quad (13)$$

where, f is the dynamic modulus test frequency. The range of DMR_{mm} values from all the test results at – 10 °C, 4 °C, 21 °C, 37 °C, 45 °C after one cycle of FT were 0.05–0.15, 0.03–0.12, 0.02–0.06, 0.05–0.15 and 0.07–0.38, respectively. It was observed that the range of DMR_{mm} at 21 °C was lowest among all the temperatures and was best suited for moisture damage evaluation. As discussed earlier, similar results were observed in literature for one-phase mixes. This indicates that, in general more variation in DMR values was observed at low and high temperatures. The variation increased with increase in FT cycles. This indicates that at low and high temperatures, modulus was more sensitive to the test frequencies as the moisture damage increased. Fig. 19 shows

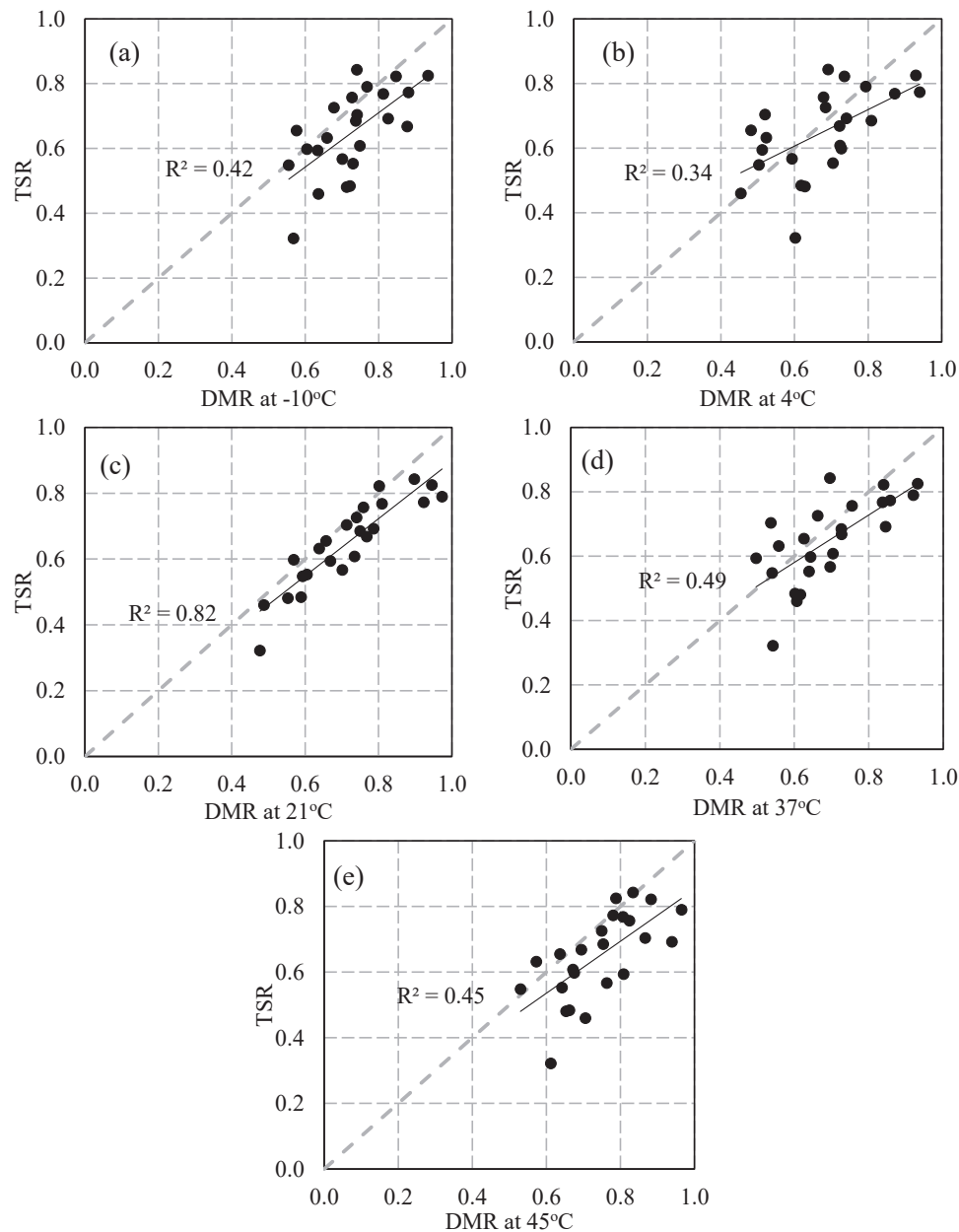


Fig. 19. Comparison of DMR and TSR: (a) DMR at -10°C , (b) DMR at 4°C , (c) DMR at 21°C , (d) DMR at 37°C , (e) DMR at 45°C .

comparison of DMR at different temperatures and TSR for all the mixes including one-phase and two-phase. For comparison, DMR at each temperature in the figure was averaged over all the test frequencies. It can be seen from the figure that DMR at 21°C exhibited the best R^2 with TSR. Further, it was observed that the moisture conditioning damage ITS more than the dynamic modulus at 21°C . In general, the difference between TSR and the DMR at 21°C reduced with increase in FT cycles except for Mix-4. Mix-4 was coarse mix with 20% aggregate retained on 19 mm sieve. Binder content in Mix-4 was lowest and the silica content was highest among all the mixes. The one-phase Mix-4 TSR after FT cycle 6 dropped to 32% indicating that the sample had significantly lost adhesion.

Based on the percentage passing at 4.75 mm sieve, except A2 rest all were coarser gradations. It can be seen from Table 1 that based on aggregate impact, crushing and LA abrasion values, A2 was the weakest aggregate among all. Comparing the aggregate strength, bond strength and TSR results it can be seen that in one-phase mixes, moisture resistance was found to be effected by these parameters. Moisture resistance

of mixes were more distinguishable after the cycle-1 moisture conditioning. In line with the aggregate-bitumen bond strength, resistance to damage over the moisture conditioning cycles was best in Mix-1 followed by Mix-3, Mix-2 and Mix-4. Better performance of Mix-3 compared to Mix-2 could be also due to the weaker and porous A2 aggregate compared to A3. Similar to one-phase mixes, in two-phase mixes also Mix-1 performed the best. Whereas, in general performance of granitic aggregates (A2, A3 and A4) were similar. This indicates that the aggregate type still had significant influence, but aggregate strength and gradation in the study had little influence on the moisture resistance of two-phase mixes. However, the distinct trend line for Mix-2, Mix-3 and Mix-4 in Fig. 8(b) shows that improvement in moisture damage from one-phase to two-phase was affected by aggregate gradation. Comparing the aggregate gradation and moisture resistance, it can be seen that gradations with higher coarse aggregate content (percentage retained on 19 mm) showed larger improvement. This collaborates with the earlier discussion that two-phase mixing with 50% bitumen was able to provide better coating for the coarse aggregates. Absolute improvement

(difference in TSR between CC and C mixes) in TSR after cycle-6 in Mix-1 to Mix-4 was 10.7%, 13.8%, 7.2% and 27.1%, respectively. Average improvement in the dynamic modulus values from one phase to two phases in Mix-1 to Mix-4 was 8%, 13%, 28% and 27%. Thus, the improvement was higher in the more moisture susceptible granitic aggregate mixes.

4. Summary and conclusion

The main objective of the study was to devise a mixing scheme which will produce better moisture damage resistant mixes. A two-phase mixing process is proposed in which 50% of the total design binder content is mixed with coarse aggregates in the first phase. After pre-coating the coarse aggregates, fines were mixed followed by mixing the remaining bitumen. Aggregates from four different sources were used to produce four two-phase and four one-phase conventional dense asphalt mixes. The following conclusions can be drawn from the results:

- (i) Increase in binder absorption was observed in two-phase mixes. This indicates that coating coarse aggregates with excess binder in the absence of filler and dust would result in better coating and higher binder absorption.
- (ii) Marshall stability, Marshall quotient and dynamic modulus of the mixes increased on using two-phase mixing. This was expected as reduction in free effective bitumen due to higher absorption will result in stiffer mix. However, better coating and absorption has resulted in better bonding and therefore higher stability and cracking resistance.
- (iii) ITS values both before and after conditioning at all the studied FT cycles were higher in two-phase mixing. The better moisture damage and cracking resistance in two-phase mixing could be because in two-phase mixing, bitumen appears to have better opportunity to fill the texture of aggregate surface, which provide additional mechanical interlocking between aggregate and bitumen.
- (iv) Strain energy and fatigue index were found to be very effective parameters in distinguishing moisture sensitivity of mixes for cracking potential. Cracking resistance was distinctly higher when two-phase mixing was used.
- (v) Loss in bond strength due to moisture was found to be affected by water absorption and chemical composition of aggregate. Aggregate-bitumen interfacial failure mode in granite and basalt aggregates were cohesive in unconditioned specimens. With moisture conditioning the failure mechanism was found to be drifting from cohesive to adhesive. Granitic aggregates had higher silica content and were more moisture susceptible as compared to basalt.
- (vi) Both TSR and DMR had similar trend, but the best correlation was observed for dynamic modulus measured at 21 °C. Also, variation in DMR with frequency was lowest at 21 °C. Generally, 21 °C is considered as typically annual average temperature. Therefore, it is recommended that dynamic modulus at 21 °C should be used to assess moisture sensitivity of dense asphalt mixes.
- (vii) Improvement in resistance to moisture damage from one phase to two phase mixing was found to be affected by parameters such as (i) percentage of coarse aggregates retained on 19 mm, (ii) aggregate type and (iii) aggregate-bitumen bond strength. Mixes with weaker aggregate-bitumen interfacial properties could produce significant improvement in TSR and damaged dynamic modulus in two phase mixing.

The test results show that the proposed two-phase mixing process can significantly improve moisture damage resistance of dense asphalt mixes. The mixing process can be implemented in batch type asphalt plants without needing any changes in the plant facility.

CRedit authorship contribution statement

Varma Sudhir: Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kumar Ajit:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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